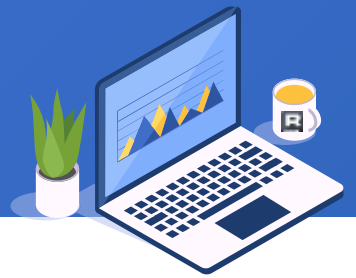
ReportLite
课程



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第1章
安装试用



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1.2 用向导制作行式报表

1.3 扩展与表达式

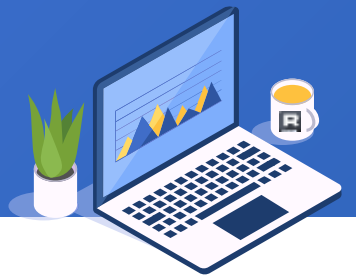
1.4 报表区域

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1.1 下载与安装

下载软件



下载软件

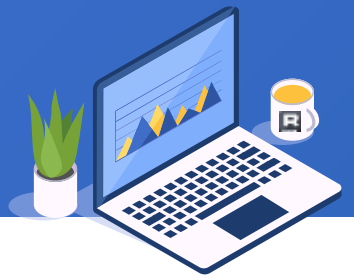
课程资料

安装

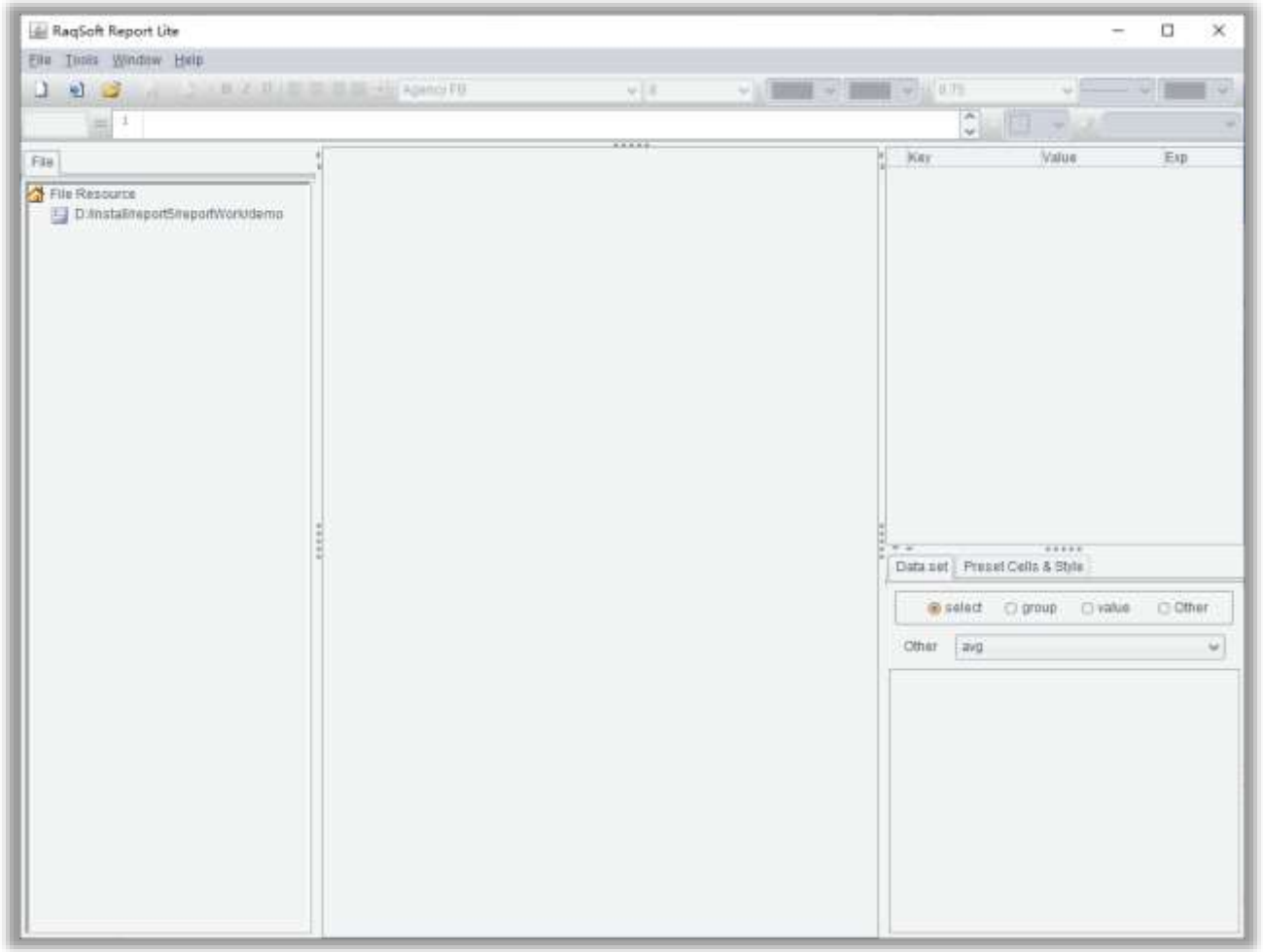


- 软件下载后，指定安装目录（可以默认）下一步，下一步就安装完成了
- 安装完成后，直接启动软件

软件界面



启动软件后，程序的初始界面

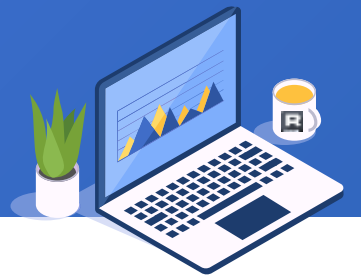


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1.2 用向导制作行式报表

1.2 用向导制作行式报表



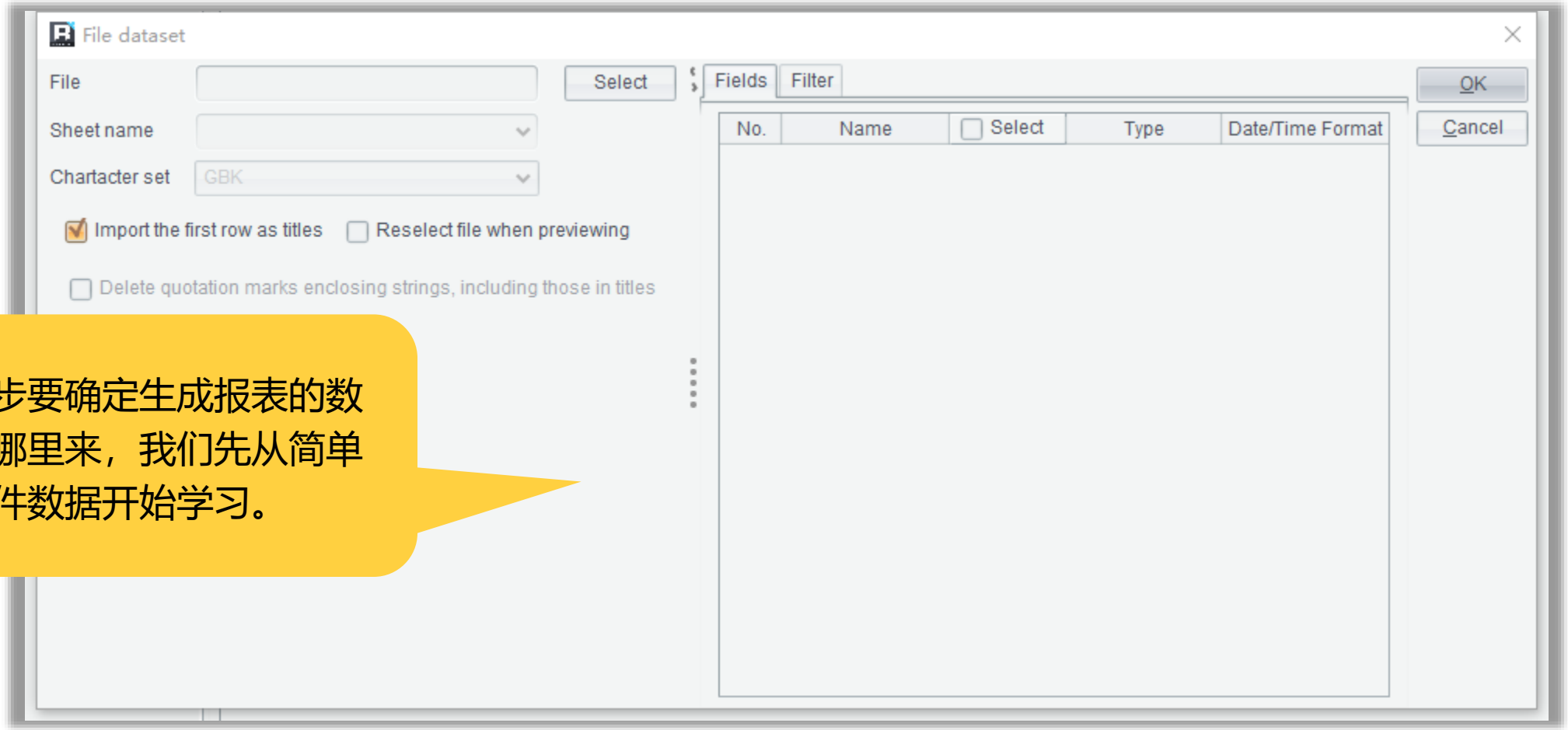
如图Excel格式的考生成绩表，看下如何用润乾报表生成简单的行式报表，

data1.2.xlsx如图：

	A	B	C	D
1	Name	Subject	Midterm	Final
2	Rose	Math	59	80
3	Rose	Physics	48	65
4	Mike	Math	87	77
5	Mike	Physics	70	86
6	Ronald	Math	36	50
7	Ronald	Physics	54	75
8	Smith	Math	25	55
9	Smith	Physics	56	75
10	Frank	Math	66	80
11	Frank	Physics	75	65
12	Aimay	Math	32	50
13	Aimay	Physics	42	65
14	James	Math	37	67
15	James	Physics	97	88
16	Jay	Math	56	56
17	Jay	Physics	88	79

1.2 用向导制作行式报表

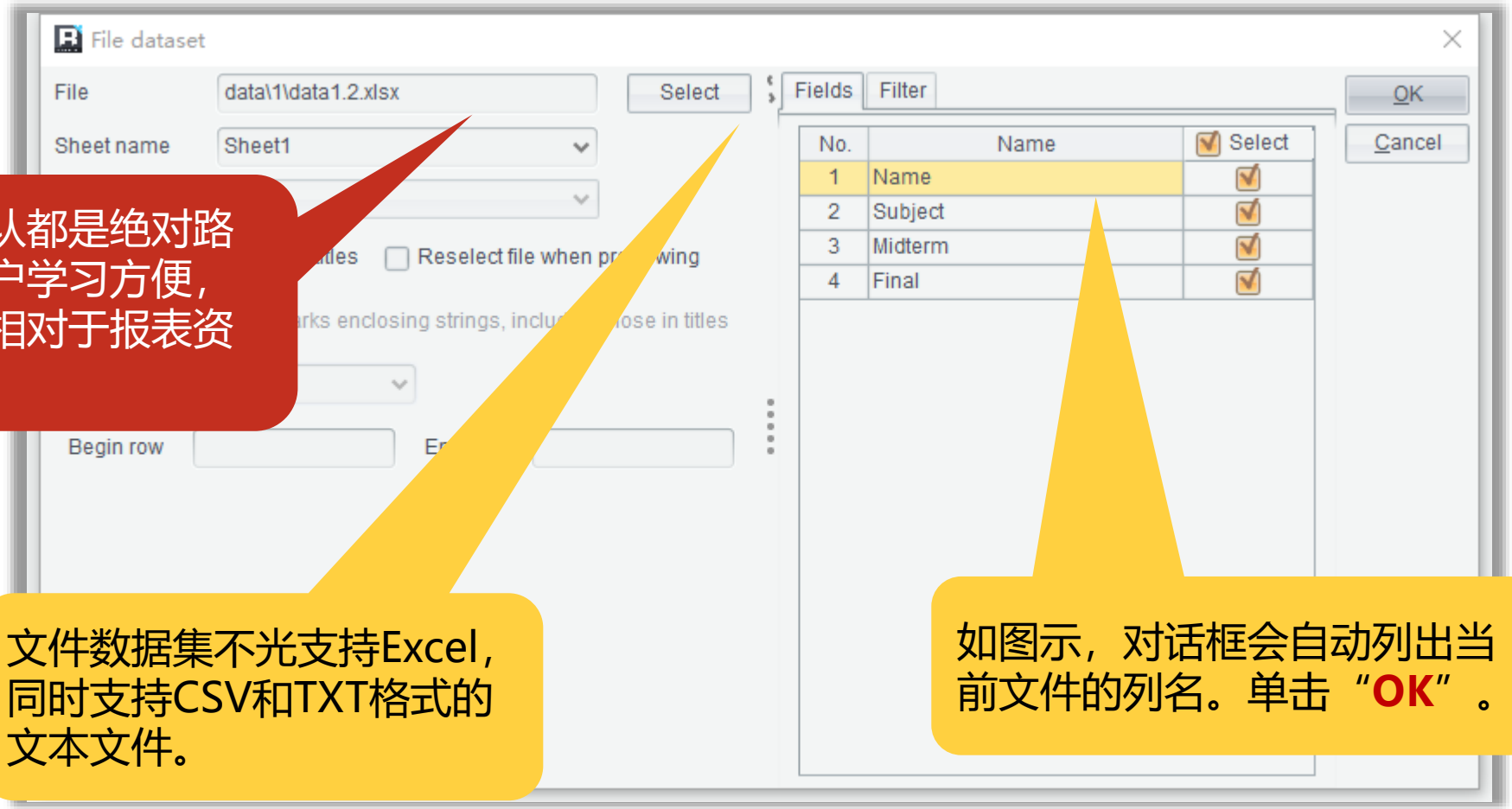
运行润乾报表后，单击菜单“File”、“Open dataset” “File”，打开数据文件设置界面：



第一步要确定生成报表的数据从哪里来，我们先从简单的文件数据开始学习。

1.2 用向导制作行式报表

在该对话框中，单击 “Select”并选出data1.2.xlsx数据文件：

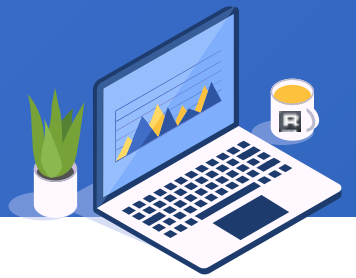


注意：选出的文件默认都是绝对路径，但为了移植和用户学习方便，本文所有路径都采用相对于报表资源路径以及主目录。

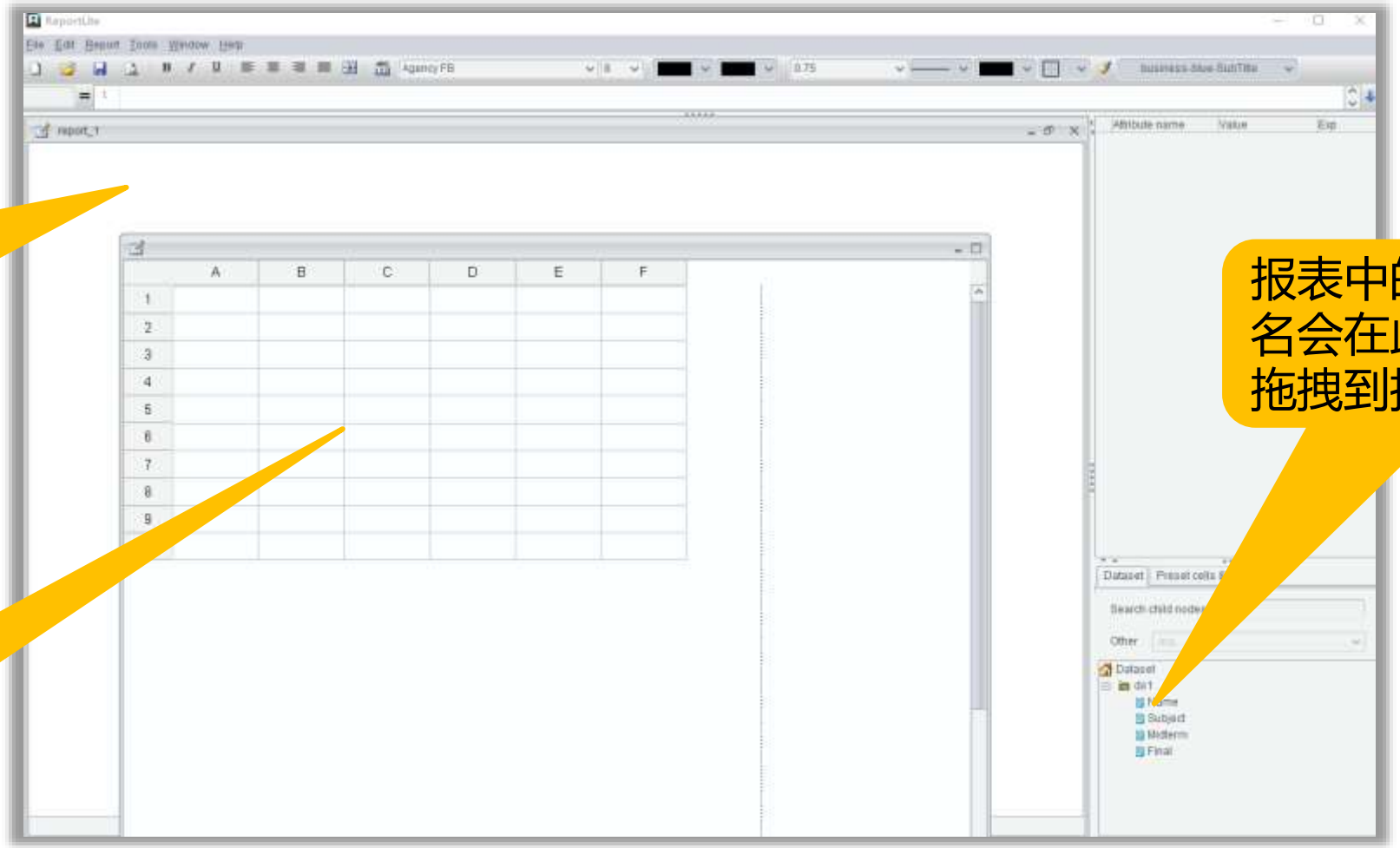
文件数据集不光支持Excel，同时支持CSV和TXT格式的文本文件。

如图示，对话框会自动列出当前文件的列名。单击“**OK**”。

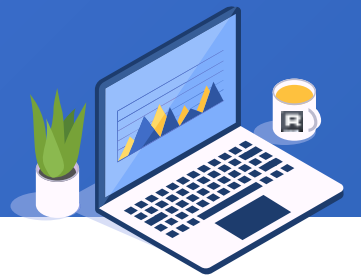
1.2 用向导制作行式报表



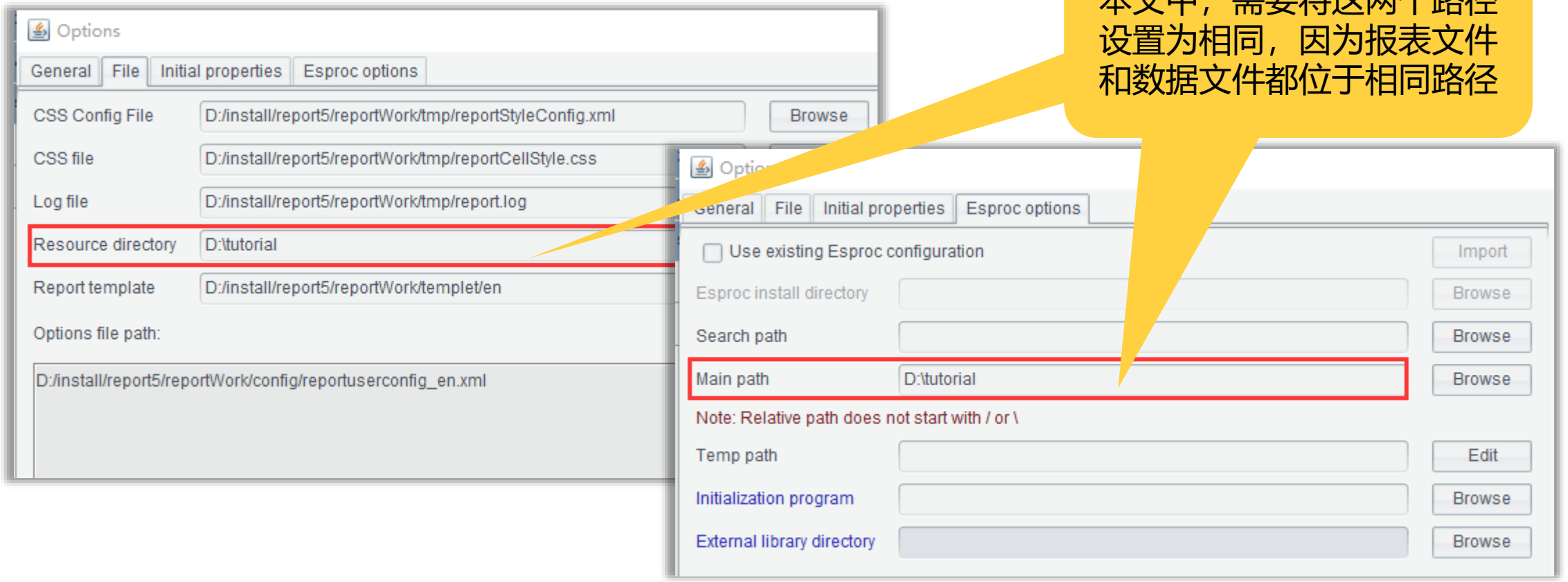
单击OK后，得到一张数据集名字缺省为ds1的空白报表：



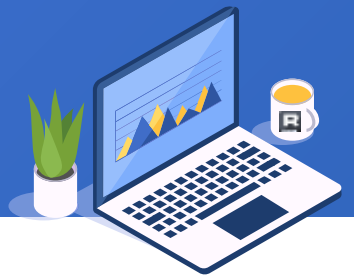
1.2 用向导制作行式报表



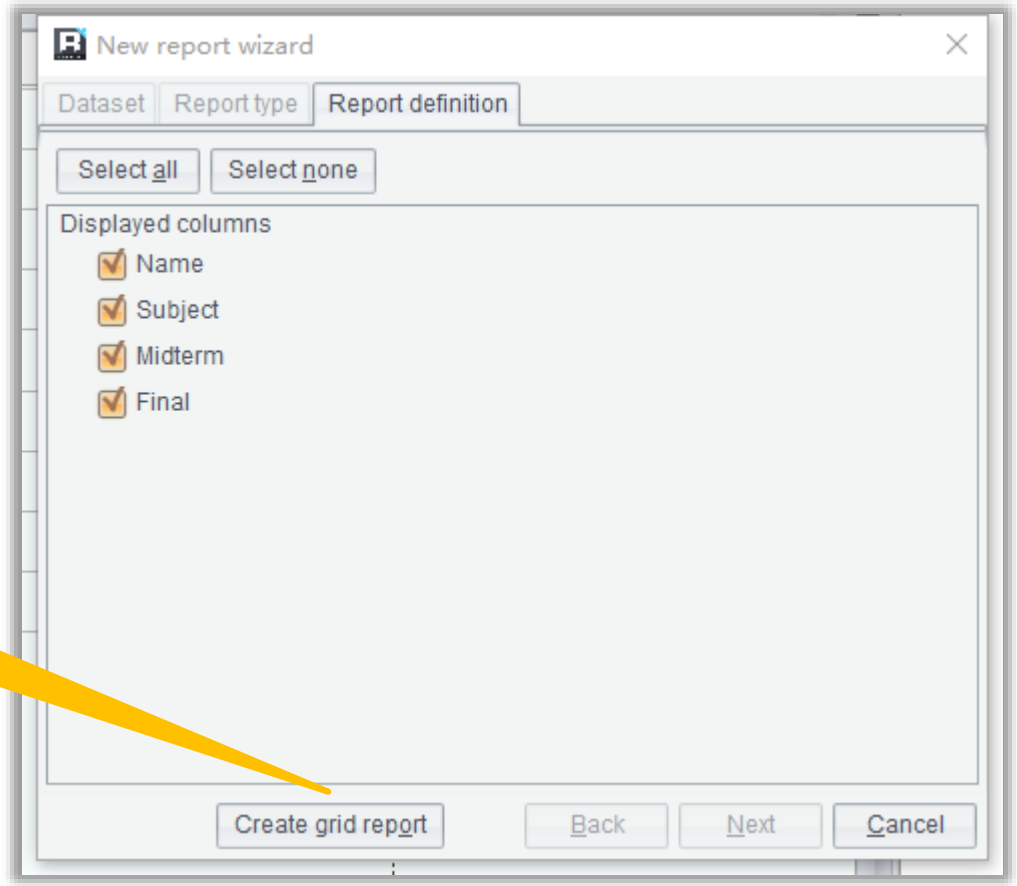
在选项对话框中，需要将如图资源路径和主路径修改为用户自己课程的根目录：



1.2 用向导制作行式报表

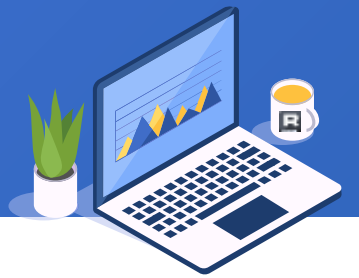


先了解最简单的网格报表，单击菜单“Edit”、“Generate report”、“Grid”，打开网格报表向导界面后，单击“Create grid report”按钮。



使用缺省选项生成网格报表。

1.2 用向导制作行式报表



这种简单的网格报表就是行式报表：

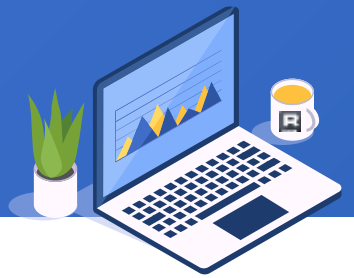
报表的设计效果，能实时预览出来。

report_1

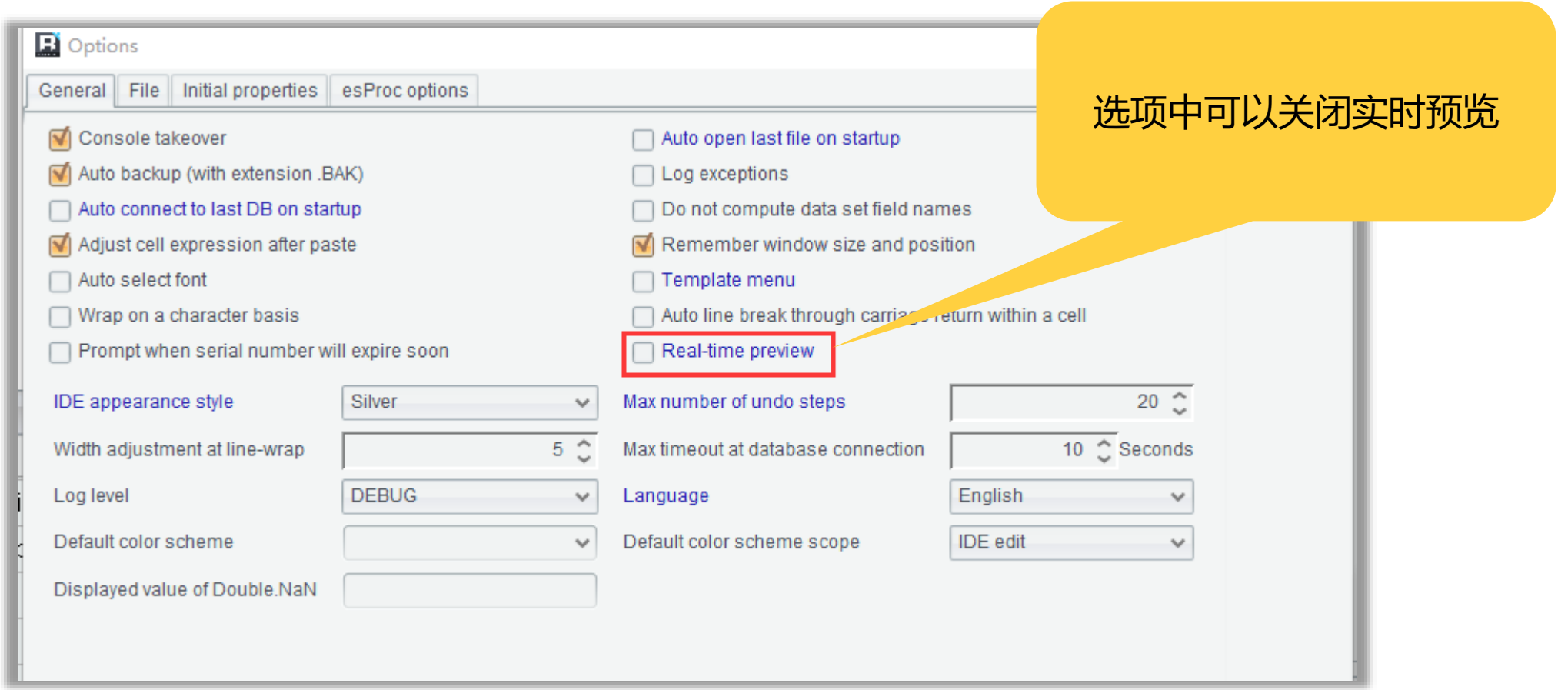
Name	Subject	Midterm	Final
Rose	Math	59	80
Rose	Physics	48	65
Mike	Math	87	77
Mike	Physics	70	86
Ronald	Math	36	50
Ronald	Physics	54	75
Smith	Math	25	55
Smith	Physics	56	75

	A	B	C	D	E
1	Name	Subject	Midterm	Final	
2	=ds1.select(	=ds1.Subject	=ds1.Midter	=ds1.Final	
3					

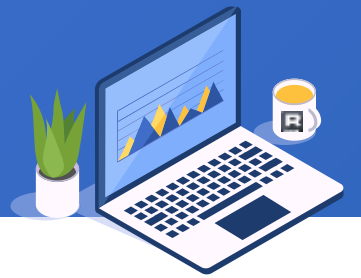
1.2 用向导制作行式报表



实时预览会在编辑状态就加载数据，如果同时编辑多个报表而又担心内存空间时，可以关闭实时预览：



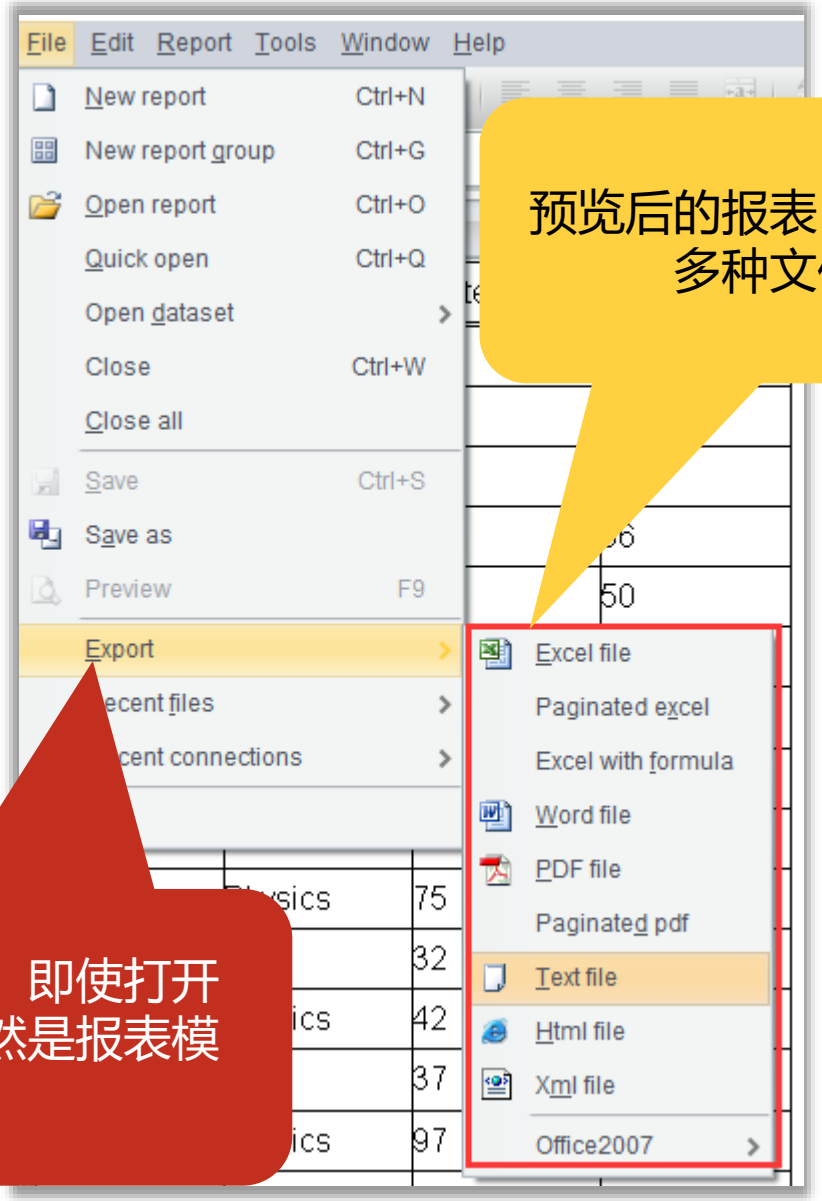
1.2 用向导制作行式报表



给报表加上边框，预览报表的效果：

Preview report:report_1

Name	Subject	Midterm	Final
Rose	Math	59	80
Rose	Physics	48	65
Mike	Math	87	77
Mike	Physics	70	86
Ronald	Math	36	50
Ronald	Physics	54	75
Smith	Math	25	55
Smith	Physics	56	75
Frank	Math	66	80
Frank	Physics	75	
Aimay	Math	32	
Aimay	Physics	42	
James	Math	37	
James	Physics	97	



预览后的报表，可以导出为多种文件格式

注意：在报表设计页面，即使打开了实时预览，导出的仍然是报表模板。

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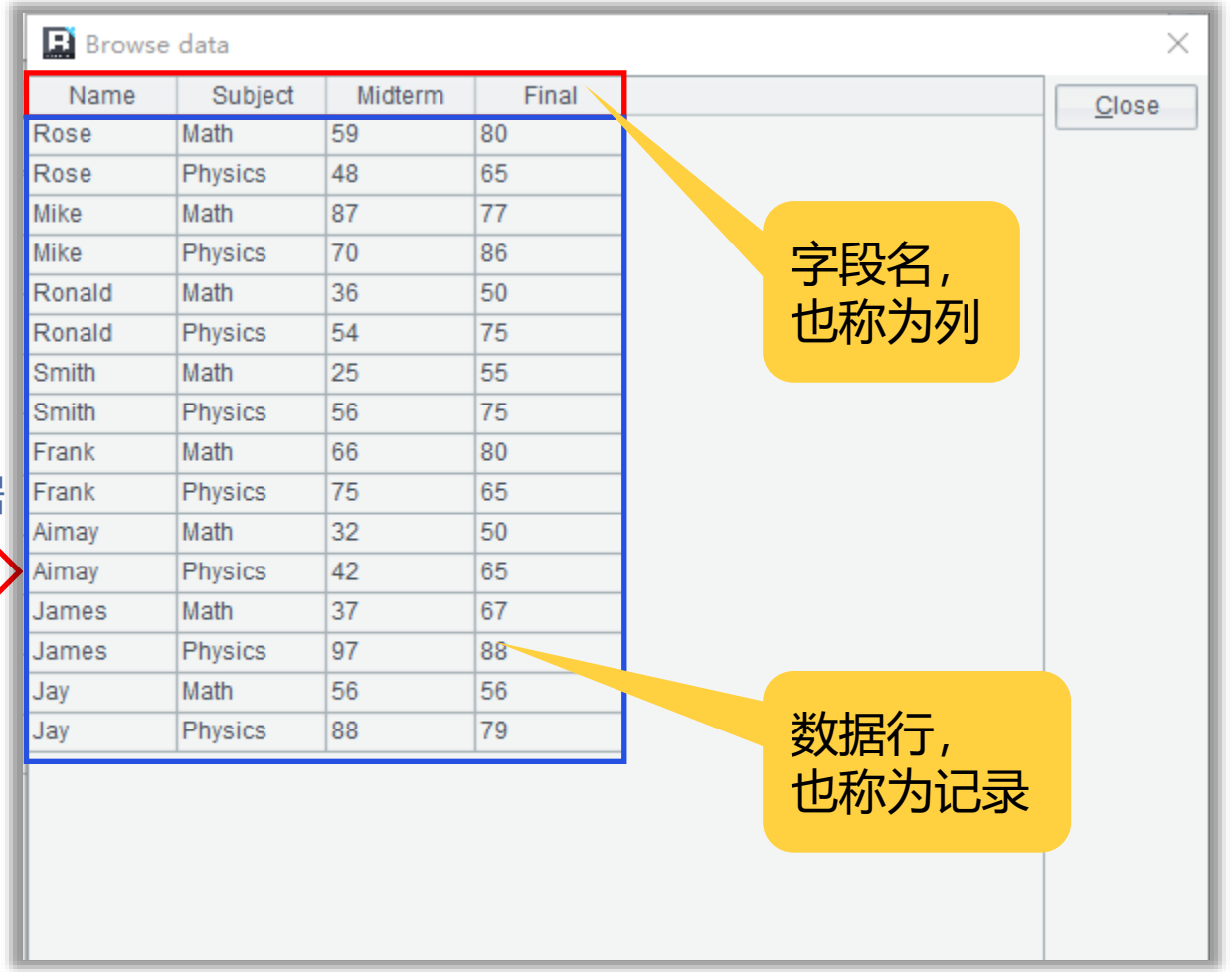
1.3 扩展与表达式

1.3 扩展与表达式

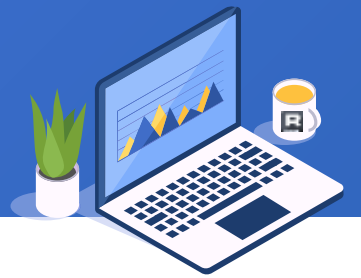
依次点击菜单“Report”，“Dataset”（或者直接按快捷键F11），打开数据集设置界面：



单击Browse浏览数据



1.3 扩展与表达式



D:\tutorial\data\1\1.2.rptx

	A	B	C	D
1(TH)	Name	Subject	Midterm	Final
2	=ds1.select(N	=ds1.Subject	=ds1.Midterm	=ds1.Final

选中第一行后，在下方所示属性面板中设置类型为报表头(TH)，报表头在滚动浏览数据时会固定不动，类似Excel的冻结窗格

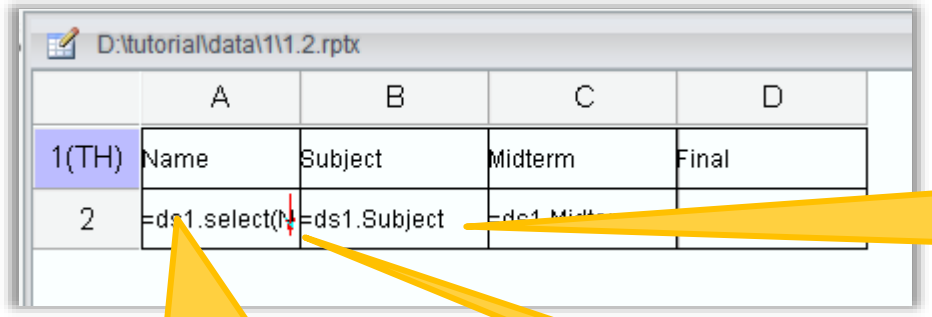
Attribute name	Value	Exp
Row		
Type	Report header	
Group header level	1	
Visible	☑	
Height	8.0	
Fill after page break	☐	
Page break after r...	☐	
Value		
Layout		
Paragraph		

报表模板中的数据区域根据数据集的记录行数纵向扩展为报表中的相应行数称为**纵向扩展**

Preview report:D:\tutorial\data\1\1.2.rptx

Name	Subject	Midterm	Final
Rose	Math	59	80
Rose	Physics	48	65
Mike	Math	87	77
Mike	Physics	70	86
Ronald	Math	36	50
Ronald	Physics	54	75
Smith	Math	25	55
Smith	Physics	56	75
Frank	Math	66	80
Frank	Physics	75	65
Aimay	Math	32	50
Aimay	Physics	42	65
James	Math	37	67
James	Physics	97	88
Jay	Math	56	56
Jay	Physics	88	79

1.3 扩展与表达式

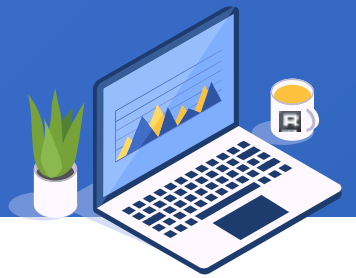


B2以及后面的表达式不再用select函数，表示科目和相应成绩分别对应于A2的选出姓名(Name)

A2中的“=”表示后面跟的是表达式，要计算后才能得出结果。
”**ds1.select(Name)**”表示从数据集ds1中选出所有姓名(Name)

A2格子后面还有个红色的竖箭头，表示当前的格子计算后会有多个数据，且会纵向扩展

1.3 扩展与表达式



下面根据期中占比0.4、期末占比0.6，计算出加权后的成绩。

D:\tutorial\data\1\1.2.rptx

	A	B	C	D	E
1(TH)	Name	Subject	Midterm	Final	WA
2	=ds1.select(N	=ds1.Subject	=ds1.Midterm	=ds1.Final	=C2*0.4+D2*0.6

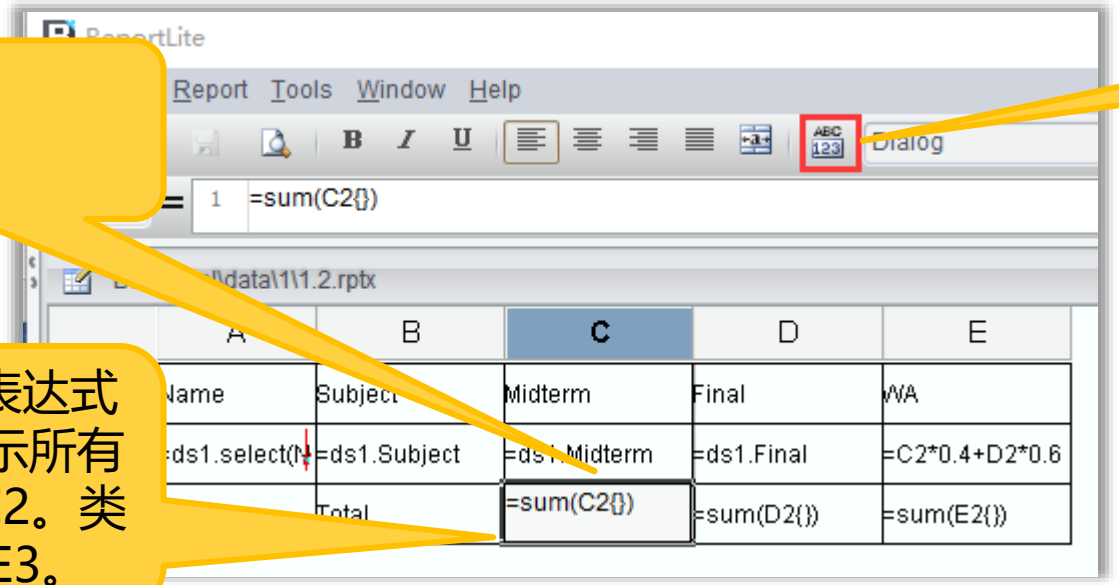
1.设置E列的标题为加权成绩WA

2.在E2中使用跟Excel完全相同的表达式计算出加权分

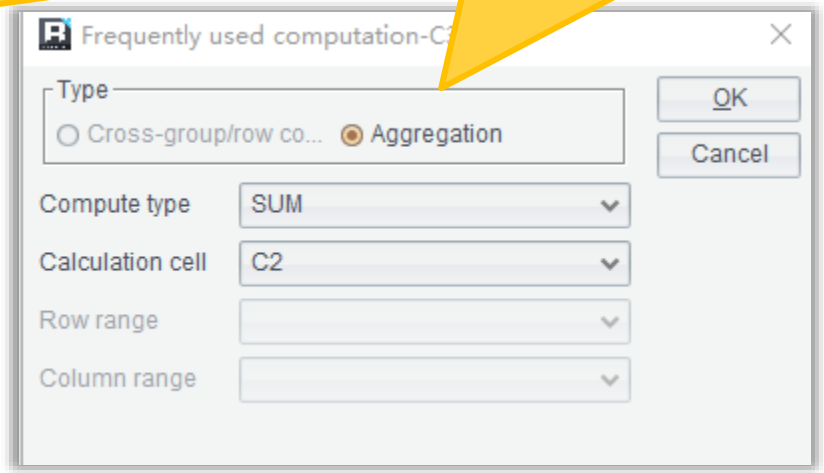
1.3 扩展与表达式

再修改A2中表达式，只选出Math成绩。然后在第三行，用表达式统计出Math的总分。

1.选中C3格



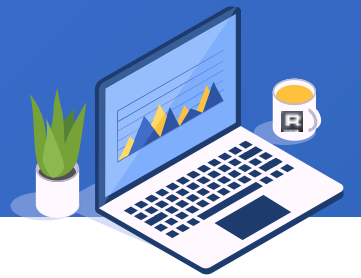
2.单击图示常用计算列，如下设置对C2求和



3.C3生成的表达式中，C2{} 表示所有扩展出来的C2。类似设置D3和E3。

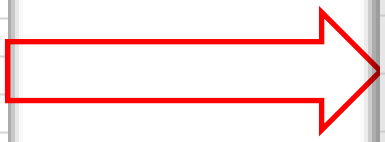
A2中表达式” =ds1.select(Name,Subject=="Math")”表示只选出Math科目。
select函数的详细说明参考： http://d.raqsoft.com.cn:6999/report/tutorial/select104.html

1.3 扩展与表达式



预览报表后，通过菜单导出为”带公式Excel”(Excel With Formula)，可以对比看出改动源文件后，只要重新预览报表并导出，便可方便地生成相应的Excel报表。

	A	B	C	D
1	Name	Subject	Midterm	Final
2	Rose	Math	59	80
3	Rose	Physics	48	65
4	Mike	Math	87	77
5	Mike	Physics	70	86
6	Ronald	Math	36	50
7	Ronald	Physics	54	75
8	Smith	Math	25	55
9	Smith	Physics	56	75
10	Frank	Math	66	80
11	Frank	Physics	75	65
12	Aimay	Math	32	50
13	Aimay	Physics	42	65
14	James	Math	37	67
15	James	Physics	97	88
16	Jay	Math	56	56
17	Jay	Physics	88	79



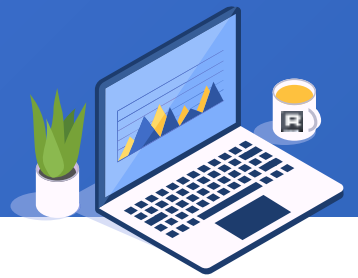
	A	B	C	D	E
1	Name	Subject	Midterm	Final	WA
2	Rose	Math	59	80	71.6
3	Mike	Math	87	77	81.0
4	Ronald	Math	36	50	44.4
5	Smith	Math	25	55	43.0
6	Frank	Math	66	80	74.4
7	Aimay	Math	32	50	42.8
8	James	Math	37	67	55.0
9	Jay	Math	56	56	56.0
10		Total			468.2

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1.4 报表区域

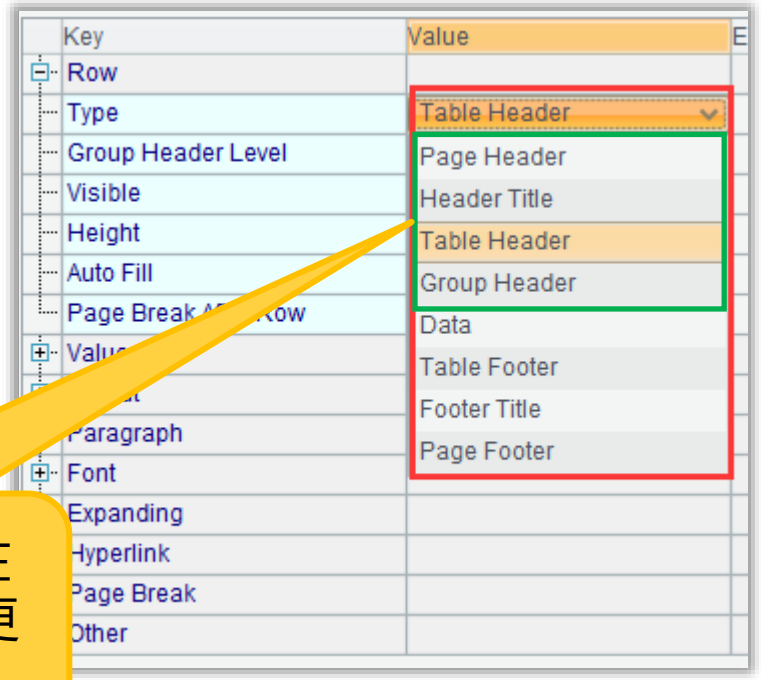
1.4 报表区域



D:\tutorial\data\1\1.2.rptx

	A	B	C	D	E
1(TH)	Name	Subject	Midterm	Final	WA
2	=ds1.select(N	=ds1.Subject	=ds1.Midterm	=ds1.Final	=C2*0.4+D2*0.6
3		Total	=sum(C2{})	=sum(D2{})	=sum(E2{})

1.报表区域只能在行首格上设置



2.选中行首格后，可以在右边属性面板的行类型更改行的区域

通常，行类型只能从上到下，顺序设置所属区域。且数据区(Data)以上的类型(绿色部分)会在生成Excel时，生成冻结窗口。如果不需要冻结窗口，则行类型都可以缺省为数据区。

第2章
外观与格式



CONTENTS



2.1 边框字体颜色

2.2 显示格式、显示值

2.3 条件格式

2.4 卡片报表

2.5 图片

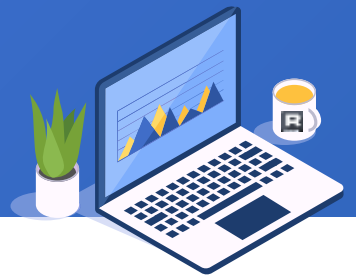
2.6 习题

CONTENTS

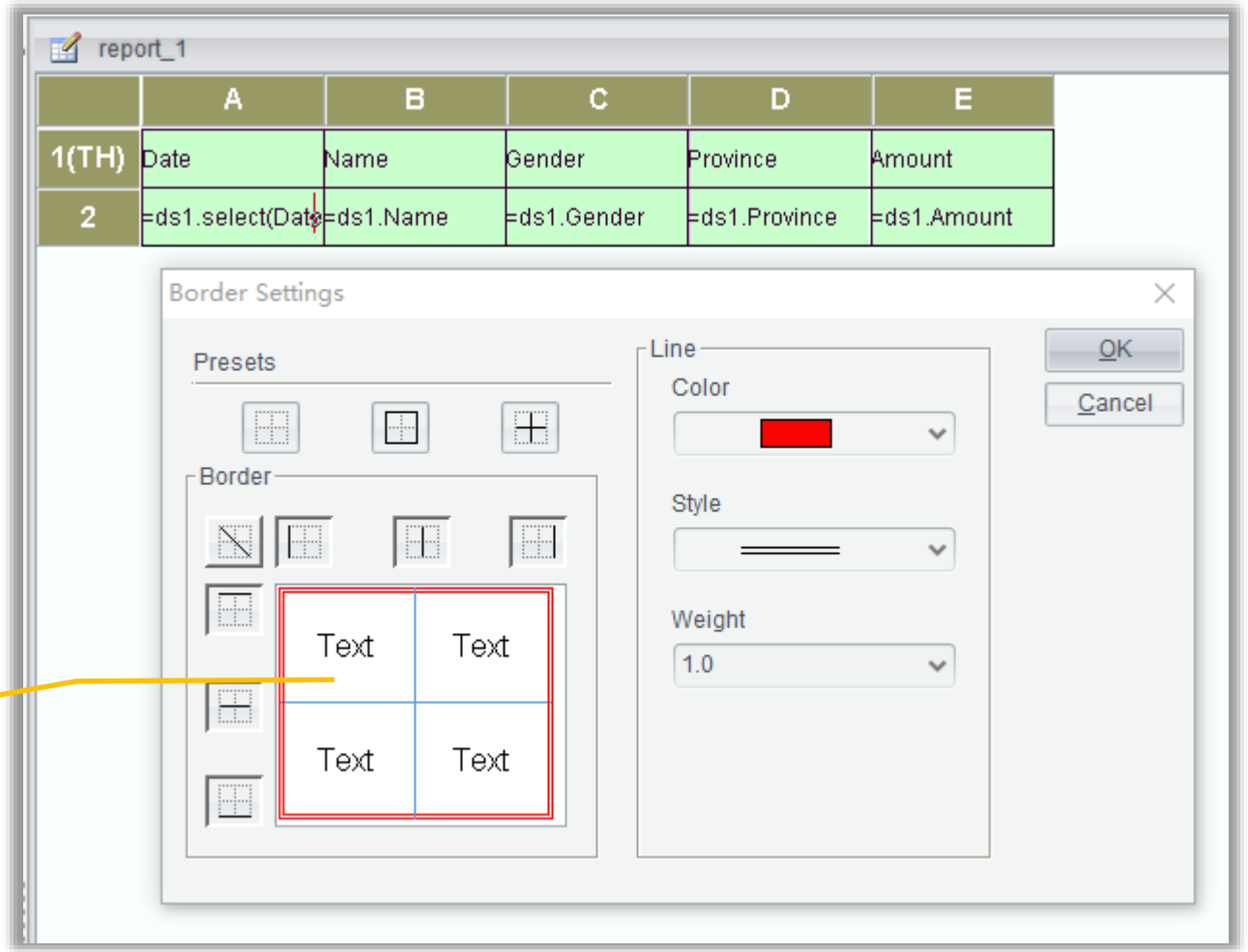


2.1 边框字体颜色

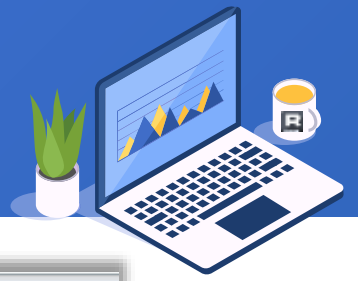
2.1 边框字体颜色



单击左上角，选中全部格子区域，然后右击菜单“边框设置”如图设置边框的颜色，风格以及粗度



2.1 边框字体颜色



Preview report:report_1

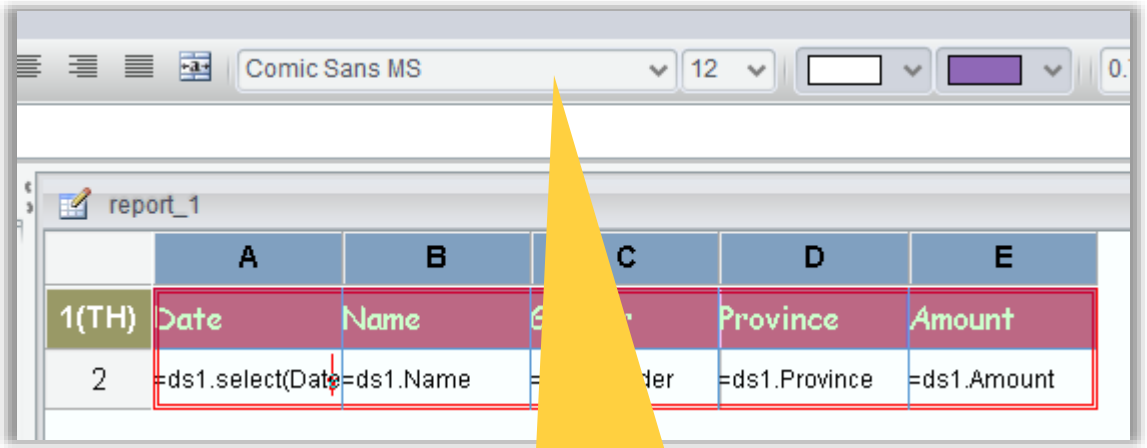
Date	Name	Gender	Province	Amount
2019-01-12	Lisa	1	1	100
2019-02-08	Ice Chan	1	2	200
2019-02-14	Tossman	0	3	50
2019-02-15	Lisa	1	1	200
2019-04-16	Amay	1	4	100
2019-04-27	Tossman	0	3	100
2019-04-28	Lisa	1	1	50
2019-05-09	Smith	0	5	200
2019-05-11	Tossman	0	3	100
2019-05-21	Amay	1	4	200
2019-06-22	Ice Chan	1	2	100
2019-06-23	Tossman	0	3	100
2019-06-24	Lisa	1	1	50
2019-06-25	Ice Chan	1	2	150
2019-06-26	Tossman	0	3	50
2019-06-27	Smith	0	5	100
2019-06-28	Ice Chan	1	2	50
2019-06-29	Tossman	0	3	50
2019-06-30	Lisa	1	1	50
2019-07-01	Amay	1	4	200
2019-07-02	Tossman	0	3	50

设置边框
对比效果

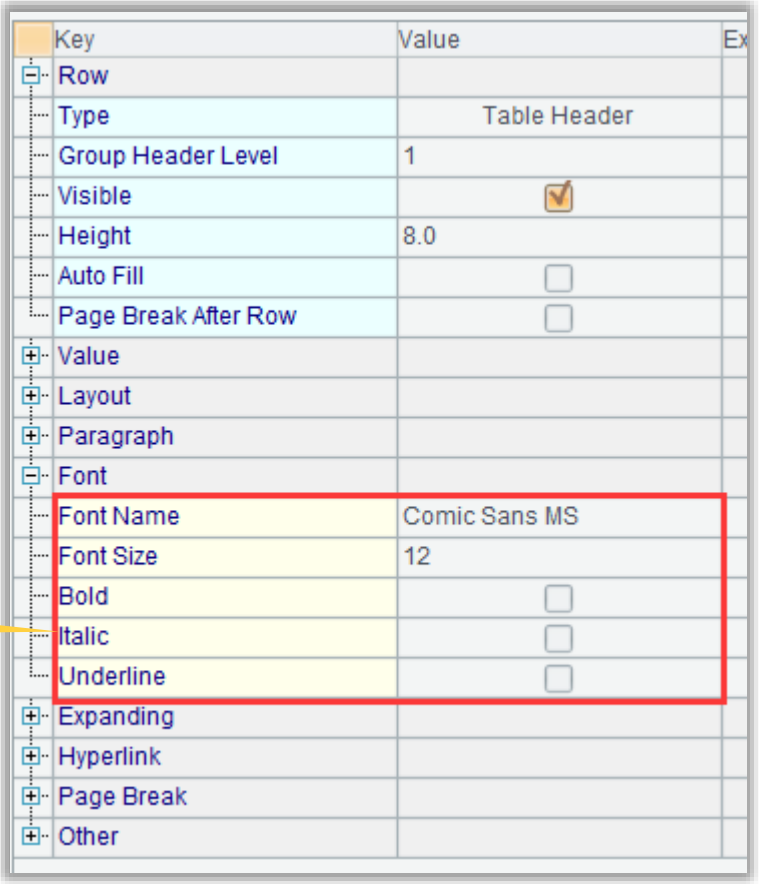
Preview report:report_1

Date	Name	Gender	Province	Amount
2019-01-12	Lisa	1	1	100
2019-02-08	Ice Chan	1	2	200
2019-02-14	Tossman	0	3	50
2019-02-15	Lisa	1	1	200
2019-04-16	Amay	1	4	100
2019-04-27	Tossman	0	3	100
2019-04-28	Lisa	1	1	50
2019-05-09	Smith	0	5	200
2019-05-11	Tossman	0	3	100
2019-05-21	Amay	1	4	200
2019-06-22	Ice Chan	1	2	100
2019-06-23	Tossman	0	3	100
2019-06-24	Lisa	1	1	50
2019-06-25	Ice Chan	1	2	150
2019-06-26	Tossman	0	3	50
2019-06-27	Smith	0	5	100
2019-06-28	Ice Chan	1	2	50
2019-06-29	Tossman	0	3	50
2019-06-30	Lisa	1	1	50
2019-07-01	Amay	1	4	200
2019-07-02	Tossman	0	3	50

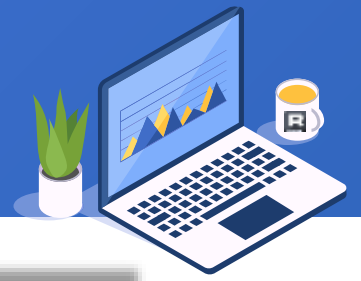
2.1 边框字体颜色



选中格子区域，通过工具条或者右侧的属性面板的字体属性，可以设置报表的字体和前、背景色



2.1 边框字体颜色



Preview report:report_1

Date	Name	Gender	Province	Amount
2019-01-12	Lisa	1	1	100
2019-02-08	Ice Chan	1	2	200
2019-02-14	Tossman	0	3	50
2019-02-15	Lisa	1	1	200
2019-04-16	Amay	1	4	100
2019-04-27	Tossman	0	3	100
2019-04-28	Lisa	1	1	50
2019-05-09	Smith	0	5	200
2019-05-11	Tossman	0	3	100
2019-05-21	Amay	1	4	200
2019-06-22	Ice Chan	1	2	100
2019-06-23	Tossman	0	3	100
2019-06-24	Lisa	1	1	50
2019-06-25	Ice Chan	1	2	150
2019-06-26	Tossman	0	3	50
2019-06-27	Smith	0	5	100
2019-06-28	Ice Chan	1	2	50
2019-06-29	Tossman	0	3	50
2019-06-30	Lisa	1	1	50
2019-07-01	Amay	1	4	200
2019-07-02	Tossman	0	3	50

设置报表头的字体以及颜色对比效果

Preview report:report_1

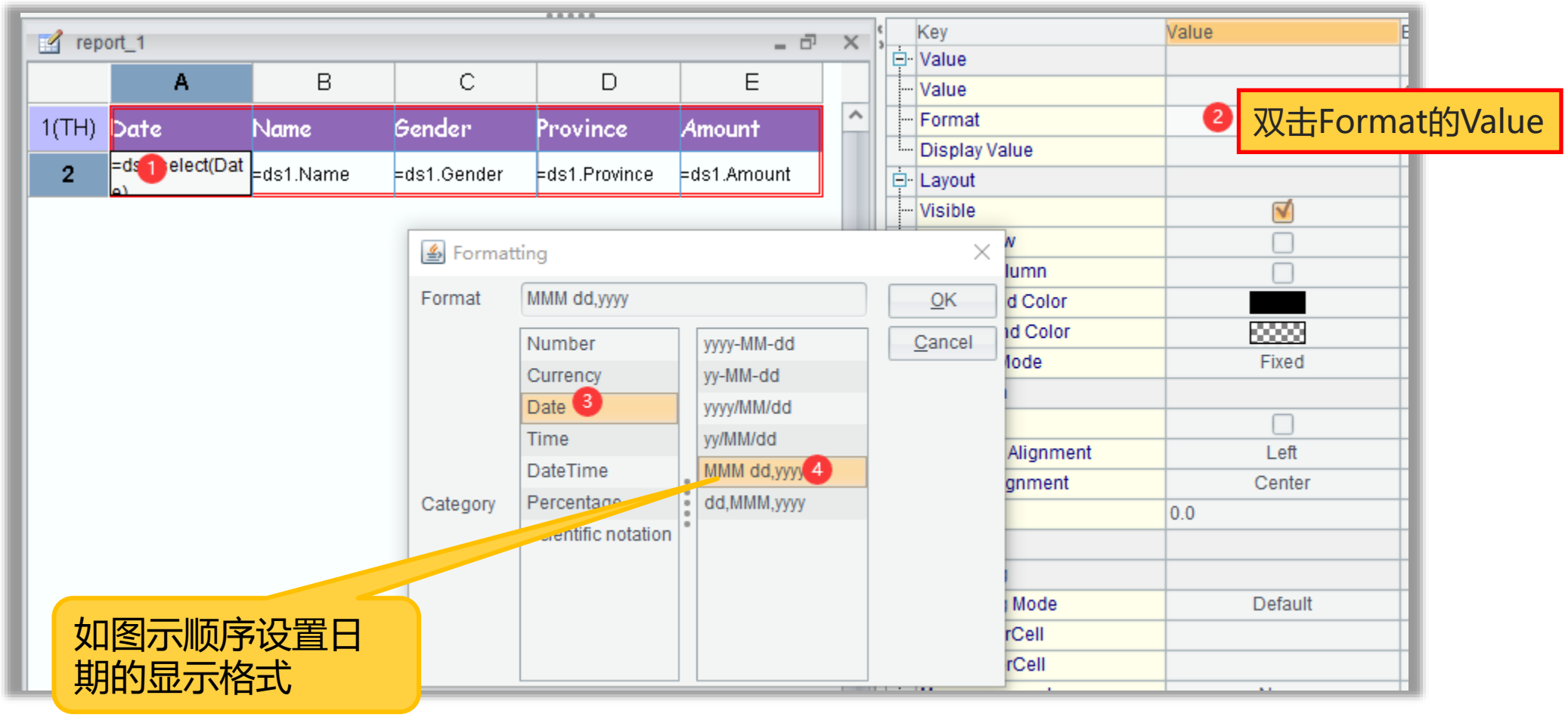
Date	Name	Gender	Province	Amount
2019-01-12	Lisa	1	1	100
2019-02-08	Ice Chan	1	2	200
2019-02-14	Tossman	0	3	50
2019-02-15	Lisa	1	1	200
2019-04-16	Amay	1	4	100
2019-04-27	Tossman	0	3	100
2019-04-28	Lisa	1	1	50
2019-05-09	Smith	0	5	200
2019-05-11	Tossman	0	3	100
2019-05-21	Amay	1	4	200
2019-06-22	Ice Chan	1	2	100
2019-06-23	Tossman	0	3	100
2019-06-24	Lisa	1	1	50
2019-06-25	Ice Chan	1	2	150
2019-06-26	Tossman	0	3	50
2019-06-27	Smith	0	5	100
2019-06-28	Ice Chan	1	2	50
2019-06-29	Tossman	0	3	50
2019-06-30	Lisa	1	1	50
2019-07-01	Amay	1	4	200
2019-07-02	Tossman	0	3	50

CONTENTS

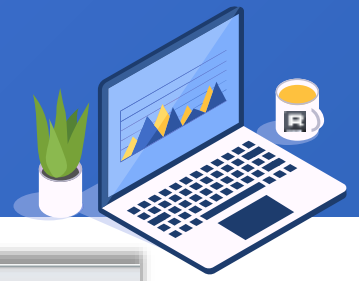


2.2 显示格式显示值

2.2 显示格式显示值


report_1
A
B
C
D
E
1(TH)
Date
Name
Gender
Province
Amount
2
=ds1.Name
=ds1.Gender
=ds1.Province
=ds1.Amount
Key
Value
Format
Display Value
Layout
Visible
Formatting
Format
MMM dd,yyyy
OK
Cancel
Number
Currency
Date
Time
DateTime
Percentage
Category
yyyy-MM-dd
yy-MM-dd
yyyy/MM/dd
yy/MM/dd
MMM dd,yyyy
dd,MMM,yyyy
Alignment
Left
Center
0.0
Mode
Default
双击Format的Value
如图示顺序设置日期的显示格式

2.2 显示格式显示值



Preview report:report_1

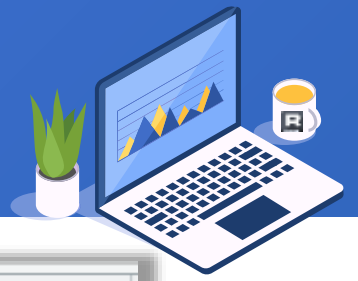
Date	Name	Gender	Province	Amount
2019-01-12	Lisa	1	1	100
2019-02-08	ce Chan	1	2	200
2019-02-14	Tossman	0	3	50
2019-02-15	Lisa	1	1	200
2019-04-16	Amay	1	4	100
2019-04-27	Tossman	0	3	100
2019-04-28	Lisa	1	1	50
2019-05-09	Smith	0	5	200
2019-05-11	Tossman	0	3	100
2019-05-21	Amay	1	4	200
2019-06-22	ce Chan	1	2	100
2019-06-23	Tossman	0	3	100
2019-06-24	Lisa	1	1	50
2019-06-25	ce Chan	1	2	150
2019-06-26	Tossman	0	3	50
2019-06-27	Smith	0	5	100
2019-06-28	ce Chan	1	2	50
2019-06-29	Tossman	0	3	50
2019-06-30	Lisa	1	1	50
2019-07-01	Amay	1	4	200
2019-07-02	Tossman	0	3	50

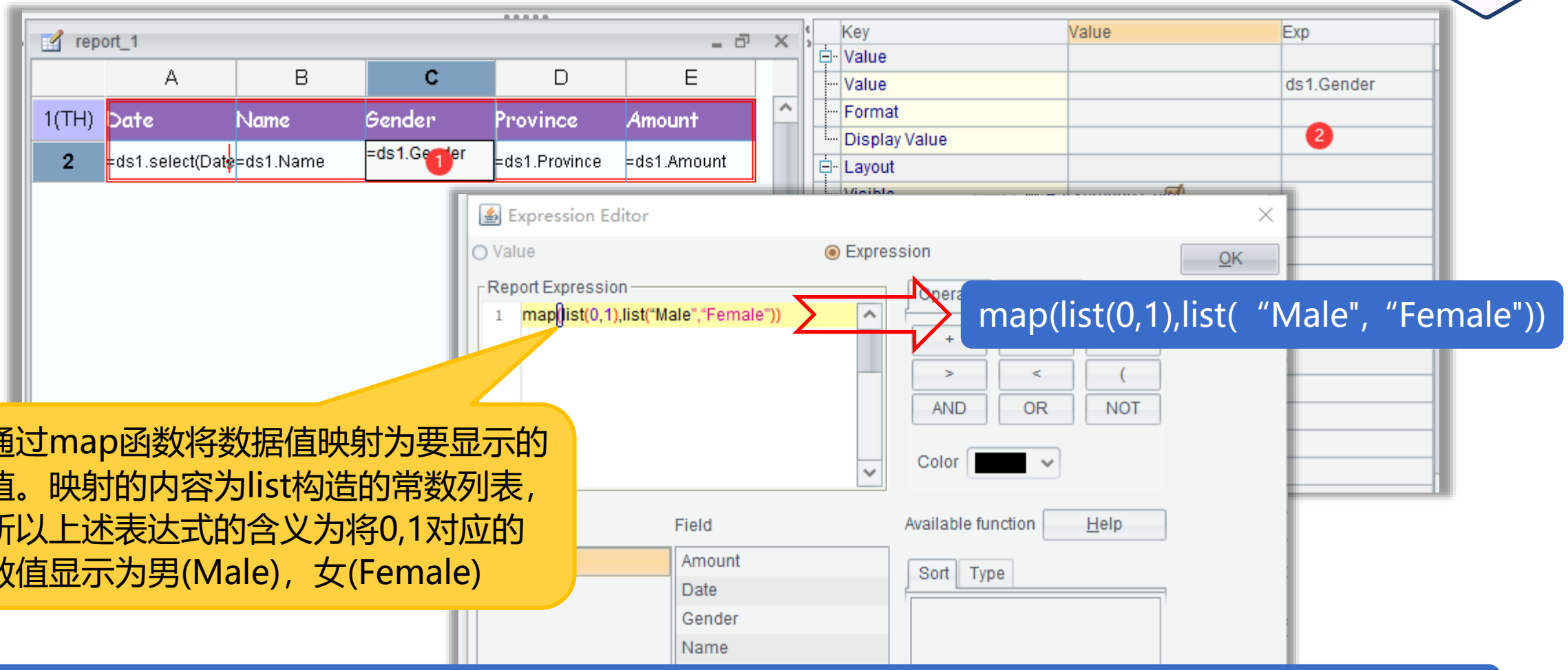
设置日期显示格式对比效果

Preview report:report_1

Date	Name	Gender	Province	Amount
Jan 12,2019	Lisa	1	1	100
Feb 08,2019	ce Chan	1	2	200
Feb 14,2019	Tossman	0	3	50
Feb 15,2019	Lisa	1	1	200
Apr 16,2019	Amay	1	4	100
Apr 27,2019	Tossman	0	3	100
Apr 28,2019	Lisa	1	1	50
May 09,2019	Smith	0	5	200
May 11,2019	Tossman	0	3	100
May 21,2019	Amay	1	4	200
Jun 22,2019	ce Chan	1	2	100
Jun 23,2019	Tossman	0	3	100
Jun 24,2019	Lisa	1	1	50
Jun 25,2019	ce Chan	1	2	150
Jun 26,2019	Tossman	0	3	50
Jun 27,2019	Smith	0	5	100
Jun 28,2019	ce Chan	1	2	50
Jun 29,2019	Tossman	0	3	50
Jun 30,2019	Lisa	1	1	50
Jul 01,2019	Amay	1	4	200
Jul 02,2019	Tossman	0	3	50

2.2 显示格式显示值

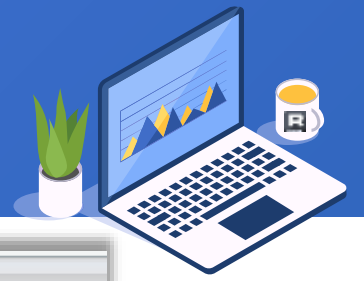




通过map函数将数据值映射为要显示的值。映射的内容为list构造的常数列表，所以上述表达式的含义为将0,1对应的数值显示为男(Male)，女(Female)

map函数更多说明请参考：http://d.raqsoft.com.cn:6999/report/preference/mapvle.html

2.2 显示格式显示值



Preview report:report_1

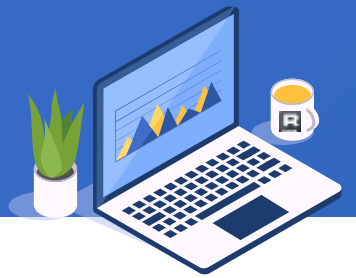
Date	Name	Gender	Province	Amount
Jan 12,2019	Lisa	1	1	100
Feb 08,2019	Ice Chan	1	2	200
Feb 14,2019	Tossman	0	3	50
Feb 15,2019	Lisa	1	1	200
Apr 16,2019	Amay	1	4	100
Apr 27,2019	Tossman	0	3	100
Apr 28,2019	Lisa	1	1	50
May 09,2019	Smith	0	5	200
May 11,2019	Tossman	0	3	100
May 21,2019	Amay	1	4	200
Jun 22,2019	Ice Chan	1	2	100
Jun 23,2019	Tossman	0	3	100
Jun 24,2019	Lisa	1	1	50
Jun 25,2019	Ice Chan	1	2	150
Jun 26,2019	Tossman	0	3	50
Jun 27,2019	Smith	0	5	100
Jun 28,2019	Ice Chan	1	2	50
Jun 29,2019	Tossman	0	3	50
Jun 30,2019	Lisa	1	1	50
Jul 01,2019	Amay	1	4	200
Jul 02,2019	Tossman	0	3	50

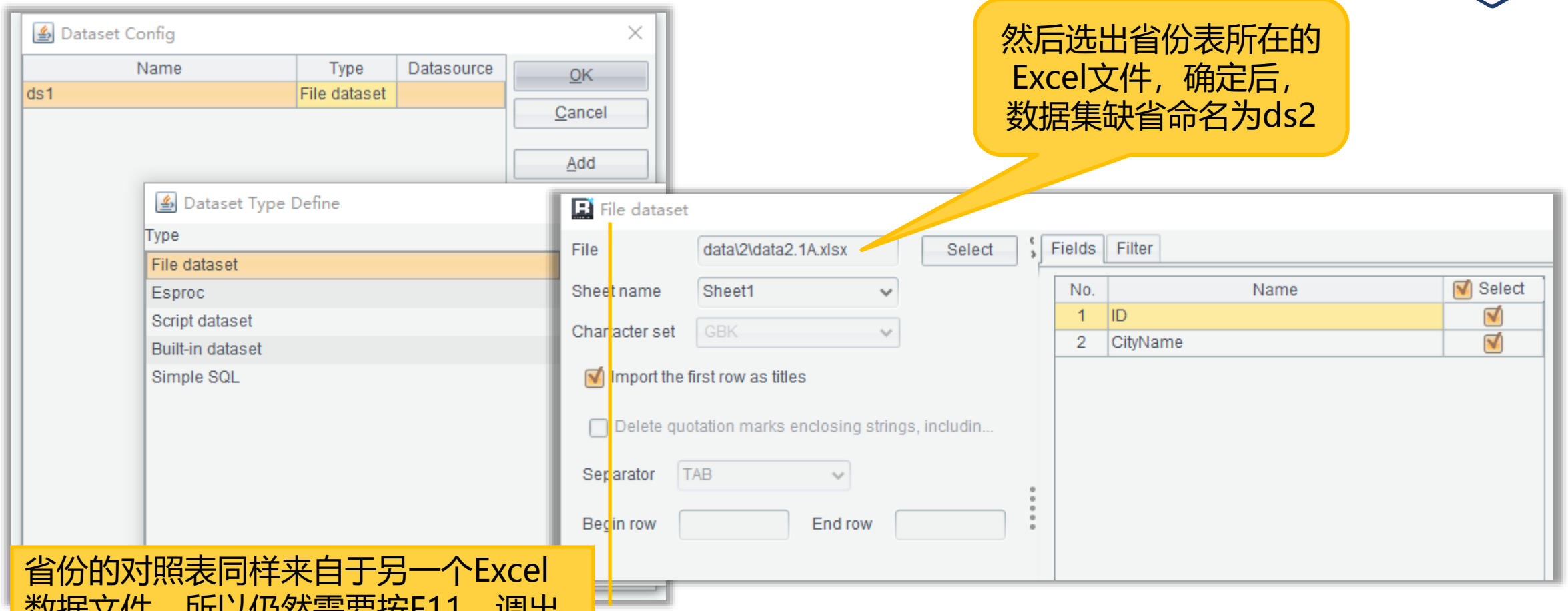
设置性别常量显示对比效果

Preview report:report_1

Date	Name	Gender	Province	Amount
Jan 12,2019	Lisa	Female	1	100
Feb 08,2019	Ice Chan	Female	2	200
Feb 14,2019	Tossman	Male	3	50
Feb 15,2019	Lisa	Female	1	200
Apr 16,2019	Amay	Female	4	100
Apr 27,2019	Tossman	Male	3	100
Apr 28,2019	Lisa	Female	1	50
May 09,2019	Smith	Male	5	200
May 11,2019	Tossman	Male	3	100
May 21,2019	Amay	Female	4	200
Jun 22,2019	Ice Chan	Female	2	100
Jun 23,2019	Tossman	Male	3	100
Jun 24,2019	Lisa	Female	1	50
Jun 25,2019	Ice Chan	Female	2	150
Jun 26,2019	Tossman	Male	3	50
Jun 27,2019	Smith	Male	5	100
Jun 28,2019	Ice Chan	Female	2	50
Jun 29,2019	Tossman	Male	3	50
Jun 30,2019	Lisa	Female	1	50
Jul 01,2019	Amay	Female	4	200
Jul 02,2019	Tossman	Male	3	50

2.2 显示格式显示值

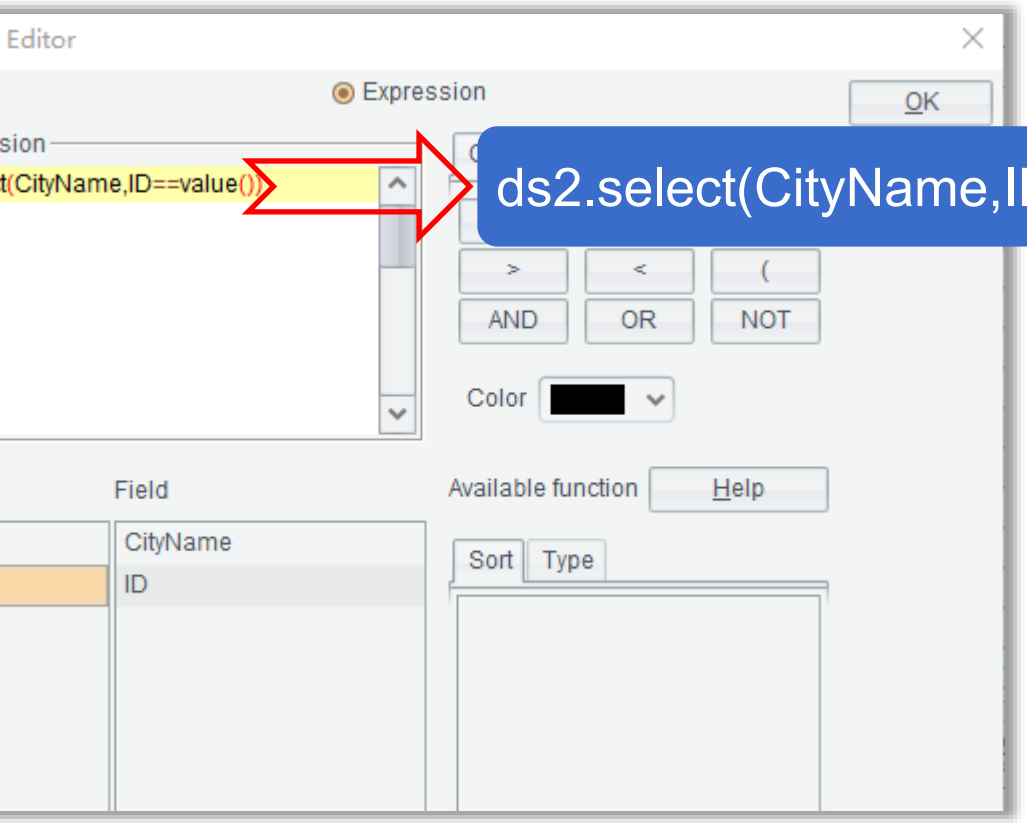




然后选出省份表所在的Excel文件，确定后，数据集缺省命名为ds2

省份的对照表同样来自于另一个Excel数据文件，所以仍然需要按F11，调出数据集设置，然后点增加按钮，并如图选中文件数据集，再按确定

2.2 显示格式显示值

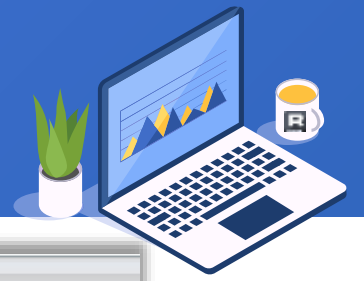


ds2.select(CityName,ID==value())

如图设置省份列的显示值表达式。

含义为从省份对照数据集ds2中选出省份的名称，对应条件为ds2中的编号等于value()，由于省份字段是扩展出来的，所以用value()函数获取扩展后的每一个具体的值。

2.2 显示格式显示值



Preview report:report_1

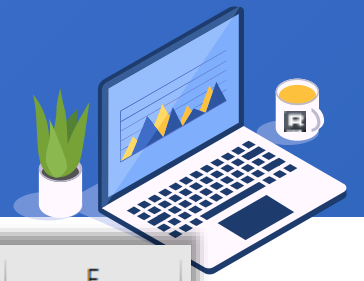
Date	Name	Gender	Province	Amount
Jan 12,2019	Lisa	1	1	100
Feb 08,2019	Ice Chan	1	2	200
Feb 14,2019	Tossman	0	3	50
Feb 15,2019	Lisa	1	1	200
Apr 16,2019	Amay	1	4	100
Apr 27,2019	Tossman	0	3	100
Apr 28,2019	Lisa	1	1	50
May 09,2019	Smith	0	5	200
May 11,2019	Tossman	0	3	100
May 21,2019	Amay	1	4	200
Jun 22,2019	Ice Chan	1	2	100
Jun 23,2019	Tossman	0	3	100
Jun 24,2019	Lisa	1	1	50
Jun 25,2019	Ice Chan	1	2	150
Jun 26,2019	Tossman	0	3	50
Jun 27,2019	Smith	0	5	100
Jun 28,2019	Ice Chan	1	2	50
Jun 29,2019	Tossman	0	3	50
Jun 30,2019	Lisa	1	1	50
Jul 01,2019	Amay	1	4	200
Jul 02,2019	Tossman	0	3	50

设置省份和金额的显示格式对比效果

Preview report:report_1

Date	Name	Gender	Province	Amount
Jan 12,2019	Lisa	Female	Washington	$100.0
Feb 08,2019	Ice Chan	Female	New York	$200.0
Feb 14,2019	Tossman	Male	Seattle	$50.0
Feb 15,2019	Lisa	Female	Washington	$200.0
Apr 16,2019	Amay	Female	Los Angeles	$100.0
Apr 27,2019	Tossman	Male	Seattle	$100.0
Apr 28,2019	Lisa	Female	Washington	$50.0
May 09,2019	Smith	Male	Detroit	$200.0
May 11,2019	Tossman	Male	Seattle	$100.0
May 21,2019	Amay	Female	Los Angeles	$200.0
[illegible]	[illegible]	[illegible]	New York	$100.0
Jun 23,2019	Tossman	Male	Seattle	$100.0
Jun 24,2019	Lisa	Female	Washington	$50.0
Jun 25,2019	Ice Chan	Female	New York	$150.0
Jun 26,2019	Tossman	Male	Seattle	$50.0
Jun 27,2019	Smith	Male	Detroit	$100.0
Jun 28,2019	Ice Chan	Female	New York	$50.0
Jun 29,2019	Tossman	Male	Seattle	$50.0
Jun 30,2019	Lisa	Female	Washington	$50.0
Jul 01,2019	Amay	Female	Los Angeles	$200.0
Jul 02,2019	Tossman	Male	Seattle	$50.0

2.2 显示格式显示值



	A	B	C	D	E
1	Date	Name	Gender	Province	Amount
2	2019/1/12	Lisa	1	1	100
3	2019/2/8	Ice Chan	1	2	200
4	2019/2/14	Tossman	0	3	50
5	2019/2/15	Lisa	1	1	200
6	2019/4/16	Amay	1	4	100
7	2019/4/27	Tossman	0	3	100
8	2019/4/28	Lisa	1	1	50
9	2019/5/9	Smith	0	5	200
10	2019/5/11	Tossman	0	3	100
11	2019/5/21	Amay	1	4	200
12	2019/6/22	Ice Chan	1	2	100
13	2019/6/23	Tossman	0	3	100
14	2019/6/24	Lisa	1	1	50
15	2019/6/25	Ice Chan	1	2	150
16	2019/6/26	Tossman	0	3	50
17	2019/6/27	Smith	0	5	100
18	2019/6/28	Ice Chan	1	2	50
19	2019/6/29	Tossman	0	3	50
20	2019/6/30	Lisa	1	1	50
21	2019/7/1	Amay	1	4	200
22	2019/7/2	Tossman	0	3	50

源数据和设置显示格式后再导出为Excel对比效果

	A	B	C	D	E
1	Date	Name	Gender	Province	Amount
2	Jan 12,2019	Lisa	Female	Washington	$100.00
3	Feb 08,2019	Ice Chan	Female	New York	$200.00
4	Feb 14,2019	Tossman	Male	Seattle	$50.00
5	Feb 15,2019	Lisa	Female	Washington	$200.00
6	Apr 16,2019	Amay	Female	Los Angeles	$100.00
7	Apr 27,2019	Tossman	Male	Seattle	$100.00
8	Apr 28,2019	Lisa	Female	Washington	$50.00
9	May 09,2019	Smith	Male	Detroit	$200.00
10	May 11,2019	Tossman	Male	Seattle	$100.00
11	May 21,2019	Amay	Female	Los Angeles	$200.00
12	Jun 22,2019	Ice Chan	Female	New York	$100.00
13	Jun 23,2019	Tossman	Male	Seattle	$100.00
14	Jun 24,2019	Lisa	Female	Washington	$50.00
15	Jun 25,2019	Ice Chan	Female	New York	$150.00
16	Jun 26,2019	Tossman	Male	Seattle	$50.00
17	Jun 27,2019	Smith	Male	Detroit	$100.00
18	Jun 28,2019	Ice Chan	Female	New York	$50.00
19	Jun 29,2019	Tossman	Male	Seattle	$50.00
20	Jun 30,2019	Lisa	Female	Washington	$50.00

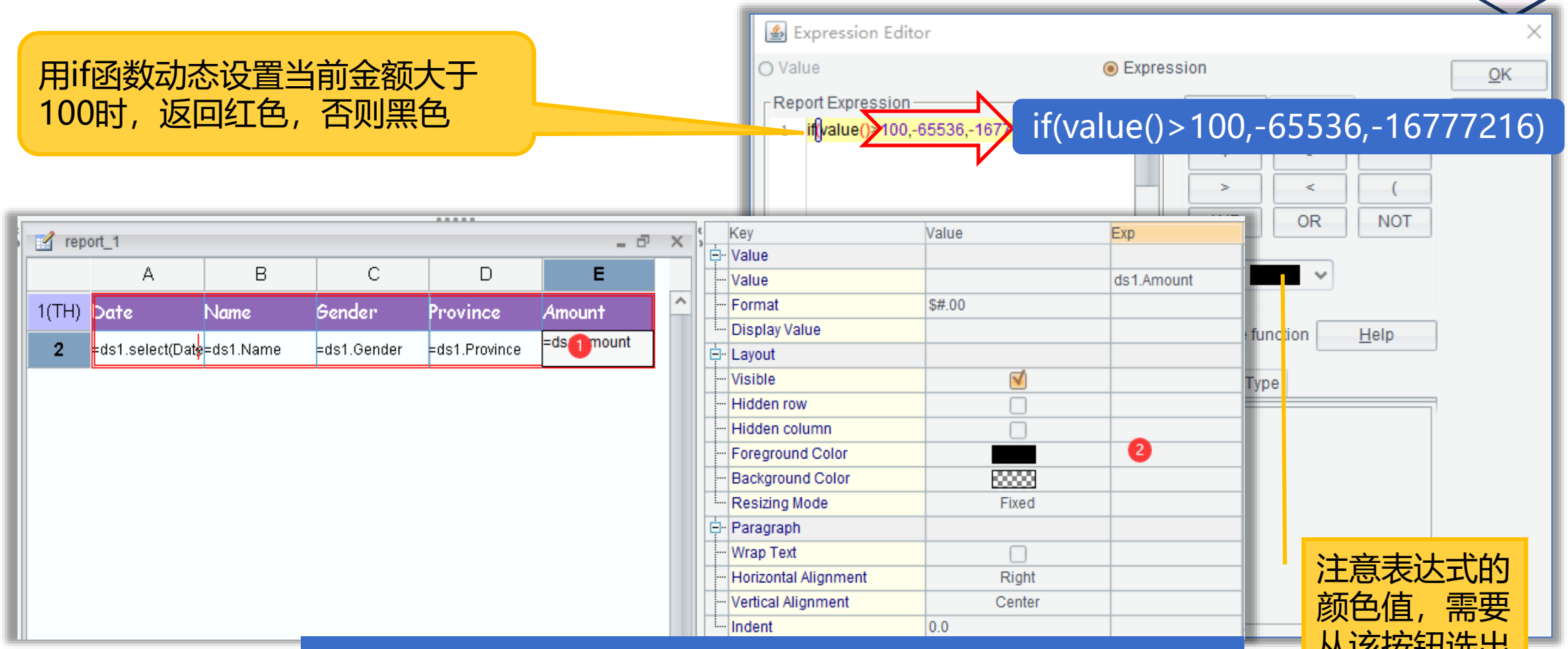
CONTENTS



2.3 条件格式

2.3 条件格式

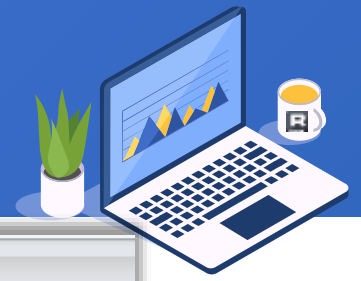
用if函数动态设置当前金额大于100时，返回红色，否则黑色



注意表达式的颜色值，需要从该按钮选出

在报表中将字段值大于某条件的数值醒目显示

2.3 条件格式



Preview report:report_1

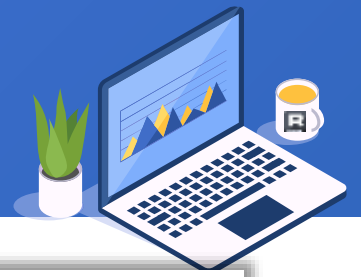
Date	Name	Gender	Province	Amount
Jan 12,2019	Lisa	Female	Washington	$100.00
Feb 08,2019	Ice Chan	Female	New York	$200.00
Feb 14,2019	Tossman	Male	Seattle	$50.00
Feb 15,2019	Lisa	Female	Washington	$200.00
Apr 16,2019	Amay	Female	Los Angeles	$100.00
Apr 27,2019	Tossman	Male	Seattle	$100.00
Apr 28,2019	Lisa	Female	Washington	$50.00
May 09,2019	Smith	Male	Detroit	$200.00
May 11,2019	Tossman	Male	Seattle	$100.00
May 21,2019	Amay	Female	Los Angeles	$200.00
Jun 22,2019	Ice Chan	Female	New York	$100.00
Jun 23,2019	Tossman	Male	Seattle	$100.00
Jun 24,2019	Lisa	Female	Washington	$50.00
Jun 25,2019	Ice Chan	Female	New York	$150.00
Jun 26,2019	Tossman	Male	Seattle	$50.00
Jun 27,2019	Smith	Male	Detroit	$100.00
Jun 28,2019	Ice Chan	Female	New York	$50.00
Jun 29,2019	Tossman	Male	Seattle	$50.00
Jun 30,2019	Lisa	Female	Washington	$50.00
Jul 01,2019	Amay	Female	Los Angeles	$200.00
Jul 02,2019	Tossman	Male	Seattle	$50.00

使用动态前景色设置预警颜色后对比效果

Preview report:report_1

Date	Name	Gender	Province	Amount
Jan 12,2019	Lisa	Female	Washington	$100.00
Feb 08,2019	Ice Chan	Female	New York	$200.00
Feb 14,2019	Tossman	Male	Seattle	$50.00
Feb 15,2019	Lisa	Female	Washington	$200.00
Apr 16,2019	Amay	Female	Los Angeles	$100.00
Apr 27,2019	Tossman	Male	Seattle	$100.00
Apr 28,2019	Lisa	Female	Washington	$50.00
May 09,2019	Smith	Male	Detroit	$200.00
May 11,2019	Tossman	Male	Seattle	$100.00
May 21,2019	Amay	Female	Los Angeles	$200.00
Jun 22,2019	Ice Chan	Female	New York	$100.00
Jun 23,2019	Tossman	Male	Seattle	$100.00
Jun 24,2019	Lisa	Female	Washington	$50.00
Jun 25,2019	Ice Chan	Female	New York	$150.00
Jun 26,2019	Tossman	Male	Seattle	$50.00
Jun 27,2019	Smith	Male	Detroit	$100.00
Jun 28,2019	Ice Chan	Female	New York	$50.00
Jun 29,2019	Tossman	Male	Seattle	$50.00
Jun 30,2019	Lisa	Female	Washington	$50.00
Jul 01,2019	Amay	Female	Los Angeles	$200.00
Jul 02,2019	Tossman	Male	Seattle	$50.00

2.3 条件格式

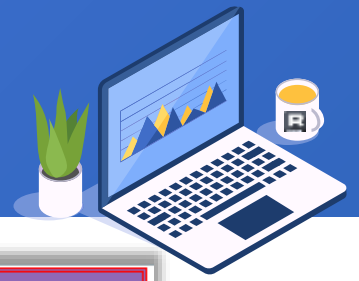




如图的if条件表达式，含义为隔行设置淡黄背景色，其中row()函数用于获取扩展格所在的行号

在报表中动态设置格子背景色

2.3 条件格式

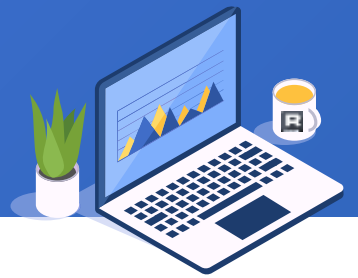


Date	Name	Gender	Province	Amount
Jan 12,2019	Lisa	Female	Washington	$100.00
Feb 08,2019	Ice Chan	Female	New York	$200.00
Feb 14,2019	Tossman	Male	Seattle	$50.00
Feb 15,2019	Lisa	Female	Washington	$200.00
Apr 16,2019	Amay	Female	Los Angeles	$100.00
Apr 27,2019	Tossman	Male	Seattle	$100.00
Apr 28,2019	Lisa	Female	Washington	$50.00
May 09,2019	Smith	Male	Detroit	$200.00
May 11,2019	Tossman	Male	Seattle	$100.00
May 21,2019	Amay	Female	Los Angeles	$200.00
Jun 22,2019	Ice Chan	Female	New York	$100.00
Jun 23,2019	Tossman	Male	Seattle	$100.00
Jun 24,2019	Lisa	Female	Washington	$50.00
Jun 25,2019	Ice Chan	Female	New York	$150.00
Jun 26,2019	Tossman	Male	Seattle	$50.00
Jun 27,2019	Smith	Male	Detroit	$100.00
Jun 28,2019	Ice Chan	Female	New York	$50.00
Jun 29,2019	Tossman	Male	Seattle	$50.00
Jun 30,2019	Lisa	Female	Washington	$50.00
Jul 01,2019	Amay	Female	Los Angeles	$200.00
Jul 02,2019	Tossman	Male	Seattle	$50.00

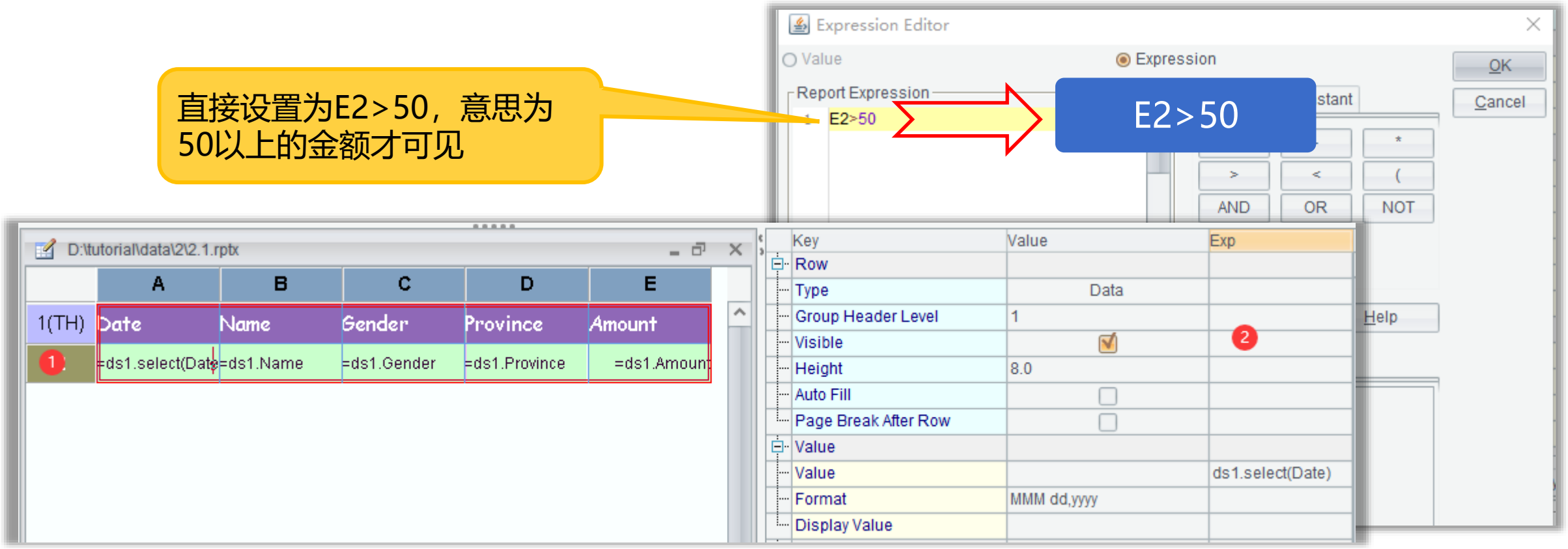
使用动态背景色设置间隔颜色后对比效果

Date	Name	Gender	Province	Amount
Jan 12,2019	Lisa	Female	Washington	$100.00
Feb 08,2019	Ice Chan	Female	New York	$200.00
Feb 14,2019	Tossman	Male	Seattle	$50.00
Feb 15,2019	Lisa	Female	Washington	$200.00
Apr 16,2019	Amay	Female	Los Angeles	$100.00
Apr 27,2019	Tossman	Male	Seattle	$100.00
Apr 28,2019	Lisa	Female	Washington	$50.00
May 09,2019	Smith	Male	Detroit	$200.00
May 11,2019	Tossman	Male	Seattle	$100.00
May 21,2019	Amay	Female	Los Angeles	$200.00
Jun 22,2019	Ice Chan	Female	New York	$100.00
Jun 23,2019	Tossman	Male	Seattle	$100.00
Jun 24,2019	Lisa	Female	Washington	$50.00
Jun 25,2019	Ice Chan	Female	New York	$150.00
Jun 26,2019	Tossman	Male	Seattle	$50.00
Jun 27,2019	Smith	Male	Detroit	$100.00
Jun 28,2019	Ice Chan	Female	New York	$50.00
Jun 29,2019	Tossman	Male	Seattle	$50.00
Jun 30,2019	Lisa	Female	Washington	$50.00
Jul 01,2019	Amay	Female	Los Angeles	$200.00
Jul 02,2019	Tossman	Male	Seattle	$50.00

2.3 条件格式

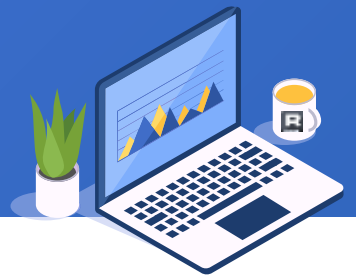


直接设置为E2>50，意思为50以上的金额才可见



在报表中动态设置行首格可视

2.3 条件格式

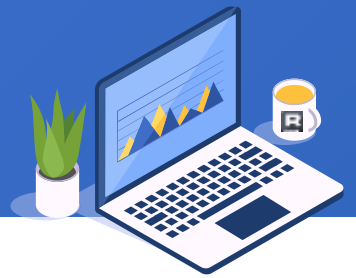


Date	Name	Gender	Province	Amount
Jan 12,2019	Lisa	Female	Washington	$100.00
Feb 08,2019	Ice Chan	Female	New York	$200.00
Feb 14,2019	Tossman	Male	Seattle	$50.00
Feb 15,2019	Lisa	Female	Washington	$200.00
Apr 16,2019	Amay	Female	Los Angeles	$100.00
Apr 27,2019	Tossman	Male	Seattle	$100.00
Apr 28,2019	Lisa	Female	Washington	$50.00
May 09,2019	Smith	Male	Detroit	$200.00
May 11,2019	Tossman	Male	Seattle	$100.00
May 21,2019	Amay	Female	Los Angeles	$200.00
Jun 22,2019	Ice Chan	Female	New York	$100.00
Jun 23,2019	Tossman	Male	Seattle	$100.00
Jun 24,2019	Lisa	Female	Washington	$50.00
Jun 25,2019	Ice Chan	Female	New York	$150.00
Jun 26,2019	Tossman	Male	Seattle	$50.00
Jun 27,2019	Smith	Male	Detroit	$100.00
Jun 28,2019	Ice Chan	Female	New York	$50.00
Jun 29,2019	Tossman	Male	Seattle	$50.00
Jun 30,2019	Lisa	Female	Washington	$50.00
Jul 01,2019	Amay	Female	Los Angeles	$200.00
Jul 02,2019	Tossman	Male	Seattle	$50.00

设置小额金额隐藏后效果：

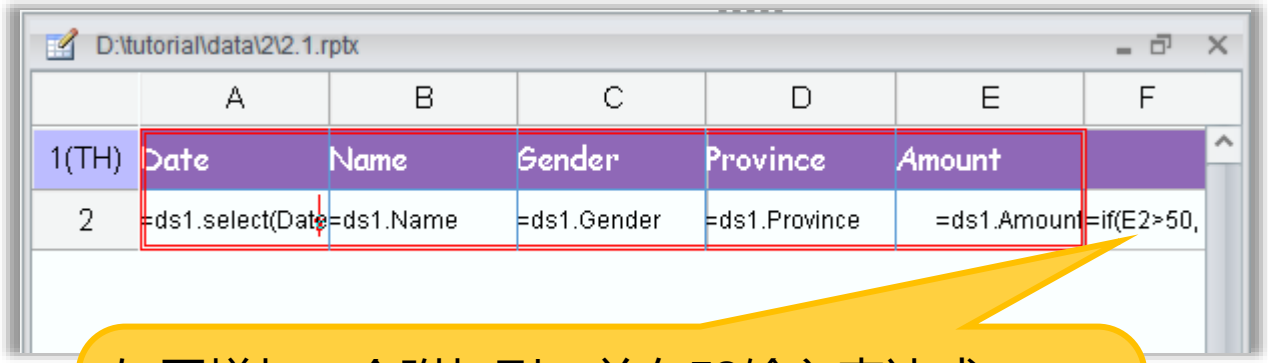
Date	Name	Gender	Province	Amount
Jan 12,2019	Lisa	Female	Washington	$100.00
Feb 08,2019	Ice Chan	Female	New York	$200.00
Feb 15,2019	Lisa	Female	Washington	$200.00
Apr 16,2019	Amay	Female	Los Angeles	$100.00
Apr 27,2019	Tossman	Male	Seattle	$100.00
May 09,2019	Smith	Male	Detroit	$200.00
May 11,2019	Tossman	Male	Seattle	$100.00
May 21,2019	Amay	Female	Los Angeles	$200.00
Jun 22,2019	Ice Chan	Female	New York	$100.00
Jun 23,2019	Tossman	Male	Seattle	$100.00
Jun 25,2019	Ice Chan	Female	New York	$150.00
Jun 27,2019	Smith	Male	Detroit	$100.00
Jul 01,2019	Amay	Female	Los Angeles	$200.00

2.3 条件格式



上一页中，正常的行间间隔色，因为有些行被隐藏了，结果间隔色乱了。

我们用附加列重新计算出隐藏小额金额后的行序号。

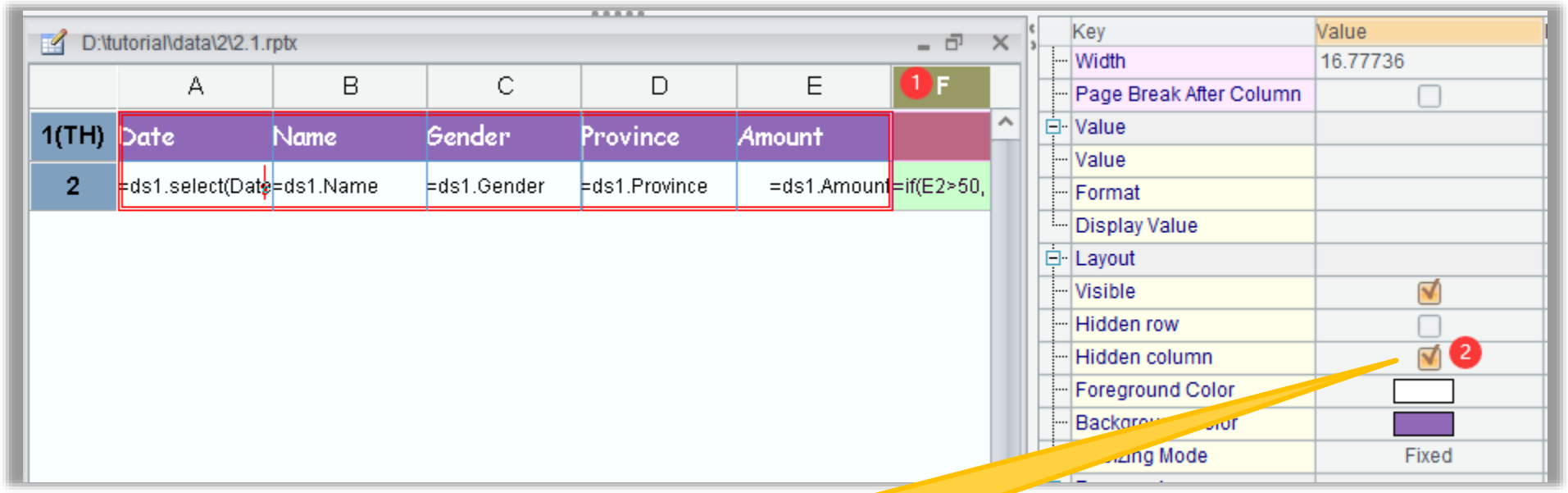


如图增加一个附加列，并在F2输入表达式“=if(E2>50, F2[-1]+1, F2[-1])”，根据过滤条件重新计算行号。该表达式用到了位移坐标，详细用法参考第6章 格间计算

使用附加列表达式重新计算的行号：

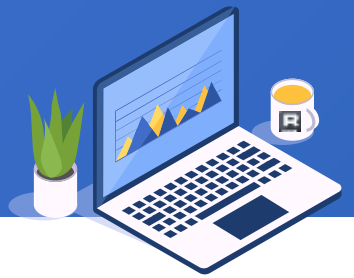
Date	Name	Gender	Province	Amount	
Jan 12,2019	Lisa	Female	Washington	$100.00	1
Feb 08,2019	Ice Chan	Female	New York	$200.00	2
Feb 15,2019	Lisa	Female	Washington	$200.00	3
Apr 16,2019	Amay	Female	Los Angeles	$100.00	4
Apr 27,2019	Tossman	Male	Seattle	$100.00	5
May 09,2019	Smith	Male	Detroit	$200.00	6
May 11,2019	Tossman	Male	Seattle	$100.00	7
May 21,2019	Amay	Female	Los Angeles	$200.00	8
Jun 22,2019	Ice Chan	Female	New York	$100.00	9
Jun 23,2019	Tossman	Male	Seattle	$100.00	10
Jun 25,2019	Ice Chan	Female	New York	$150.00	11
Jun 27,2019	Smith	Male	Detroit	$100.00	12
Jul 01,2019	Amay	Female	Los Angeles	$200.00	13

2.3 条件格式



然后如图示顺序，将多余的附加列F隐藏掉

2.3 条件格式



最后重新调整第2行所有格子的背景色表达式为： if(F2%2==0,-52,-1)

再次预览报表：

Date	Name	Gender	Province	Amount
Jan 12,2019	Lisa	Female	Washington	$100.00
Feb 08,2019	Ice Chan	Female	New York	$200.00
Feb 15,2019	Lisa	Female	Washington	$200.00
Apr 16,2019	Amay	Female	Los Angeles	$100.00
Apr 27,2019	Tossman	Male	Seattle	$100.00
May 09,2019	Smith	Male	Detroit	$200.00
May 11,2019	Tossman	Male	Seattle	$100.00
May 21,2019	Amay	Female	Los Angeles	$200.00
Jun 22,2019	Ice Chan	Female	New York	$100.00
Jun 23,2019	Tossman	Male	Seattle	$100.00
Jun 25,2019	Ice Chan	Female	New York	$150.00
Jun 27,2019	Smith	Male	Detroit	$100.00
Jul 01,2019	Amay	Female	Los Angeles	$200.00

以及导出为Excel效果：

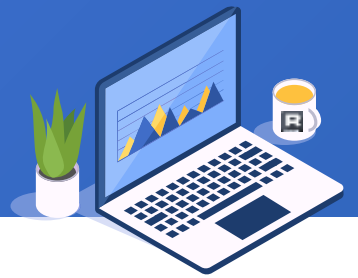
	A	B	C	D	E
1	Date	Name	Gender	Province	Amount
2	Jan 12,2019	Lisa	Female	Washington	$100.00
3	Feb 08,2019	Ice Chan	Female	New York	$200.00
5	Feb 15,2019	Lisa	Female	Washington	$200.00
6	Apr 16,2019	Amay	Female	Los Angeles	$100.00
7	Apr 27,2019	Tossman	Male	Seattle	$100.00
9	May 09,2019	Smith	Male	Detroit	$200.00
10	May 11,2019	Tossman	Male	Seattle	$100.00
11	May 21,2019	Amay	Female	Los Angeles	$200.00
12	Jun 22,2019	Ice Chan	Female	New York	$100.00
13	Jun 23,2019	Tossman	Male	Seattle	$100.00
15	Jun 25,2019	Ice Chan	Female	New York	$150.00
17	Jun 27,2019	Smith	Male	Detroit	$100.00
21	Jul 01,2019	Amay	Female	Los Angeles	$200.00

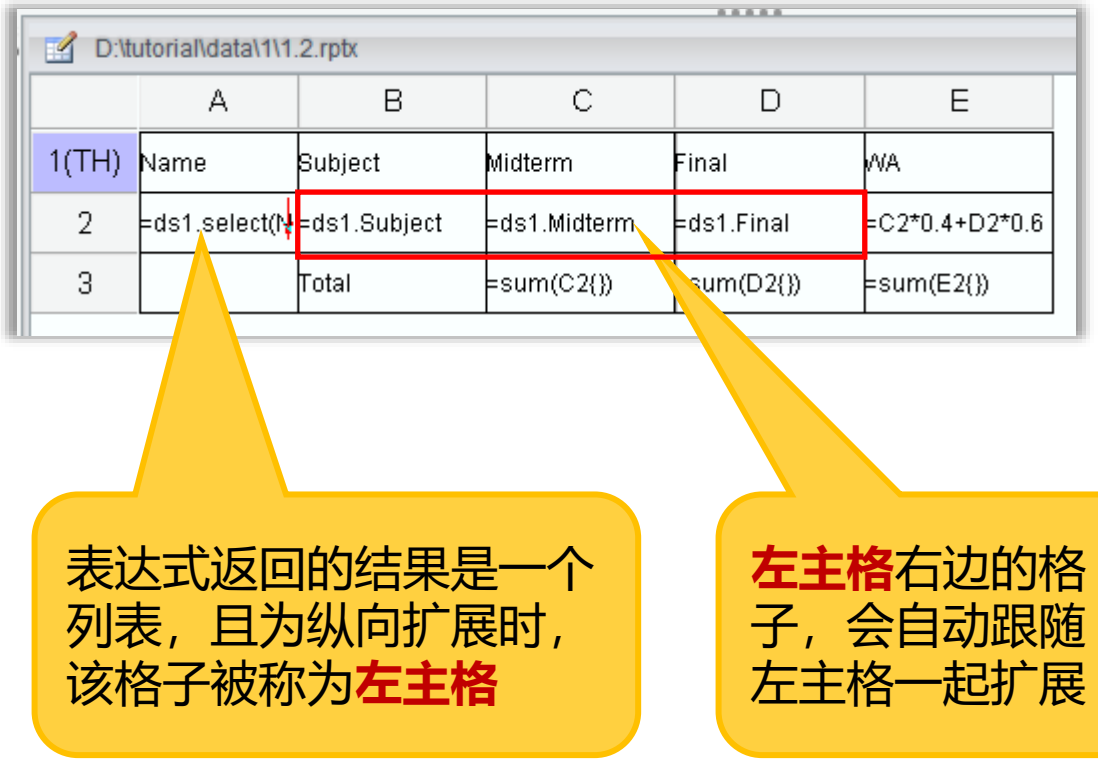
CONTENTS



2.4 卡片报表

2.4 卡片报表

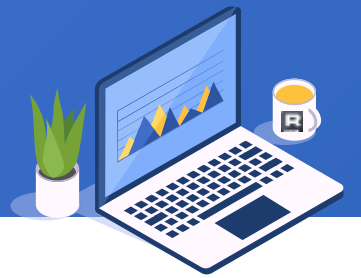




回顾前文例子，A2格选出的是一个姓名列表，然后这个列表会在展现报表时扩展出多行。而科目和相应成绩会自动跟随扩展出对应数据。

左主格的概念

2.4 卡片报表



D:\tutorial\data\2\2.4.rptx

	A	B	C	D
1	Student Information			
2	ID	=ds1.select	photo	
3	Name	=ds1.Name		
4	Gender	=ds1.Gende		
5	Age	=ds1.Age		
6	Address			=ds1.Address
7				

浏览报表

Preview report:D:\tutorial\data\2\2.4.rptx

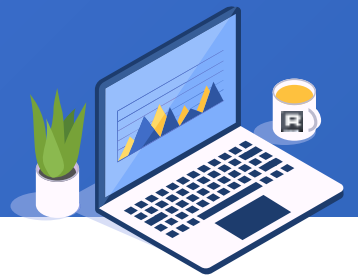
Student Information

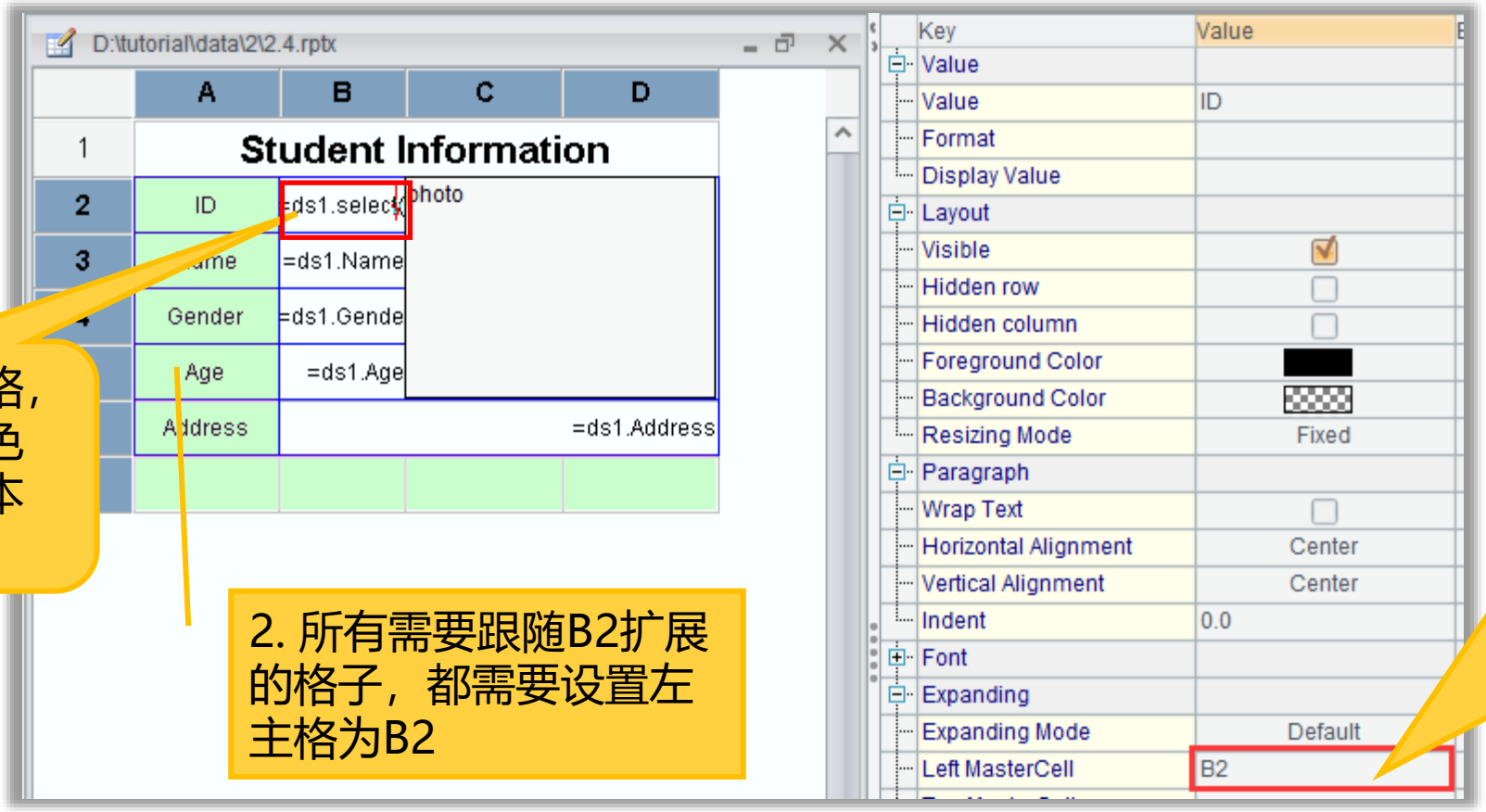
ID	1	photo
ID	2	
ID	3	
ID	4	
ID	5	
Name	Roddy	
Gender	0	
Age	17	
Address	30830 Orchard Lake Road	

依照卡片样式编辑好上述卡片报表

预览报表后，并不是期望的每个学生一张卡片，仅仅是编号扩展了，其他行并没有跟着扩展

2.4 卡片报表





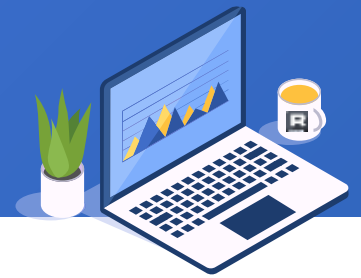
1. 找到左主格，通常是自带红色箭头的格子，本例为B2

2. 所有需要跟随B2扩展的格子，都需要设置左主格为B2

3. 格子会自动跟随左边设置了左主格的格子扩展，所以本例只要设置A2到A7以及C2的左主格为B2

设置卡片报表格子的左主格

2.4 卡片报表



Preview report:D:\tutorial\data\2\2.4.rptx

Student Information

ID	1	photo
ID	2	
ID	3	
ID	4	
ID	5	
Name	Roddy	
Gender	0	
Age	17	
Address	30830 Orchard Lake Road	

设置了格子主格后运行的对比效果

Preview report:D:\tutorial\data\2\2.4.rptx

Student Information

ID	1	photo
Name	Roddy	
Gender	0	
Age	17	
Address	30830 Orchard Lake Road	

ID	2	photo
Name	Lisa	
Gender	1	
Age	16	
Address	120 Wall St. 22nd Floor New York, NY	

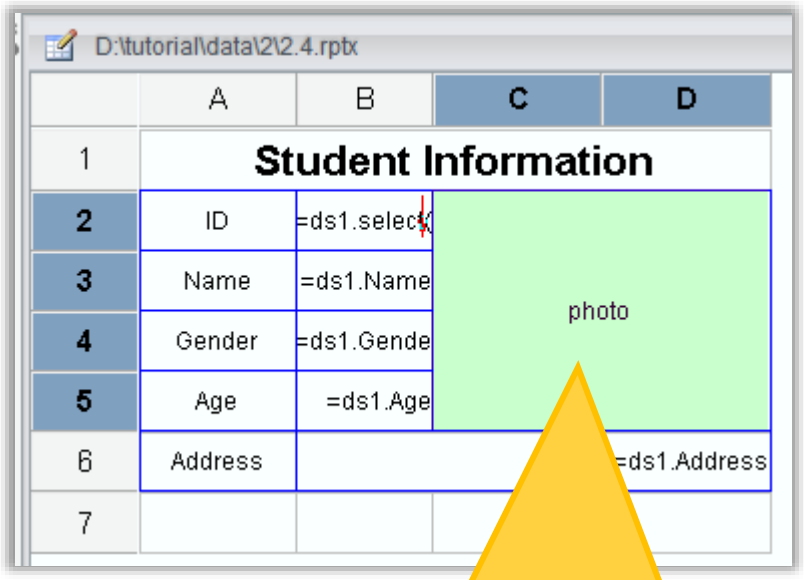
CONTENTS



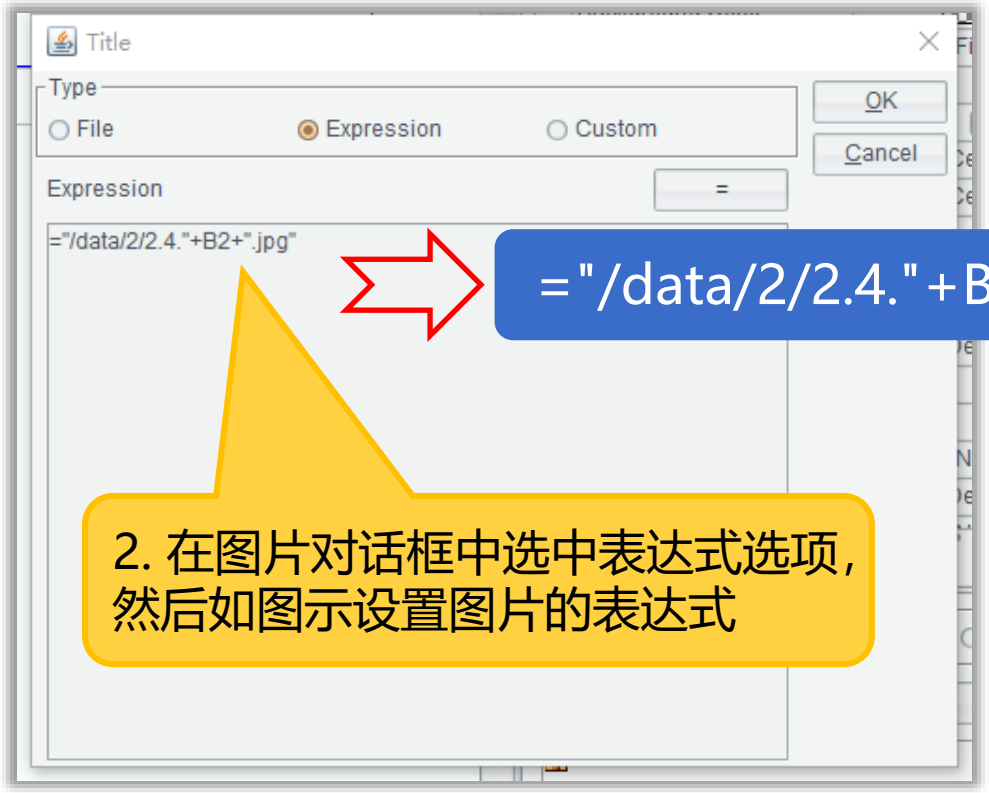
2.5 图片

2.5 图片

例子2.4中的卡片报表，相片需要用图片类型来显示。



1. 选中格子C2，右击鼠标弹出菜单，然后单击”图片”菜单项

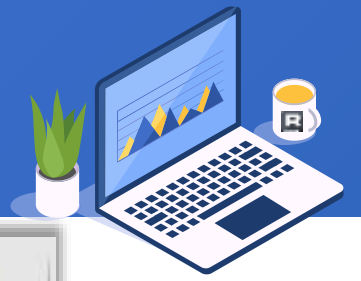


2. 在图片对话框中选中表达式选项，然后如图示设置图片的表达式

="/data/2/2.4."+B2+".jpg"

设置报表格子的动态图片

2.5 图片



Preview report:D:\tutorial\data\2\2.4.rptx

Student Information

ID	1
Name	Roddy
Gender	0
Age	17
Address	30830 Orchard Lake Road

ID	2
Name	Lisa
Gender	1
Age	16
Address	120 Wall St. 22nd Floor New York, NY

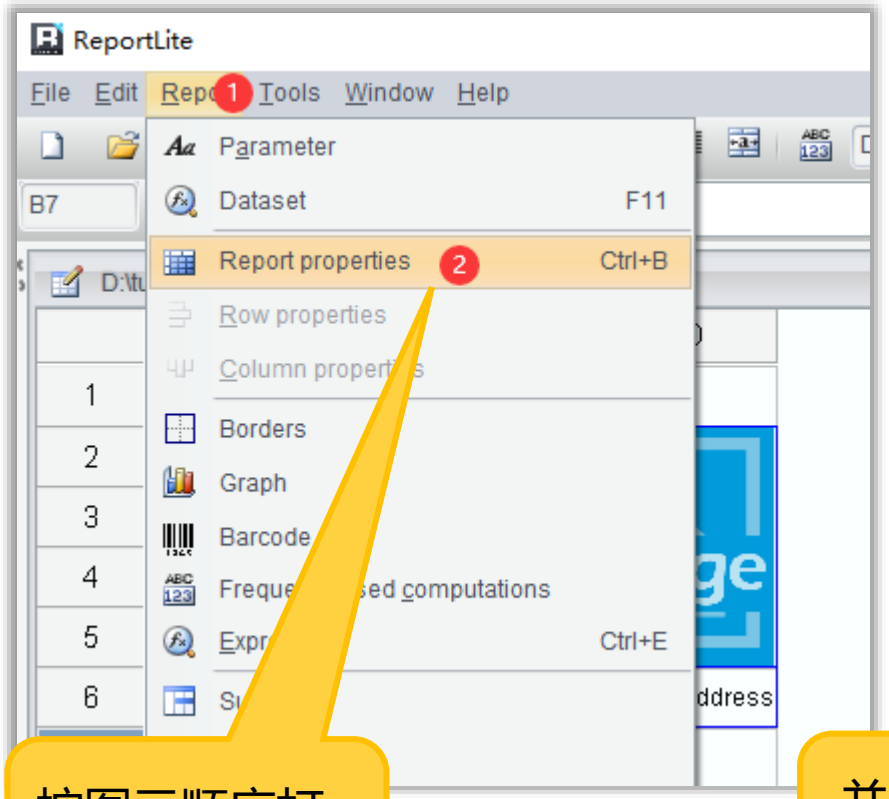
ID	3
Name	Shaw
Gender	0
Age	15
Address	4114 Sepulveda Blvd Culver City，CA

设置了动态图片的报表预览效果

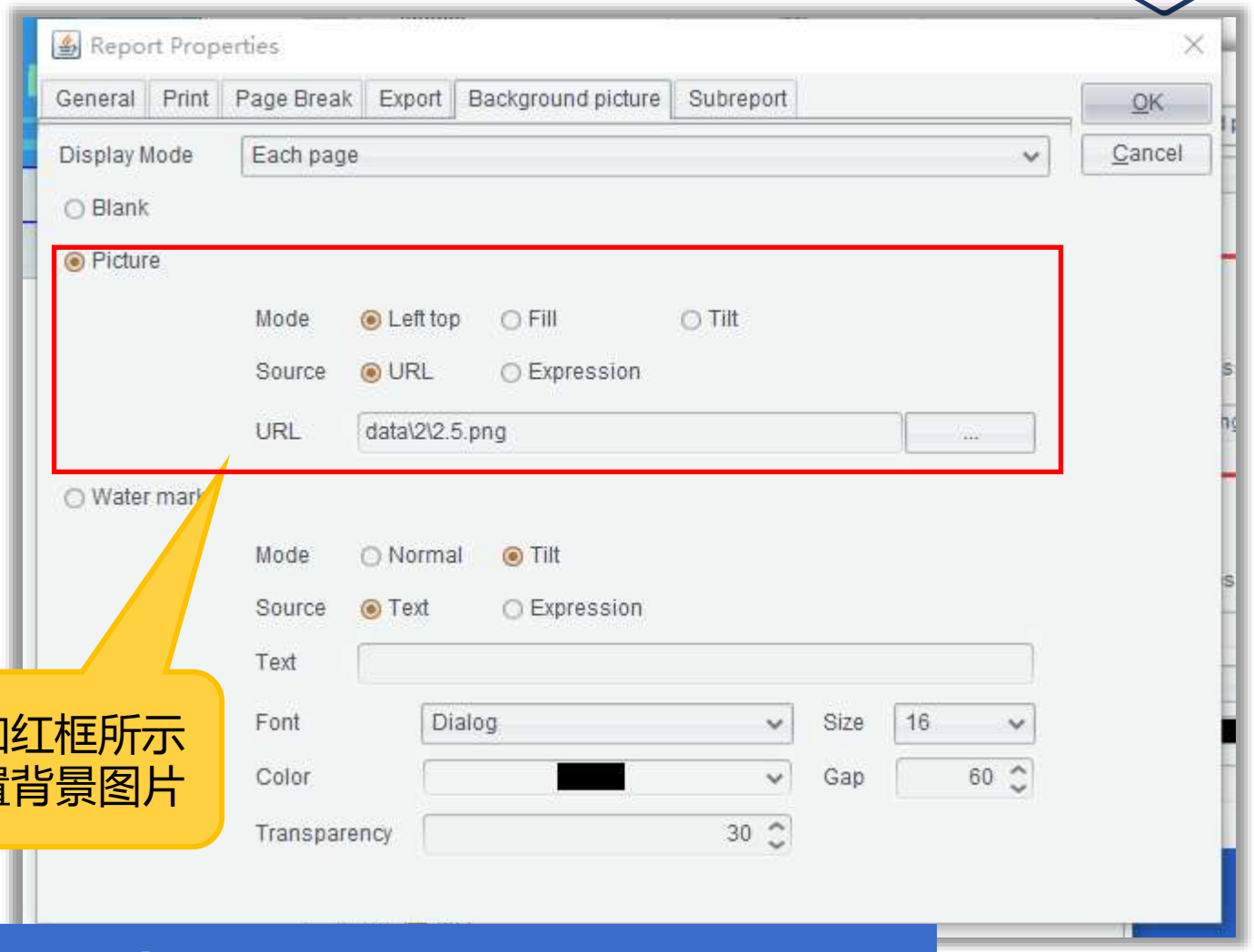
以及报表导出到Excel的效果

	A	B	C	D
1	**Student Information**			
2	ID	1		
3	Name	Roddy		
4	Gender	0		
5	Age	17		
6	Address	30830 Orchard Lake Road		
7				
8	ID	2		
9	Name	Lisa		
10	Gender	1		
11	Age	16		
12	Address	120 Wall St. 22nd Floor New York, NY		
13				
14	ID	3		
15	Name	Shaw		
16	Gender	0		
17	Age	15		
18	Address	4114 Sepulveda Blvd Culver City，CA		
19				

2.5 图片



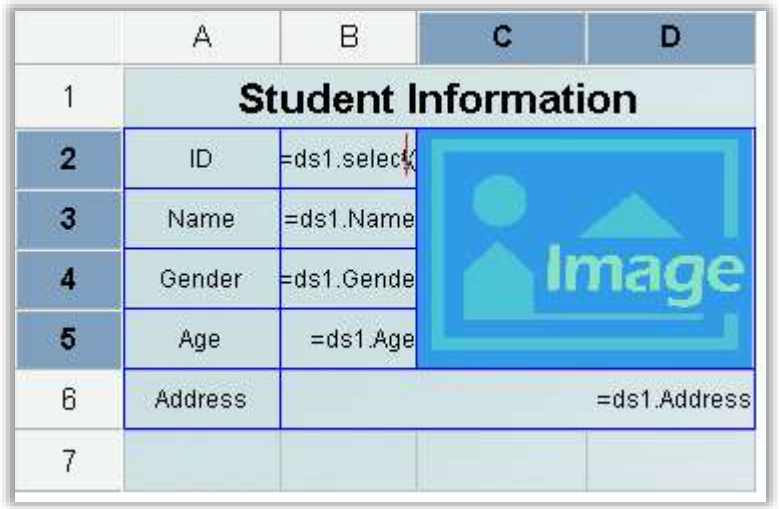
按图示顺序打开报表属性



并如红框所示设置背景图片

设置报表的背景图

2.5 图片



设置了报表背景
图后运行的效果



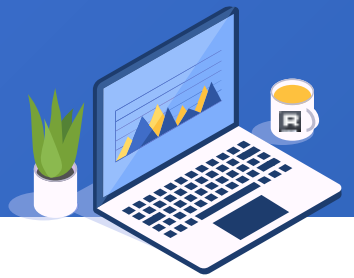
注意：
设置了背景图后，格子的背景色会自动设为透明

CONTENTS



2.6 习题

2.6 习题1



根据例子数据data1.2.xlsx，用红绿背景将期末数学成绩是否通过，明显区分出来，报表样式如下：

	A	B	C
1	Math Score		
2	Name	Final	Status
3	Rose	80	Pass
4	Mike	77	Pass
5	Ronald	50	Make-up
6	Smith	55	Make-up
7	Frank	80	Pass
8	Aimay	50	Make-up
9	James	67	Pass
10	Jay	56	Make-up
11	Average	64.375	

第3章
分组报表



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3.1 使用向导制作分组报表

3.2 常见分组报表的制作

3.3 分组报表的结构和主格

3.4 分组报表的运算

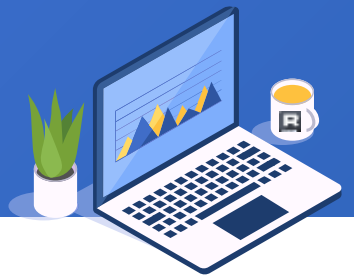
3.5 习题

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3.1 使用向导制作分组报表

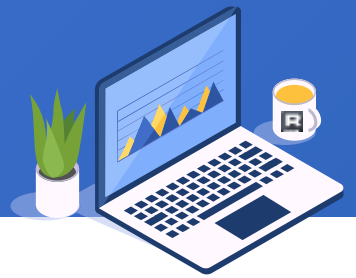
3.1 使用向导制作分组报表 - 数据源



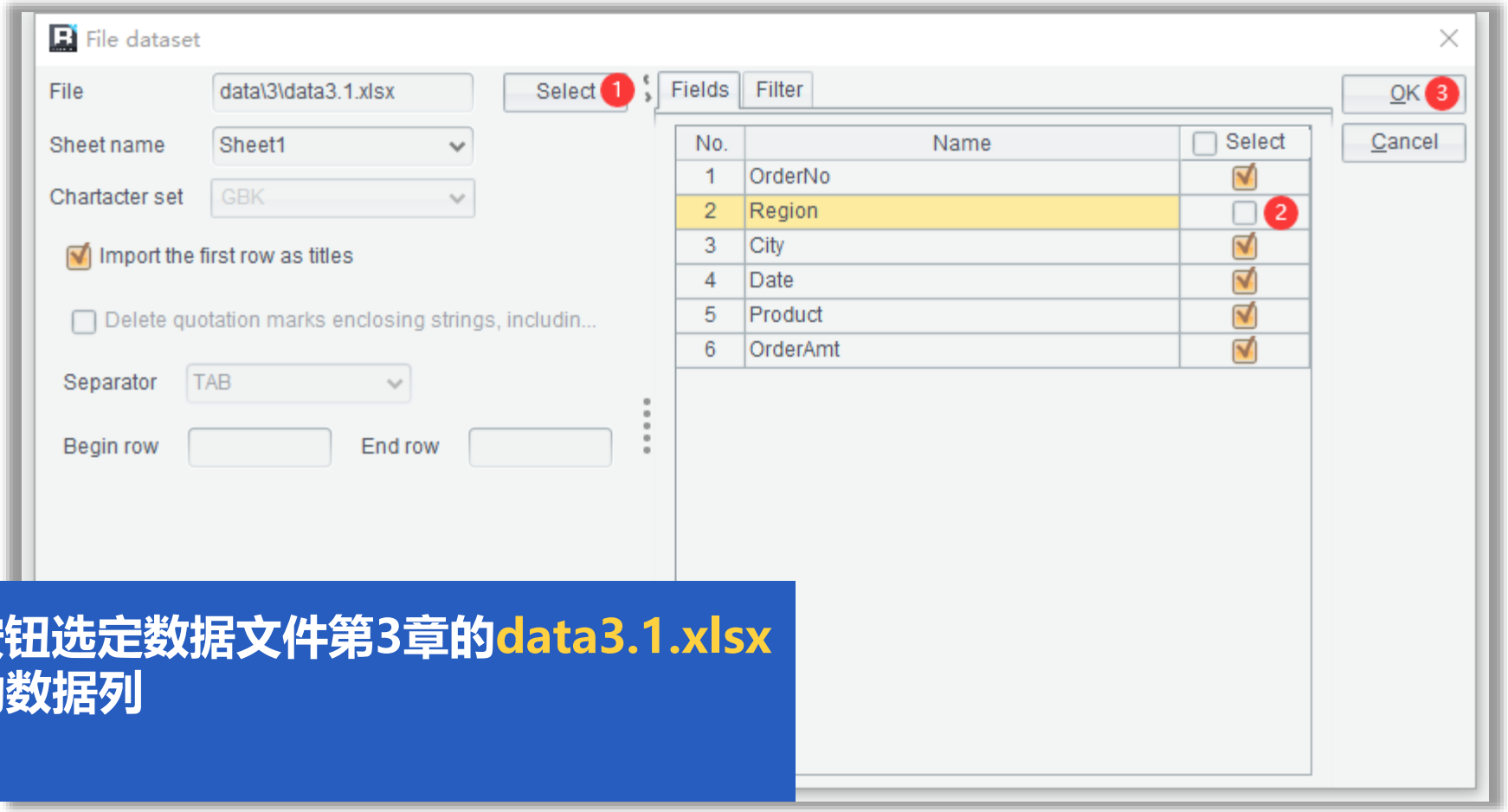
现有某公司销售订单数据表 – 例子数据 data3.1.xlsx

	A	B	C	D	E	F
1	OrderNo	Region	City	Date	Product	OrderAmt
2	1001	NE	Dalian	2018-01-02	Apple	189.73
3	1002	N	Beijing	2018-01-02	Peach	203.96
4	1003	NW	Lanzhou	2018-01-02	Banana	208.09
5	1004	NW	Xining	2018-01-02	Banana	179.19
6	1005	NW	Lanzhou	2018-01-03	Apple	342.06
7	1006	N	Beijing	2018-01-03	Apple	69.19
8	1007	NE	Shenyang	2018-01-03	Apple	231.44
9	1008	N	Tianjin	2018-01-03	Apple	256.15
10	1009	NW	Xining	2018-01-04	Peach	52.28
11	1010	NS	Guiyang	2018-01-04	Apple	290.31
12	1011	N	Shijiazhuang	2018-01-04	Banana	201.71
13	1012	NE	Changchun	2018-01-04	Peach	267.02
14	1013	NE	Shenyang	2018-01-05	Banana	83.34
15	1014	NE	Dalian	2018-01-05	Banana	204.73
16	1015	NE	Dalian	2018-01-05	Peach	229.84
17	1016	NW	Lanzhou	2018-01-05	Banana	57.38
18	1017	NW	Lanzhou	2018-01-06	Banana	232.57
19	1018	NS	Kunming	2018-01-06	Banana	262.99
20	1019	NW	Xining	2018-01-06	Apple	256.56

3.1 使用向导制作分组报表 – 简单分组

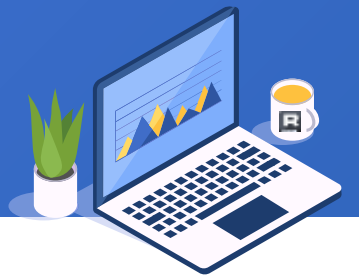


运行报表设计器，单击菜单“File”、“Open dataset” “File”，打开数据文件设置界面：

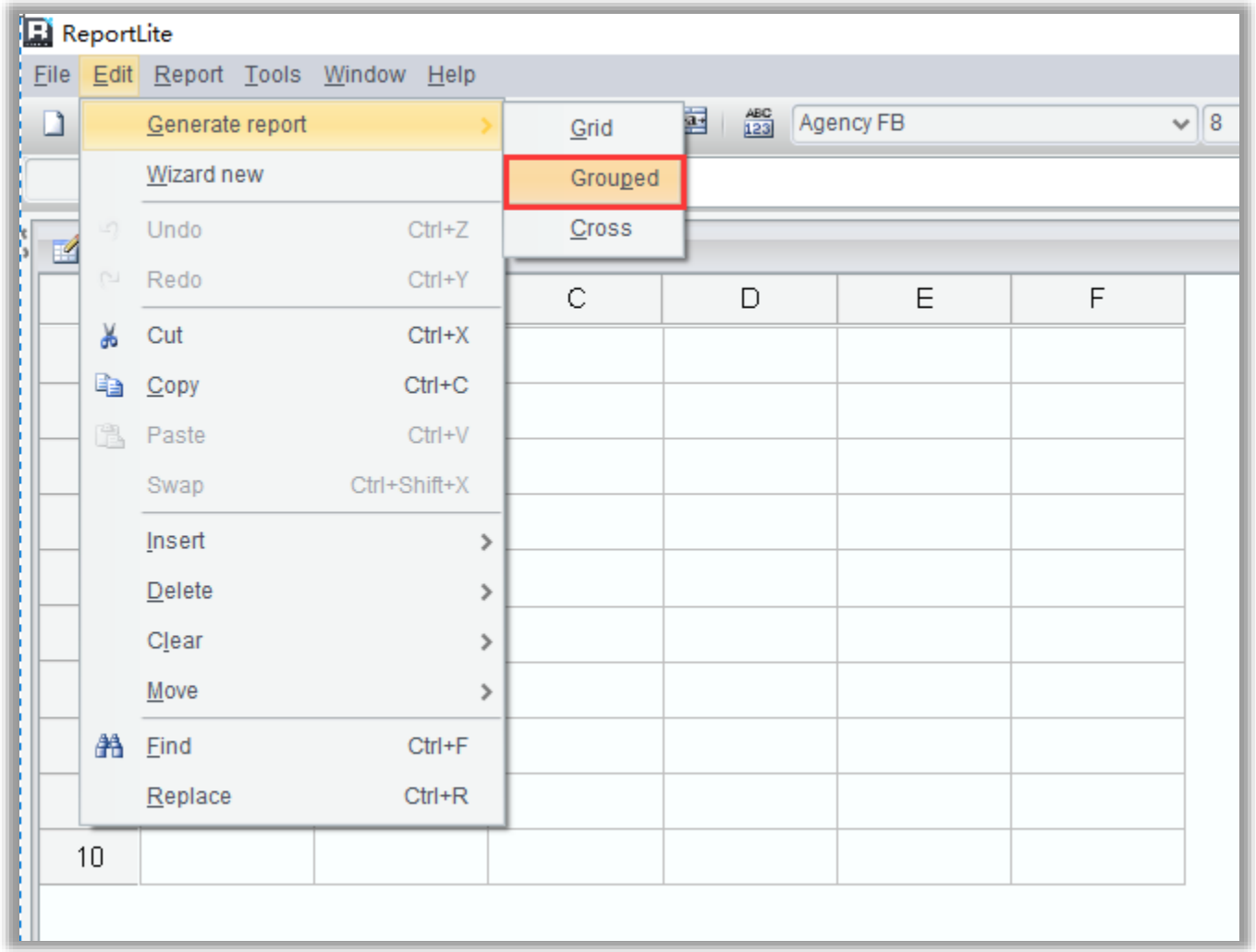


1、点击Select按钮选定数据文件第3章的data3.1.xlsx
2、去掉不需要的数据列
3、点击OK按钮

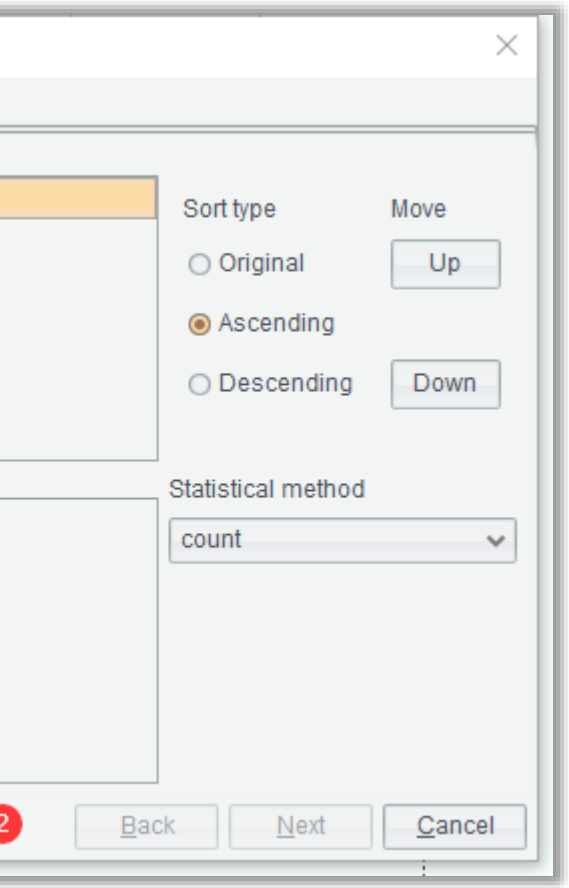
3.1 使用向导制作分组报表 – 简单分组



新建空白报表后，再如图菜单项，打开分组报表向导



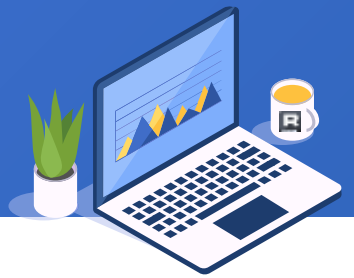
3.1 使用向导制作分组报表 – 简单分组



1、将City拖入分组定义，并确认按照升序排序

2、点击Create grouped report按钮

3.1 使用向导制作分组报表 – 简单分组



向导创建好了如下图所示的分组报表，设置各单元格为居中对齐方式，删掉多余行列后，将设计好的报表保存为3.1.rptx

D:\tutorial\data\3\3.1.rptx

	A	B	C	D	E
1	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(City	=ds1.select(Or	=ds1.Date	=ds1.Product	=ds1.OrderAmt

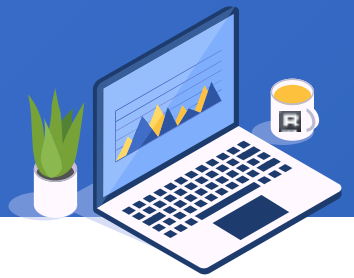
A2单元格中表达式为：=ds1.group(City;City:1)

这里group函数表示对数据集ds1按照City字段进行分组，城市相同的记录分为同一组。

第二个参数“City:1"表示按城市排序，1为升序，-1为降序

B2-E2的表达式与第1章介绍的行式报表类似，将同一组的数据记录进行纵向扩展

3.1 使用向导制作分组报表 – 简单分组



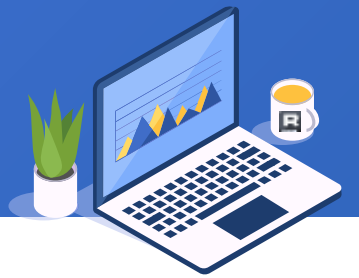
点击工具栏中的预览按钮，结果如下图所示

分组，同组合并为一格

Preview report:D:\tutorial\data\3\3.1.rptx

City	OrderNo	Date	Product	OrderAmt
Beijing	1002	2018-01-02	Peach	203.96
	1006	2018-01-03	Apple	69.19
	1021	2018-01-07	Apple	96.53
	1026	2018-01-08	Apple	325.56
	1040	2018-01-11	Banana	243.77
Changchun	1012	2018-01-04	Peach	267.02
	1038	2018-01-11	Peach	53.57
Chengdu	1027	2018-01-08	Banana	121.32
	1034	2018-01-10	Peach	311.34
Dalian	1001	2018-01-02	Apple	189.73
	1014	2018-01-05	Banana	204.73
	1015	2018-01-05	Peach	229.84

3.1 使用向导制作分组报表 – 简单分组

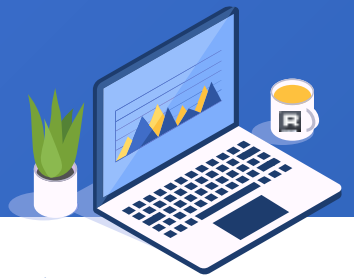


导出Excel，结果如下图所示

分组，同组合并为一格

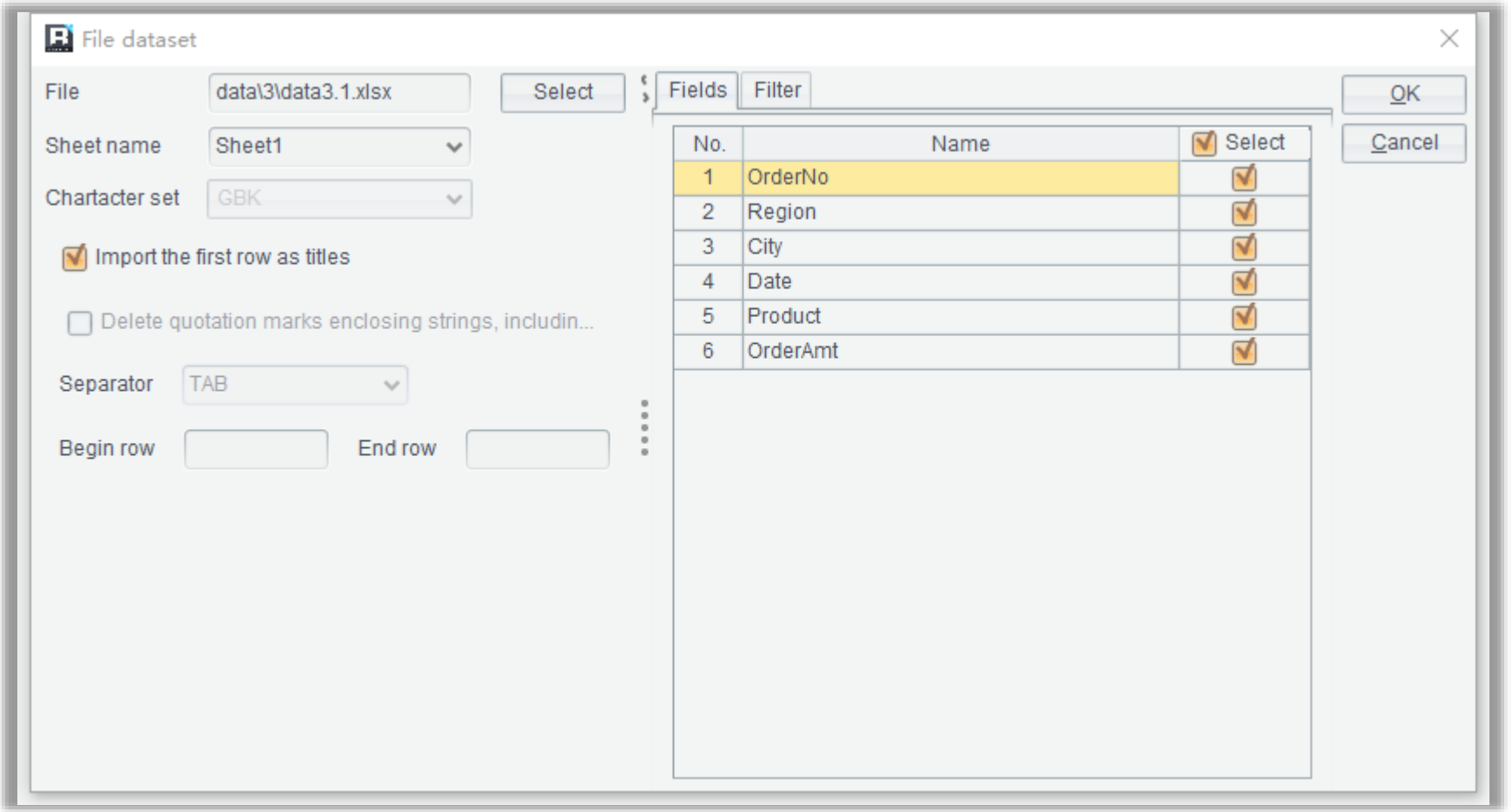
	A	B	C	D	E
1	City	OrderNo	Date	Product	OrderAmt
2	Beijing	1002	2018-01-02	Peach	203.96
3		1006	2018-01-03	Apple	69.19
4		1021	2018-01-07	Apple	96.53
5		1026	2018-01-08	Apple	325.56
6		1040	2018-01-11	Banana	243.77
7	Changchun	1012	2018-01-04	Peach	267.02
8		1038	2018-01-11	Peach	53.57
9	Chengdu	1027	2018-01-08	Banana	121.32
10		1034	2018-01-10	Peach	311.34
11	Dalian	1001	2018-01-02	Apple	189.73
12		1014	2018-01-05	Banana	204.73
13		1015	2018-01-05	Peach	229.84
14	Guiyang	1010	2018-01-04	Apple	290.31
15		1028	2018-01-08	Peach	228.96
16		1033	2018-01-10	Banana	173.26

3.1 使用向导制作分组报表 – 多层分组

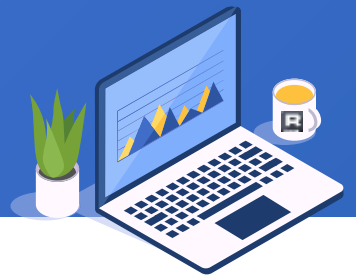


前面的分组报表是按照一个字段分组，称为单层分组。有时需要按照多个字段来分组，称为多层分组。

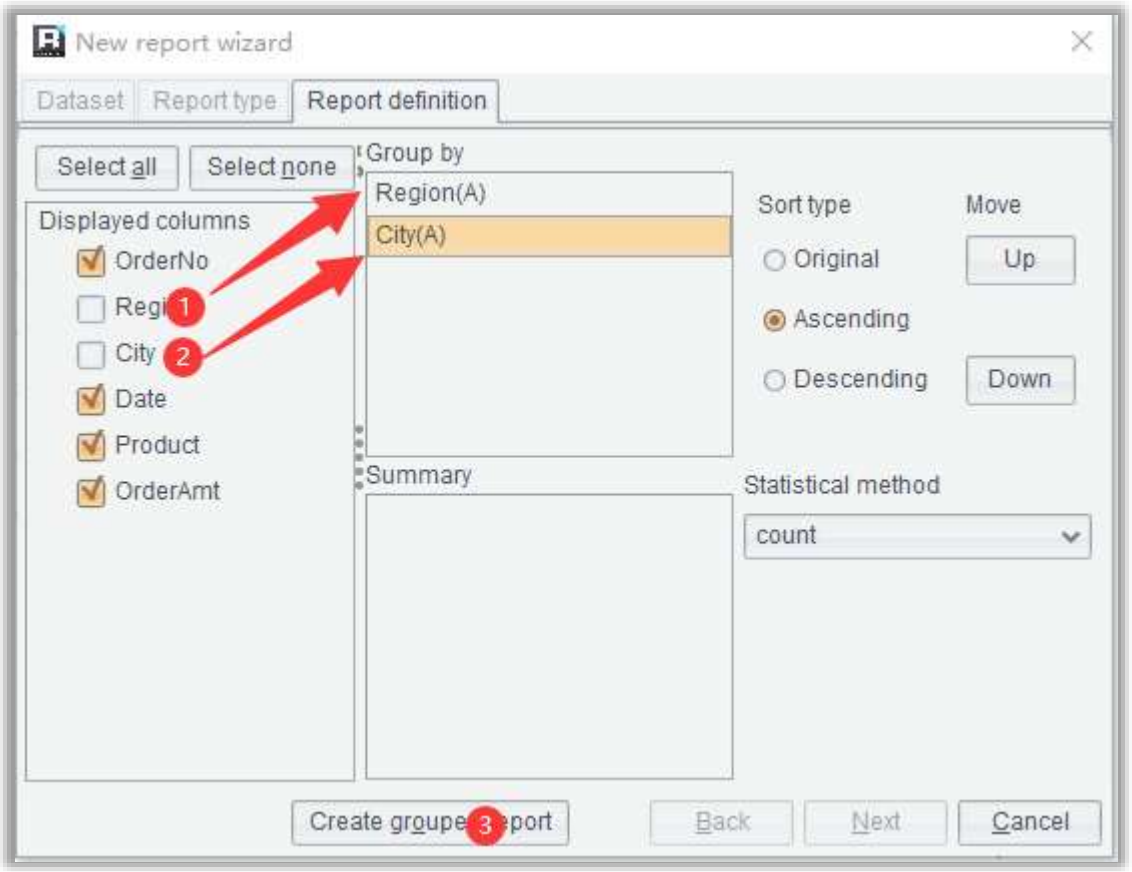
比如将这个订单表按照地区和城市进行分组。重新打开文件数据集，默认选出所有字段，点击OK按钮



3.1 使用向导制作分组报表 – 多层分组



单击同样的菜单项Grouped，打开分组报表向导



1、将地区Region拖入分组定义，默认升序

2、将城市City拖入分组定义，默认升序

3、点击生成分组报表Create grouped report按钮

3.1 使用向导制作分组报表 – 多层分组



向导创建好了如下图所示的分组报表，设置各单元格为居中对齐方式，删除多余行列，修改首行类型为Report header(TH)后，将设计好的报表保存为3.2.rptx

D:\tutorial\data\3\3.2.rptx

	A	B	C	D	E	F
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(R	=ds1.group(Ci	=ds1.select(O	=ds1.Date	=ds1.Product	=ds1.OrderAmt

A2表达式为：=ds1.group(Region;Region:1)

B2表达式为：=ds1.group(City;City:1)

报表预览如右图所示

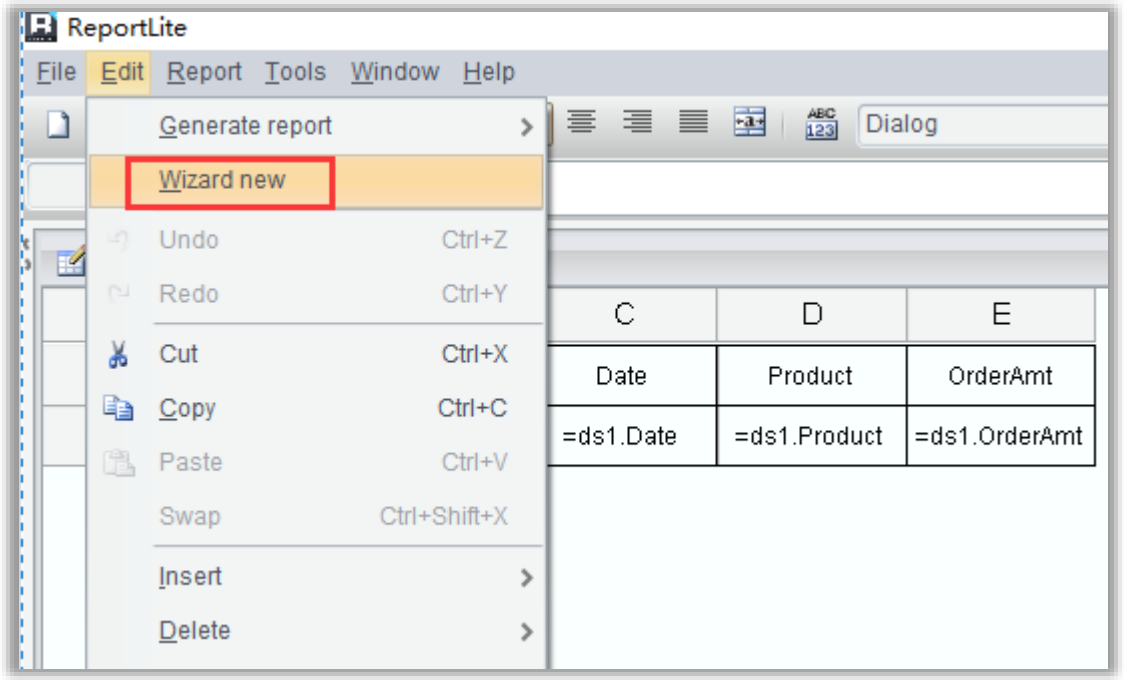
Region	City	OrderNo	Date	Product	OrderAmt
N	Beijing	1002	2018-01-02	Peach	203.96
		1006	2018-01-03	Apple	69.19
		1021	2018-01-07	Apple	96.53
		1026	2018-01-08	Apple	325.56
		1040	2018-01-11	Banana	243.77
	Shijiazhuang	1011	2018-01-04	Banana	201.71
		1022	2018-01-07	Apple	127.97
	Tianjin	1008	2018-01-03	Apple	256.15
		1023	2018-01-07	Banana	319.83
		1031	2018-01-09	Banana	61.1
NE	Changchun	1012	2018-01-04	Peach	267.02
		1038	2018-01-11	Peach	53.57
	Dalian	1001	2018-01-02	Apple	189.73
		1014	2018-01-05	Banana	204.73
		1015	2018-01-05	Peach	229.84

3.1 使用向导制作分组报表 – 汇总统计



对数据进行分组，一般是为了在组内进行数据汇总统计。如何给分组报表增加汇总统计呢？

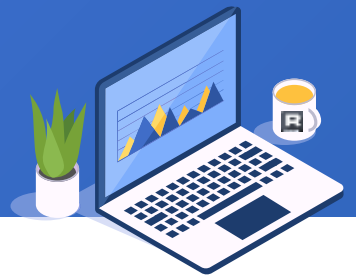
再打开报表3.1.rptx，然后单击如图菜单项Wizard new:



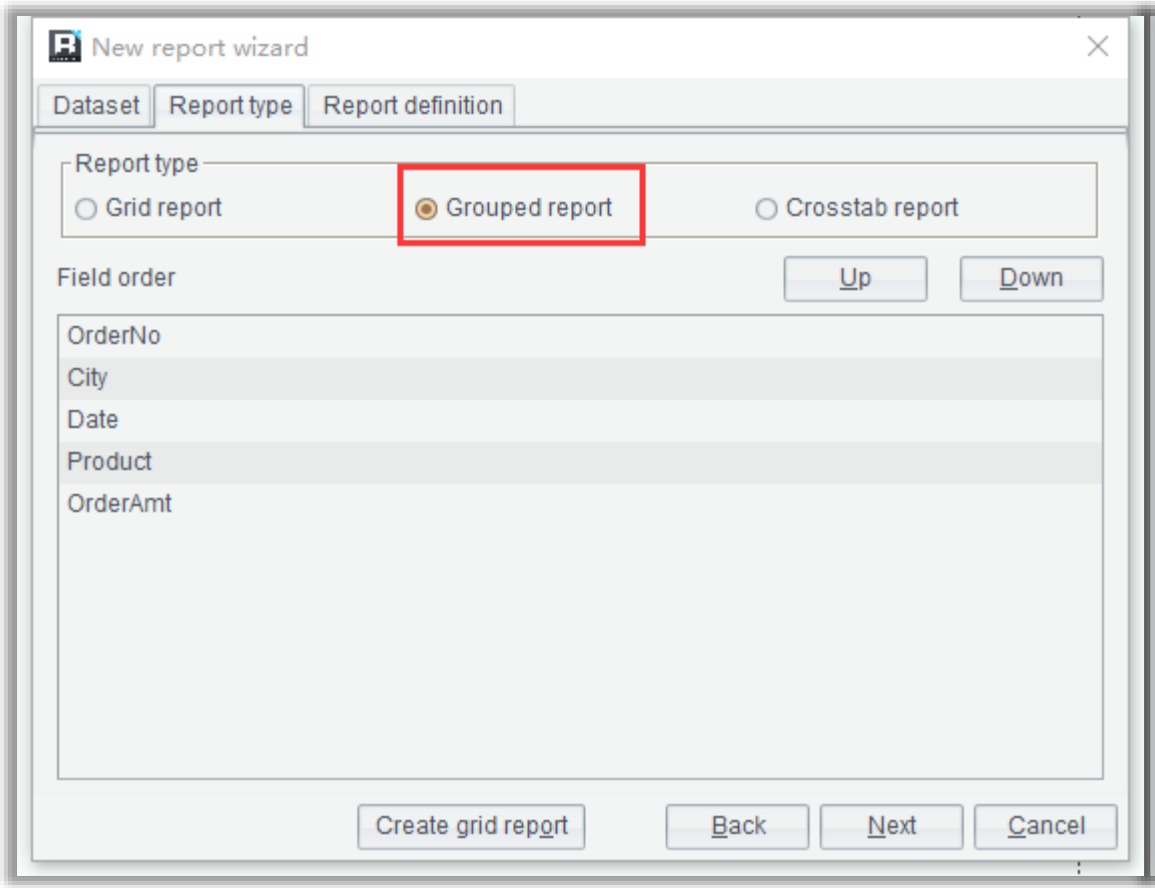
使用Wizard new可以复用当前报表的数据集，使用向导界面产生新报表。

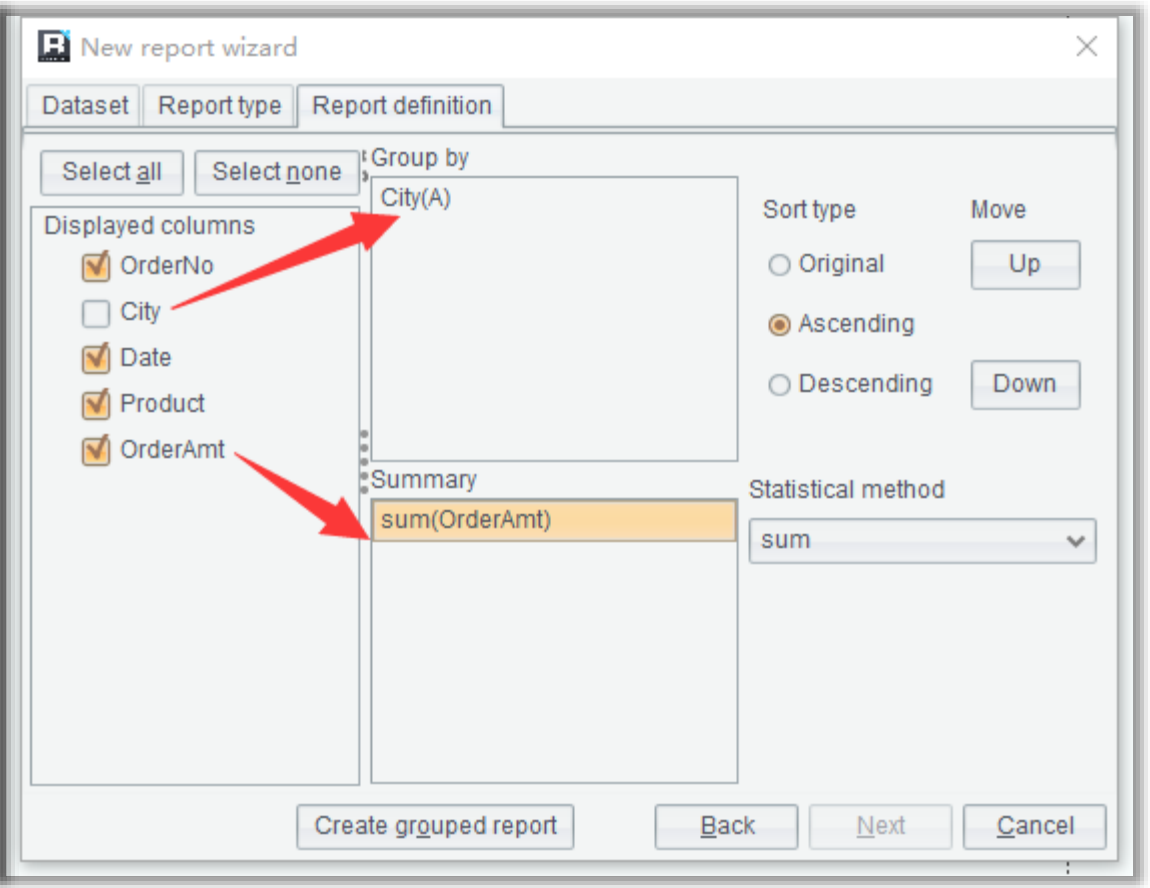
注意：Generat report与Wizard new的使用区别。
前者在当前报表区域生成，后者会产生一张新报表。

3.1 使用向导制作分组报表 – 汇总统计

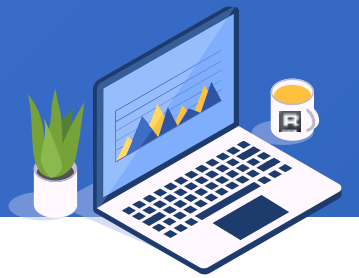


在Report type页选中分组报表Grouped report后。如右图拖拽相应的列到分组以及汇总区域：





3.1 使用向导制作分组报表 – 汇总统计



向导创建好了如下左图所示的分组报表，调整相应格式后，点击工具栏保存按钮，将设计好的报表保存为3.3.rptx

D:\tutorial\data\3\3.3.rptx

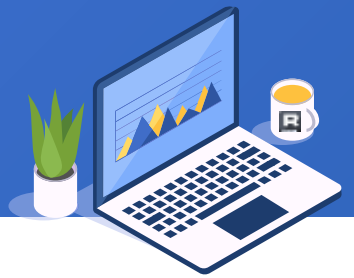
	A	B	C	D	E
1(TH)	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(Ci	=ds1.select(O	=ds1.Date	=ds1.Product	=ds1.OrderAmt
3	sum(City)				=ds1.sum(Ord
4	sum				=ds1.sum(Ord

将各单元格设置为居中对齐，E2、E3、E4单元格的显示格式设置为#.00

报表预览如右图所示

City	OrderNo	Date	Product	OrderAmt
Beijing	1002	2018-01-02	Peach	203.96
	1006	2018-01-03	Apple	69.19
	1021	2018-01-07	Apple	96.53
	1026	2018-01-08	Apple	325.56
	1040	2018-01-11	Banana	243.77
sum(City)				939.01
Changchun	1012	2018-01-04	Peach	267.02
	1038	2018-01-11	Peach	53.57
sum(City)				320.59
Chengdu	1027	2018-01-08	Banana	121.32
	1034	2018-01-10	Peach	311.34
sum(City)				432.66
Dalian	1001	2018-01-02	Apple	189.73
	1014	2018-01-05	Banana	204.73
	1015	2018-01-05	Peach	229.84
sum(City)				624.30

3.1 使用向导制作分组报表 – 汇总统计



用向导创建多层分组报表时，也可以添加汇总统计，如下图所示的报表3.4.rptx

D:\tutorial\data\3\3.4.rptx

	A	B	C	D	E	F
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(R	=ds1.group(Ci	=ds1.select(O	=ds1.Date	=ds1.Product	=ds1.OrderA
3		sum(City)				=ds1.sum(Or
4	sum(Region)					=ds1.sum(Or
5	sum					=ds1.sum(Or

将各单元格设置为居中对齐，F2、F3、F4、F5单元格的显示格式设置为#.00

报表预览如右图所示

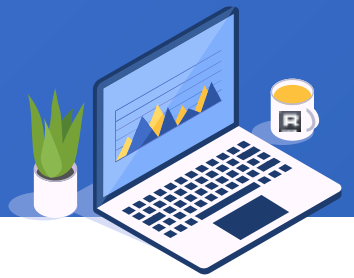
Region	City	OrderNo	Date	Product	OrderAmt
N	Beijing	1002	2018-01-02	Peach	203.96
		1006	2018-01-03	Apple	69.19
		1021	2018-01-07	Apple	96.53
		1026	2018-01-08	Apple	325.56
		1040	2018-01-11	Banana	243.77
	sum(City)				939.01
	Shijiazhuang	1011	2018-01-04	Banana	201.71
		1022	2018-01-07	Apple	127.97
	sum(City)				329.68
	Tianjin	1008	2018-01-03	Apple	256.15
		1023	2018-01-07	Banana	319.83
		1031	2018-01-09	Banana	61.10
	sum(City)				637.08
sum(Region)					1905.77

CONTENTS



3.2 常见分组报表的制作

3.2 常见分组报表的制作



1、向导创建的分组报表3.3.rptx，统计区位于分组格的下面，如果把统计区放在分组格的右边，看起来会更加直观一些，所以我们手工做些修改，然后另存为3.5.rptx

D:\tutorial\data\3\3.3.rptx

	A	B	C	D	E
1(TH)	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(Ci	=ds1.select(O	=ds1.Date	=ds1.Product	=ds1.OrderAmt
3	sum(City)				=ds1.sum(Ord
4	sum				=ds1.sum(Ord

删除A3、A4中文字

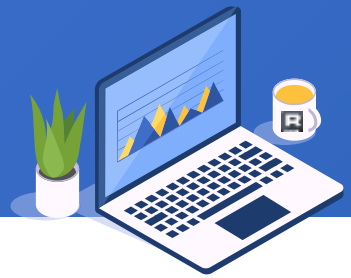
D:\tutorial\data\3\3.5.rptx

	A	B	C	D	E
1(TH)	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(City;	=ds1.select(Ord	=ds1.Date	=ds1.Product	=ds1.OrderAmt
3		=A2+" subtotal"			=ds1.sum(Order
4		Total			=ds1.sum(Order

合并A2、A3
合并B3 - D3
合并A4 - D4

1、在B3中输入：=A2+“小计”
2、在A4中输入：订单金额总计
3、修改B3、A4为右对齐，其余单元格为居中对齐

3.2 常见分组报表的制作



点击工具栏中的预览按钮，结果如下图所示

Preview report:D:\tutorial\data\3\3.5.rptx

City	OrderNo	Date	Product	OrderAmt
Tianjin	1008	2018-01-03	Apple	256.15
	1023	2018-01-07	Banana	319.83
	1031	2018-01-09	Banana	61.10
			Tianjin subtotal	637.08
Xiamen	1039	2018-01-11	Banana	231.30
			Xiamen subtotal	231.30
Xining	1004	2018-01-02	Banana	179.19
	1009	2018-01-04	Peach	52.28
	1019	2018-01-06	Apple	256.56
	1032	2018-01-09	Peach	197.31
	1035	2018-01-10	Banana	271.55
			Xining subtotal	956.89
Yinchuan	1037	2018-01-11	Peach	208.94
			Yinchuan subtotal	208.94
			Total	7872.10

分组

组内小计

各组总计

3.2 常见分组报表的制作

同样，将多层分组报表3.4.rptx的汇总统计格移到分组格右边，然后另存为3.6.rptx

D:\tutorial\data\3\3.4.rptx

	A	B	C	D	E	F
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(R	=ds1.group(Ci	=ds1.select(O	=ds1.Date	=ds1.Product	=ds1.OrderA
3		sum(City)				=ds1.sum(Or
4	sum(Region)					=ds1.sum(Or
5	sum					=ds1.sum(Or

另存为3.6.rptx并修改

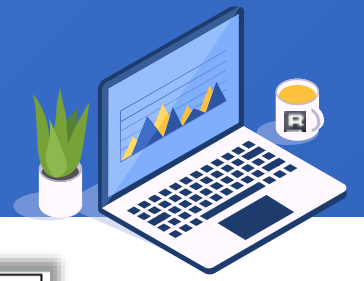
D:\tutorial\data\3\3.6.rptx

	A	B	C	D	E	F
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(	=ds1.group(C	=ds1.select(	=ds1.Date	=ds1.Product	=ds1.OrderA
3			=B2+" Subtotal"			=sum(F2{})
4		=A2+" Subtotal"				=sum(F2{})
5	Total					=ds1.sum(Or

预览

Region	City	OrderNo	Date	Product	OrderAmt
NW	Lanzhou	1003	2018-01-02	Banana	208.09
		1005	2018-01-03	Apple	342.06
		1016	2018-01-05	Banana	57.38
		1017	2018-01-06	Banana	232.57
		1036	2018-01-10	Apple	169.24
		Lanzhou Subtotal			1009.34
	Xining	1004	2018-01-02	Banana	179.19
		1009	2018-01-04	Peach	52.28
		1019	2018-01-06	Apple	256.56
		1032	2018-01-09	Peach	197.31
		1035	2018-01-10	Banana	271.55
		Xining Subtotal			956.89
	Yinchuan	1037	2018-01-11	Peach	208.94
		Yinchuan Subtotal			208.94
	NW Subtotal				2175.17
S	Shenzhen	1029	2018-01-09	Apple	278.33
		Shenzhen Subtotal			278.33
	Xiamen	1039	2018-01-11	Banana	231.30
		Xiamen Subtotal			231.30
	S Subtotal				509.63
Total					7872.10

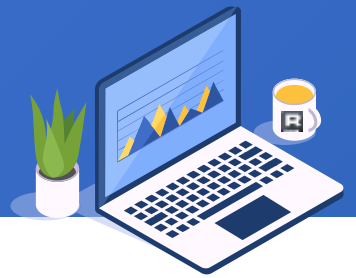
3.2 常见分组报表的制作



2、分组格独占一行，也有各组的汇总统计，如右图所示

Region	City	OrderNo	Date	Product	OrderAmt
N					
	Beijing				
		1002	2018-01-02	Peach	203.96
		1006	2018-01-03	Apple	69.19
		1021	2018-01-07	Apple	96.53
		1026	2018-01-08	Apple	325.56
		1040	2018-01-11	Banana	243.77
				Beijing Subtotal	939.01
	Shijiazhuang				
		1011	2018-01-04	Banana	201.71
		1022	2018-01-07	Apple	127.97
				Shijiazhuang Subtotal	329.68
	Tianjin				
		1008	2018-01-03	Apple	256.15
		1023	2018-01-07	Banana	319.83
		1031	2018-01-09	Banana	61.10
				Tianjin Subtotal	637.08
				N Total	1905.77

3.2 常见分组报表的制作



下面来看看如何制作上页这种分组报表

1、新建7行6列的空白报表，第一行设为表头，其它单元格如下图填写

	A	B	C	D	E	F
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt
2						
3						
4						
5						
6						
7						

	A	B	C	D	E	F
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(Reg					
3		=ds1.group(City				
4			=ds1.select(Ord	=ds1.date	=ds1.Product	=ds1.OrderAmt
5		=B3+" Subtotal"				=ds1.sum(Order
6	=A2+" Total"					=ds1.sum(Order
7	Total					=ds1.sum(Order

2、用第3章例子data3.1.xlsx新建文件数据集ds1

3、在A2格填写：=ds1.group(Region;Region:1)

4、在B3格填写：=ds1.group(City;City:1)

5、设置A2、B3为纵向扩展

6、合并A3、A4、A5三格

7、合并A6 – E6，合并A7 – E7

8、设置A3、B3、A6的左主格为A2

9、在C4格填写：=ds1.select(OrderNo)

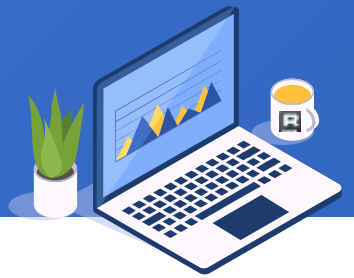
10、设置C4为纵向扩展

11、在D4格填写：=ds1.Date

12、在E4格填写：=ds1.Product

(接下页)

3.2 常见分组报表的制作



	A	B	C	D	E	F
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(Reg					
3		=ds1.group(City				
4			=ds1.select(Ord	=ds1.date	=ds1.Product	=ds1.OrderAmt
5		=B3+" Subtotal"				=ds1.sum(Order
6	=A2+" Total"					=ds1.sum(Order
7	Total					=ds1.sum(Order

13、在F4格填写：=ds1.OrderAmt

14、合并B5 – E5

15、设置B4、B5的左主格为B3

16、在B5格填写：=B3+“ Subtotal"

17、在F5、F6、F7格填写：=ds1.sum(OrderAmt)

18、在A6单元格填写：=A2+“ Total"

19、在A7单元格填写：Total

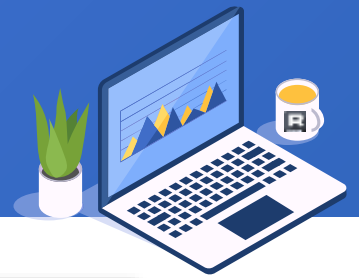
20、设置所有单元格为居中对齐

21、设置B5、A6、A7为右对齐

22、设置F4、F5、F6、F7的显示格式为#.00

23、保存报表为3.7.rptx

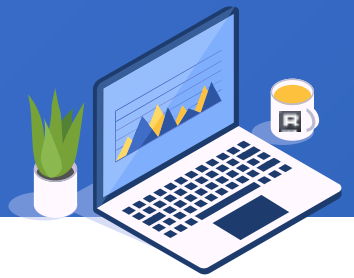
3.2 常见分组报表的制作



3、各组汇总统计在右边，以合并格方式显示，如右图所示

Region	City	OrderNo	Date	Product	OrderAmt	CitySubtotal	RegionTotal
N	Beijing	1002	2018-01-02	Peach	203.96	939.01	1905.77
		1006	2018-01-03	Apple	69.19		
		1021	2018-01-07	Apple	96.53		
		1026	2018-01-08	Apple	325.56		
		1040	2018-01-11	Banana	243.77		
	Shijiazhuang	1011	2018-01-04	Banana	201.71	329.68	
		1022	2018-01-07	Apple	127.97		
	Tianjin	1008	2018-01-03	Apple	256.15	637.08	
		1023	2018-01-07	Banana	319.83		
		1031	2018-01-09	Banana	61.10		
NE	Changchun	1012	2018-01-04	Peach	267.02	320.59	1589.60
		1038	2018-01-11	Peach	53.57		
	Dalian	1001	2018-01-02	Apple	189.73	624.30	
		1014	2018-01-05	Banana	204.73		
		1015	2018-01-05	Peach	229.84		
	Shenyang	1007	2018-01-03	Apple	231.44	644.71	
		1013	2018-01-05	Banana	83.34		
		1025	2018-01-08	Banana	243.93		
		1030	2018-01-09	Banana	86.00		

3.2 常见分组报表的制作



下面来看看如何制作上页这种分组报表。先打开报表3.2.rptx，另存为3.8.rptx

D:\tutorial\data\3\3.2.rptx

	A	B	C	D	E	F
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(R	=ds1.group(Ci	=ds1.select(Or	=ds1.Date	=ds1.Product	=ds1.OrderAmt

在F列按鼠标右键，选择追加数据列两次

D:\tutorial\data\3\3.8.rptx

	A	B	C	D	E	F	G	H
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt	CitySubtotal	RegionTotal
2	=ds1.group(Reg	=ds1.group(City	=ds1.select(Orc	=ds1.Date	=ds1.Product	=ds1.OrderAmt	=ds1.sum(Order	=ds1.sum(Order

填写：城市合计

填写：地区合计

填写：=ds1.sum(OrderAmt)
设置左主格为B2

填写：=ds1.sum(OrderAmt)
设置左主格为A2

CONTENTS



3.3 分组报表的结构和主格

3.3 分组报表的结构和主格



主格：报表计算时会主动扩展复制的单元格。主格有扩展方向，纵向扩展(格内标有红色向下箭头)的称为左主格，横向扩展(标有红色向右箭头)的称为上主格。

扩展区域：由一组单元格组成，有且仅有一个主格，主格的扩展方向决定本区域的扩展方向，区域内其它单元格跟随主格一起复制。

D:\tutorial\data\3\3.1.rptx

	A	B	C	D	E
1(TH)	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(City	=ds1.select(Or	=ds1.Date	=ds1.Product	=ds1.OrderAmt

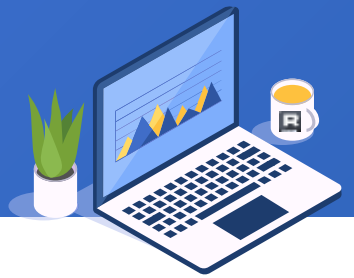
红色区域为城市分组的扩展区，主格A2

绿色区域为城市分组的明细扩展区，主格B2

左主格默认规则：
从本单元格左边开始，从右向左查找，第一个纵向扩展的单元格即为它的默认左主格；若左边某单元格指定了左主格，则那个指定的格也默认为本单元格的左主格；若找到最左边时还未找到，则默认左主格为报表首格(左上角)

左图中，C2、D2、E2的默认左主格为B2，B2的默认左主格为A2

3.3 分组报表的结构和主格



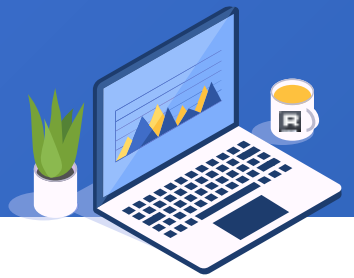
前页报表计算后：

City	OrderNo	Date	Product	OrderAmt
Beijing	1002	2018-01-02	Peach	203.96
	1006	2018-01-03	Apple	69.19
	1021	2018-01-07	Apple	96.53
	1026	2018-01-08	Apple	325.56
	1040	2018-01-11	Banana	243.77
Changchun	1012	2018-01-04	Peach	267.02
	1038	2018-01-11	Peach	53.57
Chengdu	1027	2018-01-08	Banana	121.32
	1034	2018-01-10	Peach	311.34
Dalian	1001	2018-01-02	Apple	189.73
	1014	2018-01-05	Banana	204.73
	1015	2018-01-05	Peach	229.84

城市主格为每个城市复制一个红色区域，区域内的绿色明细区域跟随复制

明细区域主格B2根据本组的订单数进行扩展，复制出5个绿色区域

3.3 分组报表的结构和主格



报表3.6.rptx的结构分析

洋红区域为城市分组区域，主格B2，区域内的C2、C3、F3的默认左主格也正好是B2

D:\tutorial\data\3\3.6.rptx

	A	B	C	D	E	F
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(Re	=ds1.group(Cit	=ds1.select(Or	=ds1.Date	=ds1.Product	=ds1.OrderAmt
3			=B2+" Subtotal"			=sum(F2{})
4		=A2+" Subtotal"				=sum(F2{})
5	Total					=ds1.sum(Ord

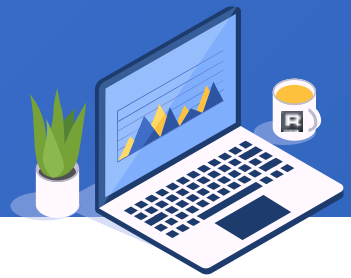
青色明细区域，同前

左主格为报表首格，不会扩展

黄色区域为地区分组区域，主格A2，区域内的B2、B4、F4的默认左主格也正好是A2

知识点：单元格的默认左主格正好是扩展区域的主格时，无需指定左主格

3.3 分组报表的结构和主格



前页报表计算后：

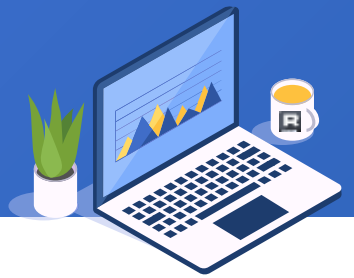
地区主格为每个地区复制一个黄色区域，区域内的洋红城市区域跟随复制

城市主格为每个城市复制一个洋红区域，区域内的青色明细区域跟随复制

明细区域主格C2根据本组的订单数进行扩展，复制出相应的青色区域

NW	Lanzhou	1003	2018-01-02	Banana	208.09
		1005	2018-01-03	Apple	342.06
		1016	2018-01-05	Banana	57.38
		1017	2018-01-06	Banana	232.57
		1036	2018-01-10	Apple	169.24
		Lanzhou Subtotal			1009.34
	Xining	1004	2018-01-02	Banana	179.19
		1009	2018-01-04	Peach	52.28
		1019	2018-01-08	Apple	256.56
		1032	2018-01-09	Peach	197.31
		1035	2018-01-10	Banana	271.55
		Xining Subtotal			956.89
	Yinchuan	1037	2018-01-11	Peach	208.94
		Yinchuan Subtotal			208.94
	NW Subtotal				2175.17
S	Shenzhen	1029	2018-01-09	Apple	278.33
		Shenzhen Subtotal			278.33
	Xiamen	1039	2018-01-11	Banana	231.30
		Xiamen Subtotal			231.30
	S Subtotal				509.63
Total					7872.10

3.3 分组报表的结构和主格



报表3.7.rptx的结构分析

红色区域为城市分组区域，主格B3，区域内的B4、C4、B5的默认左主格不是B3，所以必须指定它们的左主格为B3，才能跟随B3一起复制

绿色明细区域，同前

D:\tutorial\data\3\3.7.rptx

	A	B	C	D	E	F
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(Reg					
3		=ds1.group(Ci				
4			=ds1.select(Ord	=ds1.date	=ds1.Product	=ds1.OrderAmt
5					=B3+" Subtotal"	=ds1.sum(Order
6					=A2+" Total"	=ds1.sum(Order
7					Tot	ds1.sum(Order

F5未指定左主格，其默认左主格为B5指定的左主格B3

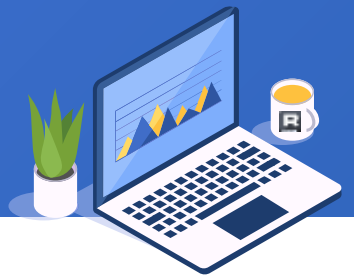
左主格为报表首格，不会扩展

橙色区域为地区分组区域，主格A2，区域内的A3、B3、A6的默认左主格不是A2，所以必须指定它们的左主格为A2，才能跟随A2一起复制

F6未指定左主格，其默认左主格为A6指定的左主格A2

知识点：单元格的默认左主格不是扩展区域的主格时，必须指定本扩展区主格为它的左主格

3.3 分组报表的结构和主格



前页报表计算后：

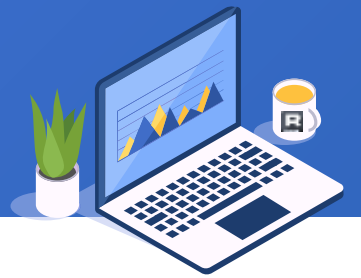
Region	City	OrderNo	Date	Product	OrderAmt
N					
	Beijing				
		1002	2018-01-02	Peach	203.96
		1006	2018-01-03	Apple	69.19
		1021	2018-01-07	Apple	96.53
		1026	2018-01-08	Apple	325.56
		1040	2018-01-11	Banana	243.77
				Beijing Subtotal	939.01
	Shijiazhuang				
		1011	2018-01-04	Banana	201.71
		1022	2018-01-07	Apple	127.97
				Shijiazhuang Subtotal	329.68
	Tianjin				
		1008	2018-01-03	Apple	256.15
		1023	2018-01-07	Banana	319.33
		1031	2018-01-09	Banana	61.10
				Tianjin Subtotal	637.08
				N Total	1905.77

城市主格为每个城市复制一个红色区域，区域内的绿色明细区域跟随复制

明细区域主格C4根据本组的订单数进行扩展，复制出相应个数的绿色区域

地区主格为每个地区复制一个橙色区域，区域内的红色城市区域跟随复制

3.3 分组报表的结构和主格



报表3.8.rptx的结构分析

D:\tutorial\data\3\3.8.rptx

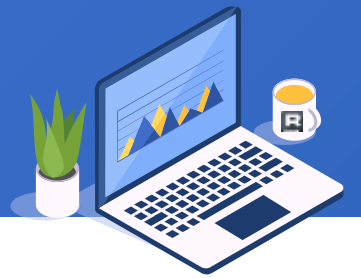
	A	B	C	D	E	F	G	H
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt	CitySubtotal	RegionTotal
2	=ds1.group(Reg	=ds1.group(City	=ds1.select(Or	=ds1.Date	=ds1.Product	=ds1.OrderAmt	=ds1.sum(Order	=ds1.sum(Order

H2属于蓝色地区区域，但默认左主格为C2，所以需要指定左主格为所在区域主格A2

G2属于红色城市区域，但默认左主格为C2，所以需要指定左主格为所在区域主格B2

Region	City	OrderNo	Date	Product	OrderAmt	CitySubtotal	RegionTotal
N	Beijing	1002	2018-01-02	Peach	203.96	939.01	1905.77
		1006	2018-01-03	Apple	69.19		
		1021	2018-01-07	Apple	96.53		
		1026	2018-01-08	Apple	325.56		
		1040	2018-01-11	Banana	243.77		
	Shijiazhuang	1011	2018-01-04	Banana	201.71	329.68	
		1022	2018-01-07	Apple	127.97		
	Tianjin	1008	2018-01-03	Apple	256.15	637.08	
		1023	2018-01-07	Banana	319.83		
		1031	2018-01-09	Banana	61.10		

3.3 分组报表的结构和主格



作为主格知识的应用，看看如何制作左图所示的报表？先打开报表3.1.rptx，另存为3.9.rptx

ID	City	OrderNo	Date	Product	OrderAmt
1	Beijing	1002	2018-01-02	Peach	203.96
2		1006	2018-01-03	Apple	69.19
3		1021	2018-01-07	Apple	96.53
4		1026	2018-01-08	Apple	325.56
5		1040	2018-01-11	Banana	243.77
6	Changchun	1012	2018-01-04	Peach	267.02
7		1038	2018-01-11	Peach	53.57
8	Chengdu	1027	2018-01-08	Banana	121.32
9		1034	2018-01-10	Peach	311.34
10	Dalian	1001	2018-01-02	Apple	189.73
11		1014	2018-01-05	Banana	204.73
12		1015	2018-01-05	Peach	229.84
13	Guiyang	1010	2018-01-04	Apple	290.31
14		1028	2018-01-08	Peach	228.96
15		1033	2018-01-10	Banana	173.26

在A列按鼠标右键，选择插入列

	A	B	C	D	E
1(TH)	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(City	=ds1.select(Orc	=ds1.Date	=ds1.Product	=ds1.OrderAmt

A1填写：ID

	A	B	C	D	E	F
1(TH)	ID	City	OrderNo	Date	Product	OrderAmt
2	=row()-1	=ds1.group(City	=ds1.select(Orc	=ds1.Date	=ds1.Product	=ds1.OrderAmt

A2填写：=row()-1
从表样可以看出，序号是与明细订单对应的，所以设置左主格为C2

设置C2的左主格为B2

CONTENTS



3.4 分组报表的运算

3.4 分组报表的运算 – 汇总统计的范围



分组报表中，汇总表达式写在不同的分组层次中，参与统计的数据范围不同。以sum为例，其他avg、max、min、count类似。

	A	B	C	D	E	F
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(Re	=ds1.group(Cit	=ds1.select(Or	=ds1.Date	=ds1.Product	=ds1.OrderAmt
3			=B2+" Subtotal"			=ds1.sum(Ord
4		=A2+" Subtotal"				=ds1.sum(Ord
	N	Beijing	1021	2018-01-07	Apple	96.53
			1026	2018-01-08	Apple	325.56
			1040	2018-01-11	Banana	243.77
			Beijing Subtotal			939.01
		Shijiazhuang	1011	2018-01-04	Banana	201.7
			1022	2018-01-07	Apple	127.97
			Shijiazhuang Subtotal			329.68
		Tianjin	1008	2018-01-03	Apple	256.15
			1023	2018-01-07	Banana	319.83
			1031	2018-01-09	Banana	61.10
			Tianjin Subtotal			637.08
		N Subtotal				1905.77

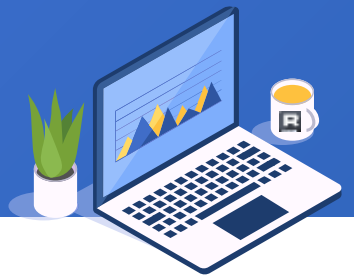
ds1.sum(OrderAmt)
第二层分组统计，主格为B2，统计B2组内所有订单的金额之和。

ds1.sum(OrderAmt)
第一层分组统计，主格为A2，统计A2组内所有订单的金额之和。

主格为Beijing，只统计Beijing组内所有订单的金额之和。

主格为华北(N)，只统计华北组内所有订单。

3.4 分组报表的运算 – 汇总统计的范围



除了用数据集的汇总函数外，还可以用单元格的值来计算，计算范围的确认方式相同。

	A	B	C	D	E	F
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(Re	=ds1.group(Cit	=ds1.select(Or	=ds1.Date	=ds1.Product	=ds1.OrderAmt
3					=B2+" Subtotal"	=sum(F2{})
4					=A2+" Subtotal"	=sum(F2{})
	N	Beijing	1002	2018-01-02	Peach	203.96
			1006	2018-01-03	Apple	69.19
			1021	2018-01-07	Apple	96.53
			1026	2018-01-08	Apple	325.56
			1040	2018-01-11	Banana	243.77
					Beijing Subtotal	939.01
		Shijiazhuang	1011	2018-01-04	Banana	201.71
			1022	2018-01-07	Apple	127.97
					Shijiazhuang Subtotal	329.68
		Tianjin	1008	2018-01-03	Apple	256.15
			1023	2018-01-07	Banana	319.83
			1031	2018-01-09	Banana	61.10
					Tianjin Subtotal	637.08
					N Subtotal	1905.77

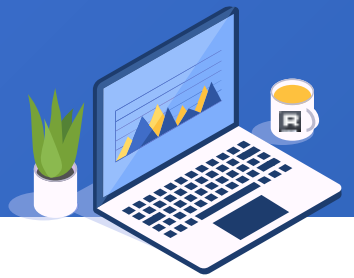
sum(F2{})
第二层分组统计，主格为B2，统计B2组内所有扩展出的F2之和。

sum(F2{})
第一层分组统计，主格为A2，统计A2组内所有扩展出的F2之和。

主格为Shijiazhuang，只统计Shijiazhuang组内所有F2(红框内)之和。

主格为华北（N），只统计华北组内所有F2之和。

3.4 分组报表的运算 – 汇总统计的范围



用单元格的值计算时，导出带公式的Excel文件里，汇总统计格会有计算公式；用数据集计算时，则不会带公式。

	A	B	C	D	E	F
1	Region	City	OrderNo	Date	Product	OrderAmt
2	N	Beijing	1002	2018-01-02	Peach	203.96
3			1006	2018-01-03	Apple	69.19
4			1021	2018-01-07	Apple	96.53
5			1026	2018-01-08	Apple	325.56
6			1040	2018-01-11	Banana	243.77
7			Beijing Subtotal			939.01
8		Shijiazhuang	1011	2018-01-04	Banana	201.71
9			1022	2018-01-07	Apple	127.97
10			Shijiazhuang Subtotal			329.68
11		Tianjin	1008	2018-01-03	Apple	256.15
12			1023	2018-01-07	Banana	319.83
13			1031	2018-01-09	Banana	61.1
14			Tianjin Subtotal			637.08
15		N Subtotal				1905.77

=SUM(F2:F6)

=SUM(F8:F9)

=SUM(F11:F13)

=SUM(F2:F6,F8:F9,F11:F13)

3.4 分组报表的运算 – group函数

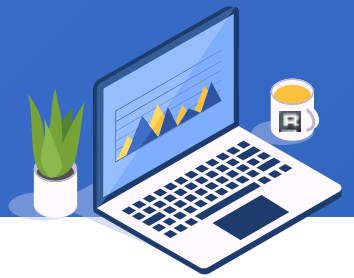


在分组报表中，总是在分组主格使用数据集的分组函数group来进行分组

group函数的使用说明可以在浏览器中访问

http://d.raqsoft.com.cn:6999/report/preference/%20dngroupsof.html 去查阅

3.4 分组报表的运算 – group函数



下面来看看group函数使用不同参数时的分组效果 – 分组前排序

	A	B	C
1	OrderNo	Region	City
2	1001	NE	Dalian
3	1002	N	Beijing
4	1003	NW	Lanzhou
5	1004	NW	Xining
6	1005	NW	Lanzhou
7	1006	N	Beijing
8	1007	NE	Shenyang
9	1008	N	Tianjin
10	1009	NW	Xining
11	1010	NS	Guiyang
12	1011	N	Shijiazhuang
13	1012	NE	Changchun
14	1013	NE	Shenyang
15	1014	NE	Dalian
16	1015	NE	Dalian
17	1016	NW	Lanzhou
18	1017	NW	Lanzhou
19	1018	NS	Kunming
20	1019	NW	Xining

原始数据

City	OrderNo
Dalian	1001
Beijing	1002
Lanzhou	1003
Xining	1004
Lanzhou	1005
Beijing	1006
Shenyang	1007
Tianjin	1008
Xining	1009
Guiyang	1010
Shijiazhuang	1011
Changchun	1012
Shenyang	1013
Dalian	1014
	1015
Lanzhou	1016
	1017
Kunming	1018
Xining	1019

ds1.group(City)
分组前不排序，原始顺序邻行同城为一组

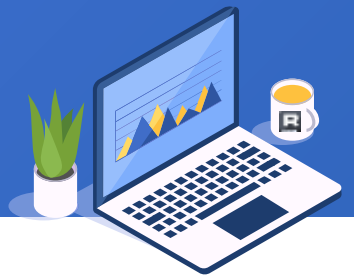
ds1.group(City;City:1)
分组前升序，同城皆为一组

ds1.group(City;City:-1)
分组前降序，同城皆为一组

City	OrderNo
Beijing	1002
	1006
	1021
	1026
	1040
Changchun	1012
	1038
Chengdu	1027
	1034
Dalian	1001
	1014
Guiyang	1010
	1028
	1033
Kunming	1018
	1020
	1024

City	OrderNo
Yinchuan	1037
Xining	1004
	1009
	1019
	1032
	1035
Xiamen	1039
Tianjin	1008
	1023
	1031
Shijiazhuang	1011
	1022
Shenzhen	1029
Shenyang	1007
	1013
	1025
	1030

3.4 分组报表的运算 – group函数



下面来看看group函数使用不同参数时的分组效果 – 分组前过滤

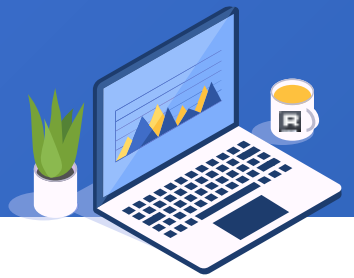
City	OrderNo	Date	Product	OrderAmt
Beijing	1002	2018-01-02	Peach	203.96
	1006	2018-01-03	Apple	69.19
	1021	2018-01-07	Apple	96.53
	1026	2018-01-08	Apple	325.56
	1040	2018-01-11	Banana	243.77
Changchun	1012	2018-01-04	Peach	267.02
	1038	2018-01-11	Peach	53.57
Chengdu	1027	2018-01-	Banana	121.32
	1034			
Dalian	1001			
	1014			
	1015			

ds1.group(City;City:1)
没有过滤

City	OrderNo	Date	Product	OrderAmt
Beijing	1006	2018-01-03	Apple	69.19
	1021	2018-01-07	Apple	96.53
	1026	2018-01-08	Apple	325.56
Dalian	1001	2018-01-02	Apple	189.73
Guiyang	1010	2018-01-04	Apple	290.31
Kunming	1024	2018-01-07	Apple	238.61
Lanzhou	1005	2018-01	Apple	342.06
	1036	2	Apple	169.24
				231.44
				278.33
				127.97
				256.15
Xining	1019	2018-01-06	Apple	256.56

ds1.group(City,Product=="Apple";City:1)
只使用产品名称为Apple的数据记录来分组

3.4 分组报表的运算 – group函数



下面来看看group函数使用不同参数时的分组效果 – 分组后排序

Preview report:D:\tutorial\data\3\3.10.rptx

City	OrderNo	Date	Product	OrderAmt
Beijing	1002	2018-01-02	Peach	203.96
	1006	2018-01-03	Apple	69.19
	1021	2018-01-07	Apple	96.53
	1026	2018-01-08	Apple	325.56
	1040	2018-01-11	Banana	243.77
			Beijing subtotal	939.01
Changchun	1012	2018-01-04	Peach	267.02
		2018-01-11	Peach	53.57
			Changchun subtotal	320.59
Chengdu	102	8-01-08	Banana	121.32

ds1.group(City;City:1)
未指定分组后排序，按城市升序显示

Preview report:D:\tutorial\data\3\3.10.rptx

City	OrderNo	Date	Product	OrderAmt
Lanzhou	1003	2018-01-02	Banana	208.09
	1005	2018-01-03	Apple	342.06
	1016	2018-01-05	Banana	57.38
	1017	2018-01-06	Banana	232.57
	1036	2018-01-10	Apple	169.24
			Lanzhou subtota	1009.34
Xining	1004	2018-01-02	Banan	179.19
	1009	2018-01-04	ach	52.28
	1019	2018-01-	Apple	256.56
	1032	20	Peach	197.31
	1035		Banana	271.55
			Xining subtotal	956.89
			Peach	203.96
			Apple	69.19
			Apple	96.53
			Apple	325.56
			Banana	243.77
			eijing subtotal	939.01

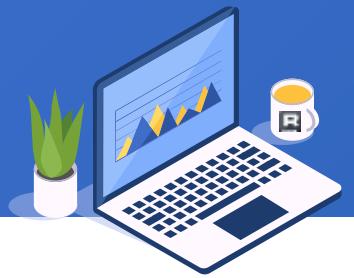
ds1.group(City;City:1 ;
ds1.sum(OrderAmt):-1)
指定了分组后排序，按各城市销售额从大到小倒序显示

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3.5 习 题

3.5 习题

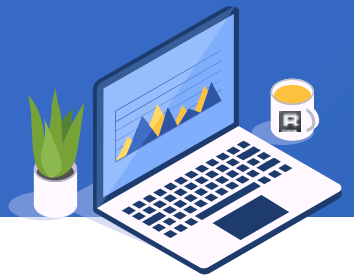


1、使用practice3.1.xlsx文件数据，分别用向导和手工制作按产品分组的分组报表，列出各产品订单明细并统计各产品的最大销售额。

2、对习题1的报表，增加过滤条件：只统计Beijing、Tianjin、Shenzhen3个城市。并按产品最大销售额从大到小的顺序显示。

3、使用practice3.2.xlsx文件数据，制作按地区、城市、产品分组的三层分组报表，列出各组订单明细，统计各地区销售额总计、各地区中城市销售额合计、各城市中产品销售额小计。

3.5 习题



4、分组要求同习题3，但使用报表3.7.rptx所示的格式：分组主格独占一行。

3.7.rptx

	A	B	C	D	E	F
1(TH)	Region	City	OrderNo	Date	Product	OrderAmt
2	=ds1.group(Reg					
3		=ds1.group(City				
4			=ds1.select(Ord	=ds1.date	=ds1.Product	=ds1.OrderAmt
5					=B3+" Subtotal"	=ds1.sum(Order
6					=A2+" Total"	=ds1.sum(Order
7					Total	=ds1.sum(Order

5、使practice3.1.xlsx文件数据， 制作按Region分组的报表，要求第一列是序号，分组格Region放在最右边，只需列出各组订单明细，不用汇总统计。

第4章
交叉报表



CONTENTS



4.1 使用向导制作交叉报表

4.2 常见交叉报表的制作

4.3 交叉报表的结构和主格

4.4 交叉报表的运算

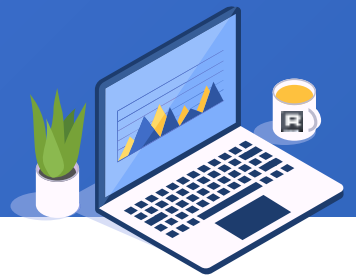
4.5 习题

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4.1 使用向导制作交叉报表

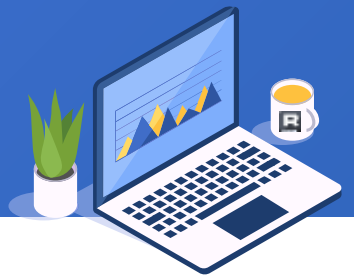
4.1 使用向导制作交叉报表 - 数据源



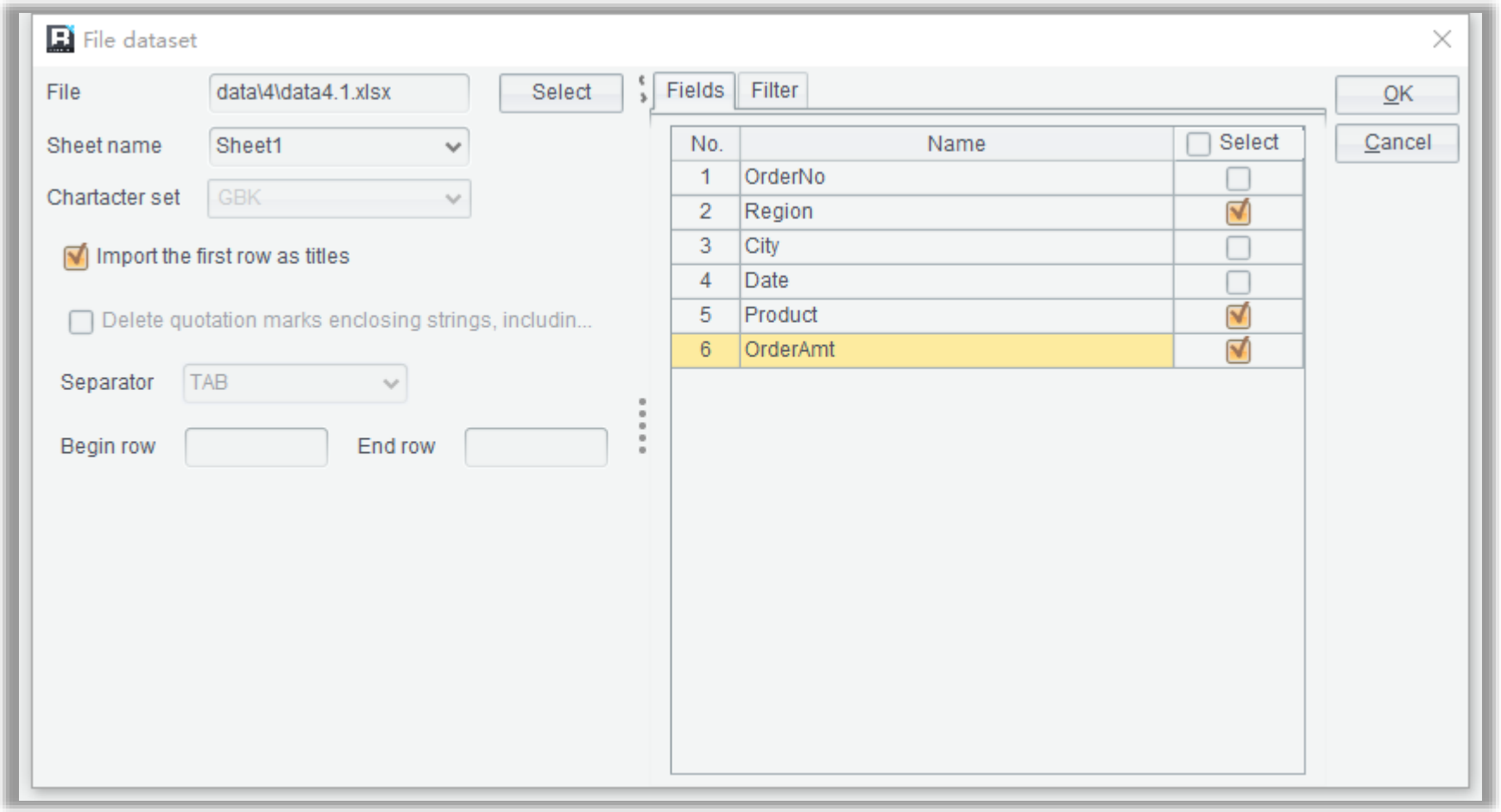
现有某公司销售订单数据表 – 第4章例子数据 data4.1.xlsx，与第3章数据结构相同，有2016、2017、2018三年的订单

	A	B	C	D	E	F
1	OrderNo	Region	City	Date	Product	OrderAmt
2	1001	SW	Guiyang	2016-01-02	Apple	118.11
3	1002	N	Beijing	2016-01-02	Apple	64.44
4	1003	NE	Shenyang	2016-01-02	Banana	183.19
5	1004	SW	Guiyang	2016-01-02	Pear	133.56
6	1005	NW	Lanzhou	2016-01-03	Banana	115.5
7	1006	NE	Changchun	2016-01-03	Pear	82.78
8	1007	N	Beijing	2016-01-03	Pear	176.87
9	1008	SW	Guiyang	2016-01-03	Banana	73.57
10	1009	S	Xiamen	2016-01-04	Apple	344.26
11	1010	S	Guangzhou	2016-01-04	Apple	236.14
12	1011	SW	Kunming	2016-01-04	Apple	338.59
13	1012	SW	Guiyang	2016-01-04	Banana	156.35
14	1013	N	ıijiazhuaı	2016-01-05	Banana	74.46
15	1014	N	Beijing	2016-01-05	Apple	321.07
16	1015	S	Shenzhen	2016-01-05	Banana	82.26
17	1016	NE	Dalian	2016-01-05	Apple	145.64
18	1017	SW	Kunming	2016-01-06	Pear	291.04
19	1018	SW	Kunming	2016-01-06	Apple	333.48
20	1019	SW	Chengdu	2016-01-06	Apple	76.03

4.1 使用向导制作交叉报表 – 简单交叉



运行报表设计器，跟3.1节同样的方式打开文件数据集(File dataset)，弹出如下窗口

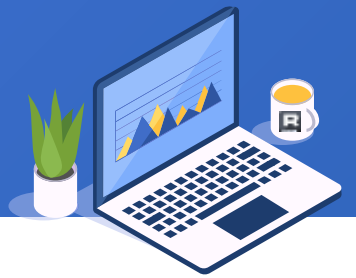


1、点击选择按钮选定数据文件第4章data4.1.xlsx

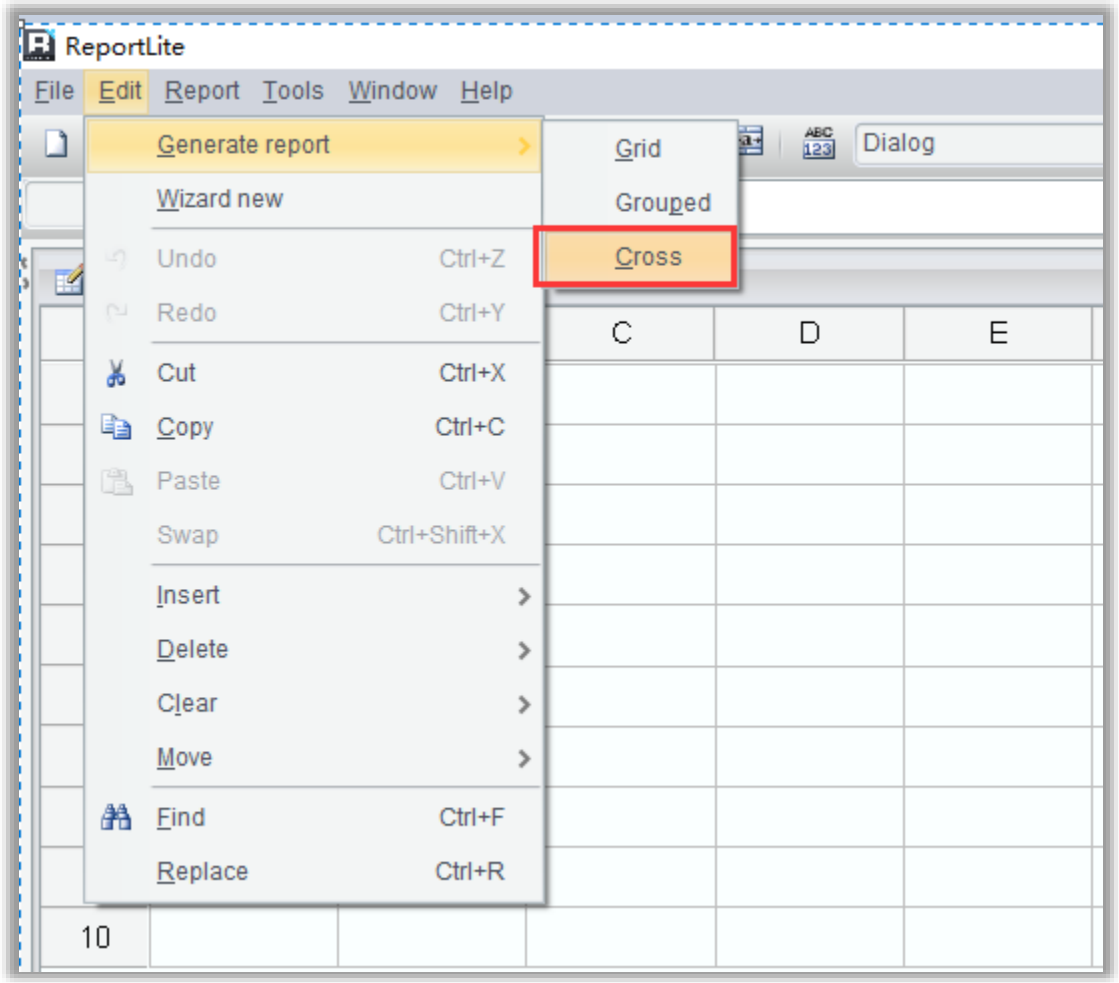
2、如图选出需要的数据列

3、点击确定按钮

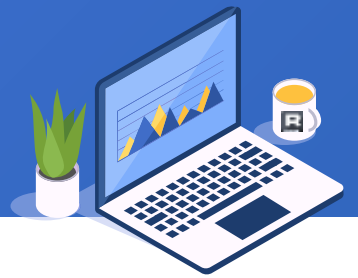
4.1 使用向导制作交叉报表 – 简单交叉

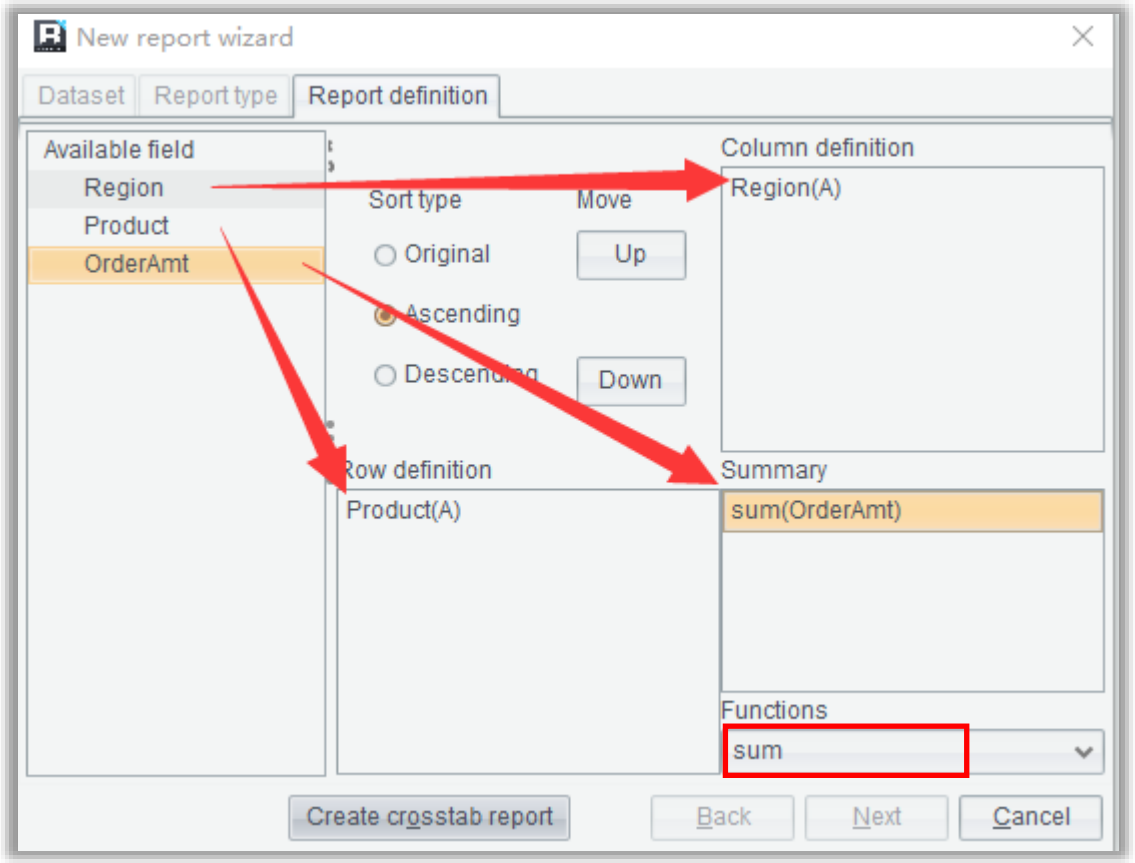


如图打开菜单项Cross：



4.1 使用向导制作交叉报表 – 简单交叉





1、将地区拖入列定义

2、将产品拖入行定义

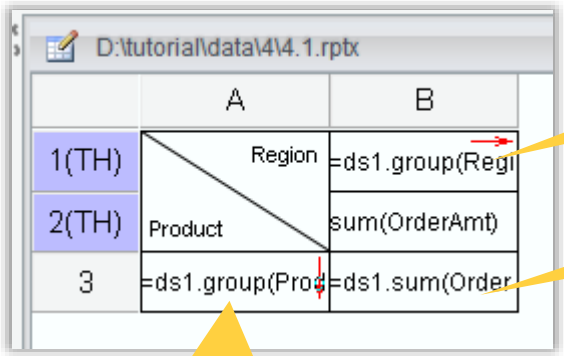
3、将金额拖入汇总

4、默认求和函数

5、点击生成交叉报表

4.1 使用向导制作交叉报表 – 简单交叉

向导创建好了如下图所示的交叉报表，设置各单元格为居中对齐方式和行表头，B3格的显示格式设为#.00，将设计好的报表保存为4.1.rptx



=ds1.group(Region; Region :1)

=ds1.sum(OrderAmt)

=ds1.group(Product;Product:1)

预览

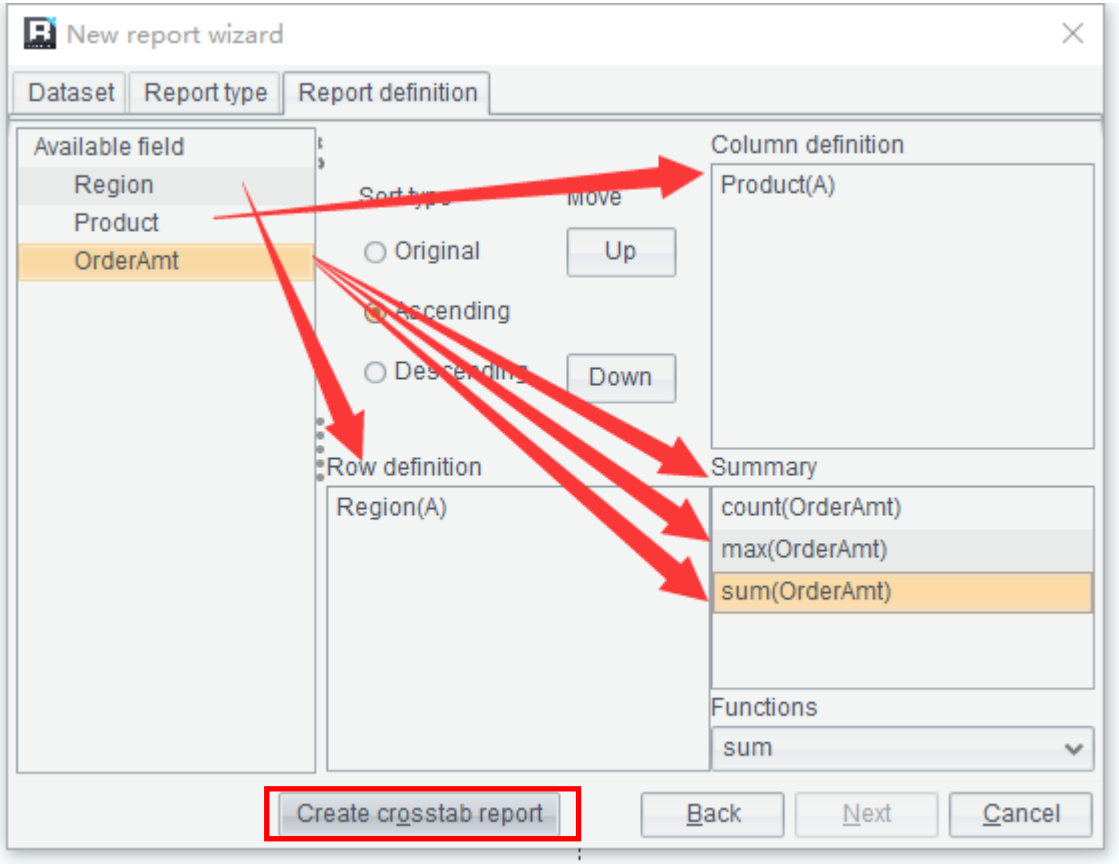
西北地区(NW)的苹果销售额之和

Region / Product	N	NE		S	SW
	sum(OrderAmt)	sum(OrderAmt)	sum(Or rAmt)	sum(OrderAmt)	sum(OrderAmt)
Apple	5354.74	6296.86	6214.17	4063.84	7056.92
Banana	4732.21	3648.91	5657.35	4255.41	5668.79
Pear	4313.48	5619.03	5770.16	5380.94	4535.99

4.1 使用向导制作交叉报表 – 多个统计值



在4.1.rptx报表界面中，使用菜单Wizard new，选择Cross报表，然后在报表定义页面增加多种统计值



1. 将地区拖入行定义
2. 将产品拖入列定义
3. 将金额拖入汇总，汇总函数改成计数;再将金额拖入汇总，汇总函数改成最大；再将金额拖入汇总，汇总函数默认为求和
4. 点击生成交叉报表

4.1 使用向导制作交叉报表 – 多个统计值

向导创建好了如下图所示的交叉报表，设置各单元格为居中对齐方式以及行表头，C3、D3格的显示格式设为#.00，将设计好的报表保存为4.2.rptx

D:\tutorial\data\4\4.2.rptx

	A	B	C	D
1(TH)	Product	=ds1.group(Product;Product:1)		
2(TH)	Region	count(OrderAmt)	max(OrderAmt)	sum(OrderAmt)
3	=ds1.group(Reg	=ds1.count()	=ds1.max(Order	=ds1.sum(Order

=ds1.sum(OrderAmt)

=ds1.max(OrderAmt)

=ds1.count()

=ds1.group(Region;Region:1)

预览

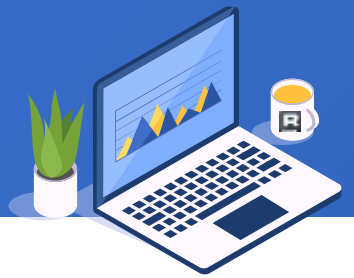
华北地区苹果订单金额总和

东北地区香蕉最大订单金额

东北地区梨的订单数量

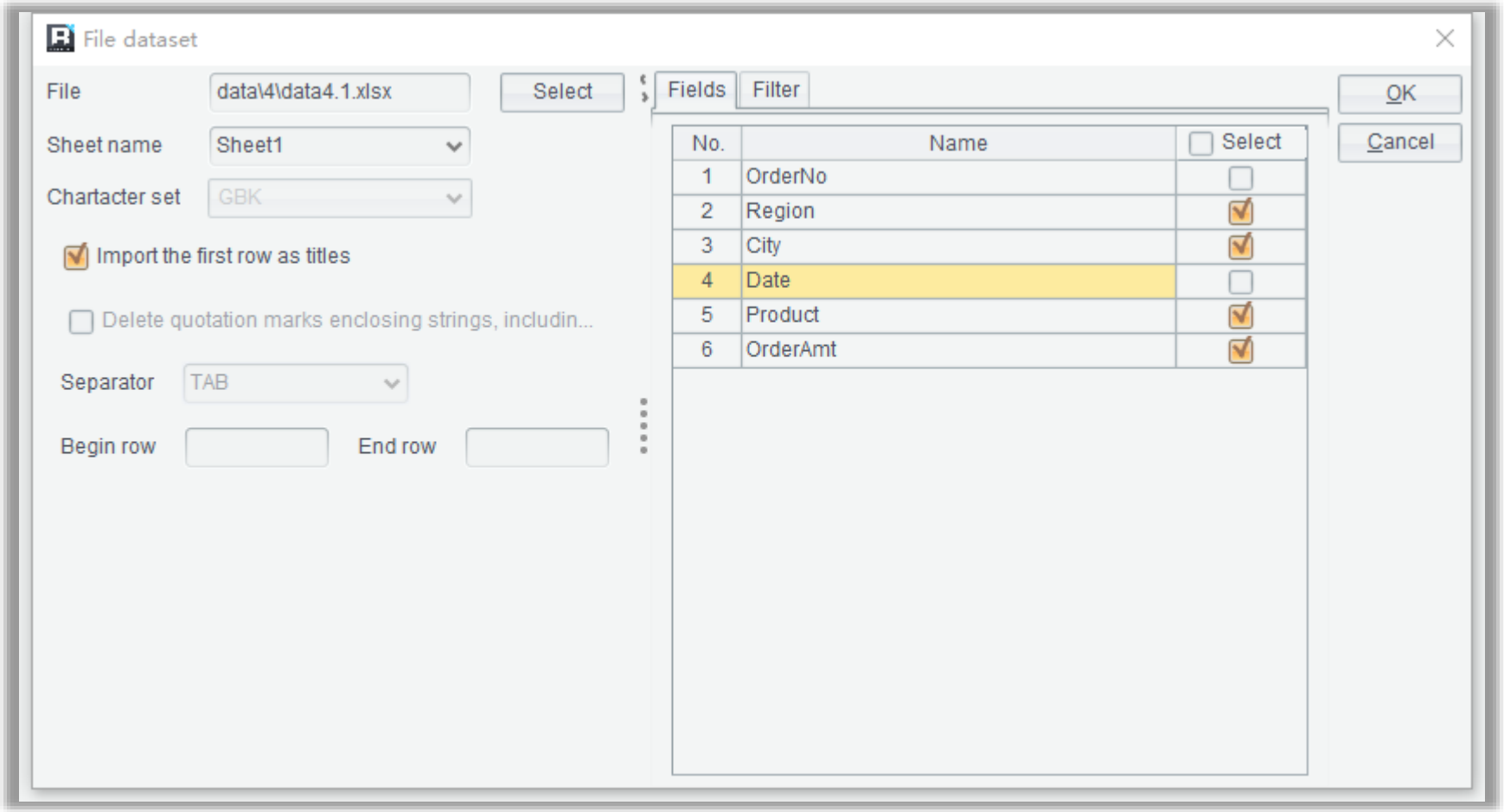
Product / Region	Apple			Banana			Pear		
	count(Order	max(OrderA	su OrderA	count(Ora	max(OrderA	sum(OrderA	cou Order	max(OrderA	sum(OrderA
N	28	329.56	5354.74	23	347.55	4732.21	2	339.80	4313.48
NE	32	334.63	6296.86	21	313.64	3648.91	27	341.37	5619.03
NW	29	336.48	6214.17	28	349.87	5657.35	26	344.84	5770.16
S	20	346.74	4063.84	23	335.23	4255.41	30	346.06	5380.94
SW	34	338.59	7056.92	33	344.34	5668.79	24	320.83	4535.99

4.1 使用向导制作交叉报表 – 多层交叉



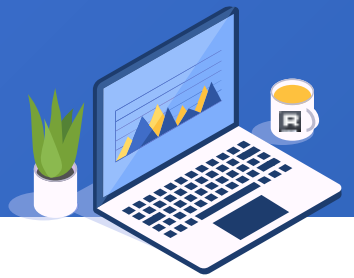
行定义或列定义上有多层分组时，称为多层交叉。

运行报表设计器，打开文件数据集，如下图设置选出字段

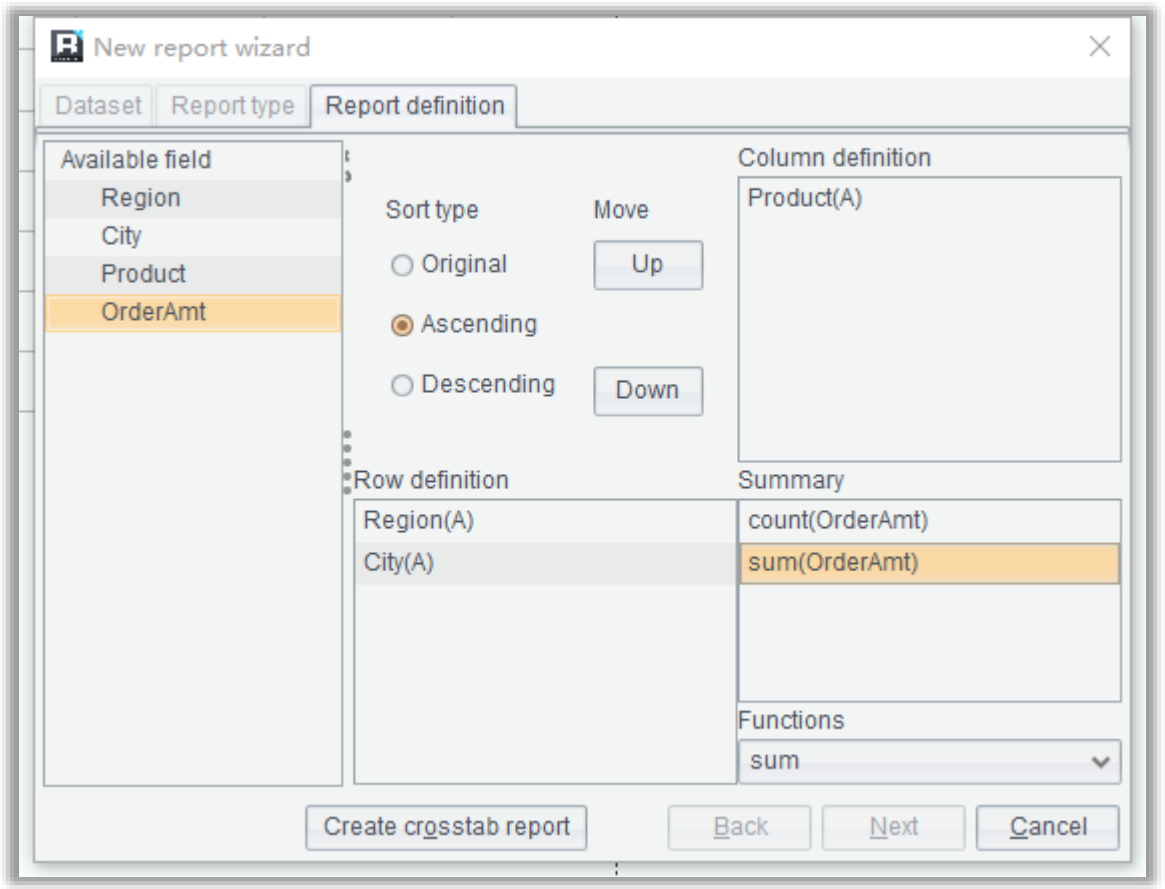


1. 点击选择按钮选定数据文件第4章data4.1.xlsx
2. 选择需要的数据列
3. 点击确定按钮

4.1 使用向导制作交叉报表 – 多层交叉

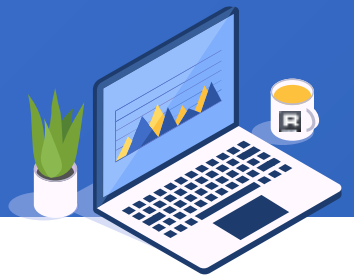


菜单项上选择交叉报表Cross，并按右边步骤设置图示定义

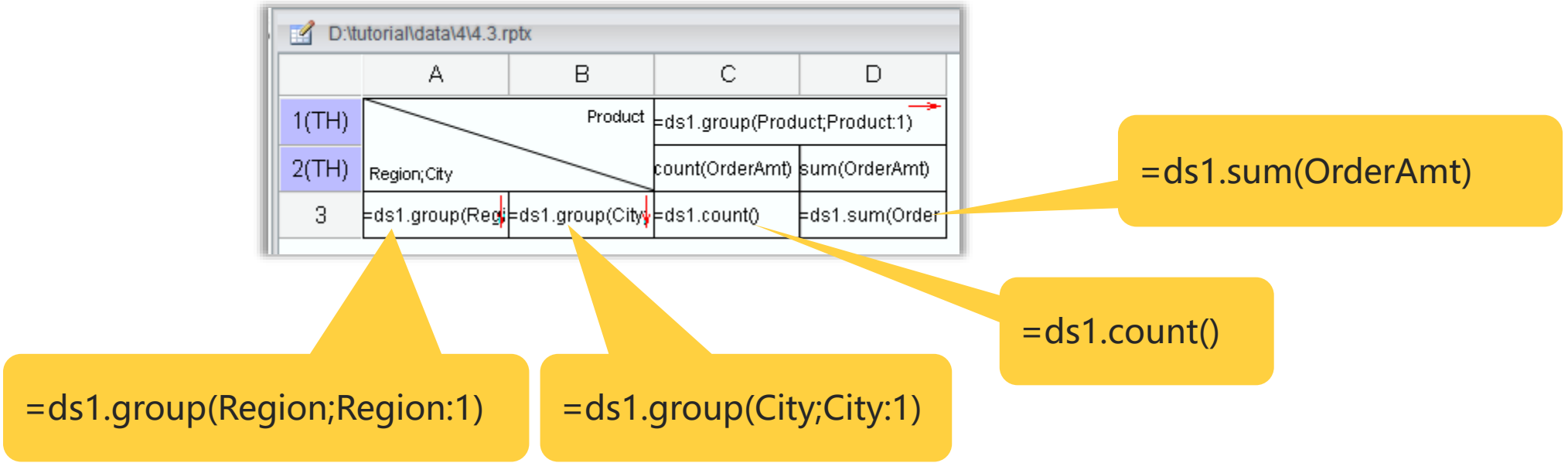


1. 将地区拖入行定义
2. 将城市拖入行定义
3. 将产品拖入列定义
4. 将金额拖入汇总
5. 修改汇总函数为计数
6. 再将金额拖入汇总，汇总函数默认为求和
7. 点击生成交叉报表

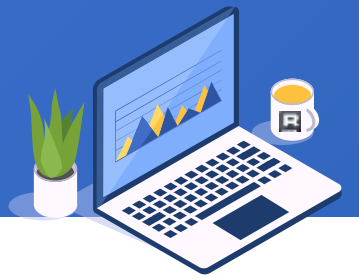
4.1 使用向导制作交叉报表 – 多层交叉



向导创建了如下图所示的交叉报表，设置各单元格为居中对齐方式以及行表头，C3格的显示格式设为#.00，将设计好的报表保存为4.3.rptx



4.1 使用向导制作交叉报表 – 多层交叉



点击预览报表，如下图

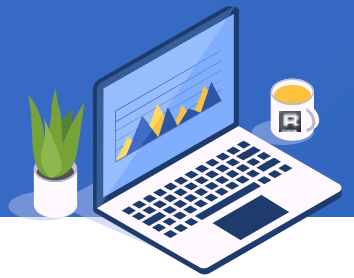
Region;City		Product: Apple		Banana		Pear	
		count(OrderA	sum(OrderA	count(OrderA	sum(OrderA	count(OrderA	sum(OrderA
N	Beijing	10	2227.93	8	2049.68	8	1432.25
	Shijiazhuang	9	1595.69	7	1246.39	5	1246.26
	Tianjin	9	1531.12	8	1436.14	9	1634.97
NE	Changchun	13	2771.91	4	663.41	10	1992.59
	Dalian	13	2376.10	10	1721.67	10	2147.70
	Shenyang	6	1148.85	7	1263.83	7	1478.74
NW	Lanzhou	9	2026.33	11	2229.90	5	789.28
	Xining	3	621.34	11	2425.09	11	2242.21
	Yinchuan	17	3566.50	6	1002.36	10	2738.67
S	Guangzhou	10	1969.51	4	857.02	10	1572.92
	Shenzhen	5	964.31	9	1544.93	10	1814.28
	Xiamen	5	1130.02	10	1853.46	10	1993.74
SW	Chengdu	10	1944.38	13	1987.25	11	2089.25
	Guiyang	14	3081.82	9	1605.42	8	1385.13
	Kunming	10	2030.72	11	2076.12	5	1061.61

CONTENTS

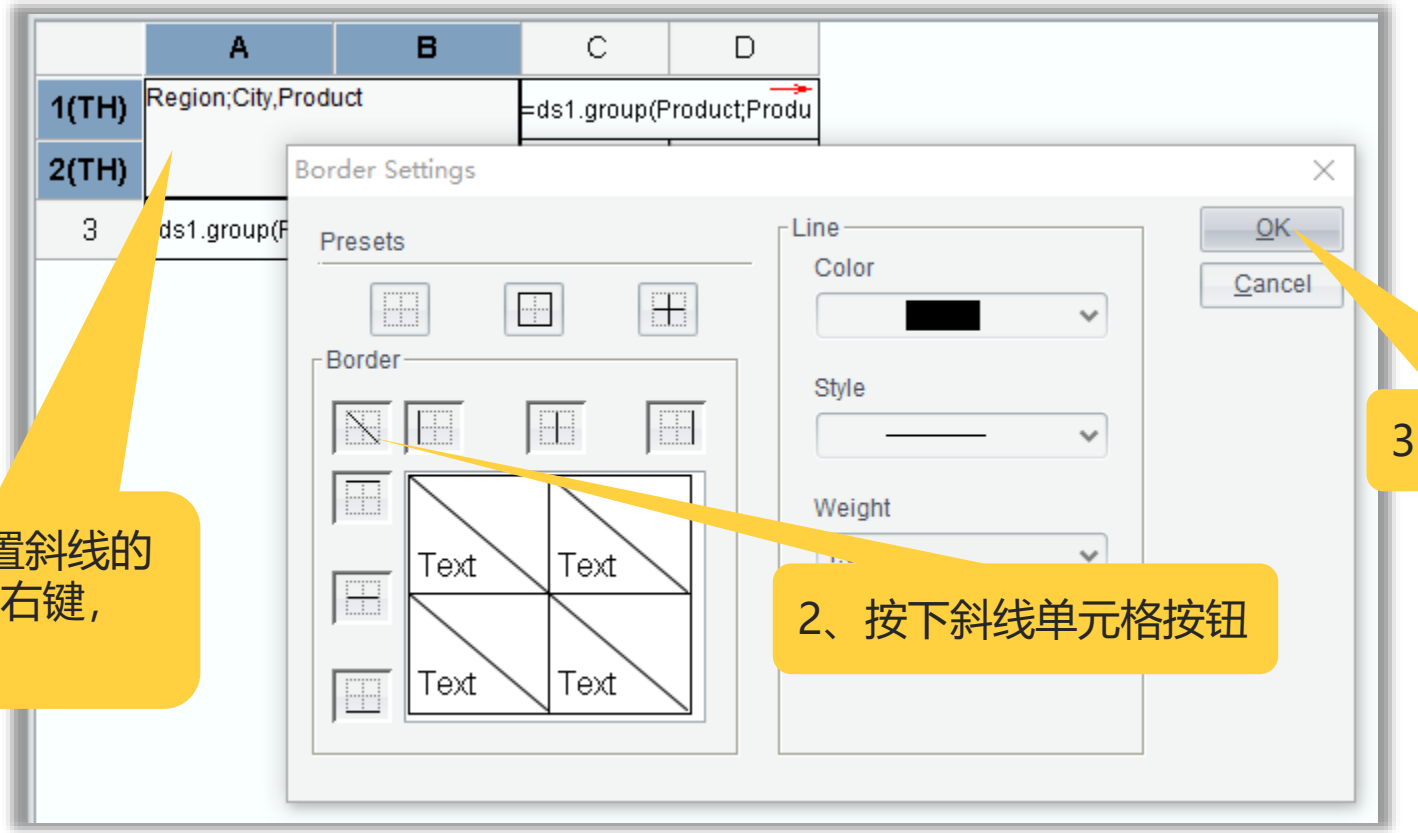


4.2 常见交叉表制作

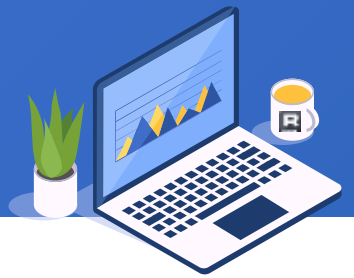
4.2 常见交叉报表的制作 – 斜线单元格



交叉报表中，行列定义交汇的格子(通常是左上角)，需要同时填写行列定义的描述，并用斜线将它们分隔开。我们看看斜线单元格是如何制作的：

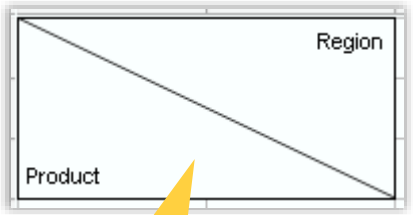


4.2 常见交叉报表的制作 – 斜线单元格

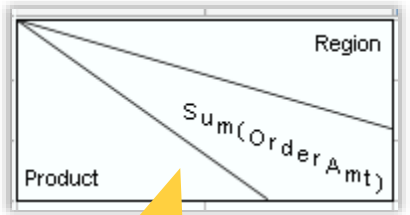


斜线单元格的值为：行描述,列描述,汇总描述

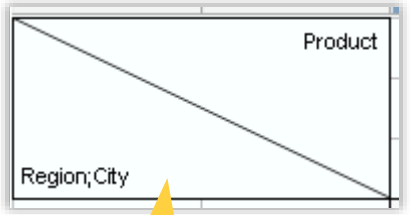
分隔符为半角逗号，可以没有汇总描述



格值：Product,Region

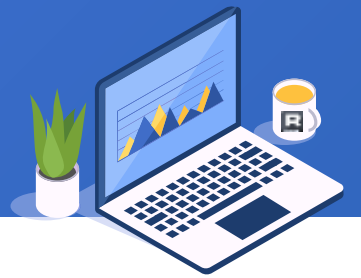


格值：
Product,Region,Sum(OrderAmt)



格值：
Region;City,Product

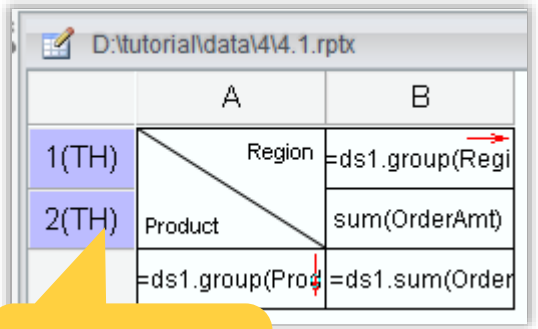
4.2 常见交叉报表的制作



当只有一个统计值时，习惯将统计描述放到左上角斜线单元格，如下图。打开4.1.rptx，另存为4.4.rptx

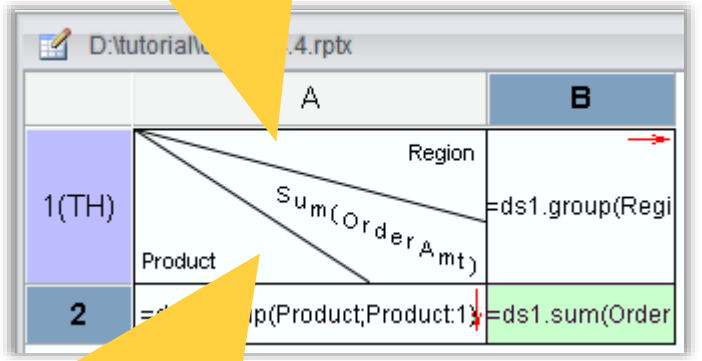
Region / Sum(OrderAmt) / Product	N	NE	NW	S	SW
Apple	5354.74	6296.86	6214.17	4063.84	7056.92
Banana	4732.21	3648.91	5657.35	4255.41	5668.79
Pear	4313.48	5619.03	5770.16	5380.94	4535.99

4.1.rptx



点鼠标右键选择删除行

另存为4.4.rptx



格值修改为：Product,Region,Sum(OrderAmt)

调整第一行行高、第一列列宽至合适大小

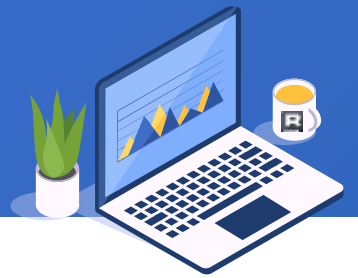
4.2 常见交叉报表的制作 – 多层交叉带汇总



多层交叉报表，上层分组带小计，如右图。

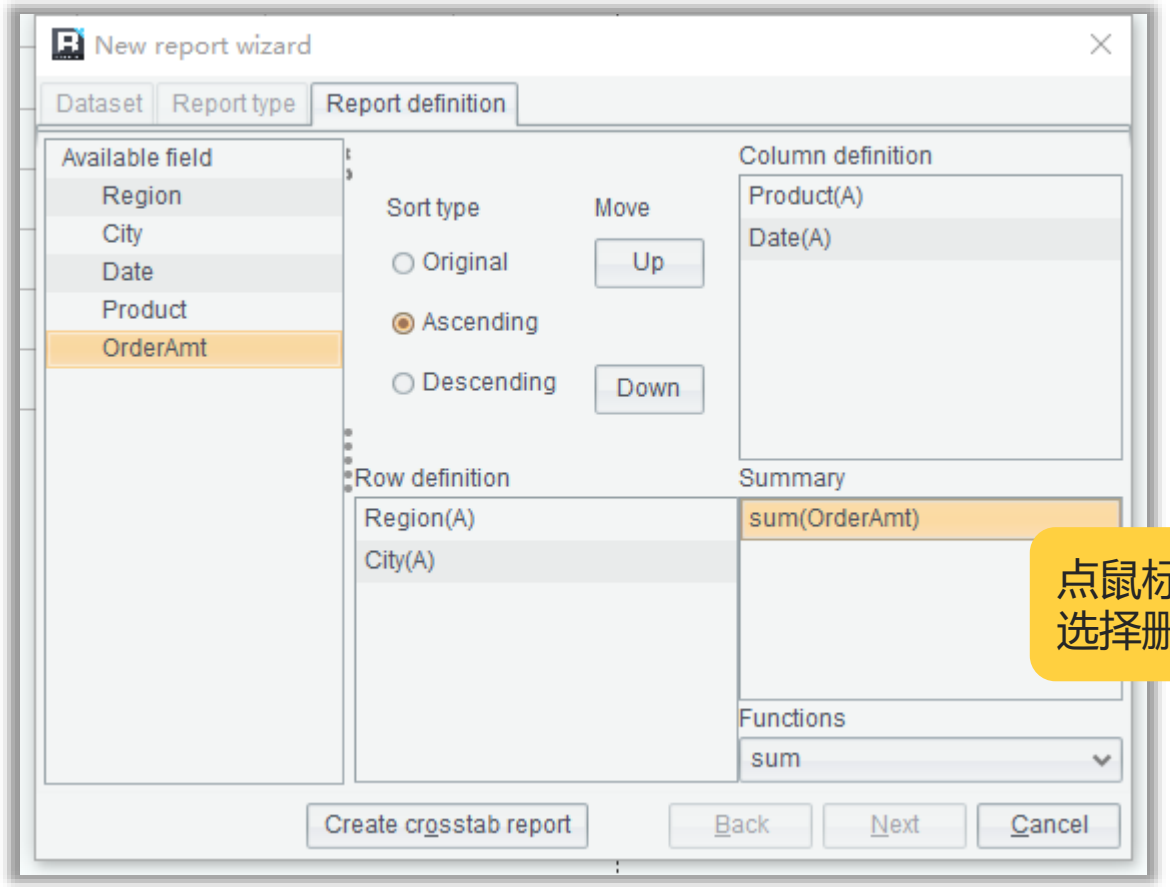
Region;City \ Product;Year / Sum(OrderAmt)		Apple				Banana				Pear				Total
		2016	2017	2018	Subtotal	2016	2017	2018	Subtotal	2016	2017	2018	Subtotal	
N	Beijing	385.51	1783.87	58.55	2227.93	0.00	2049.68	0.00	2049.68	176.87	678.25	577.13	1432.25	5709.86
	Shijiazhuang	581.09	943.54	71.06	1595.69	519.39	621.09	105.91	1246.39	0.00	1019.19	227.07	1246.26	4088.34
	Tianjin	417.82	914.77	198.53	1531.12	183.51	1252.63	0.00	1436.14	0.00	992.93	642.04	1634.97	4602.23
	Subtotal	1384.42	3642.18	328.14	5354.74	702.90	3923.40	105.91	4732.21	176.87	2690.37	1446.24	4313.48	14400.43
NE	Changchun	1079.08	1366.23	326.60	2771.91	174.60	488.81	0.00	663.41	596.57	1130.20	265.82	1992.59	5427.91
	Dalian	457.98	1358.01	560.11	2376.10	642.03	791.12	288.52	1721.67	174.07	1973.63	0.00	2147.70	6245.47
	Shenyang	469.36	504.87	174.62	1148.85	486.44	568.64	208.75	1263.83	368.33	955.69	154.72	1478.74	3891.42
	Subtotal	2006.42	3229.11	1061.33	6296.86	1303.07	1848.57	497.27	3648.91	1138.97	4059.52	420.54	5619.03	15564.80
NW	Lanzhou	213.45	1592.62	220.26	2026.33	592.32	1637.58	0.00	2229.90	99.73	689.55	0.00	789.28	5045.51
	Xining	0.00	621.34	0.00	621.34	153.36	1882.07	389.66	2425.09	649.91	1592.30	0.00	2242.21	5288.64
	Yinchuan	547.02	2323.09	696.39	3566.50	0.00	777.38	224.98	1002.36	316.73	2421.94	0.00	2738.67	7307.53
	Subtotal	760.47	4537.05	916.65	6214.17	745.68	4297.03	614.64	5657.35	1066.37	4703.79	0.00	5770.16	17641.68
S	Guangzhou	489.58	1111.86	368.07	1969.51	0.00	98.08	758.94	857.02	263.25	929.80	379.87	1572.92	4399.45
	Shenzhen	0.00	906.74	57.57	964.31	336.87	872.83	335.23	1544.93	497.07	1046.75	270.46	1814.28	4323.52
	Xiamen	739.74	390.28	0.00	1130.02	71.27	1590.39	191.80	1853.46	336.61	966.50	690.63	1993.74	4977.22
	Subtotal	1229.32	2408.88	425.64	4063.84	408.14	2561.30	1285.97	4255.41	1096.93	2943.05	1340.96	5380.94	13700.19
SW	Chengdu	76.03	1319.73	548.62	1944.38	399.65	948.21	639.39	1987.25	282.70	1213.50	593.05	2089.25	6020.88
	Guiyang	735.60	1980.90	365.32	3081.82	688.42	804.14	112.86	1605.42	299.13	1086.00	0.00	1385.13	6072.37
	Kunming	923.92	1106.80	0.00	2030.72	387.20	1556.20	132.72	2076.12	579.94	320.83	160.84	1061.61	5168.45
	Subtotal	1735.55	4407.43	913.94	7056.92	1475.27	3308.55	884.97	5668.79	1161.77	2620.33	753.89	4535.99	17261.70
Total		7116.18	18224.65	3645.70	28986.53	4635.06	15938.85	3388.76	23962.67	4640.91	17017.06	3961.63	25619.60	78568.80

4.2 常见交叉报表的制作 – 多层交叉带汇总

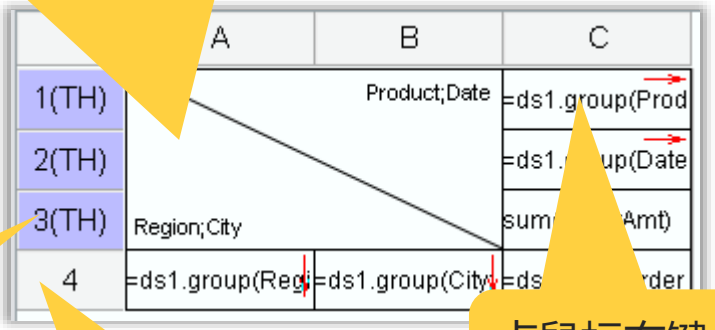


下面看看如何制作前页所示的交叉报表

类似于用向导制作4.3.rptx，在选择字段时加上日期列，并如左图设置，得到右图报表



修改单元格值为：
Region;City,Product;Year,Sum(OrderAmt)

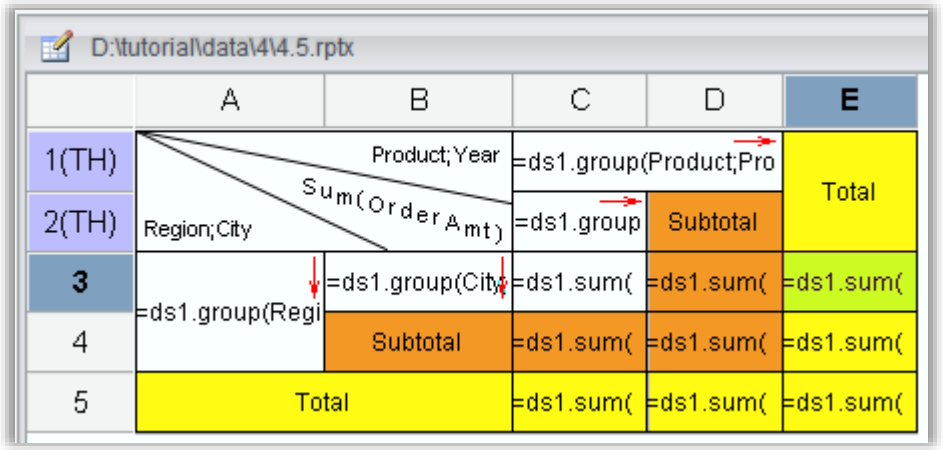


点鼠标右键
选择删除第三行

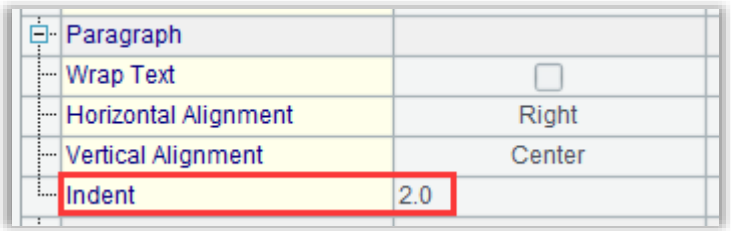
点鼠标右键
选择追加数据行2次

点鼠标右键
选择追加数据列2次

4.2 常见交叉报表的制作 – 多层交叉带汇总



选中C3到E5设置缩进



接着再进行以下步骤：

1、所有单元格设置边框线，居中对齐

2、合并C1、D1，在D2填写小计

3、合并E1、E2，并填写总计

4、合并A3、A4，在B4填写小计

5、合并A5、B5，并填写总计

6、C3到E5单元格均填写：=ds1.sum(OrderAmt)

7、C3到E5单元格均设置显示格式#0.00，右对齐

8、 C3到E5单元格均设置缩进2mm

9、按图中样式设置单元格背景色

10、C2格按年份分组，表达式修改为：

=ds1.group(year(Date);Date:1)

11、保存报表为4.5.rptx

4.2 常见交叉报表的制作 – 表头表尾



在报表4.5.rptx的基础上，增加表头标题区和表尾标题区，如右图。

Product sales crosstab

Unit: 10,000$ Date: 2022-10-11

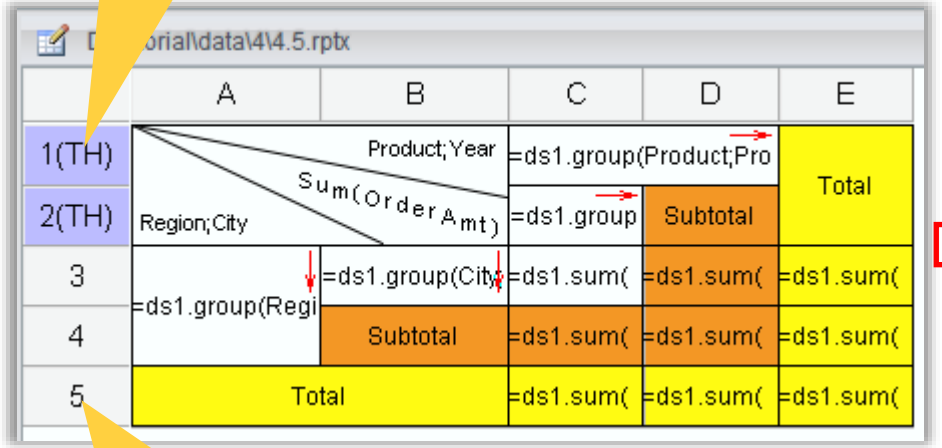
Region;City \ Product;Year Sum(Amount)		Apple				Banana				Pear				Total
		2016	2017	2018	Subtotal	2016	2017	2018	Subtotal	2016	2017	2018	Subtotal	
N	Beijing	385.51	1783.87	58.55	2227.93	0.00	2049.68	0.00	2049.68	176.87	678.25	577.13	1432.25	5709.86
	Shijiazhuang	581.09	943.54	71.06	1595.69	519.39	621.09	105.91	1246.39	0.00	1019.19	227.07	1246.26	4088.34
	Tianjin	417.82	914.77	198.53	1531.12	183.51	1252.63	0.00	1436.14	0.00	992.93	642.04	1634.97	4602.23
	Subtotal	1384.42	3642.18	328.14	5354.74	702.90	3923.40	105.91	4732.21	176.87	2690.37	1446.24	4313.48	14400.43
NE	Changchun	1079.08	1366.23	326.60	2771.91	174.60	488.81	0.00	663.41	596.57	1130.20	265.82	1992.59	5427.91
	Dalian	457.98	1358.01	560.11	2376.10	642.03	791.12	288.52	1721.67	174.07	1973.63	0.00	2147.70	6245.47
	Shenyang	469.36	504.87	174.62	1148.85	486.44	568.64	208.75	1263.83	368.33	955.69	154.72	1478.74	3891.42
	Subtotal	2006.42	3229.11	1061.33	6296.86	1303.07	1848.57	497.27	3648.91	1138.97	4059.52	420.54	5619.03	15564.80
NW	Lanzhou	213.45	1592.62	220.26	2026.33	592.32	1637.58	0.00	2229.90	99.73	689.55	0.00	789.28	5045.51
	Xining	0.00	621.34	0.00	621.34	153.36	1882.07	389.66	2425.09	649.91	1592.30	0.00	2242.21	5288.64
	Yinchuan	547.02	2323.09	696.39	3566.50	0.00	777.38	224.98	1002.36	316.73	2421.94	0.00	2738.67	7307.53
	Subtotal	760.47	4537.05	916.65	6214.17	745.68	4297.03	614.64	5657.35	1066.37	4703.79	0.00	5770.16	17641.68
S	Guangzhou	489.58	1111.86	368.07	1969.51	0.00	98.08	758.94	857.02	263.25	929.80	379.87	1572.92	4399.45
	Shenzhen	0.00	906.74	57.57	964.31	336.87	872.83	335.23	1544.93	497.07	1046.75	270.46	1814.28	4323.52
	Xiamen	739.74	390.28	0.00	1130.02	71.27	1590.39	191.80	1853.46	336.61	966.50	690.63	1993.74	4977.22
	Subtotal	1229.32	2408.88	425.64	4063.84	408.14	2561.30	1285.97	4255.41	1096.93	2943.05	1340.96	5380.94	13700.19
SW	Chengdu	76.03	1319.73	548.62	1944.38	399.65	948.21	639.39	1987.25	282.70	1213.50	593.05	2089.25	6020.88
	Guiyang	735.60	1980.90	365.32	3081.82	688.42	804.14	112.86	1605.42	299.13	1086.00	0.00	1385.13	6072.37
	Kunming	923.92	1106.80	0.00	2030.72	387.20	1556.20	132.72	2076.12	579.94	320.83	160.84	1061.61	5168.45
	Subtotal	1735.55	4407.43	913.94	7056.92	1475.27	3308.55	884.97	5668.79	1161.77	2620.33	753.89	4535.99	17261.70
Total		7116.18	18224.65	3645.70	28986.53	4635.06	15938.85	3388.76	23962.67	4640.91	17017.06	3961.63	25619.60	78568.80

Prepared by: Joancy

4.2 常见交叉报表的制作 – 表头表尾

下面看看如何制作前页所示的交叉报表？打开报表4.5.rptx，另存为4.6.rptx

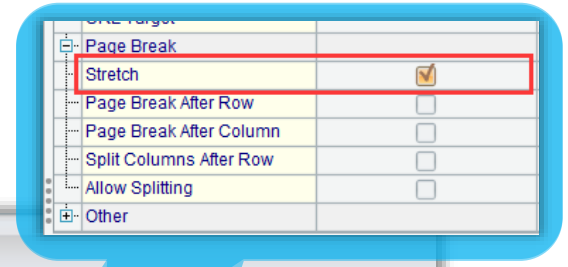
点鼠标右键
选择插入行2次

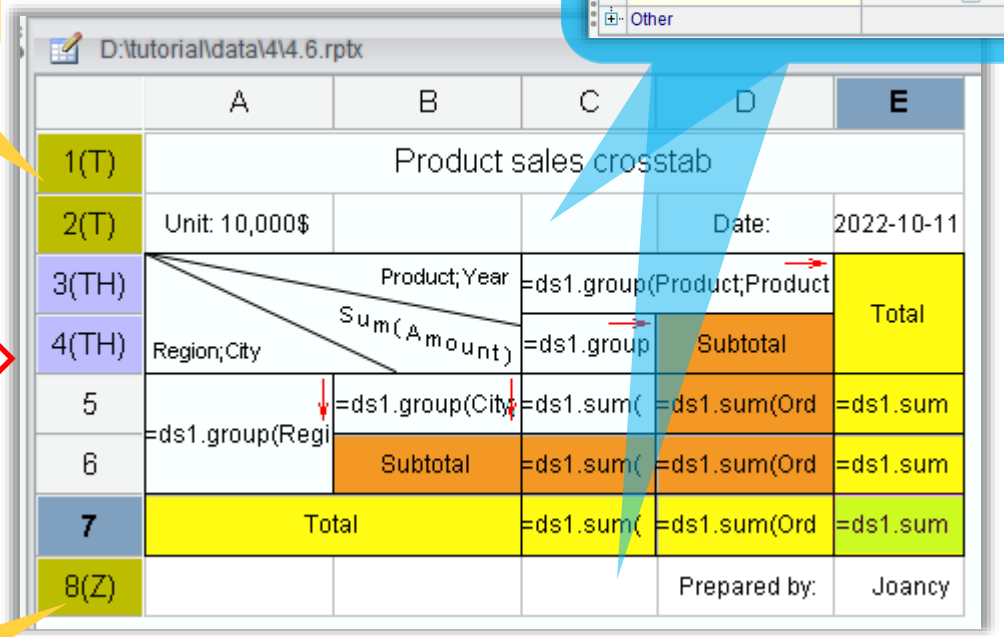


点鼠标右键
选择追加数据行

前2行设为头标题

C2、C8设置：
分页时伸缩

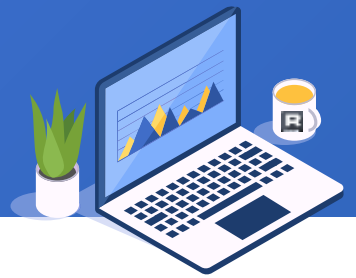




末行设为尾标题

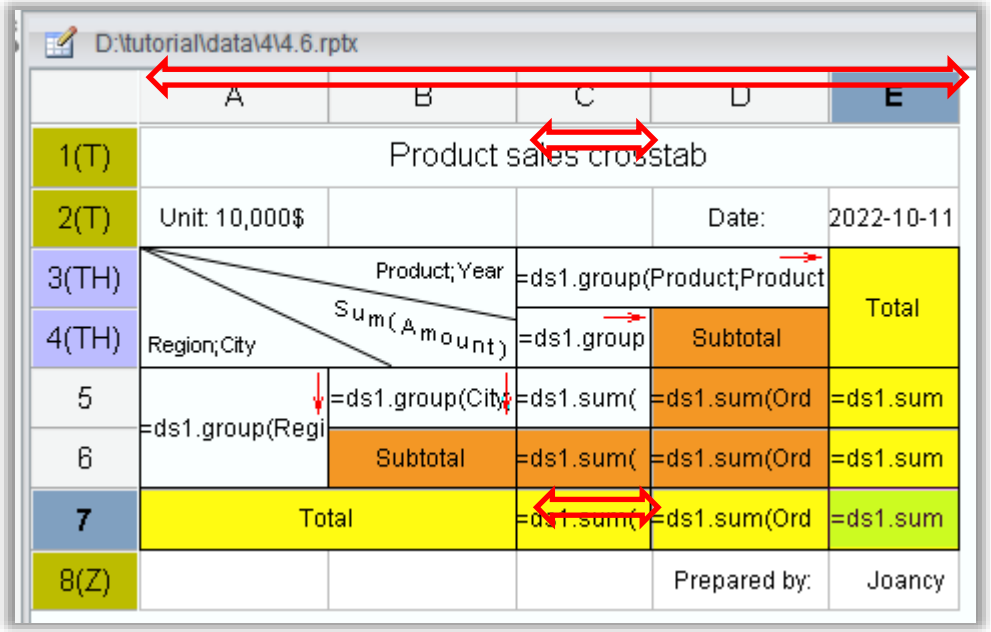
其余新增行的单元格如图中所示设计

4.2 常见交叉报表的制作 – 表头表尾



知识点：单元格定位拉伸规则

在报表横向扩展时报表宽度会增加，为了保持报表整体方正美观，头标题、尾标题、页眉、页脚四个区域的宽度，也需要跟随增加，增加的规则是：每行只有一个单元格会拉伸扩宽，其余单元格的大小和位置保持不变。行内从左到右查找，如果某单元格勾选了“分页时伸缩”属性，则它为本行的拉伸格，如果没有找到，则拉伸最右边的单元格。



在报表4.6.rptx中：

第1行没有单元格勾选“分页时伸缩”，所以A1为拉伸格。

第2行C2单元格勾选了“分页时伸缩”，C2为拉伸格。

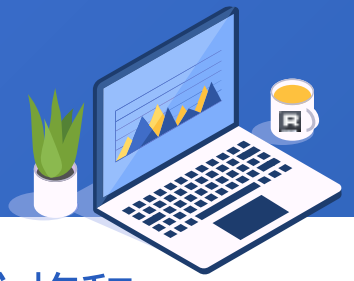
同理，第8行中C8为拉伸格

CONTENTS



4.3 交叉报表的结构和主格

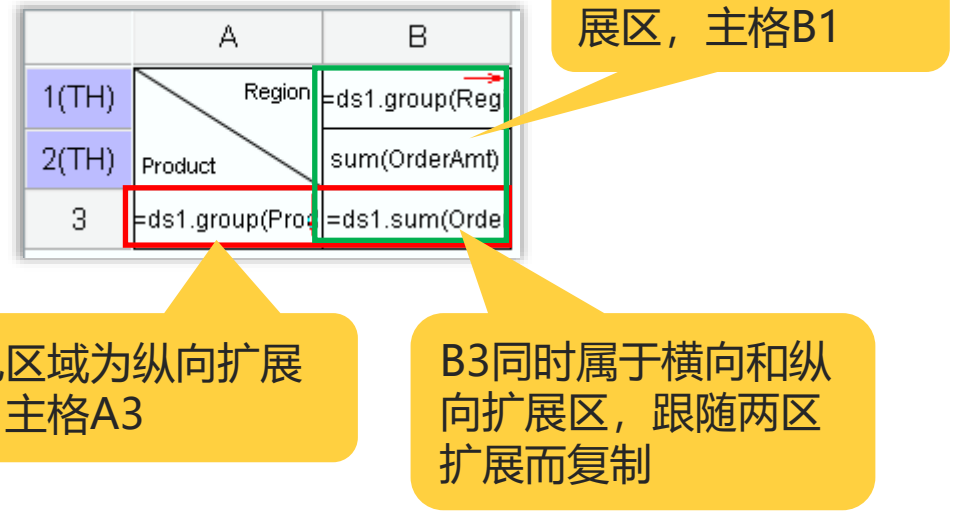
4.3 交叉报表的结构和主格



交叉报表由至少一个横向扩展区域和至少一个纵向扩展区域构成，因此至少有一个左主格和一个上主格。复杂的多层交叉报表可能会有多个横向扩展或多个纵向扩展区域。

交叉报表左主格的默认规则 及设置原理与分组报表相同。

4.1.rptx

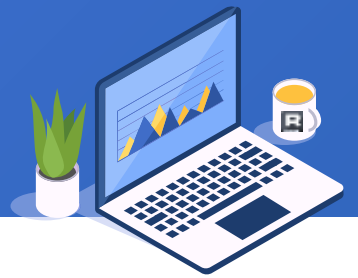


上主格默认规则：
从本单元格上边开始，从下向上查找，第一个横向扩展的单元格即为它的默认上主格；若上边某单元格指定了上主格，则那个指定的格也默认为本单元格的上主格；若找到最上边时还未找到，则默认上主格为报表首格(左上角)

左图中，B3的默认左主格为A3，

B2、B3的默认上主格为B1

4.3 交叉报表的结构和主格



4.1.rptx

D:\tutorial\data\4\4.1.rptx

	A	B
1(TH)	Region	=ds1.group(Regi
2(TH)	Product	sum(OrderAmt)
3	=ds1.group(Pro	=ds1.sum(Order

计算时遇到B1
先横向扩展

	A	B	C	D	E	F
1(TH)	Region	N	NE	NW	S	SW
2(TH)	Product	sum(OrderAmt)	sum(OrderAmt)	sum(OrderAmt)	sum(OrderAmt)	sum(OrderAmt)
3	=ds1.group(Pro	=ds1.sum(Order	=ds1.sum(Order	=ds1.sum(Order	=ds1.sum(Order	=ds1.sum(Orde

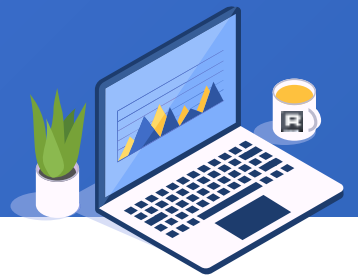
有5个地区，复制出
5个绿色区域

计算时遇到A3
再纵向扩展

有3个产品，复制出
3个红色区域

Region / Product	N	NE	NW	S	SW
	sum(OrderAmt)	sum(OrderAmt)	sum(OrderAmt)	sum(OrderAmt)	sum(OrderAmt)
Apple	5354.74	6296.88	6214.17	4063.84	7056.92
Banana	4732.21	3648.91	5657.35	4255.41	5668.79
Pear	4313.48	5619.03	5770.16	5380.94	4535.99

4.3 交叉报表的结构和主格



报表4.2.rptx的结构分析

D:\tutorial\data\4\4.2.rptx

	A	B	C	D
1(TH)	Product	=ds1.group(Product;Product:1)		
2(TH)	Region	count(Order	max(OrderA	sum(OrderA
3	=ds1.group(Reg	=ds1.count()	=ds1.max(O	=ds1.sum(O

绿色框内为横向扩展区域，主格B1，区域内的其他单元格的默认上主格也正好是B1

知识点：单元格的默认上主格正好是横向扩展区域的主格时，无需指定上主格

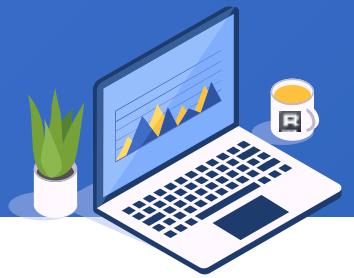
红色框内为纵向扩展区域，主格A3

有3个产品，复制出3个绿色区域

有5个地区，复制出5个红色区域

Product / Region	Apple			Banana			Pear		
	count(Order	max(OrderA	sum(OrderA	count(Order	max(OrderA	sum(OrderA	count(Order	max(OrderA	sum(OrderA
N	28	329.56	5354.74	23	347.55	4732.21	22	339.80	4313.48
NE	32	334.63	6296.86	21	313.64	3648.91	27	341.37	5619.03
NW	29	336.48	6214.17	28	349.87	5657.35	26	344.84	5770.16
S	20	346.74	4063.84	23	335.23	4255.41	30	346.06	5380.94
SW	34	338.59	7056.92	33	344.34	5668.79	24	320.83	4535.99

4.3 交叉报表的结构和主格



报表4.5.rptx的结构分析

绿色粗框内为一层横向扩展区域产品，主格C1

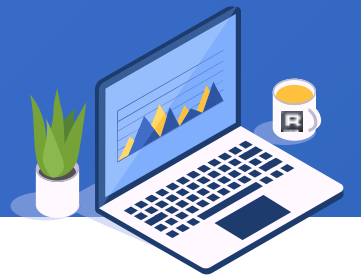
绿色细框内为二层横向扩展区域年份，主格C2

红色粗框内为一层纵向扩展区域地区，主格A3

红色细框内为二层纵向扩展区域城市，主格B3

	A	B	C	D	E
1(TH)	Product;Year		=ds1.group(Product;Pro		Total
2(TH)	Region;City	Sum(OrderAmt)	=ds1.group	Subtotal	
3	ds1.group(Regi	=ds1.group(City	=ds1.sum(	=ds1.sum(	=ds1.sum(
4		Subtotal	=ds1.sum(	=ds1.sum(	=ds1.sum(
5			=ds1.sum(	=ds1.sum(	=ds1.sum(

4.3 交叉报表的结构和主格



报表4.5.rptx预览后

苹果有3年销售数据，复制出3个绿色细框区域

有3种产品，复制出3个绿色粗框区域

华北有3个城市，复制出3个红色细框区域

有5个地区，复制出5个红色粗框区域

Region;City	Product;Year / Sum(OrderAmt)	Apple				Banana				Pear				Total
		2016	2017	2018	Subtotal	2016	2017	2018	Subtotal	2016	2017	2018	Subtotal	
N	Beijing	385.51	1783.87	58.55	2227.93	0.00	2049.68	0.00	2049.68	176.87	678.25	577.13	1432.25	5709.86
	Shijiazhuang	581.09	943.54	71.06	1595.69	519.39	621.09	105.91	1246.39	0.00	1019.19	227.07	1246.26	4088.34
	Tianjin	417.82	914.77	198.53	1531.12	183.51	1252.63	0.00	1436.14	0.00	992.93	642.04	1634.97	4602.23
	Subtotal	1384.42	3642.18	328.14	5354.74	702.90	3923.40	105.91	4732.21	176.87	2690.37	1446.24	4313.48	14400.43
NE	Changchun	1079.08	1366.23	326.60	2771.91	174.60	488.81	0.00	663.41	596.57	1130.20	265.82	1992.59	5427.91
	Dalian	457.98	1358.01	560.11	2376.10	642.03	791.12	288.52	1721.67	174.07	1973.63	0.00	2147.70	6245.47
	Shenyang	469.36	504.87	174.62	1148.85	486.44	568.64	208.75	1263.83	368.33	955.69	154.72	1478.74	3891.42
	Subtotal	2006.42	3229.11	1061.33	6296.86	1303.07	1848.57	497.27	3648.91	1138.97	4059.52	420.54	5619.03	15564.80
NW	Lanzhou	213.45	1592.62	220.26	2026.33	592.32	1637.58	0.00	2229.90	99.73	689.55	0.00	789.28	5045.51
	Xining	0.00	621.34	0.00	621.34	153.36	1882.07	389.66	2425.09	649.91	1592.30	0.00	2242.21	5288.64
	Yinchuan	547.02	2323.09	696.39	3566.50	0.00	777.38	224.98	1002.36	316.73	2421.94	0.00	2738.67	7307.53
	Subtotal	760.47	4537.05	916.65	6214.17	745.68	4297.03	614.64	5657.35	1066.37	4703.79	0.00	5770.16	17641.68
S	Guangzhou	489.58	1111.86	368.07	1969.51	0.00	98.08	758.94	857.02	263.25	929.80	379.87	1572.92	4399.45
	Shenzhen	0.00	906.74	57.57	964.31	336.87	872.83	335.23	1544.93	497.07	1046.75	270.46	1814.28	4323.52
	Xiamen	739.74	390.28	0.00	1130.02	71.27	1590.39	191.80	1853.46	336.61	966.50	690.63	1993.74	4977.22
	Subtotal	1229.32	2408.88	425.64	4063.84	408.14	2561.30	1285.97	4255.41	1096.93	2943.05	1340.96	5380.94	13700.19
SW	Chengdu	76.03	1319.73	548.62	1944.38	399.65	948.21	639.39	1987.25	282.70	1213.50	593.05	2089.25	6020.88
	Guiyang	735.60	1980.90	365.32	3081.82	688.42	804.14	112.86	1605.42	299.13	1086.00	0.00	1385.13	6072.37
	Kunming	923.92	1106.80	0.00	2030.72	387.20	1556.20	132.72	2076.12	579.94	320.83	160.84	1061.61	5168.45
	Subtotal	1735.55	4407.43	913.94	7056.92	1475.27	3308.55	884.97	5668.79	1161.77	2620.33	753.89	4535.99	17261.70
Total		7116.18	18224.65	3645.70	28986.53	4635.06	15938.85	3388.76	23962.67	4640.91	17017.06	3961.63	25619.60	78568.80

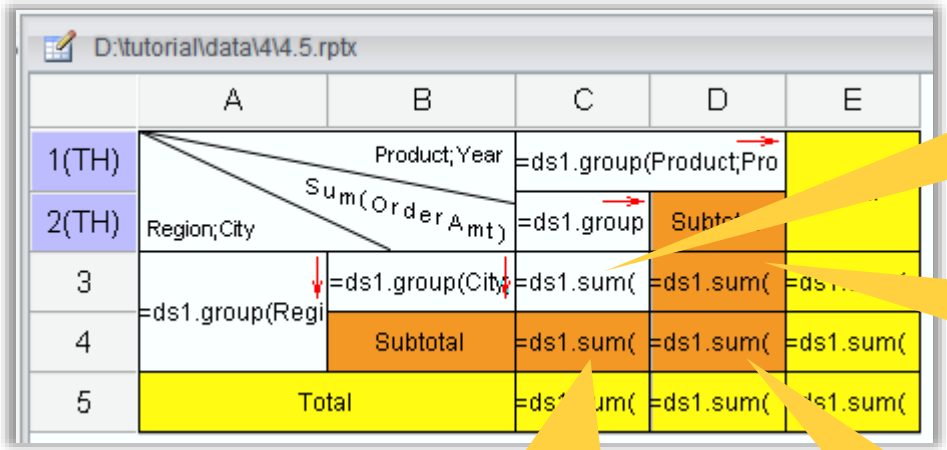
CONTENTS



4.4 交叉报表的运算

4.4 交叉报表的运算 – 汇总统计的范围

交叉报表中，汇总表达式写在不同的分组层次中，参与统计的数据范围不同，由表达式所在单元格的左主格链条与上主格链条范围共同决定。



=ds1.sum(OrderAmt)
C3左主格链条为A3、B3，上主格链条为C1、C2，统计某地区某城市中某产品某一年的所有订单的金额之和。

=ds1.sum(OrderAmt)
D3左主格链条为A3、B3，上主格链条为C1，统计某地区某城市中某产品的所有订单的金额之和。

=ds1.sum(OrderAmt)
C4左主格链条为A3，上主格链条为C1、C2，统计某地区中某产品某一年的所有订单的金额之和。

=ds1.sum(OrderAmt)
D4左主格链条为A3，上主格链条为C1，统计某地区中某产品的所有订单的金额之和。

其余汇总单元格可依此类推

4.4 交叉报表的运算 – 汇总统计的范围



统计华北地区北京市 2016年苹果订单金额总和

统计华北地区天津市香蕉订单金额总和

统计东北地区长春市订单金额总和

统计华北地区香蕉订单金额总和

统计东北地区2016年苹果订单金额总和

统计东北地区订单金额总和

统计2016年苹果订单金额总和

统计香蕉订单金额总和

统计所有订单金额总和

Sum(OrderAmt)	Product;Year	Apple				Banana				Pear				
		2016	2017	2018	Subtotal	2016	2017	2018	Subtotal	2016	2017	2018	Subtotal	
	Beijing	385.51	1783.87	58.55	2227.93	0.00	2049.68	0.00	2049.68	176.87	678.25	577.13		
	Shijiazhuang	581.09	943.54	71.06	1595.69	519.39	621.09	105.91	1246.39	0.00	1019.19	227.07	124	4088.34
	Tianjin	417.82	914.77	198.53	1531.12	183.51	1252.63	0.00	1436.14	0.00	992.93	642.04	1634.9	4602.23
	Subtotal	1384.42	3642.18	328.14	5354.74	702.90	3923.40	105.91	4732.21	176.87	2690.37	1446.24	4313.48	14400.43
NE	Changchun	1079.08	1366.23	326.60	2771.91	174.60	488.81	[illegible]	663.41	596.57	1130.20	265.82	1992.59	5427.91
	Dalian	457.98	1358.01	560.11	[illegible]	[illegible]	[illegible]	[illegible]	1721.67	174.07	1973.63	0.00	2147.70	6245.47
	Shenyang	469.36	504.87	174.62	[illegible]	[illegible]	[illegible]	[illegible]	1263.83	368.33	955.69	154.72	1478.74	3891.42
	[illegible]	2006.42	3229.11	1061.33	[illegible]	[illegible]	[illegible]	[illegible]	3648.91	1138.97	4059.52	420.54	5619.03	15564.80
	Lanzhou	213.45	1592.62	220.26	[illegible]	[illegible]	[illegible]	0.00	2229.90	99.73	689.55	0.00	789.28	5045.51
	Xining	0.00	621.34	0.00	621.34	153.36	1882.07	389.66	2425.09	649.91	1592.30	0.00	2242.21	5288.64
	Yinchuan	547.02	2323.09	696.39	3566.50	0.00	777.38	224.98	1002.36	316.73	2421.94	0.00	2738.67	7307.53
	Subtotal	760.47	4537.05	916.65	6214.17	745.68	4297.03	614.64	5657.35	1066.37	4703.79	0.00	5770.16	17641.68
S	Guangzhou	489.58	1111.86	368.07	1969.51	0.00	98.08	758.94	857.02	263.25	929.80	379.87	1572.92	4399.45
	Shenzhen	0.00	906.74	57.57	964.31	336.87	872.83	335.23	1544.93	497.07	1046.75	270.46	1814.28	4323.52
	Xiamen	739.74	390.28	0.00	[illegible]	[illegible]	[illegible]	[illegible]	1853.46	336.61	966.50	690.63	1993.74	4977.22
	Subtotal	1229.32	2408.88	425.64	[illegible]	[illegible]	[illegible]	[illegible]	4255.41	1096.93	2943.05	1340.96	5380.94	13700.19
	Chengdu	76.03	1319.73	548.62	[illegible]	[illegible]	[illegible]	[illegible]	1987.25	282.70	1213.50	593.05	2089.25	6020.88
	Guiyang	735.60	1980.90	365.32	[illegible]	[illegible]	[illegible]	[illegible]	1605.42	299.13	1086.00	0.00	1385.13	6072.37
	Kunming	923.92	1106.80	0.00	2030.72	387.20	1556.20	[illegible]	2076.12	579.94	320.83	160.84	1061.61	5168.45
	[illegible]	1735.55	4407.43	913.94	7056.92	1475.27	3308.55	[illegible]	5668.79	1161.77	2620.33	753.89	4535.99	17261.70
Total		7116.18	18224.65	3645.70	28986.53	4635.06	15938.85	3388.76	23962.67	4640.91	17017.06	3961.63	25619.60	78568.80

4.4 交叉报表的运算 – 汇总统计的范围



在汇总统计上层分组的小计及总表的总计时，也可以用单元格值来汇总，统计方式的确定原理也是相同的。将4.5.rptx另存为4.7.rptx，改成如下图：

D:\tutorial\data\4\4.7.rptx

	A	B	C	D	E
1(TH)	Product;Year / Sum(OrderAmt) / Region;City		=ds1.group(Product;Pro		Total
2(TH)			=ds1.group	Subtotal	
3	=ds1.group(Regi	=ds1.group(City	=ds1.sum(	=sum(C3{}	=sum(C3{}
4		Subtotal	=sum(C3{}	=sum(C3{}	=sum(C3{}
5	Total		um(C3{}	=sum(3{}	=sum(C3{}

=sum(C3{})
D3左主格链条为A3、B3，上主格链条为C1，统计某地区某城市中某产品的所有C3之和。

=sum(C3{})
E3左主格链条为A3、B3，上主格是报表首格，统计某地区某城市的所有C3之和。

=sum(C3{})
C4左主格链条为A3，上主格链条为C1、C2，统计某地区中某产品某一年的所有C3之和。

=sum(C3{})
D4左主格链条为A3，上主格链条为C1，统计某地区中某产品的所有C3之和。

=sum(C3{})
E5左主格是报表首格，上主格是报表首格，统计所有C3之和。

4.4 交叉报表的运算 – 汇总统计的范围



用单元格的值汇总时，导出带公式的Excel文件里，汇总统计格会有计算公式；用数据集汇总时则不会带公式

	C	D	E	F	G	H	I	J	K	L	M	N	O
			Apple			Banana			Pear				Total
			2016	2017	2018	Sub total	2016	2017	2018	Sub total	2016	2017	2018	Sub total	
3			385.51	1783.87	58.55	2227.93	0.00	2049.68	0.00	2049.68	176.87	678.25			
4	N	Shijiazhuang	581.09	943.54	71.06	1595.69	519.39	621.09	105.91	1246.39	0.00	1019.19			
5		Tianjin	417.82	914.77	198.53	1531.12	183.51	1252.63	0.00	1436.14	0.00	992.93			
6		Sub total	1384.42	3642.18	328.14	5354.74	702.90	3923.40	105.91	4732.21	176.87	2690.37			
7	NE	Changchun	1079.08	1366.23	326.60	2771.91	174.60	488.81	0.00	663.41	596.57	1130.20	265.82	199	5427.91
8		Dalian	457.98	1358.01	560.11	2376.10	642.03	791.12	28	1721.67	174.07	1973.63	0.00	2147.70	6245.47
9		Shenyang	469.36	504.87	174.62	1148.85	4			63.83	368.33	955.69	154.72	1478.74	3891.42
10		Sub total	2006.42	3229.11	1061.33	6296.86	13			48.91	1138.97	4059.52	420.54	5619.03	15564.80
11	NW	La			220.26	2026.33	592.32	1637.58	0.00	2229.90	99.73	689.55	0.00		
12		Xi			0.00	621.34	153.36	1882.07	389.66	2425.09	649.91	1592.30	0.00		
13		Yinchuan	547.02	2323.09	696.39	3566.50	0.00	777.38	224.98	1002.36	316.73	2421.94	0.00		
14		Sub total	760.47	4537.05	916.65	6214.17	745.68	4297.03	614.64	5657.35	1066.37	4703.79	0.00		
15	S	Guangzhou	489.58	1111.86	368.07	1969.51	0.00	98.08	758.94	857.02	263.25	929.80	379.87		
16		Shenzhen	0.00	906.74	57.57	964.31	336.87	872.83	335.23	1544.93	497.07	1046.75	270.46	1814.28	
17		Xiamen	739.74	390.28	0.00	1130.02	71.27	1590.39	191.80	1853.46	336.61	966.50	690.63	1993.74	.22
18		Sub total	1229.32	2408.88	425.64	4063.84	408.14	2561.30	1285.97	4255.41	1096.93	2943.05	1340.96	5380.94	13700.19
19	WS	Ch			.62	1944.38						1213.50	593.05	2089.25	6020.88
20		G			65.32	3081.82				3		1086.00	0.00	1385.13	6072.37
21		Kunming	923.92	1106.80	0.00	2030.72					94	320.83	160.84	1061.61	5168.45
22		Sub total	1735.55	4407.43	913.94	7056.92	1	3308.55	884.97	5668.79	1161.77	2620.33	753.89	4535.99	17261.70
23	Total		7116.18	18224.65	3645.70	28986.53	4635.06	15938.85	3388.76	23962.67	4640.91	17017.06	3961.63	25619.60	78568.80

=SUM(C8:E8,G8:I8,K8:M8)

=SUM(G7:I7)

=SUM(E7:E9)

=SUM(C15:E15,G15:I15,K15:M15,C16:E16,G16:I16,K16:M16,C17:E17,G17:I17,K17:M17)

=SUM(C15:E17)

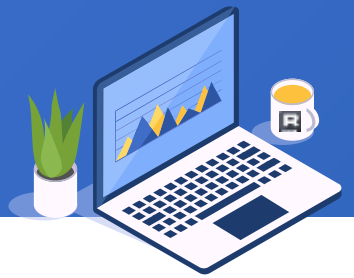
=SUM(G3:G5,G7:G9,G11:G13,G15:G17,G19:G21)

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4.5 习 题

4.5 习题



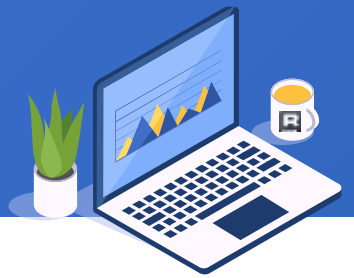
习题数据文件practice4.1.xlsx部分数据如下图：

	A	B	C	D	E	F
1	District	School	Grade	Class	Name	Math
2	Xicheng	Yumin primary school	4	1	Name1	82
3	Xicheng	Yumin primary school	4	1	Name2	84
4	Xicheng	Yumin primary school	4	1	Name3	97
5	Xicheng	Yumin primary school	4	1	Name4	68
6	Xicheng	Yumin primary school	4	1	Name5	81
7	Xicheng	Yumin primary school	4	1	Name6	78
8	Xicheng	Yumin primary school	4	1	Name7	73
9	Xicheng	Yumin primary school	4	1	Name8	94
10	Xicheng	Yumin primary school	4	1	Name9	90

习题1、用向导制作交叉报表：以城区(District)为行定义、年级(Grade)为列定义，统计各年级学生数学平均成绩。

习题2、用向导制作交叉报表：以城区为行定义、年级为列定义，统计各年级学生数学平均成绩、最高分、最低分及学生人数。

4.5 习题



习题3、用向导制作交叉报表：以城区和学校为行定义、年级和班号为列定义，统计各班学生数学平均成绩。

习题4、为习题3的报表按照4.6.rptx的样式修改：汇总标题移到斜线单元格，加上头标题、统计日期，尾标题加上些制表说明、制表人等。

第5章
多片关联



CONTENTS



5.1 从交叉到分片

5.2 扩展与静态混合

5.3 行列对称

5.4 利用分片关联做主子表

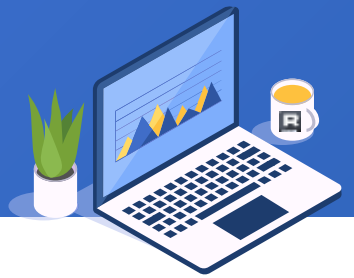
5.5 习题

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5.1 从交叉到分片

5.1 从交叉到分片



常见的交叉报表

Dimension		Item	Sales Stats						TrialCount		WebTFC
			Esproc			Report			Esproc	Report	
			OrderCount	OrderAmt	ReceivedPmt	OrderCount	OrderAmt	ReceivedPmt			
Domestic	2019	**Total**	**3197**	**¥96,936,00**	**¥96,450,00**	**3131**	**¥87,064,00**	**¥74,820,00**	**21266**	**21711**	**116929**
		Q1	329	¥10,648,00	¥9,820,000	308	¥7,360,000	¥5,170,000	5446	5384	24383
		Q2	677	¥18,672,00	¥18,150,00	531	¥17,216,00	¥15,100,00	5407	5112	33186
		Q3	757	¥25,944,00	¥24,770,00	794	¥24,664,00	¥20,850,00	5330	5015	23193
		Q4	1434	¥41,672,00	¥43,710,00	1498	¥37,824,00	¥33,700,00	5083	6200	36167
	2020	**Total**	**4493**	**¥119,600,0**	**¥111,680,0**	**4130**	**¥116,150,0**	**¥106,630,0**	**23746**	**20881**	**141675**
		Q1	418	¥10,860,00	¥9,410,000	339	¥11,320,00	¥10,490,00	6189	5943	25138
		Q2	902	¥23,840,00	¥22,450,00	791	¥21,290,00	¥20,480,00	6576	4544	40336
		Q3	1191	¥33,700,00	¥30,880,00	1090	¥34,260,00	¥30,700,00	4769	4832	42268
		Q4	1982	¥51,200,00	¥48,940,00	1910	¥49,280,00	¥44,960,00	6212	5412	33932
	Total		**7690**	**¥216,536,0**	**¥208,130,0**	**7261**	**¥203,214,0**	**¥181,450,0**	**45012**	**42592**	**258604**
	Among Beijing		**1552**	**¥36,180,00**	**¥53,140,00**	**1314**	**¥34,960,00**	**¥40,300,00**	**8183**	**7101**	**47452**
	E		1183	¥30,514,00	¥24,530,00	783	¥28,578,00	¥22,840,00	7292	7647	58112
	N		3402	¥94,016,00	¥106,760,0	3515	¥88,196,00	¥83,710,00	14736	14437	86291
	S		1799	¥50,794,00	¥43,430,00	1836	¥49,598,00	¥42,510,00	16060	14458	74226
	SW		1306	¥41,212,00	¥33,410,00	1127	¥36,842,00	¥32,390,00	6924	6050	39975
Overseas	**Total**		**9434**	**$27,636,800**	**$24,092,000**	**9350**	**$26,470,400**	**$23,758,000**	**36076**	**35332**	**209832**
	Among USA		**962**	**$3,311,200**	**$3,001,000**	**1050**	**$3,614,400**	**$3,219,000**	**8635**	**7266**	**44682**
	Asia		3316	$9,247,800	$7,781,000	3164	$8,651,200	$7,891,000	10198	10852	61460
	Europe		3272	$9,087,400	$8,070,000	2989	$8,505,200	$7,489,000	10717	10469	63923
	North America		1833	$6,369,800	$5,652,000	2255	$6,539,400	$6,078,000	11738	10350	64913
	Oceania		1013	$2,931,800	$2,589,000	942	$2,774,600	$2,300,000	3423	3661	19536

来自不同的数据表
按照相同的统计维度关联在一起

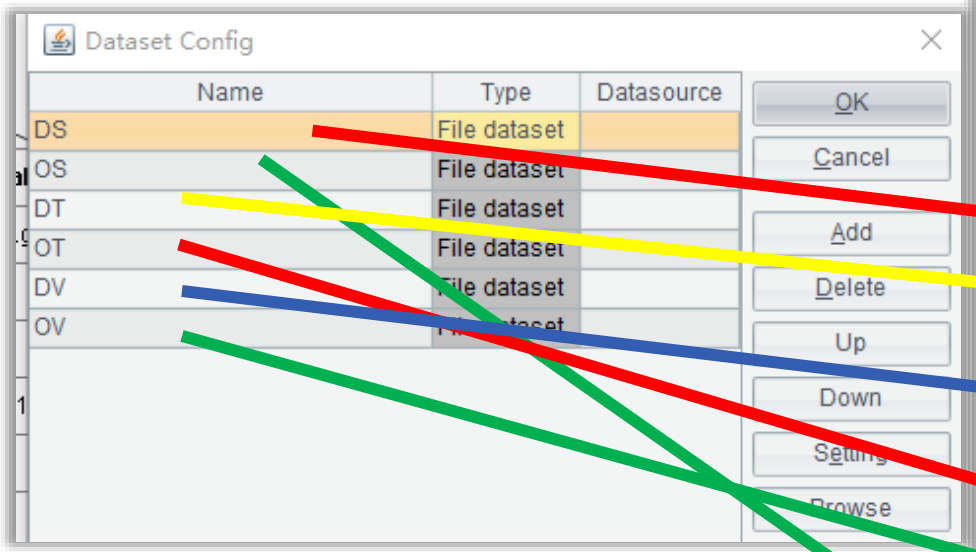
想同时看看按地区统计的数据

还想看看国外的数据

5.1 制作过程——定义数据集



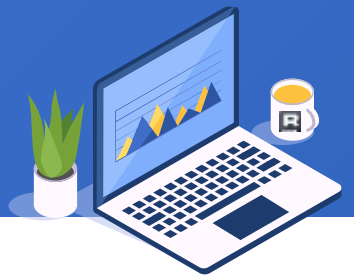
根据报表统计数据的需要，先定义多个数据集



Dimension	Item		Sales Stats						TrialCount		WebTFC
			Esproc			Report			Esproc	Report	
			OrderCount	OrderAmt	ReceivedPmt	OrderCount	OrderAmt	ReceivedPmt			
		Total	**3197**	**¥96,936,00**	**¥96,450,00**	**3131**	**¥87,064,00**	**¥74,820,00**	**21266**	**21711**	**116929**
	2019	Q1	329	¥10,648,00	¥9,820,000	308	¥7,360,000	¥5,170,000	5446	5384	24383
		Q2	677	¥18,672,00	¥18,150,00	531	¥17,216,00	¥15,100,00	5407	5112	33186
		Q3	757	¥25,944,00	¥24,770,00	794	¥24,664,00	¥20,850,00	5330	5015	23193
			1434	¥41,672,00	¥43,710,00	1498	¥37,824,00	¥33,700,00	5083	6200	36167
		Total	**4493**	**¥119,600,0**	**¥111,680,0**	**4130**	**¥116,150,0**	**¥106,630,0**	**23746**	**20881**	**141675**
	2020	Q1	418		¥9,410,000	339	¥11,320,00	¥10,490,00	6189	5943	25138
		Q2	902	¥23,840,00	¥22,450,00	791			6576	4544	40336
			1191	¥33,700,00	¥30,880,00	1090	¥34,260,00	¥30,700,00	4769	4982	42268
		Q4	1982	¥51,200,00		1910	¥49,280,00	¥44,960,00	6212	5412	33933
	Total		**7690**	**¥216,536,0**	**¥208,130,0**	**7261**	**¥203,214,0**		**45012**	**42592**	**258604**
	Among Beijing		**1552**	**¥36,180,00**	**¥53,140,00**	**1314**	**¥34,960,00**	**¥40,300,00**	**8183**	**7101**	**47452**
	E		1183	¥30,514,00	¥24,530,00	783	¥28,578,00	¥22,840,00	7292	7647	58112
	N		3402	¥94,0	¥106,760,0	3515	¥88,196,00	¥83,710,00	14736	14437	86291
	S		1750	¥50,794,00	¥43,430,00	1836	¥49,598,00	¥42,510,00	16060	14458	74226
	SW		1306	¥41,212,00	¥ 410,00	1127	842,00	¥32,390,00	6924	6050	39975
Overseas	**Total**		**9434**	**$27,636,800**	**$24,092,000**	3650	**$26,470,400**	**$23,7 800**	**36076**	**35332**	**209832**
	Among USA		**962**	**$3,311,200**	**$3,001,000**	**1080**	**$3,614,4**	**$3,219,000**	3236	**7268**	**44682**
	Asia		3316	$9,247,800	$7,781,000	3164	$8,651,200	$7,891,000	10198	10852	61460
	Europe		3272	$9,087,400	$8,070,000	2989	$8,505,200	$7,489,000	10717	10469	63923
	North America		1833	$6,369,800	$5,652,000	2255	$6,539,400	$6,078,000	11738	10350	313
	Oceania		1013	$2,931,800	$2,589,000	942	$2,774,600	$2,300,000	3423	3661	19536

每个数据集对应一片数据

5.1 制作过程——逐片定义表达式



第一片：先定义国内销售数据统计

横向扩展

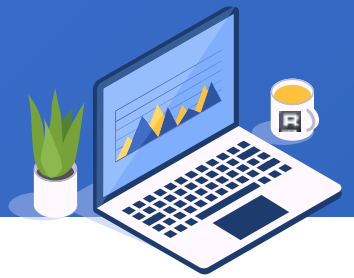
纵向扩展

	A(CH)	B(CH)	C(CH)	D	E	F
1(TH)			Item	Sales Stats		
2(TH)				= DS.group(Product:1)		
3(TH)	Dimension			OrderCount	OrderAmt	ReceivedPmt
4	Domestic	= DS.gro	Total	=sum(D5{})	=sum(E5{})	=sum(F5{})
5			= DS.gr	= DS.sum(Or	= DS.sum(O	= DS.sum(R
6		Total		=sum(D8{})	=sum(E8{})	=sum(F8{})
7		Among Beijing		= DS.sum(Or	= DS.sum(Or	= DS.sum(R
8		= DS.group(Regi		= DS.sum(Or	= DS.sum(Or	= DS.sum(R

按照汇总维度分组扩展

和上/左主格同一个数据集，不需要加过滤条件，缺省统计该维度内数据

5.1 制作过程——逐片定义表达式

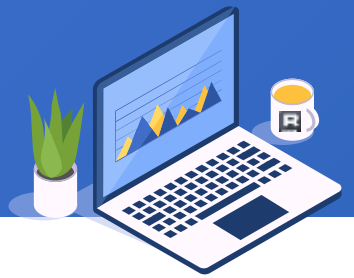


第二步：定义国外销售数据统计

	A(CH)	B(CH)	C(CH)	D	E	F
1(TH)	Item			Sales Stats		
2(TH)				= DS.group(Product:1)		
3(TH)	Dimension			OrderCount	OrderAmt	ReceivedPmt
4	Domestic	= DS.gro	Total	=sum(D5{})	=sum(E5{})	=sum(F5{})
5			= DS.gro	= DS.sum(Or	= DS.sum(Or	= DS.sum(R
6		Total		=sum(D8{})	=sum(E8{})	=sum(F8{})
7		Among Beijing		= DS.sum(Or	= DS.sum(Or	= DS.sum(R
8		= DS.group(Regio		= DS.sum(Or	= DS.sum(Or	= DS.sum(R
9	Overseas	Total		=sum(D11{})	=sum(E11{})	=sum(F11{})
10		Among USA		= OS.sum(O	= OS.sum(O	= OS.sum(R
11		= OS.group(Conti		= OS.sum(Or	= OS.sum(Or	= OS.sum(R

和上主格不是同一个数据集，需要用过滤条件关联

5.1 制作过程——逐片定义表达式



第三步：增加试用人数和好多乾访问量统计

	A(CH)	B(CH)	C(CH)	D	E	F	G	H
1(TH)	Item			Sales Stats			TrialCount	
2(TH)				= DS.group(Product:1)			= DT.group(Product:1)	WebTFC
3(TH)	Dimension			OrderCount	OrderAmt	ReceivedPmt		
4		= DS.gro	**Total**	**=sum(D5{})**	**=sum(E5{})**	**=sum(F5{})**	**=sum(G5{})**	**=sum(H5{})**
5			= DS.gro	= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(TrialCount,Year==B5 && Quarter==C5)	= DV.sum(WebTFC,Year==B5 && Quarter==C5)
6	Domestic	**Total**		**=sum(D8{})**	**=sum(E8{})**	**=sum(F8{})**	**=sum(G8{})**	**=sum(H8{})**
7		**Among Beijing**		**= DS.sum(Or**	**= DS.sum(Or**	**= DS.sum(R**	**= DT.sum(TrialCount,City=="Beijing")**	**= DV.sum(WebTFC,City=="Beijing")**
8		= DS.group(Regio		= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(TrialCount,Region==B8)	= DV.sum(WebTFC,Region==B8)
9		**Total**		**=sum(D11{})**	**=sum(E11{})**	**=sum(F11{})**	**=sum(G11{})**	**=sum(H11{})**
10	Overseas	**Among USA**		**= OS.sum(O**	**= OS.sum(O**	**= OS.sum(R**	**= OT.sum(TrialCount,Country=="USA" && Product==G2)**	**= OV.sum(WebTFC,Country=="USA")**
11		= OS.group(Conti		= OS.sum(Or	= OS.sum(Or	= OS.sum(R	= OT.sum(TrialCount,Continent==B11 && Product==G2)	= OV.sum(WebTFC,Continent==B11)

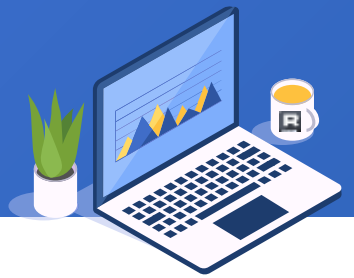
和左主格不是一个数据集，需要增加过滤条件关联

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5.2 扩展与静态混合

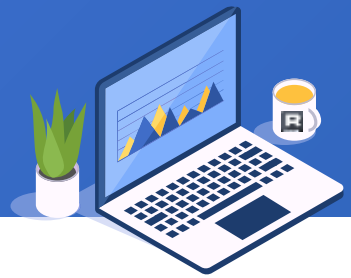
5.2 扩展区域与静态区域混合



纵向扩展

	A(CH)	B(CH)	C(CH)	D	E	F	G	H
1(TH)	Item			Sales Stats			TrialCount	
2(TH)				= DS.group(Product:1)			= DT.group(Prod	WebTFC
3(TH)	Dimension			OrderCount	OrderAmt	ReceivedPmt		
4	Domestic	= DS.group(Year:1)	Total	=sum(D5{})	=sum(E5{})	=sum(F5{})	=sum(G5{})	=sum(H5{})
5			= DS.group(Quarter:1)	= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(Trial	= DV.sum(Web
6		Total		=sum(D8{})	=sum(E8{})	=sum(F8{})	=sum(G8{})	=sum(H8{})
7		Among Beijing		= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(Trial	= DV.sum(Web
8		= DS.group(Region:1)		= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(Trial	= DV.sum(Web
9	Overseas	Total		=sum(D11{})	=sum(E11{})	=sum(F11{})	=sum(G11{})	=sum(H11{})
10		Among USA		= OS.sum(O	= OS.sum(O	= OS.sum(R	= OT.sum(Trial	= OV.sum(Web
11		= OS.group(Continent:1)		= OS.sum(Or	= OS.sum(Or	= OS.sum(R	= OT.sum(Trial	= OV.sum(Web

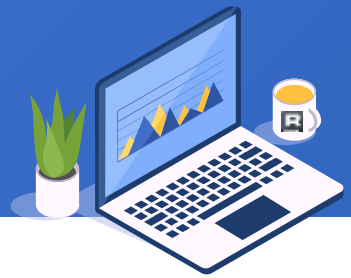
5.2 扩展区域与静态区域混合



横向扩展

	A(CH)	B(CH)	C(CH)	D	E	F	G	H
1(TH)	Item			Sales Stats			TrialCount	
2(TH)				= DS.group(Product:1)			= DT.group(Product:1)	WebTFC
3(TH)	Dimension			OrderCount	OrderAmt	ReceivedPmt		
4	Domestic	= DS.gro	Total	=sum(D5{})	=sum(E5{})	=sum(F5{})	=sum(G5{})	=sum(H5{})
5			= DS.gro	= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(TrialCount,Year=	= DV.sum(Web
6		Total		=sum(D8{})	=sum(E8{})	=sum(F8{})	=sum(G8{})	=sum(H8{})
7		Among Beijing		= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(TrialCount,City==	= DV.sum(Web
8		= DS.group(Regio		= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(TrialCount,Regio	= DV.sum(Web
9	Overseas	Total		=sum(D11{})	=sum(E11{})	=sum(F11{})	=sum(G11{})	=sum(H11{})
10		Among USA		= OS.sum(O	= OS.sum(O	= OS.sum(R	= OT.sum(TrialCount,Count	= OV.sum(Web
11		= OS.group(Contin		= OS.sum(Or	= OS.sum(Or	= OS.sum(R	= OT.sum(TrialCount,Contin	= OV.sum(Web

5.2 扩展区域与静态区域混合



静态格

	A(CH)	B(CH)	C(CH)	D	E	F	G	H
			Item	Sales Stats			TrialCount	
2(TH)				= DS.group(Product:1)			= DT.group(Product:1)	WebTFC
	Dimension			OrderCount	OrderAmt	ReceivedPmt		
4		= DS.gro	**Total**	**=sum(D5{})**	**=sum(E5{})**	**=sum(F5{})**	**=sum(G5{})**	**=sum(H5{})**
5			= DS.gro	= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(TrialCount,Year=	= DV.sum(Web
6	Domestic	**Total**		**=sum(D8{})**	**=sum(E8{})**	**=sum(F8{})**	**=sum(G8{})**	**=sum(H8{})**
7		**Among Beijing**		**= DS.sum(Or**	**= DS.sum(Or**	**= DS.sum(R**	**= DT.sum(TrialCount,City==**	**= DV.sum(Web**
8		= DS.group(Regio		= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(TrialCount,Regio	= DV.sum(Web
9		**Total**		**=sum(D11{})**	**=sum(E11{})**	**=sum(F11{})**	**=sum(G11{})**	**=sum(H11{})**
10	Overseas	**Among USA**		**= OS.sum(O**	**= OS.sum(O**	**= OS.sum(R**	**= OT.sum(TrialCount,Count**	**= OV.sum(Web**
11		= OS.group(Conti		= OS.sum(Or	= OS.sum(Or	= OS.sum(R	= OT.sum(TrialCount,Contin	= OV.sum(Web

5.2 扩展区域与静态区域混合



Dimension		Item	Sales Stats						TrialCount		WebTFC
			Esproc			Report			Esproc	Report	
			OrderCount	OrderAmt	ReceivedPmt	OrderCount	OrderAmt	ReceivedPmt			
Domestic		**Total**	**3197**	**￥96,936,00**	**￥96,450,00**	**3131**	**￥87,064,00**	**￥74,820,00**	**21266**	**21711**	**116929**
	2019	Q1	329	￥10,648,00	￥9,820,000	308	￥7,360,000	￥5,170,000	5446	5384	24383
		Q2	677	￥18,672,00	￥18,150,00	531	￥17,216,00	￥15,100,00	5407	5112	33186
		Q3	757	￥25,944,00	￥24,770,00	794	￥24,664,00	￥20,850,00	5330	5015	23193
		Q4	1434	￥41,672,00	￥43,710,00	1498	￥37,824,00	￥33,700,00	5083	6200	36167
	2020	**Total**	**4493**	**￥119,600,0**	**￥111,680,0**	**4130**	**￥116,150,0**	**￥106,630,0**	**23746**	**20881**	**141675**
		Q1	418	￥10,860,00	￥9,410,000	339	￥11,320,00	￥10,490,00	6189	5943	25138
		Q2	902	￥23,840,00	￥22,450,00	791	￥21,290,00	￥20,480,00	6576	4544	40336
		Q3	1191	￥33,700,00	￥30,880,00	1090	￥34,260,00	￥30,700,00	4769	4982	42268
		Q4	1982	￥51,200,00	￥48,940,00	1910	￥49,280,00	￥44,960,00	6212	5412	33933
	Total		**7690**	**￥216,536,0**	**￥208,130,0**	**7261**	**￥203,214,0**	**￥181,450,0**	**45012**	**42592**	**258604**
	Among Beijing		**1552**	**￥36,180,00**	**￥53,140,00**	**1314**	**￥34,960,00**	**￥40,300,00**	**8183**	**7101**	**47452**
	E		1183	￥30,514,00	￥24,530,00	783	￥28,578,00	￥22,840,00	7292	7647	58112
	N		3402	￥94,016,00	￥106,760,0	3515	￥88,196,00	￥83,710,00	14736	14437	86291
	S		1799	￥50,794,00	￥43,430,00	1836	￥49,598,00	￥42,510,00	16060	14458	74226
	SW		1306	￥41,212,00	￥33,410,00	1127	￥36,842,00	￥32,390,00	6924	6050	39975
Overseas	**Total**		**9434**	**$27,636,800**	**$24,092,000**	**9350**	**$26,470,400**	**$23,758,000**	**36076**	**35332**	**209832**
	Among USA		**962**	**$3,311,200**	**$3,001,000**	**1080**	**$3,614,400**	**$3,219,000**	**8635**	**7268**	**44682**
	Asia		3316	$9,247,800	$7,781,000	3164	$8,651,200	$7,891,000	10198	10852	61460
	Europe		3272	$9,087,400	$8,070,000	2989	$8,505,200	$7,489,000	10717	10469	63923
	North America		1833	$6,369,800	$5,652,000	2255	$6,539,400	$6,078,000	11738	10350	64913
	Oceania		1013	$2,931,800	$2,589,000	942	$2,774,600	$2,300,000	3423	3661	19536

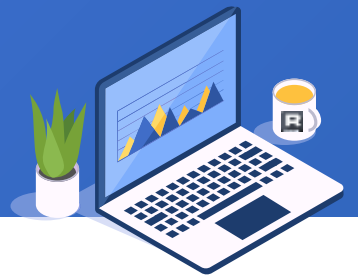
扩展格和静态格和谐共存，天然形成动静态混合报表

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5.3 行列对称

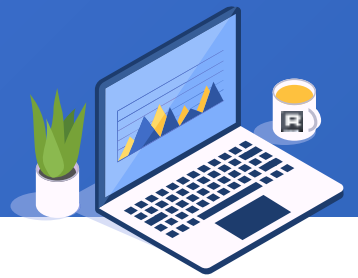
5.3 行列对称



纵向扩展

	A(CH)	B(CH)	C(CH)	D	E	F	G	H
1(TH)	Item			Sales Stats			TrialCount	
2(TH)				= DS.group(Product:1)			= DT.group(Prod	WebTFC
3(TH)	Dimension			OrderCount	OrderAmt	ReceivedPmt		
4		= DS.group(Year:1)	**Total**	**=sum(D5{})**	**=sum(E5{})**	**=sum(F5{})**	**=sum(G5{})**	**=sum(H5{})**
5			= DS.group(Quarter:1)	= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(Trial	= DV.sum(Web
6	Domestic	**Total**		**=sum(D8{})**	**=sum(E8{})**	**=sum(F8{})**	**=sum(G8{})**	**=sum(H8{})**
7		**Among Beijing**		**= DS.sum(Or**	**= DS.sum(Or**	**= DS.sum(R**	**= DT.sum(Trial**	**= DV.sum(Web**
8		= DS.group(Region:1)		= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(Trial	= DV.sum(Web
9		**Total**		**=sum(D11{})**	**=sum(E11{})**	**=sum(F11{})**	**=sum(G11{})**	**=sum(H11{})**
10	Overseas	**Among USA**		**= OS.sum(O**	**= OS.sum(O**	**= OS.sum(R**	**= OT.sum(Trial**	**= OV.sum(Web**
11		= OS.group(Continent:1)		= OS.sum(Or	= OS.sum(Or	= OS.sum(R	= OT.sum(Trial	= OV.sum(Web

5.3 行列对称

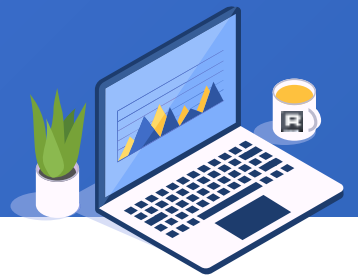


横向扩展

	A(CH)	B(CH)	C(CH)	D	E	F	G	H
1(TH)	Item			Sales Stats			TrialCount	
2(TH)				= DS.group(Product:1)			= DT.group(Product:1)	WebTFC
3(TH)	Dimension			OrderCount	OrderAmt	ReceivedPmt		
4	Domestic	= DS.gro	**Total**	**=sum(D5{})**	**=sum(E5{})**	**=sum(F5{})**	**=sum(G5{})**	**=sum(H5{})**
5			= DS.gro	= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(TrialCount,Year==B	= DV.sum(Web
6		**Total**		**=sum(D8{})**	**=sum(E8{})**	**=sum(F8{})**	**=sum(G8{})**	**=sum(H8{})**
7		**Among Beijing**		**= DS.sum(Or**	**= DS.sum(Or**	**= DS.sum(R**	**= DT.sum(TrialCount,City==''**	**= DV.sum(Web**
8		= DS.group(Regio		= DS.sum(Or	= DS.sum(Or	= DS.sum(R	= DT.sum(TrialCount,Region=	= DV.sum(Web
9	Overseas	**Total**		**=sum(D11{})**	**=sum(E11{})**	**=sum(F11{})**	**=sum(G11{})**	**=sum(H11{})**
10		**Among USA**		**= OS.sum(O**	**= OS.sum(O**	**= OS.sum(R**	**= OT.sum(TrialCount,Country**	**= OV.sum(Web**
11		= OS.group(Conti		= OS.sum(Or	= OS.sum(Or	= OS.sum(R	= OT.sum(TrialCount,Contine	= OV.sum(Web

横向扩展和纵向扩展的能力一样

5.3 行列对称



Preview report:D:\tutorial\data\5\5.1.rptx

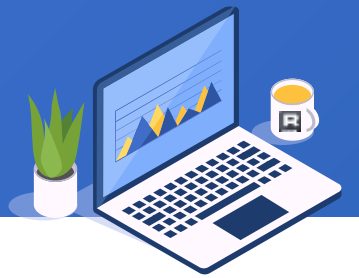
Dimension		Item	Sales Stats						TrialCount	
			Esproc			Report			Esproc	Report
			OrderCount	OrderAmt	ReceivedPmt	OrderCount	OrderAmt	ReceivedPmt		
Domestic	2019	Total	3197	￥96,936,00	￥96,450,00	3131	￥87,064,00	￥74,820,00	21266	21711
		Q1	329	￥10,648,00	￥9,820,000	308	￥7,360,000	￥5,170,000	5446	5384
		Q2	677	￥18,672,00	￥18,150,00	531	￥17,216,00	￥15,100,00	5407	5112
		Q3					￥24,664,00			
		Q4					37,824,00			
	2020	Total					116,150,0			
							11,320,00			
		Q2	902	￥23,840,00	￥22,450,00	791	￥21,290,00	￥20,480,00	6576	4544
		Q3	1191	￥33,700,00	￥30,880,00	1090	￥34,260,00	￥30,700,00	4769	4982
		Q4	1982	￥51,200,00	￥48,940,00	1910	￥49,280,00	￥44,960,00	6212	5412
	Total		7690	￥216,536,0	￥208,130,0	7261	￥203,214,0	￥181,450,0	45012	42592
	Among Beijing		1552	￥36,180,00	￥53,140,00	1314	￥34,960,00	￥40,300,00	8183	7101
	SW		1306	￥41,212,00	￥33,410,00	1127	￥36,842,00	￥32,390,00	6924	6050
	Total		9434	$27,636,800	$24,092,000	9350	$26,470,400	$23,758,000	36076	35332
	Among USA		962	$3,311,200	$3,001,000	1080	$3,614,400	$3,219,000	8635	7268

左表头固定，报表横向滚动

上表头固定，报表纵向滚动

横向和纵向的固定表头滚动表体的能力一样

5.3 行列对称特征总结



- 横向扩展和纵向扩展能力一样
- 左表头和上表头均可固定，表体滚动
- 分页时左表头和上表头均可每页重复

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5.4 用分片关联做主子表

5.4 利用分片关联做主子表



静态区域

纵向扩展区域

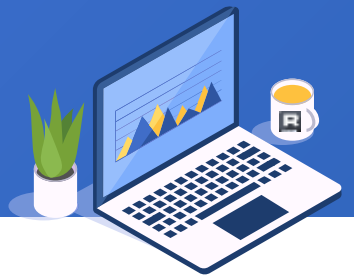
Customer Orders

Order ID	10248	Ship Date	Jul 16,2012	Delivery Date	Aug 01,2012
Customer ID	VINET			CneeName	Rose
Carrier	3	Freight:	￥32.38	CneeCity	Washington

ProductID	UnitPrice	Discount	Quantity	TotalAmt
17	￥14.00	0.0	12.0	￥168.00
42	￥9.00	0.0	10.0	￥90.00
72	￥34.00	0.0	5.0	￥170.00

两片数据来自不同的表，通过OrderID关联，一对多的关系

5.4 利用分片关联做主子表



Preview report:D:\tutorial\data\5\5.2.rptx

Customer Orders

Order ID	10248	Ship Date	Jul 16,2012	Delivery Date	Aug 01,2012
Customer ID	VINET			CneeName	Rose
Carrier	3	Freight:	￥32.38	CneeCity	Washington

ProductID	UnitPrice	Discount	Quantity	TotalAmt
17	￥14.00	0.0	12.0	￥168.00
42	￥9.00	0.0	10.0	￥90.00
72	￥34.00	0.0	5.0	￥170.00

Customer Orders

Order ID	10249	Ship Date	Jul 10,2012	Delivery Date	Aug 16,2012
Customer ID	TOMSP			CneeName	Alice
Carrier	1	Freight:	￥11.61	CneeCity	Phoenix

ProductID	UnitPrice	Discount	Quantity	TotalAmt
14	￥18.00	0.0	9.0	￥162.00
51	￥42.00	0.0	40.0	￥1680.00

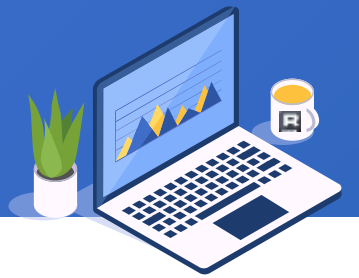
Customer Orders

Order ID	10250	Ship Date	Jul 12,2012	Delivery Date	Aug 05,2012
Customer ID	HANAR			CneeName	Alice
Carrier	2	Freight:	￥65.83	CneeCity	SanJose

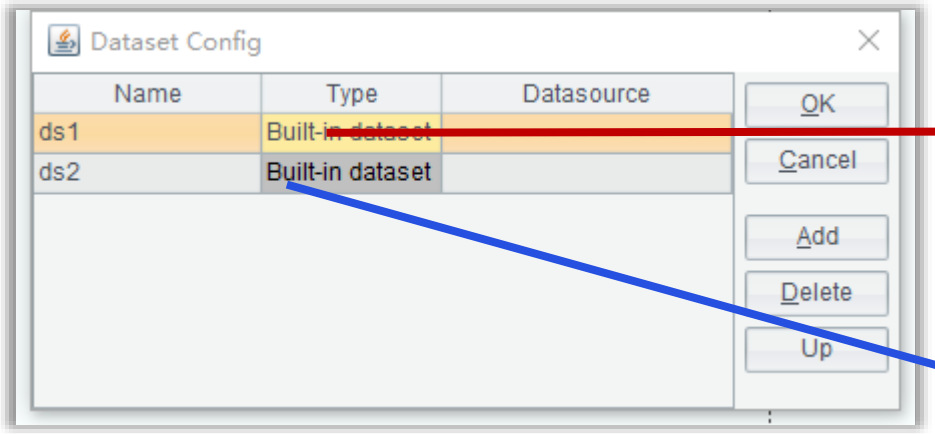
ProductID	UnitPrice	Discount	Quantity	TotalAmt
41	￥7.00	0.0	10.0	￥70.00
51	￥42.00	0.2	35.0	￥1470.00
65	￥16.00	0.2	15.0	￥240.00

扩展后形成多个主子表

5.4 利用分片关联做主子表



第一步：定义数据集



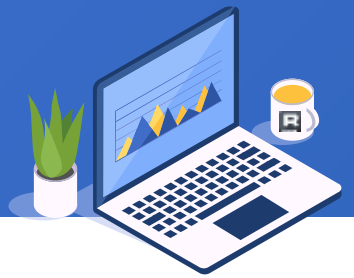
Customer Orders

Order ID	10248	Ship Date	Jul 16,2012	Delivery Date	Aug 01,2012
Customer ID	VINET			CneeName	Rose
Carrier	3	Freight:	￥32.38	CneeCity	Washington

ProductID	UnitPrice	Discount	Quantity	TotalAmt
17	￥14.00	0.0	12.0	￥168.00
42	￥9.00	0.0	10.0	￥90.00
72	￥34.00	0.0	5.0	￥170.00

两个数据集，分别对应主表和子表的数据

5.4 利用分片关联做主子表



第二步：定义主表

B2为主格，纵向扩展

其它所有红框中单元格的左主格都设置为B2，属性名称如下图：

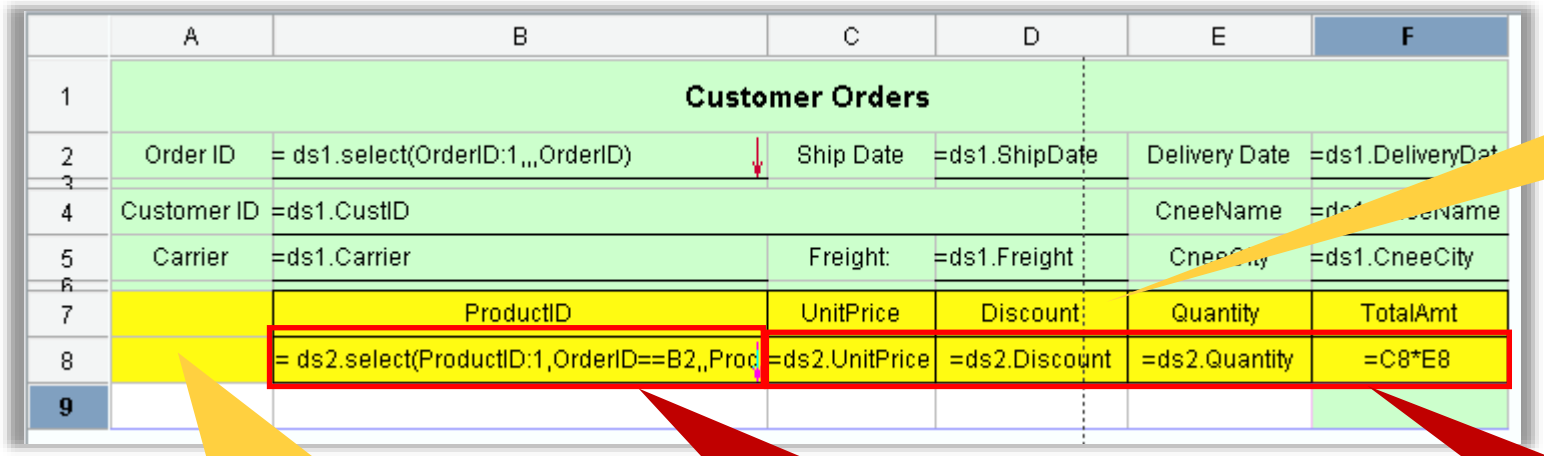
	A	B	C	D	E	F
1	Customer Orders					
2	Order ID	= ds1.select(OrderID:1,,,OrderID)	Ship Date	=ds1.ShipDate	Delivery Date	=ds1.Deliver
3						
4	Customer ID	=ds1.CustID			CneeName	=ds1.CneeName
5	Carrier	=ds1.Carrier	Freight:	=ds1.Freight	CneeCity	=ds1.CneeCity
6						

Key	Value
Value	
Layout	
Paragraph	
Font	
Expanding	
Expanding Mode	Default
Left MasterCell	B2
Top MasterCell	
Merge same value	None
The mode of merge same va...	Default
Merge null values	None

主格扩展时，会带动所有红框中格子跟随扩展

5.4 利用分片关联做主子表

第三步：定义子表



	A	B	C	D	E	F
1	Customer Orders					
2	Order ID	= ds1.select(OrderID:1,,,OrderID)	Ship Date	=ds1.ShipDate	Delivery Date	=ds1.DeliveryDat
4	Customer ID	=ds1.CustID			CneeName	=ds1.CneeName
5	Carrier	=ds1.Carrier	Freight:	=ds1.Freight	CneeCity	=ds1.CneeCity
7		ProductID	UnitPrice	Discount	Quantity	TotalAmt
8		= ds2.select(ProductID:1,OrderID==B2,,Prod	=ds2.UnitPrice	=ds2.Discount	=ds2.Quantity	=C8*E8
9						

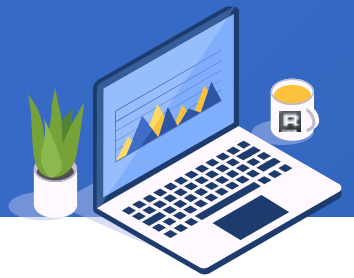
背景为黄色的区域为子表区域

子表中除了红框以外的其它单元格都设成B2的附属格，跟随B2扩展

子表的主格，本身是B2的附属格，纵向扩展

右侧单元格缺省为B8的附属格，跟随B8扩展

5.4 主格认定规则



缺省主格认定

单元格横向扩展时，上方横向扩展单元格缺省为它的上主格，下方单元格缺省为它的附属格；如果上方没有横向扩展格，则上主格缺省为 `0 格

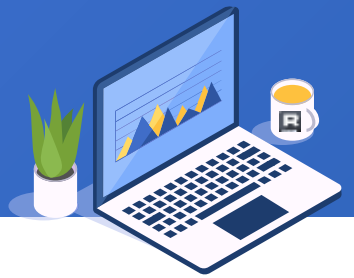
单元格纵向扩展时，左边纵向扩展单元格缺省为它的左主格，右边单元格缺省为它的附属格；如果左边没有纵向扩展格，则左主格缺省为 `0 格

人为改变主格规则

为了符合扩展变化的规则，允许人为设置主格：

- 左主格必须是纵向扩展格，上主格必须是横向扩展格，否则设置无效。
- 不允许出现循环设置的情况，即设置A的主格是B，B的主格是C，C的主格又是A，出现循环设置时认为设置有误，报表无法计算。显然，在缺省的情况下是不可能出现循环设置的，而在人为设置时必须避免这种情况的出现。
- 横向扩展格不允许有左主格，纵向扩展格不允许有上主格。

5.4 分片关联的概念与特征总结



多源是指一个报表的数据来源有多个（文件，也可能是数据库里的数据）。这里的“多个”常常不是两个三个，而是七八个乃至十几个。

多源往往带来分片，正是由于分片，使得报表设计必须直接基于多源进行，而不能先将多源转成单源进行。有相当一部分分片报表无论如何也不可能转化成单源处理，部分能转成单源的报表处理也非常繁琐。

分片是指报表的纵向或横向或双向同时被分成了多个区域，每个区域独立扩展或者其中一个扩展区域为另一个扩展区域的子扩展区域，也可能是扩展区域和静态区域的混合。

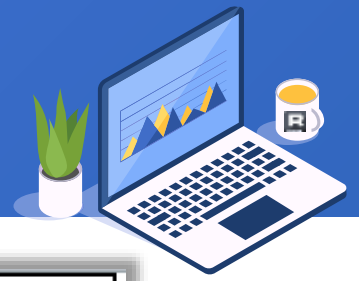
关联是指不同的扩展区域或者扩展与静态区域之间，数据存在着关联，不同区域之间的数据还可能进行运算。

CONTENTS



习 题

5.5 习题1



设计如下样式的报表，数据为附件中的practice5.1.xls

Resume

ID: 1 **Name:** Zhang Ying **Birthday:** 1988-12-08

Title: Sale **TitleofHonor:** Lady **Tel:** (010) 65559857

Country: China **Region:** North China **City:** Beijing

Note: Zhang Ying - Bachelor's degree in Psychology at Peking University while being a member of International Academy of Gastronomy with her "The Art of Cold Dishes".

ID	Education	School	From	To
1	Primary	Shangdi Primary school	1994-09-01	2000-09-01
1	Junior	Shangdi Junior school	2000-09-01	2006-07-01
1	Undergraduate	Beijing University	2006-09-01	2010-07-01

Resume

ID: 2 **Name:** Wang Wei **Birthday:** 1972-02-19

Title: CEO **TitleofHonor:** Doctor **Tel:** (010) 65559482

Country: China **Region:** North China **City:** Beijing

Note: Wang Wei - Bachelor of Business and PhD in International Marketing at Nanjing University; speak French and Italian fluently, and can read German; joined company as a sales rep. and promoted to sales manager and then VP of sales; now a member of Sales Manager Round Table, Beijing Chamber of Commerce and Pan-Pacific Trade Association.

ID	Education	School	From	To
2	Junior	Shangdi Junior school	1984-09-01	1987-07-01
2	Senior	Affiliated high school of Tsinghua university	1987-09-01	1990-07-01
2	Undergraduate	Nanjing University	1990-09-01	1995-07-01
2	Doctor	Nanjing University	1995-09-01	2000-07-01

5.5 习题2



设计如下样式的报表，数据为附件中的pratice5.2.xls

Item / Total		China							USA							Zhong tong
Item		NE	NW	S	SE	SW	W	T_OrderAmt	NE	NW	S	SE	SW	W	T_OrderAmt	
Zhang Ying		￥868.3	￥0	￥1323.3	￥6781.7	￥2048.5	￥8335.4	￥19357.34	$881.17	$0	$2470.26	$3549.47	$2239.56	$8988.72	$18129.17	￥2218.67
Wang Wei		￥257.6	￥0	￥2080.1	￥3535.8	￥588.99	￥5664.0	￥12126.67	$943.81	$0	$2634.68	$1844.18	$1883.07	$8395.79	$15701.53	￥2281.09
Li Fang		￥822.6	￥0	￥1663.2	￥5081.1	￥2098.6	￥8896.8	￥18562.5	$736.89	$721.11	$333.24	$7652.85	$1764.44	$5756.8	$16965.33	￥2491.74
Zheng Jianjie		￥2562.0	￥0	￥4397.6	￥4616.7	￥3891.6	￥10426.	￥25894.16	$2008.06	$117.47	$1059.06	$4042.6	$1854.28	$9268.81	$18350.27	￥3105.45
Zhao Jun		￥113.53	￥0	￥671.09	￥1911.2	￥0	￥959.54	￥3655.4	$776.39	$0	$782.85	$2377.87	$1565.73	$3269.84	$8772.68	￥1218.27
Sun Lin		￥1110.4	￥0	￥1944.3	￥1611.3	￥1192	￥4877.3	￥10735.48	$296.72	$0	$1382.49	$3295.6	$2565.73	$5800.65	$13341.19	￥1049.83
Jin Shipeng		￥1286.2	￥427.87	￥1077.5	￥3118.2	￥525	￥2923.1	￥9358.05	$769.46	$0	$2335.57	$837.24	$2084.39	$6561.62	$12588.28	￥1204.6
Liu YingMei		￥1463.8	￥0	￥1873.6	￥2785.5	￥1749.8	￥6527.4	￥14400.21	$1465.34	$249.67	$694.05	$2787.62	$2108.55	$4986.04	$12291.26	￥1889.27
Zhang XueMei		￥556.01	￥0	￥23.13	￥1638.3	￥1287.6	￥1767.0	￥5272.2	$0	$335.67	$1646.86	$3394.51	$98.33	$2778.91	$8254.28	￥726.41
Year 1996	Month 7	￥0	￥0	￥0	￥0	￥0	￥0	￥0	$0	$0	$0	$0	$444.55	$0	$444.55	￥0
	Month 8	￥33.41	￥0	￥215.04	￥672.75	￥0	￥1535.3	￥2456.56	$0	$0	$297.8	$2046.96	$0	$1351.63	$3696.39	￥656.45
	Month 9	￥0	￥0	￥0	￥1392.3	￥192.22	￥1321.5	￥2906.17	$0	$0	$1876.78	$0	$1434.92	$57.49	$3369.19	￥218.32
	Month 10	￥197.74	￥0	￥330.28	￥282.19	￥0	￥2542.0	￥3352.22	$199.73	$0	$0	$0	$3174.84	$2083.86	$5458.43	￥60.11
	Month 11	￥2.17	￥0	￥54.08	￥186.26	￥1765.3	￥2384.4	￥4392.24	$400.78	$0	$0	$0	$367.41	$2443.61	$3211.8	￥415.82
	Month 12	￥901.62	￥0	￥392.81	￥55.38	￥504.2	￥662.63	￥2516.64	$756.71	$0	$0	$610.01	$155.42	$1468.11	$2990.25	￥329.04
Year 1997	Month 1	￥270.37	￥0	￥491.4	￥2932.0	￥469.55	￥2044.0	￥6207.4	$981.8	$0	$0	$1256.46	$350.39	$1158.95	$3747.6	￥786.16
	Month 2	￥177.5	￥0	￥325.41	￥1830.0	￥1115.9	￥2611.8	￥6060.8	$356.42	$0	$0	$736.69	$1378.55	$1478.57	$3950.22	￥737.4
	Month 3	￥145.04	￥0	￥99.63	￥502.32	￥1193.6	￥516.67	￥2457.29	$696.19	$0	$464.76	$2399.22	$127.18	$462.69	$4150.04	￥331.4
	Month 4	￥98.49	￥0	￥116.92	￥1870.1	￥586.24	￥1100.2	￥3772.01	$989.31	$0	$0	$2322.42	$0	$1858.53	$5170.26	￥207.72
	Month 5	￥397.57	￥0	￥569.07	￥485.58	￥461.15	￥2168.8	￥4082.27	$159.38	$0	$0	$1466.48	$1160.82	$2075.45	$4862.13	￥537.83
	Month 6	￥332.12	￥0	￥0	￥670.54	￥777.23	￥1975.7	￥3755.67	$214.84	$0	$428.47	$1037.11	$503.94	$1927.37	$4111.72	￥707.37
	Month 7	￥323.15	￥0	￥372.15	￥922.22	￥1442.8	￥1653.7	￥4714.15	$346.88	$0	$502.21	$917.7	$389.18	$2256.21	$4412.18	￥580.62

第6章
格间计算



CONTENTS



6.1 占比、累积、简单同期比

6.2 比值，复杂比值

6.3 条件汇总，排名，复杂同期比

6.4 组内序号，组间序号

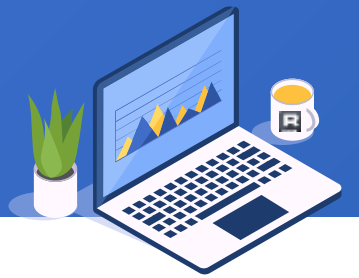
6.5 习题

CONTENTS



6.1 占比、累积、简单同期比

6.1 占比

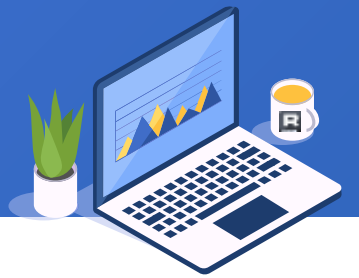


当前地区销售额占本国总销售额的比例

Country	Region	Amount	DomesticProp	CountryAmt	TotalProp
China	Central-West	47568.05	41.6%	114345.55	52.19%
	North-East	9078.40	7.94%		
	[illegible]	399.38	0.35%		
	South	18226.13	15.94%		
	South-East	27923.31	24.42%		
	South-West	11150.27	9.75%		
US	Central-West	44700.53	42.68%	104729.96	47.81%
	North-East	8113.28	7.75%		
	North-West	843.39	0.81%		
	South	10815.11	10.33%		
	South-East	27822.24	26.57%		
	South-West	12435.40	11.87%		

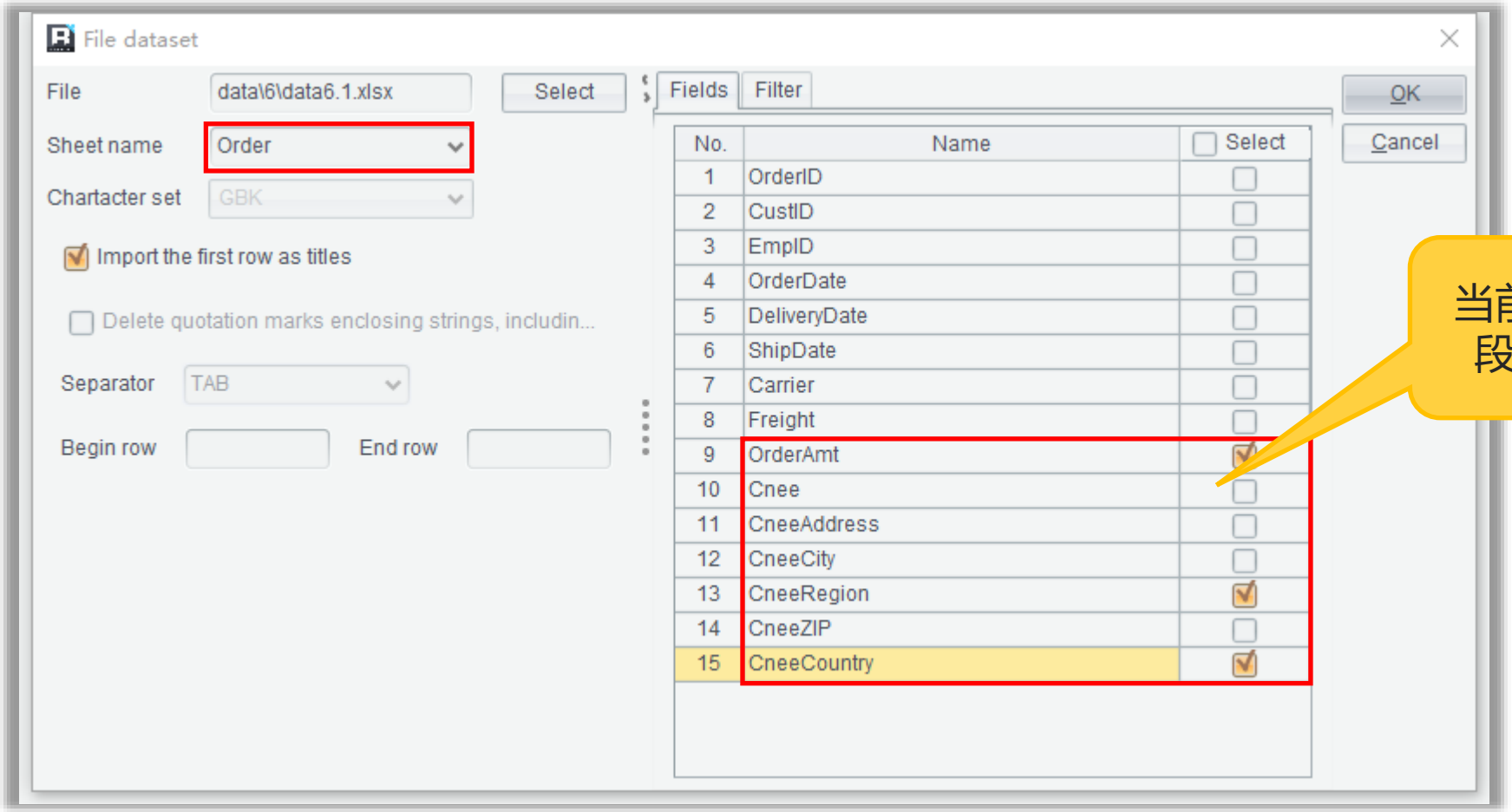
本国总销售额占全部销售额的比例

6.1 占比



制作过程——定义数据集

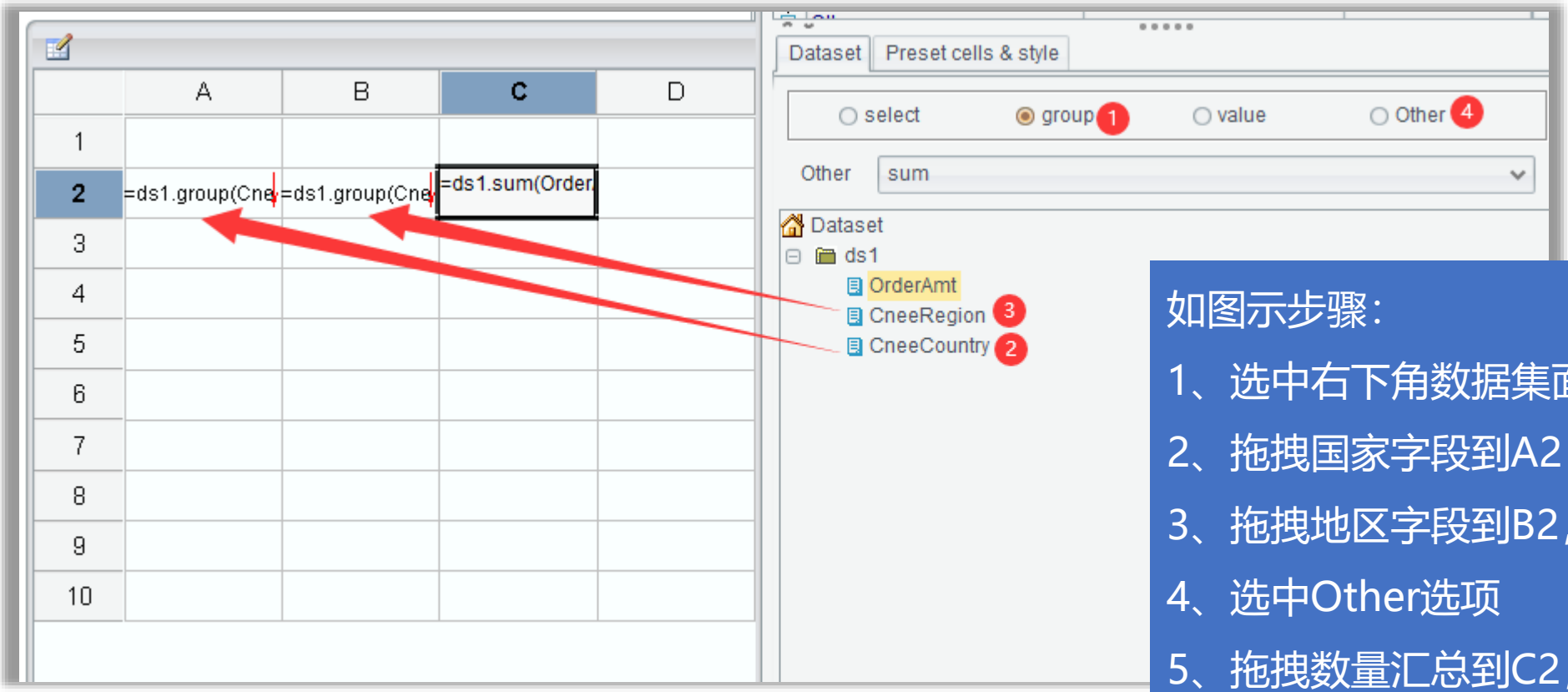
根据报表统计数据的需要，打开文件数据集时只选部分字段



当前例子只用到三个字段，所以只勾选三个

6.1 占比

制作过程——定义表达式



如图示步骤：

1、选中右下角数据集面板中group

2、拖拽国家字段到A2，先按国家分组

3、拖拽地区字段到B2，再按地区分组

4、选中Other选项

5、拖拽数量汇总到C2，统计每组销售额

6.1 占比

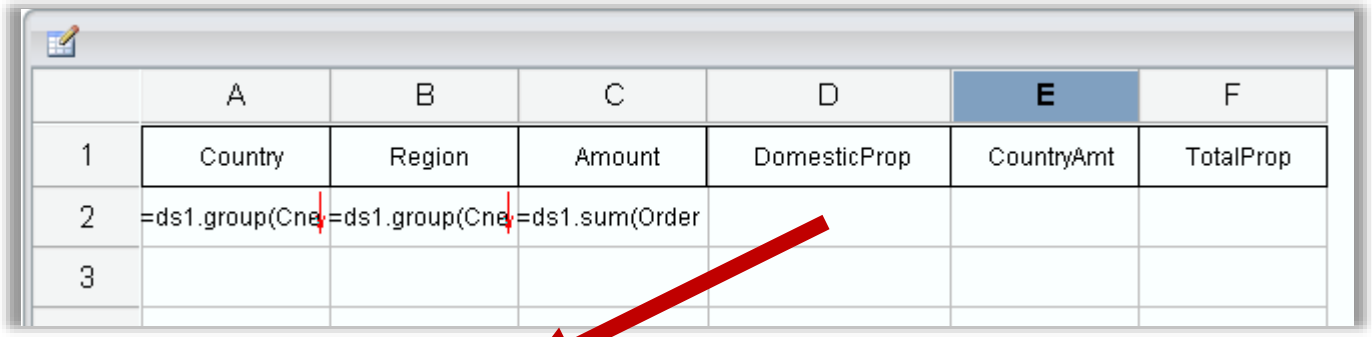
制作过程——定义表达式

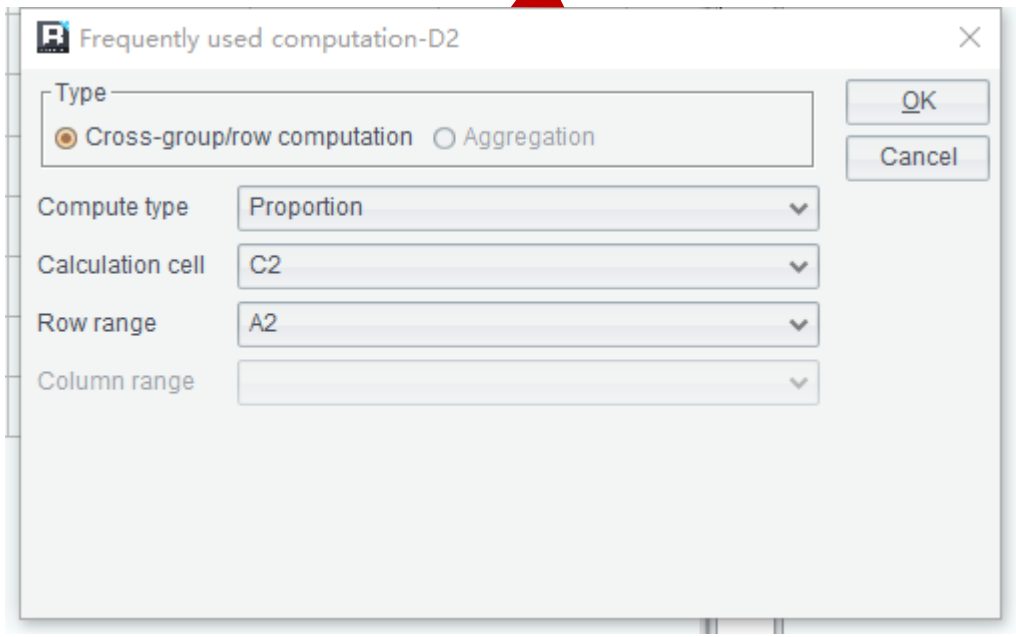
如图设置好第一行的标题，标识国内占比(DomesticProp)、国内总额(CountryAmt)、国家占比(TotalProp)

1、选中D2

2、单击工具条常用计算列

3、如右下图设置国内占比，注意选行统计范围为A2格，意思是将C2计算格的值与A2范围内的总和做占比





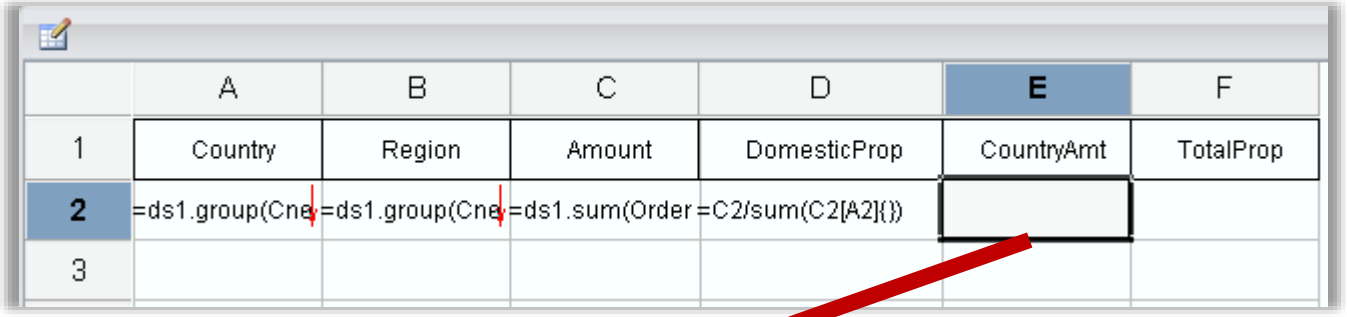
6.1 占比

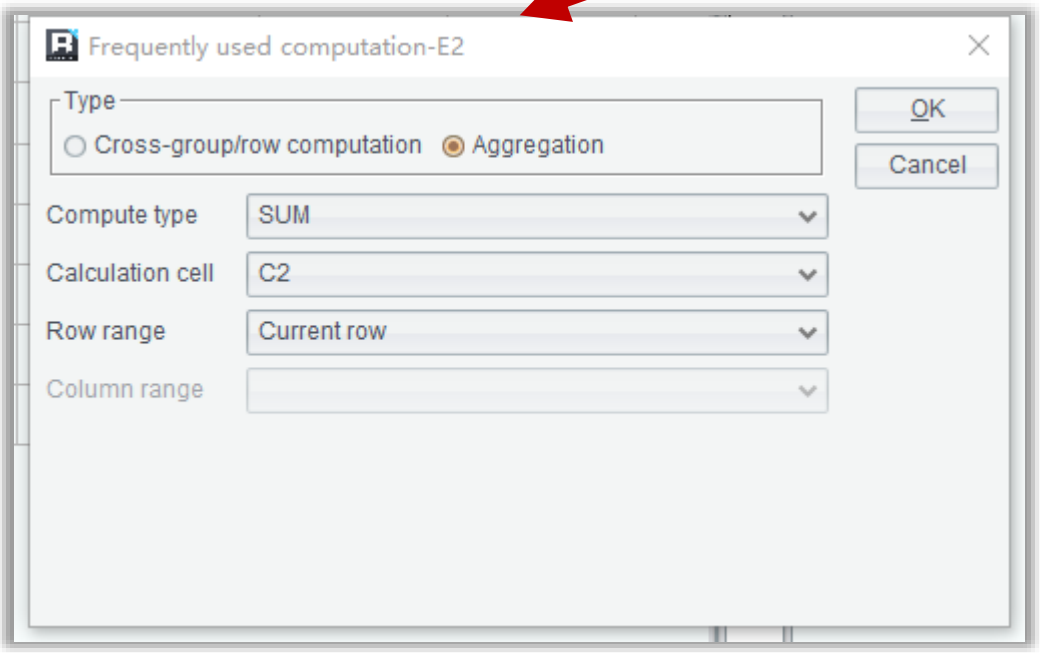
制作过程——定义表达式

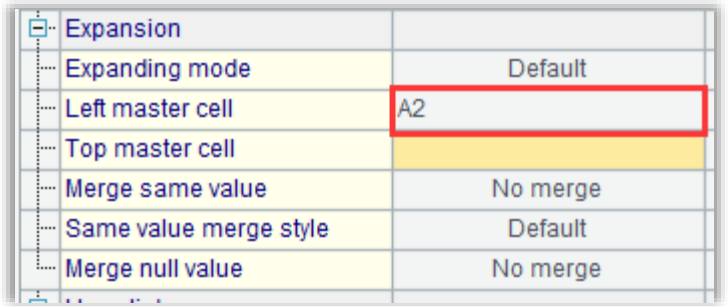
1、选中E2，为了在E2计算出国家总额，需要修改它的左主格为A2，如下图

2、单击工具条常用计算列

3、如右下图设置国内总额







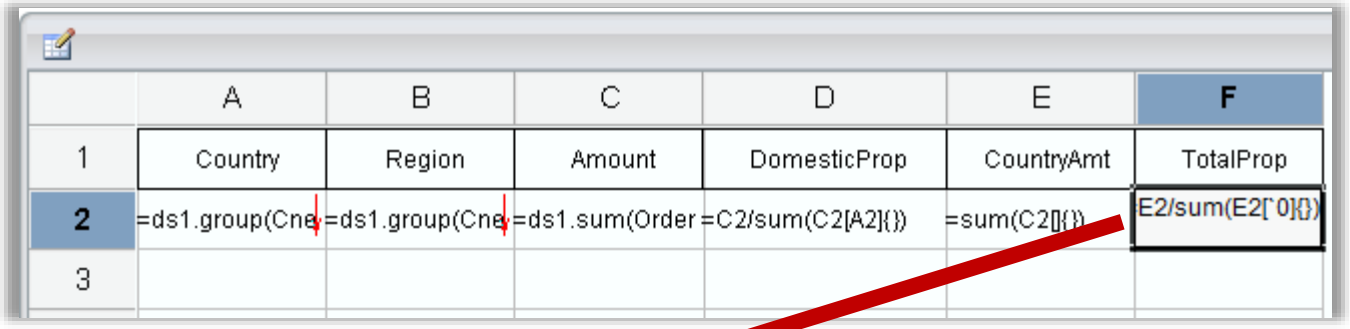
6.1 占比

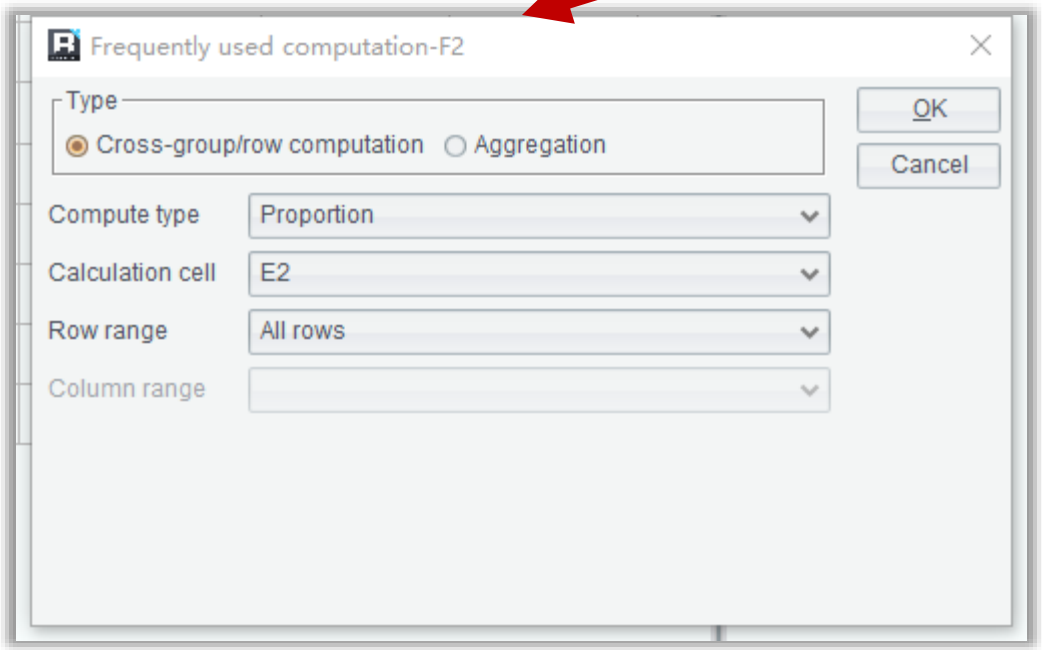
制作过程——定义表达式

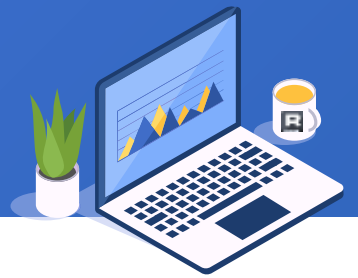
1、选中F2

2、单击工具条常用计算列

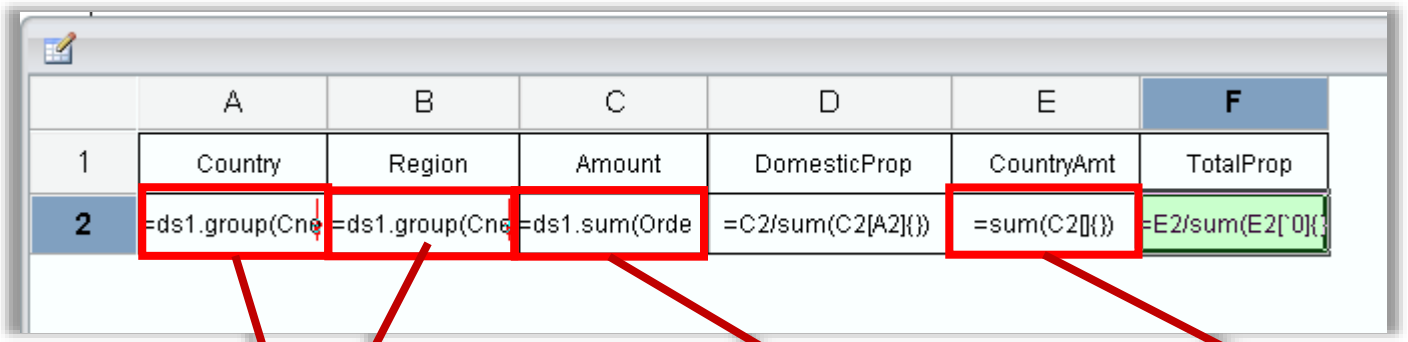
3、如右下图设置国家占比







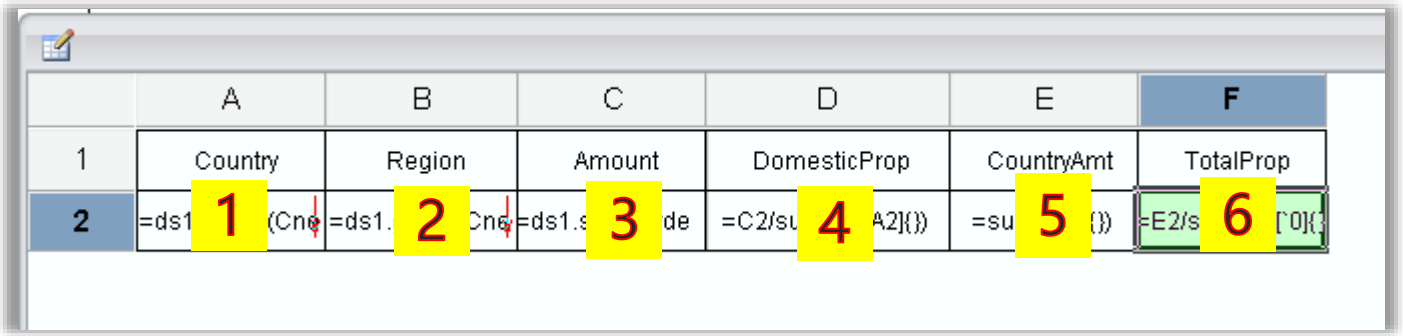
制作过程——定义表达式


A
B
C
D
E
F
1
Country
Region
Amount
DomesticProp
CountryAmt
TotalProp
2
=ds1.group(Cne
=ds1.group(Cne
=ds1.sum(Orde
=C2/sum(C2[A2]{})
=sum(C2[]{})
=E2/sum(E2[`0]{}
先按国家、地区分组
然后统计销售额
接着算本国的总销售额

6.1 占比



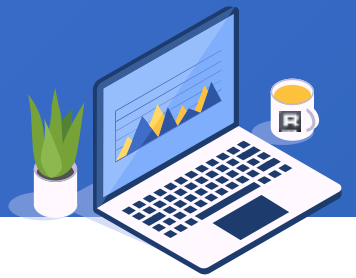
制作过程——定义表达式



	A	B	C	D	E	F
1	Country	Region	Amount	DomesticProp	CountryAmt	TotalProp
2	=ds1 1 (Cne	=ds1. 2 Cne	=ds1.s 3 de	=C2/su 4 A2]{})	=su 5 {})	=E2/s 6 [*0]{}

报表引擎会自动判断表达式的依赖关系，先算总销售额，后算占比

6.1 累积



Year	Month	Amount	CumAmount
1996	7	6137.15	6137.15
	8	7046.94	13184.09
	9	4728.29	17912.37
	10	5716.42	23628.80
	11	6839.97	30468.76
	12	8272.08	38740.84
1997	1	8814.51	8814.51
	2	7439.41	16253.92
	3	8411.89	24665.81
	4	8549.87	33215.68
	5	7399.09	40614.77
	6	8604.61	49219.38
	9	10506.78	75112.69
	10	10748.40	85861.09
	11	8945.32	94806.41
	12	12224.65	107031.06

新的一年重新开始累积

所谓的累积，就是从第一行起累加到当前行的数值之和

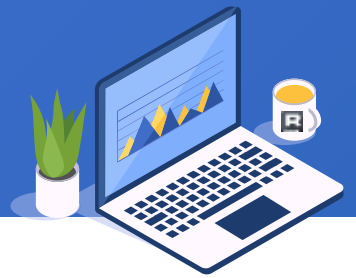
也等于上一行的累积值+本行的被累积值

17912.37=6137.15+7046.94+4728.29

Amount	CumAmount
6137.15	6137.15
7046.94	13184.09
4728.29	sum 17912.37
5716.42	23628.80

也可以是17912.37=13184.09+4728.29

Month	Amount	CumAmount
7	6137.15	6137.15
8		13184.09
9	4728.29	= 17912.37
10	5716.42	23628.80

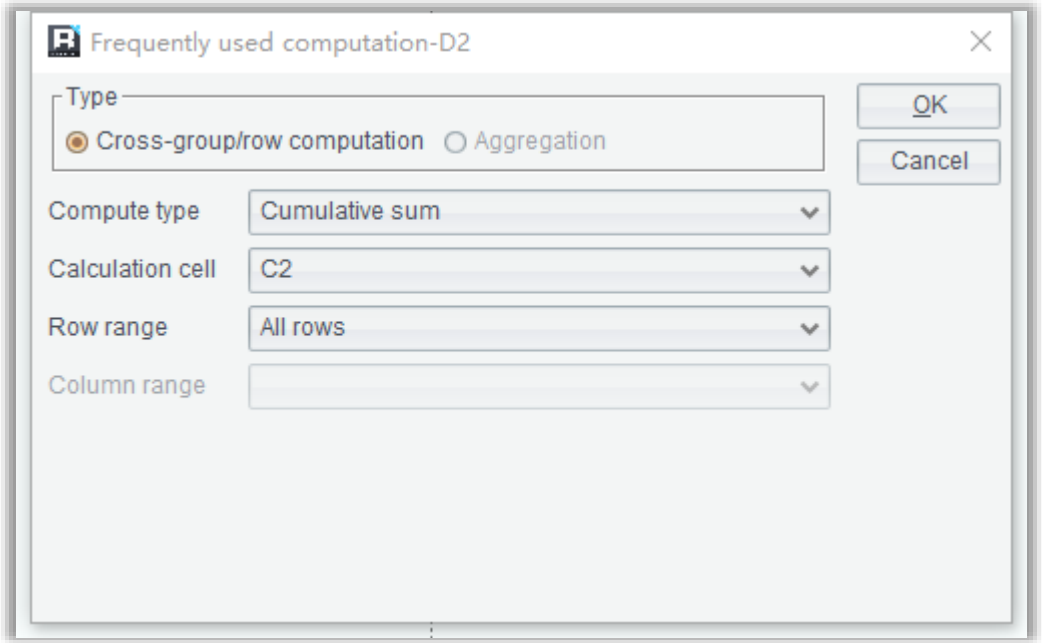


制作过程——定义表达式

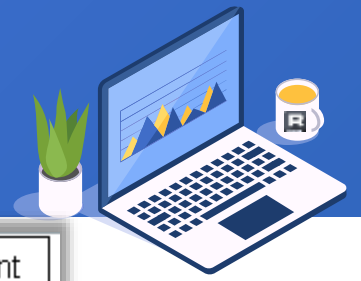
报表生成后，选中D2，如右图设置计算列

D:\tutorial\data\6\6.2.rptx

	A	B	C	D
1	Year	Month	Amount	CumAmount
2	= ds1.group(y	= ds1.group(m	=ds1.sum(Orde	=sum(C2[`0]{seq



6.1累积



制作过程——定义表达式

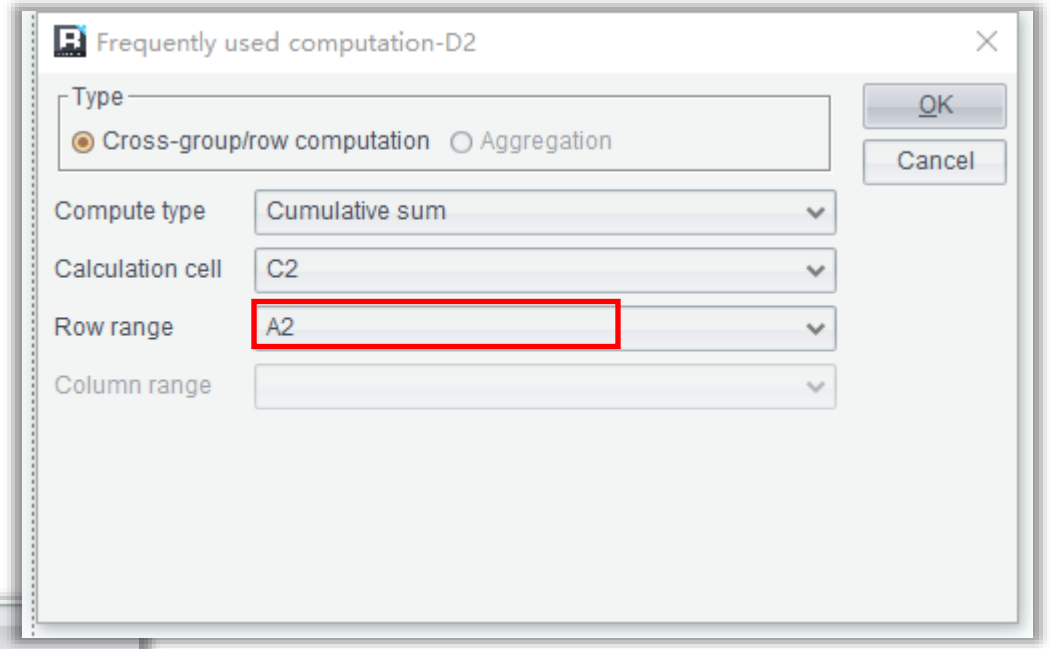
Year	Month	Amount	CumAmount
1996	7	3951.38	3951.38
	8	6433.69	10385.07
	9	5951.56	16336.64
	10	7793.37	24130.01
	11	8178.79	32308.80
	12	8656.21	40965.01
1997	1	8396.23	49361.24
	2	7468.77	56830.01
	3	9884.35	66714.36
	4	5475.53	72189.88
	5	8914.86	81104.75
	6	6808.37	87913.12
	7	9189.06	97102.18
	8	8996.63	106098.80
	9	9226.50	115325.31
	10	9322.54	124647.85
	11	8748.01	133395.85
	12	14903.59	148299.44

定义计算范围为全部行时，会累积所有数量，新的一年没有重新累积

6.1累积

制作过程——定义表达式

选中D2，调整统计范围，如右图设置计算列



D:\tutorial\data\6\6.2.rptx

	A	B	C	D
1	Year	Month	Amount	CumAmount
2	= ds1.group(y	= ds1.group(m	=ds1.sum(Orde	=sum(C2[A2]{seq($B2)>=seq(B2)})

6.1累积

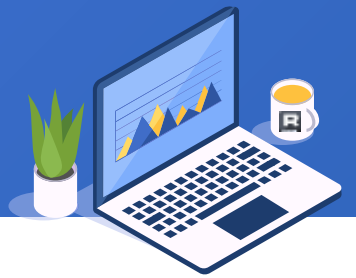


制作过程——定义表达式

Year	Month	Amount	CumAmount
1996	7	3951.38	3951.38
	8	6433.69	10385.07
	9	5951.56	16336.64
	10	7793.37	24130.01
	11	8178.79	32308.80
	12	8656.21	40965.01
1997	1	8396.23	8396.23
	2	7468.77	15865.00
	3	9884.35	25749.35
	4	5475.53	31224.88
	5	8914.86	40139.74
	6	6808.37	46948.11
	7	9189.06	56137.17
	8	8996.63	65133.79
	9	9226.50	74360.30
	10	9322.54	83682.84
	11	8748.01	92430.85
	12	14903.59	107334.43

在A2范围内计算时，
新的一年重新累积

6.1 简单同期比



这张报表的特点：

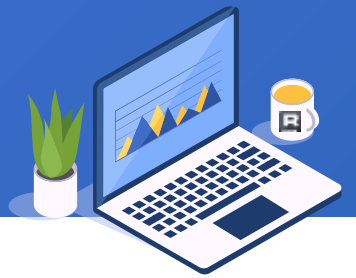
1、计算当前月份销售额和去年同月销售额的比值

2、当去年同月的数据不存在时不计算

3、年、月的数据是连续的，除了第一组找不到去年同月的数据，其它都是默认存在

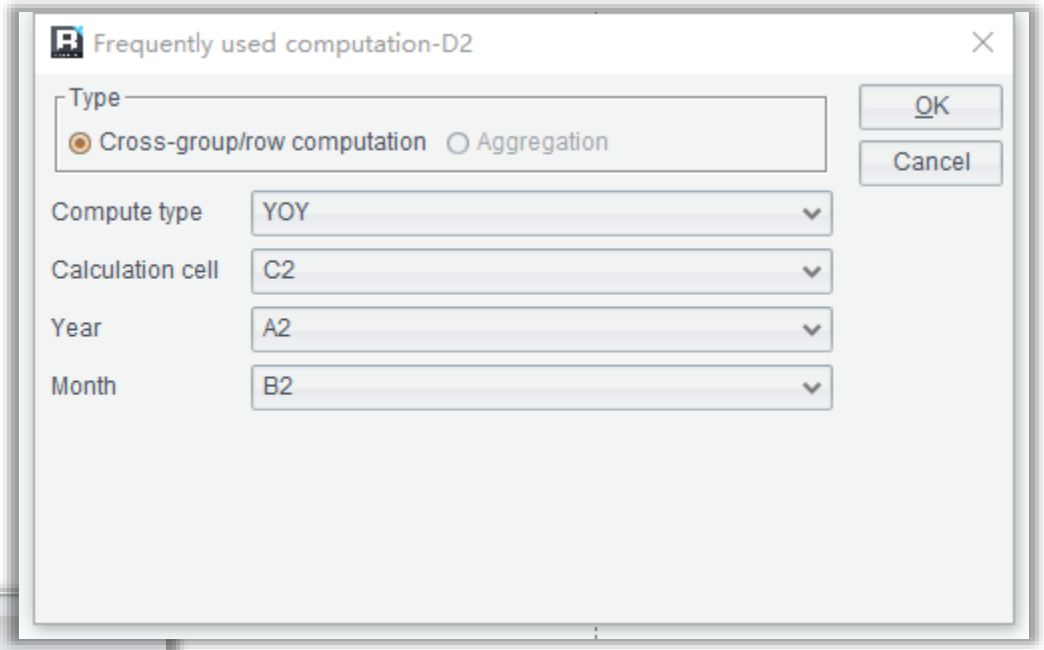
Year	Month	Amount	YOY
2020	1	844846.00	
	2	1033295.00	
	3	1120638.00	
	4	275893.00	
	5	990796.00	
	6	625331.00	
	7	834588.00	
	8	991944.00	
	9	722160.00	
	10	394600.00	
	11	554423.00	
	12	787181.00	
2021	1	664344.00	78.63%
	2	882967.00	85.45%
	3	639448.00	57.06%
	4	335189.00	121.49%
	5	688693.00	69.51%
	6	320009.00	51.17%
	7	967875.00	115.97%
	8	479270.00	48.32%
	9	344558.00	47.71%
	10	411905.00	104.39%
	11	321601.00	58.01%
	12	1077495.00	136.88%
2022	1	324602.00	48.86%
	2	1066043.00	120.73%
	3	501643.00	78.45%
	4	723386.00	215.81%
	5	765019.00	111.08%
	6	338229.00	105.69%
	7	1131292.00	116.88%
	8	543077.00	113.31%
	9	531612.00	154.29%
	10	993271.00	241.14%
	11	338070.00	105.12%
	12	788014.00	73.13%

6.1简单同期比



制作过程——定义表达式

选中D2，单击常用计算列，如右图设置计算列



D:\tutorial\data\6\6.3.rptx

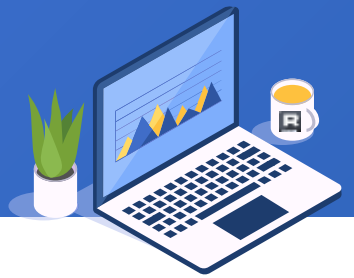
	A	B	C	D
1	Year	Month	Amount	YOY
2	= ds1.group(Year)	= ds1.select(Mon	=ds1.Amount	=C2/C2[`0]{A2==$A2-1 && B2==$B2}

CONTENTS



6.2 比值、复杂比值

6.4 比值



比值报表

这张报表的特点：

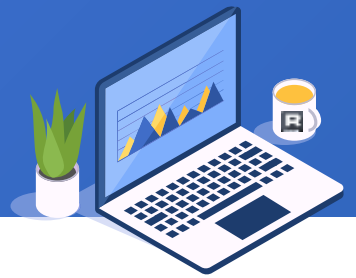
1、按销售统计销售额，并按销售额逆序排序

2、算出和总销售额第一名的销售的差距

Preview report:D:\tutorial\data\6\6.4.rptx

Rank	Name	Amount	Diff_#1
1	ZhengJianjie	40539.44	0.00
2	ZhangYing	34103.92	6435.52
3	LiFang	31728.79	8810.65
4	WangWei	29042.54	11496.90
5	LiuYingMei	24588.60	15950.84
6	JinShipeng	19430.65	21108.79
7	SunLin	18859.75	21679.68
8	ZhaoJun	12848.21	27691.22
9	ZhangXueMei	9090.68	31448.75

6.4 比值



制作过程

	A	B	C	D
1	Rank	Name	Amount	Diff_#1
2	=&B2	= ds1.group(EmplD;EmplD:1; ds1.sum(OrderAmt):-1)	=ds1.sum(OrderAmt)	=C2[1]-C2

- 首先B2使用group函数分组的时候，按照ds1.sum(OrderAmt)逆序排序，从而保证第一名的销售在第一行

- C2[1] 表示C2扩展后的第一个

6.2 复杂比值

这张报表的特点：

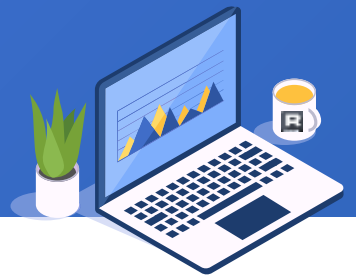
1、按地区、销售分组统计销售额，并按销售额逆序排序

2、算出和总销售额第二名的地区中，排名第三的销售的差距

3、业务背景：总销售额最高的地区也许存在较明显的地区差异；销售额最高的两名销售也许天赋异禀，也许背景深厚，不具备可比性；所以和第二名的地区中第三名的销售比差距，是比较常见且合理的需求。

Rank	Region	Name	TotalAmt	Diff_#3_RunnerUpRGN
1	Central-West	ZhengJianjie	17432.53	-8115.76
2		LiFang	16980.10	-7663.34
3		ZhangYing	14925.82	-5609.06
4		WangWei	13311.59	-3994.83
5		LiuYingMei	12360.69	-3043.92
6		JinShipeng	8606.31	710.45
7		SunLin	7825.35	1491.41
8		ZhaoJun	5904.30	3412.46
9		ZhangXueMei	2878.83	6437.93
		Region total:	100225.52	
1		ZhangYing	11544.87	-2228.11
2		LiFang	9776.16	-459.39
3		ZhengJianjie	9316.76	0.00
4	South-East	WangWei	7082.97	2233.79
5		SunLin	5303.31	4013.45
6		LiuYingMei	5213.94	4102.83
7		ZhangXueMei	3508.25	5808.51
8		ZhaoJun	3260.72	6056.05
9		JinShipeng	2158.63	7158.13
		Region total:	57165.61	
1	South	WangWei	4125.20	5191.56
2		ZhangYing	3637.53	5679.23
3		ZhengJianjie	3494.92	5821.84
4		JinShipeng	3360.64	5956.12
5		LiuYingMei	1789.22	7527.54
6		ZhaoJun	1616.56	7700.21
7		LiFang	1414.35	7902.41
8		SunLin	1215.02	8101.74
9		ZhangXueMei	643.86	8672.90

6.2 复杂比值



制作过程

D:\tutorial\data\6\6.5.rptx

	A	B	C	D	E
1	Rank	Region	Name	TotalAmt	Diff_#3_RunnerUpRGN
2	=&C2	= ds1.group(CneeRegion;CneeRegion:1; ds1.sum(OrderAmt)	= ds1.group(EmpID;EmpID:1; ds1.sum(OrderAmt)	=ds1.sum(Orde	=D2[B2:2,C2:3]-D2
3			Region total:	=ds1.sum(Orde	

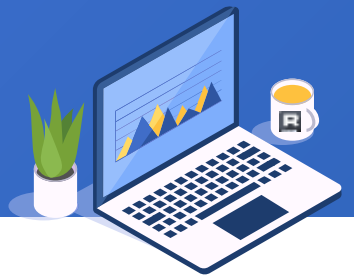
- 首先B2和C2使用group函数分组的时候，按照ds1.sum(OrderAmt)逆序排序，从而保证亚军地区是在第二组，第三名的销售在组内第三行
- D2[B2:2,C2:3] 表示主格B2扩展后的第二个，主格C2扩展后的第三个对应的D2单元格的值
- 这里需要配置A2的主格是C2，A3的主格是B3，B3的主格是`0(即根格，也称根坐标)

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6.3 条件汇总、排名、复杂同期比

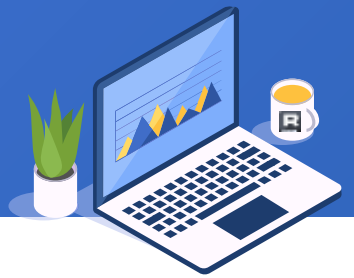
6.3 条件汇总



希望在报表中统计出销售额大于2.5万的人数

Name	Amount
ZhangYing	34103.92
WangWei	29042.54
LiFang	31728.79
ZhengJianjie	40539.44
ZhaoJun	12848.21
SunLin	18859.75
JinShipeng	19430.65
LiuYingMei	24588.60
ZhangXueMei	9090.68
Number of sellers with amounts above 25,000	**4**

6.3 条件汇总



制作过程

D:\tutorial\data\6\6.6.rptx

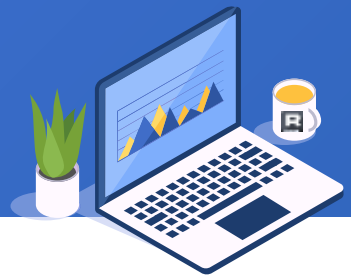
	A	B
1	Name	Amount
2	= ds1.group(EmplD;EmplD:1)	=ds1.sum(OrderAmt)
3	**Number of sellers with amounts above 25,000**	**=count(B2{B2>25000})**

B2{} 表示B2扩展后的单元格的集合

B2{B2>25000} 表示B2扩展后值大于25000的单元格的集合

Count(B2{B2>25000}) 表示对单元格集合B2{B2>25000}进行计数

6.3 排名



这张报表的特点：

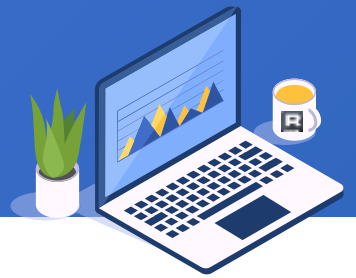
1、按照销售ID排序

2、统计每个销售的　销售额

3、算出销售额的排名

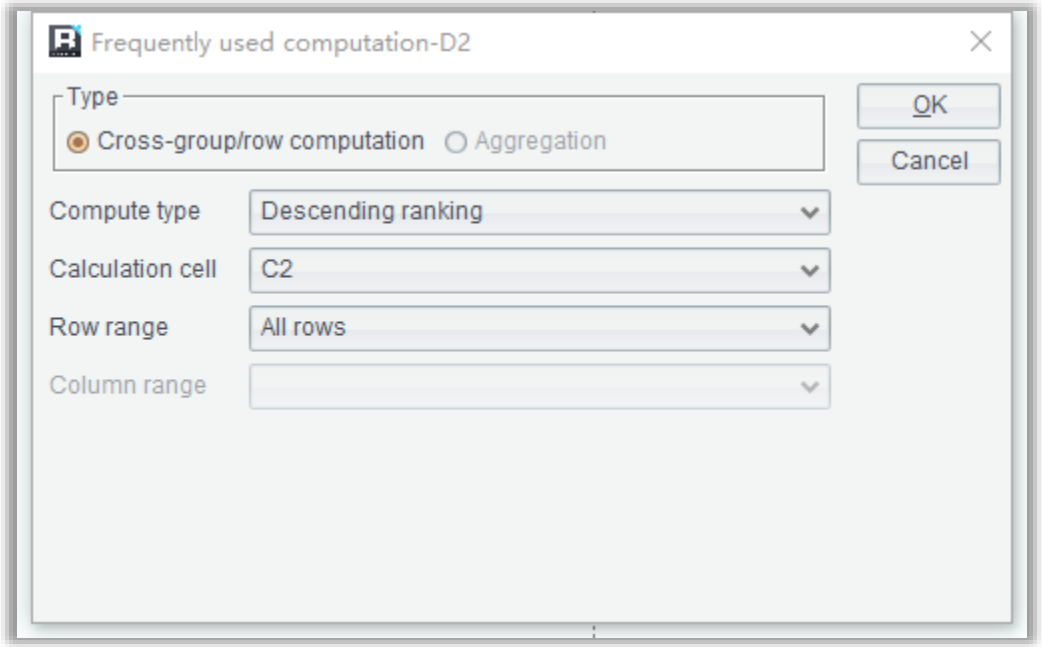
ID	Name	Amount	Rank
1	ZhangYing	34103.92	2
2	WangWei	29042.54	4
3	LiFang	31728.79	3
4	ZhengJianjie	40539.44	1
5	ZhaoJun	12848.21	8
6	SunLin	18859.75	7
7	JinShipeng	19430.65	6
8	LiuYingMei	24588.60	5
9	ZhangXueMei	9090.68	9

6.3 排名



制作过程——定义表达式

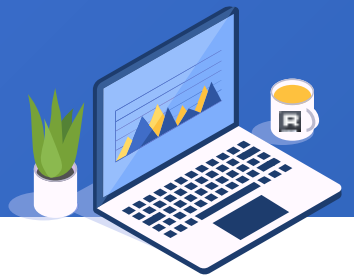
选中D2，单击常用计算列，如右图设置计算列



D:\tutorial\data\6\6.7.rptx

	A	B	C	D
1	ID	Name	Amount	Rank
2	= ds1.g	= ds2.select@1	=ds1.sum(OrderA	=count(C2[`0]{$C2<C2})+1

6.3 排名



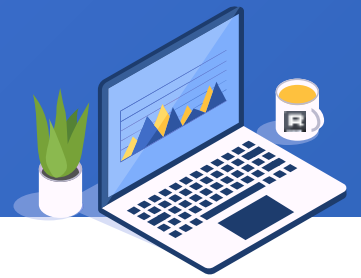
制作过程——表达式释义

D:\tutorial\data\6\6.7.rptx

	A	B	C	D
1	ID	Name	Amount	Rank
2	= ds1.group(EmpID;EmpID:1)	= ds2.select@1(FirstName+LastName,ID==A2)	=ds1.sum(OrderAmt)	=count(C2[`0]{$C2<C2})+1

- [`0]表示根坐标，对应于全部行范围
- C2[`0]{}表示根坐标下C2扩展后的所有单元格，即不受主格局限
- $C2在条件表达式中表示当前行的C2
- C2[`0]{$C2<C2}表示根坐标下C2扩展后的所有单元格中，比当前行的C2的值更大的单元格的集合

6.3 复杂同期比



这张报表的特点：

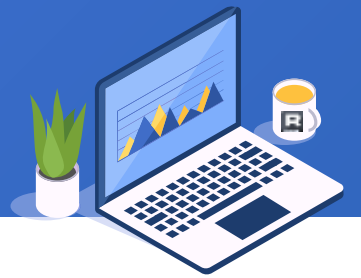
1、计算当前月份和去年同月的比值

2、当去年同月的数据不存在时不计算

3、年、月的数据可能不连续

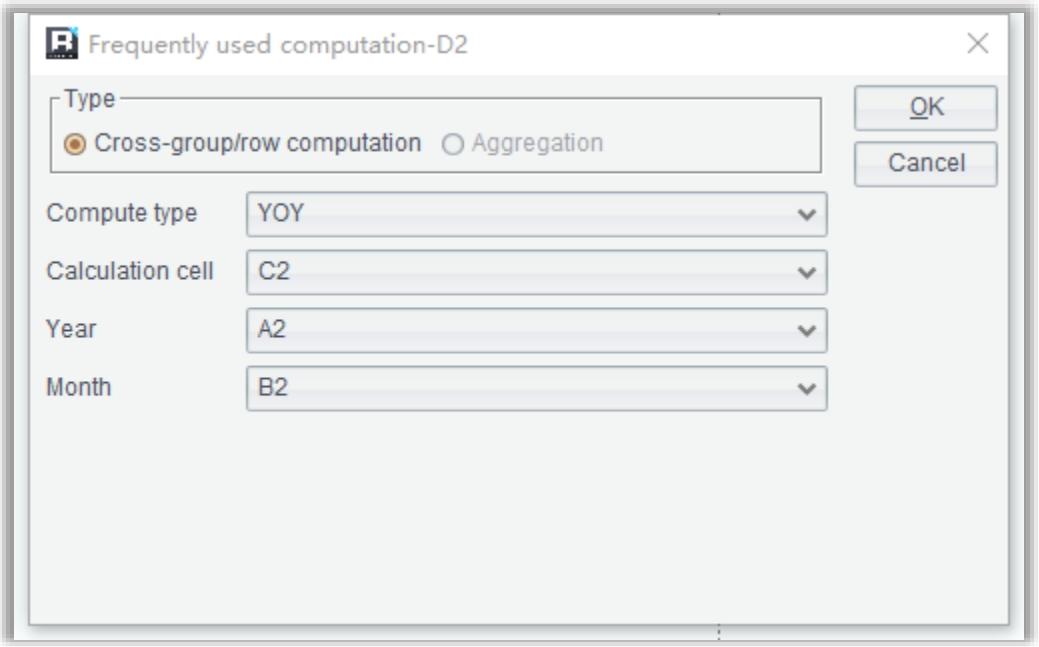
Year	Month	Amount	YOY
1996	7	3951.38	
	8	6433.69	
	9	5951.56	
	10	7793.37	
	11	8178.79	
	12	8656.21	
1997	1	8396.23	
	2	7468.77	
	3	9884.35	
	4	5475.53	
	5	8914.86	
	6	6808.37	
	7	9189.06	232.55%
	8	8996.63	139.84%
	9	9226.50	155.03%
	10	9322.54	119.62%
	11	8748.01	106.96%
	12	14903.59	172.17%
1998	1	16558.58	197.21%
	2	12453.86	166.75%
	3	17158.35	173.59%
	4	21484.24	392.37%
	5	3121.04	35.01%

6.3 复杂同期比



制作过程——定义表达式

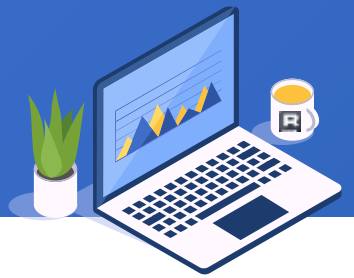
选中D2，单击常用计算列，如右图设置计算列



D:\tutorial\data\6\6.8.rptx

	A	B	C	D
1	Year	Month	Amount	YOY
2	= ds1.grou	= ds1	=ds1.sum(Orde	=C2/C2[`0]{A2==$A2-1 && B2==$B2}

6.3 复杂同期比



制作过程——释义表达式

D:\tutorial\data\6\6.8.rptx

	A	B	C	D
1	Year	Month	Amount	YOY
2	= ds1.group(year(OrderDate);year(OrderDate):1	= ds1.group(month(OrderDate);month(OrderDate):1)	=ds1.sum(OrderAmt	=C2/C2[`0]{A2==$A2-1 && B

- C2[`0]{}表示根坐标下C2扩展后的所有单元格
- C2[`0]{A2==$A2-1 && B2==$B2}表示A2的值等于当前行A2-1，B2和当前行B2相等的C2，也就是去年同月的销售额

CONTENTS



6.4 组内序号、组间序号

6.4 组内序号



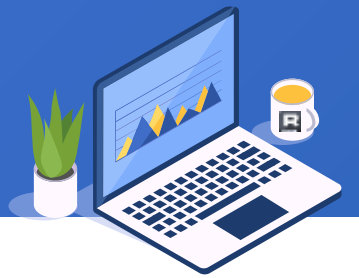
这张报表的特点：

1、按国家、地区分组

2、第一列为序号列，序号在组内独立编号

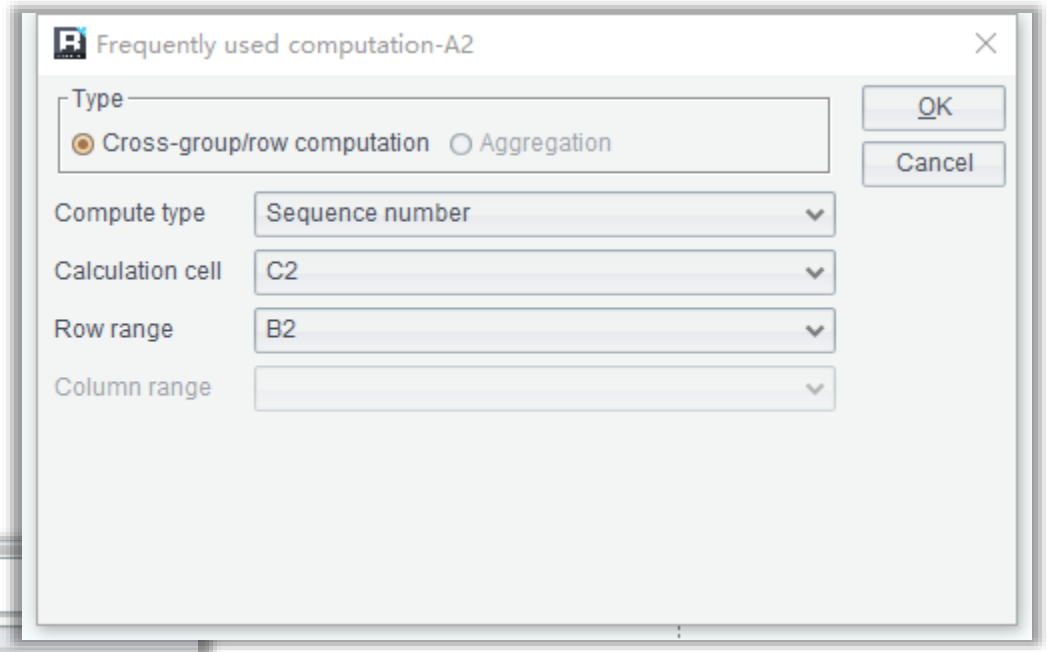
ID	Country	Region	Amount
1	China	Central-West	50627.65
2		North-East	9759.53
3		North-West	956.11
4		South	12670.96
5		South-East	29677.38
6		South-West	8644.94
1	US	Central-West	49597.87
2		North-East	9519.73
3		North-West	1399.69
4		South	8626.35
5		South-East	27488.23
6		South-West	11264.15

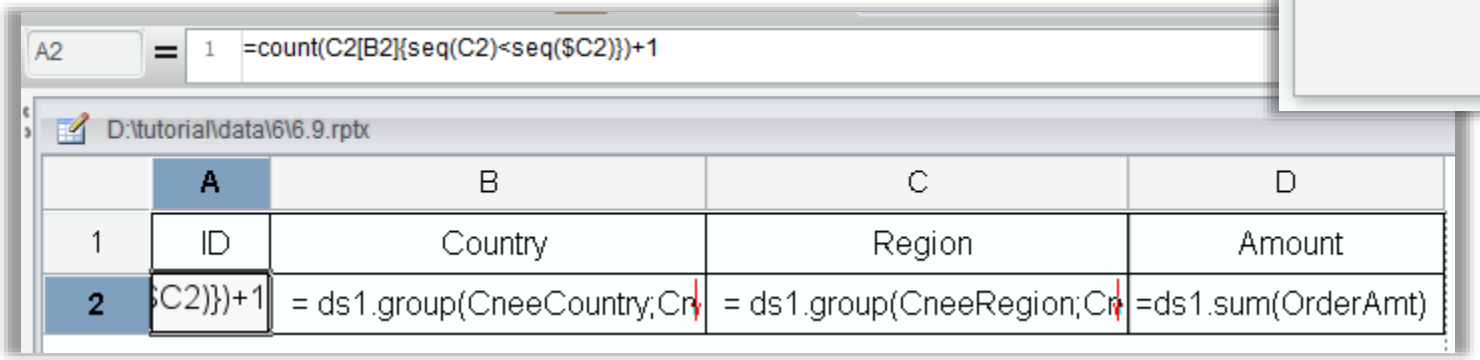
6.4 组内序号



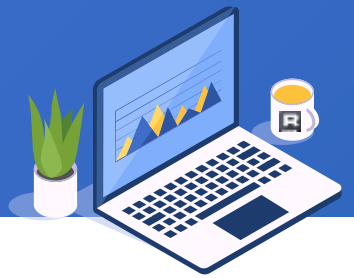
制作过程——定义表达式

选中B2，设置它的左主格为`0
选中A2，设置它的左主格为C2，再
单击常用计算列，如右图设置计算列





6.4 组内序号



制作过程——定义表达式

D:\tutorial\data\6\6.9.rptx

	A	B	C	D
1	ID	Country	Region	Amount
2	=count(C2[B2]{seq(C2)<seq($C2)})+1	= ds1.group(CneeCountry;Cn	= ds1.group(CneeRe	1.sum(OrderAmt)

A2的主格是C2

B2的主格是`0

A2也可以使用&C2表达式，直接获得当前格所属的某主格在所有扩展出来的格子中的序号，这种运算我们也称为层次坐标的逆运算。

但使用对话框生成的表达式更通用，可以用在更复杂的双向扩展方向中

6.4 组间序号



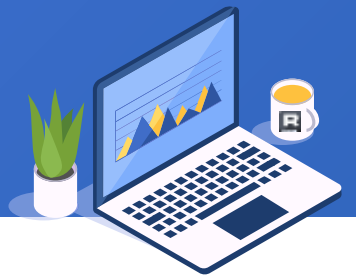
这张报表的特点：

1、按国家、地区分组

2、第一列为序号列，序号跨组连续编号

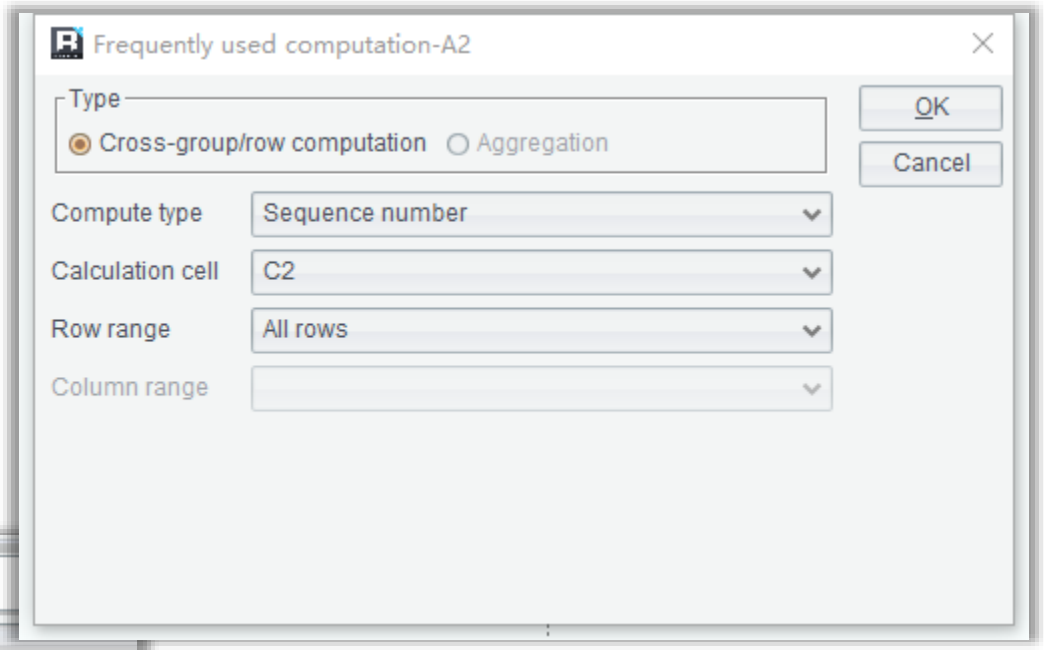
ID	Country	Region	Amount
1	China	Central-West	50627.65
2		North-East	9759.53
3		North-West	956.11
4		South	12670.96
5		South-East	29677.38
6		South-West	8644.94
7	US	Central-West	49597.87
8		North-East	9519.73
9		North-West	1399.69
10		South	8626.35
11		South-East	27488.23
12		South-West	11264.15

6.4 组间序号



制作过程——定义表达式

选中B2，设置它的左主格为`0
选中A2，设置它的左主格为C2，再单击常用计算列，与组内序号的区别为行范围不同，如右图设置

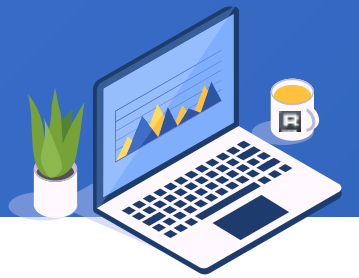


A2 = 1 =count(C2[`0]{seq(C2)<seq($C2)})+1

D:\tutorial\data\6\6.10.rptx

	A	B	C	D
1	ID	Country	Region	Amount
2	=count(C2	= ds1.group(CneeCountry;	= ds1.group(CneeRegi	=ds1.sum(OrderAmt)

6.4 组间序号



制作过程——定义表达式

D:\tutorial\data\6\6.10.rptx

	A	B	C	D
1	ID	Country	Region	Amount
2	=count(C2[`0]{seq(C2)<seq($C2)})+1	= ds1.group(CneeCountry	= ds1.group(CneeReg	=ds1.sum(OrderAmt)

A2的主格是C2

B2的主格是`0

类似上一节，A2也可以使用seq(C2)表达式，直接取得指定扩展单元格的同源号，即扩展格扩展后，将此单元格的同源格按行（列）号从小到大排序后，此单元格所在的序号。

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习 题

6.5 习题1



设计如右样式的报表，数据为附件中的data6.1.xlsx中的OrderSummary

报表特点说明：

1、计算当前月份的销售额和上个月的销售额的比值

2、跨年份的情况，也照样算，即1月份和去年12月比

3、默认年、月的数据是连续的

Year	Month	Amount	LRR
2020	1	831300.00	
	2	352825.00	42.44%
	3	1037630.00	294.09%
	4	698713.00	67.34%
	5	268115.00	38.37%
	6	614630.00	229.24%
	7	515708.00	83.91%
	8	1130095.00	219.13%
	9	650920.00	57.60%
	10	513230.00	78.85%
	11	951065.00	185.31%
	12	637152.00	66.99%
2021	1	410845.00	64.48%
	2	831134.00	202.30%
	3	798206.00	96.04%
	4	868427.00	108.80%
	5	1024625.00	117.99%
	6	869321.00	84.84%
	7	487168.00	56.04%
	8	687312.00	141.08%
	9	671822.00	97.75%
	10	1164486.00	173.33%
	11	1058133.00	90.87%
	12	865862.00	81.83%

6.5 习题2



设计如右样式的报表，数据为附件中的data6.1.xlsx中的OrderSummary

报表特点说明：

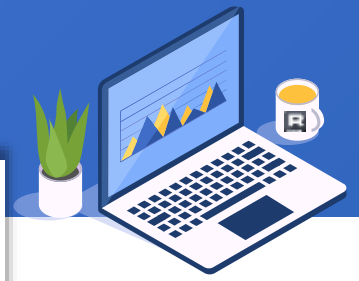
1、计算当前月份的销售额和上个月的销售额的比值

2、跨年份的情况，1月份就不比了，因为和去年12月没有可比性

3、默认年、月的数据是连续的

Year	Month	Amount	LRR
2020	1	831300.00	
	2	352825.00	42.44%
	3	1037630.00	294.09%
	4	698713.00	67.34%
	5	268115.00	38.37%
	6	614630.00	229.24%
	7	515708.00	83.91%
	8	1130095.00	219.13%
	9	650920.00	57.60%
	10	513230.00	78.85%
	11	951065.00	185.31%
	12	637152.00	66.99%
2021	1	410845.00	
	2	831134.00	202.30%
	3	798206.00	96.04%
	4	868427.00	108.80%
	5	1024625.00	117.99%
	6	869321.00	84.84%
	7	487168.00	56.04%
	8	687312.00	141.08%
	9	671822.00	97.75%
	10	1164486.00	173.33%
	11	1058133.00	90.87%
	12	865862.00	81.83%

6.5 习题3



设计如右样式的报表，数据为附件中的 data6.1.xlsx 中的Score

Class	Stu_No	Name	Score	Rank_Class	Rank_Grade
Class 1	101	Canmore	565	5	6
	102	Kim	635	1	2
	103	Oscar	511	7	19
	104	Juliet	537	6	10
	105	Jack	587	4	5
	106	William	494	8	23
	107	Olive	634	2	3
	108	George	448	9	27
	109	Lily	603	3	4
Class 2	201	Sella	549	2	9
	202	James	476	9	25
	203	Robert	478	8	24
	204	Lucas	523	3	14
	205	Ella	658	1	1
	206	Hazel	522	4	15
	207	Ivy	519	5	17
	208	Henry	508	6	20
	209	Lucy	497	7	22
Class 3	301	Charlie	515	7	18
	302	Mae	553	2	8
	303	Miles	533	3	11
	304	Audrey	532	4	12
	305	Daisy	469	9	26
	306	Edith	503	8	21
	307	Sula	554	1	7
	308	Suki	524	5	13
	309	Max	520	6	16

6.5 习题4

设计如右样式的报表，数据为附件中的data6.1.xlsx 中的Score

报表特点：

在习题3的基础上增加每个班级排名前十的人数统计

Class	Stu_No	Name	Score	Rank_Class	Rank_Grade
Class 1	101	Canmore	565	5	6
	102	Kim	635	1	2
	103	Oscar	511	7	19
	104	Juliet	537	6	10
	105	Jack	587	4	5
	106	William	494	8	23
	107	Olive	634	2	3
	108	George	448	9	27
	109	Lily	603	3	4
Class 2	201	Sella	549	2	9
	202	James	476	9	25
	203	Robert	478	8	24
	204	Lucas	523	3	14
	205	Ella	658	1	1
	206	Hazel	522	4	15
	207	Ivy	519	5	17
	208	Henry	508	6	20
	209	Lucy	497	7	22
Class 3	301	Charlie	515	7	18
	302	Mae	553	2	8
	303	Miles	533	3	11
	304	Audrey	532	4	12
	305	Daisy	469	9	26
	306	Edith	503	8	21
	307	Sula	554	1	7
	308	Suki	524	5	13
	309	Max	520	6	16

	Grade Top10 Count
Class 1	6
Class 2	2
Class 3	2

第7章
特殊排序和分组



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7.1 不完全分组

7.2 固定排序分组

7.3 按段分组

7.4 固定条件分组

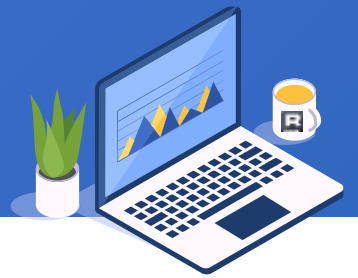
7.5 习题

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7.1 不完全分组

7.1 不完全分组

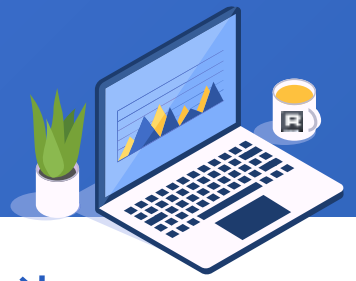


不完全分组：仅对数据集中的部分记录进行分组，而把其它记录都划分为同一组，命名为“其他”。如下图：

只关注东北和华北地区的销售金额统计，别的地区都划分为其他组

Region	City	Amount
North China	Beijing	939.01
	Shijiazhuang	329.68
	Tianjin	637.08
	North China total:	1905.77
Northeast China	Changchun	320.59
	Dalian	624.30
	Shenyang	644.71
	Northeast China total:	1589.60
Others	Chengdu	432.66
	Guiyang	692.53
	Kuming	65.14
	Kunming	501.60
	Lanzhou	1009.34
	Shenzhen	278.33
	Xiamen	231.30
	Xining	956.89
	Yinchuan	208.94
	Others total:	4376.73

7.1 不完全分组



前页的不完全分组报表制作过程如下图所示，使用7.1.xlsx文件，报表存为7.1.rptx

=ds1.group(City;City:1)

=ds1.group(Region,Region=='Northeast China' or Region=='North China';Region:1)选出东北和华北的订单进行分组

D:\tutorial\data\7\7.1.rptx

	A	B	C
1(TH)	Region	City	Amount
2	=ds1.group(Regio	=ds1.group(City;Ci	=ds1.sum(Amount)
		=A2+" total:"	=ds1.sum(Amount)
4	Others	=ds1.group(City,	=ds1.sum(Amount)
5		Others total:	=ds1.sum(Amount,R

C5的表达式也可写成：
=sum(C4{})
所有扩展出的C4求和

=ds1.group(City,Region!='Northeast China' and Region!='North China';City:1)选出既不是东北又不是华北的订单按城市进行分组

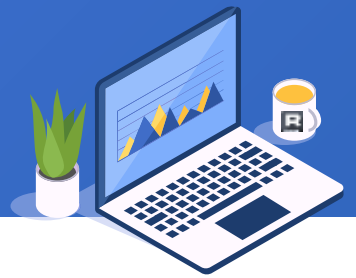
=ds1.sum(Amount,Region!='Northeast China' and Region!='North China')选出既不是东北又不是华北的订单算出金额之和

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7.2 固定排序分组

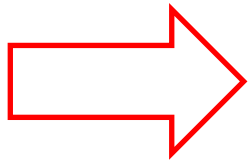
7.2 固定排序和分组



固定排序：按照数据记录中某字段值的固定顺序来显示数据

如下左图中是学生某次考试的成绩，需要按右图列出本小组5人的成绩报表，且显示为小组人员的固定顺序： Mason,Mia,Jack,Nora,Luke

	A	B	C	D	E
1	Stu_No	Name	Chinese	Math	English
2	2001	David	68	87	91
3	2002	Emma	82	88	78
4	2003	Daniel	87	92	85
5	2004	Mason	77	84	83
6	2005	Mia	89	93	95
7	2006	Zoe	72	75	76
8	2007	Michael	79	94	86
9	2008	Nora	81	96	88
10	2009	Luke	78	83	82
11	2010	Sophia	79	81	68

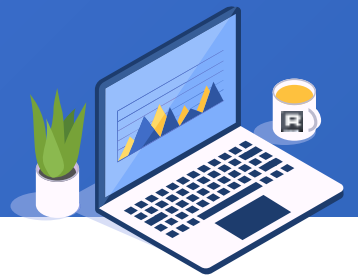


Preview report:D:\tutorial\data\7\7.2.rptx

Name	Student_no	Chinese	Math	English
Mason	2004	77	84	83
Mia	2005	89	93	95
Jack				
Nora	2008	81	96	88
Luke	2009	78	83	82

Jack因病缺考

7.2 固定排序和分组



前页的固定排序报表制作过程如下图所示，使用7.2.xlsx文件，报表存为7.2.rptx

D:\tutorial\data\7\7.2.rptx

	A	B	C	D	E
1(TH)	Name	Student_no	Chinese	Math	English
2	=list("Mason",...	=ds1.select(S...	=ds1.Chinese	=ds1.Math	=ds1.English

=list("Mason","Mia","Jack","Nora","Luke")
用list按小组人员姓名顺序列出

=ds1.select(Stu_No,Name==A2)
加过滤条件“Name==A2"选出主格A2学生的成绩

7.2 固定排序和分组

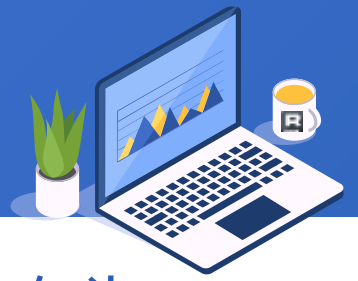


固定排序分组：有时按照升降序排列的分组不是我们想要的，而是想要按要求的固定顺序来显示。例如下图的报表，要求按照South China,Southwest China, North China,Northwest China,Northeast China的地区顺序显示。

从南到北的顺序显示

Region	Product	Amount
South China	dish soap	88912.39
	powder laundry det	87749.34
	soap	82660.64
Southwest China	dish soap	87514.58
	powder laundry det	90399.02
	soap	84147.52
North China	dish soap	86480.96
	powder laundry det	93098.99
	soap	87883.39
Northwest China	dish soap	82168.32
	powder laundry det	87320.34
	soap	89458.07
Northeast China	dish soap	85529.33
	powder laundry det	78992.82
	soap	89452.21

7.2 固定排序和分组



前页的固定排序分组报表制作过程如下图所示，使用7.3.xlsx文件，报表存为7.3.rptx

D:\tutorial\data\7\7.3.rptx

	A	B	C
1(TH)	Region	Product	Amount
2	=list("South China	=ds1.group(Produ	=ds1.sum(Amount)

=list("South China","Southwest China","North China","Northwest China","Northeast China")
在分组主格用list按固定顺序列出分组字段值

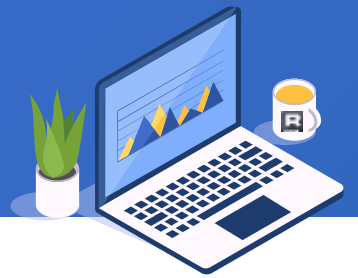
=ds1.group(Product,Region==A2;Product:1)
加过滤条件"地区==A2"选出主格A2所在的地区订单

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7.3 按段分组

7.3 按段分组

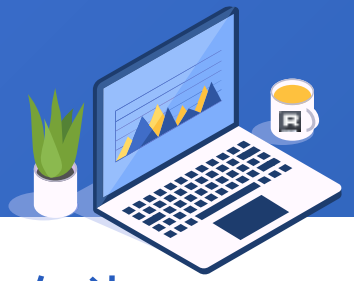


按段分组：把分组字段值划分为几个区间段，按字段值所属的段进行分组。比如常见的按学习成绩分为优、良、中、及格、不及格，再比如下图的例子：

把订单日期按照节日划分为不同的区间，进行分组统计

Date / Total amt / Region	Before May,2017	Between May,2017 to Oct,2017	Between Oct,2017 to Jan,2018	After Jan,2018
North China	124860.36	34293.42	20143.77	88165.79
Northeast China	111141.83	34587.00	20150.62	88094.91
Northwest China	115778.21	37185.40	21463.38	84519.74
South China	108753.29	36949.93	24809.01	88810.14
Southwest China	116549.48	41136.33	23678.30	80697.01

7.3 按段分组



前页的按段分组报表制作过程如下图所示，使用data7.4.xlsx文件，报表存为7.4.rptx

D:\tutorial\data\7\7.4.rptx

	A	B
1(TH)	Date / Total amt / Region	=(dates=list(date("2017-05-01"),date("2017-10-01")),ds1.plot(Date,dates))
2	=ds1.group(Region;Region:1)	=ds1.sum(Amount)

=(dates=list(date("2017-05-01"),date("2017-10-01")),ds1.plot(Date,dates))
由于字段名是大小写不敏感的，所以此处的Date字段跟date函数冲突了，解决办法便是让dates先算，再将dates变量传给plot函数，让plot按照订单日期所在区间分组

=ds1.group(Region;Region:1)

B1显示值表达式：
map(list(0,1,2,3),list("Before May,2017","Between May,2017 to Oct,2017","Between Oct,2017 to Jan,2018","After Jan,2018"))
plot函数返回分组的序号值，共有4组，分别是0、1、2、3，map函数指定了这4个组号对应的显示值

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7.4 固定条件分组

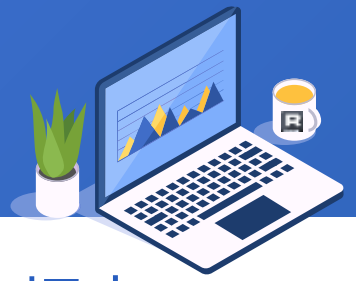
7.4 固定条件分组



固定条件分组：用几个固定的条件对数据进行分组，满足同一个条件的分为同一组。如右图的例子：用“无折扣订单、5万元以上订单、未付款订单”3个固定条件对订单分组统计

Condition	OrderNo	Order date	Product	Amount	Discount	Actual amount	Paid
No discount	10009	2018-01-08	RaqReport	10000	0	10000	No
	10012	2018-01-09	RaqReport	10000	0	10000	Yes
	10013	2018-01-09	RaqReport	10000	0	10000	Yes
	10015	2018-01-10	esProc	60000	0	57000	No
	10016	2018-01-10	YModel	25000	0	25000	No
	10020	2018-01-13	RaqReport	10000	0	10000	Yes
					Total	122000	
Over 50K amount	10001	2018-01-02	esProc	60000	0.1	54000	Yes
	10003	2018-01-04	YModel	80000	0.08	73600	Yes
	10004	2018-01-05	esProc	60000	0.05	57000	Yes
	10006	2018-01-06	esProc	60000	0.05	57000	Yes
	10008	2018-01-07	YModel	80000	0.1	72000	Yes
	10015	2018-01-10	esProc	60000	0	57000	No
	10017	2018-01-11	YModel	80000	0.1	72000	Yes
	10021	2018-01-13	esProc	150000	0.1	135000	Yes
					Total	577600	
No paid	10002	2018-01-03	YModel	50000	0.05	47500	No
	10009	2018-01-08	RaqReport	10000	0	10000	No
	10015	2018-01-10	esProc	60000	0	57000	No
	10016	2018-01-10	YModel	25000	0	25000	No
					Total	139500	

7.4 固定条件分组



前页的固定条件分组报表制作过程如下图所示，使用data7.5.xlsx文件，报表存为7.5.rptx

=ds1.select(OrderNo,Discount==0)
用过滤条件选出折扣率为0的订单

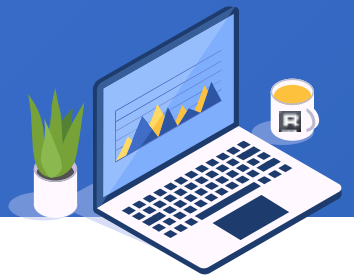
D:\tutorial\data\7\7.5.rptx

	A	B	C	D	E	F	G	H
1(TH)	Condition	OrderNo	Order date	Product	Amount	Discount	Actual amount	Paid
2	No discount	=ds1.select(	=ds1.OrderDat	=ds1.Produc	=ds1.Amoun	=ds1.Discou	=ds1.ActualAmt	=ds1.Paid
3						Total	=sum(G2{})	
4	Over 50K amount	=ds1.select(	=ds1.OrderDat	=ds1.Produc	=ds1.Amoun	=ds1.Discou	=ds1.ActualAmt	=ds1.Paid
5						Total	=sum(G4{})	
6	No paid	=ds1.select(	=ds1.OrderDat	=ds1.Produc	=ds1.Amoun	=ds1.Discou	=ds1.ActualAmt	=ds1.Paid
7						Total	=sum(G6{})	

=ds1.select(OrderNo,ActualAmt>50000)
用过滤条件选出应付金额大于5万的订单

=ds1.select(OrderNo,Paid=="No")
用过滤条件选出未付款的订单

7.4 固定条件分组



按固定条件分组时，同一条记录可以出现在多个组中

Condition	OrderNo	Order date	Product	Amount	Discount	Actual amount	Paid
No discount	10009	2018-01-08	RaqReport	10000	0	10000	No
	10012	2018-01-09	RaqReport	10000	0	10000	Yes
	10013	2018-01-09	RaqReport	10000	0	10000	Yes
	10015	2018-01-10	esProc	60000	0	57000	No
	10016	2018-01-10	YModel	25000	0	25000	No
	10020	2018-01-13	RaqReport	10000	0	10000	Yes
					Total	122000	
Over 50K amount	10001	2018-01-02	esProc	60000	0.1	54000	Yes
	10003	2018-01-04	YModel	80000	0.08	73600	Yes
	10004	2018-01-05	esProc	60000	0.05	57000	Yes
	10006	2018-01-06	esProc	60000	0.05	57000	Yes
	10008	2018-01-07	YModel	80000	0.1	72000	Yes
	10015	2018-01-10	esProc	60000	0	57000	No
	10017	2018-01-11	YModel	80000	0.1	72000	Yes
	10021	2018-01-13	esProc	150000	0.1	135000	Yes
					Total	577600	
No paid	10002	2018-01-03	YModel	50000	0.05	47500	No
	10009	2018-01-08	RaqReport	10000	0	10000	No
	10015	2018-01-10	esProc	60000	0	57000	No
	10016	2018-01-10	YModel	25000	0	25000	No
					Total	139500	

出现在2个组中

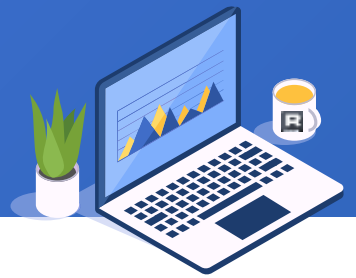
出现在3个组中

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习 题

7.5 习题



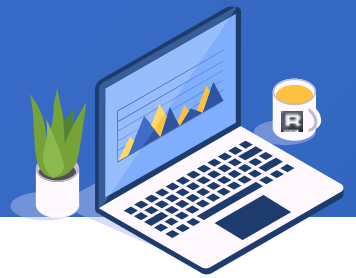
习题1、使用data7.1.xlsx制作下图所示的不完全分组，按城市分组，重点关注北京、成都、大连的分组统计，其余城市都归于其他组。

City	Amount
Beijing	939.01
Chengdu	432.66
Dalian	624.30
Others	5876.13

习题2、使用data7.1.xlsx制作右图所示的不完全分组，按城市、产品分组，重点关注北京、成都、大连的分组统计，其余城市都归于其他组。

City	Product	Amount
Beijing	dish soap	203.96
	powder laundry det	243.77
	soap	491.28
	Beijing total:	939.01
Chengdu	dish soap	311.34
	powder laundry det	121.32
	Chengdu total:	432.66
Dalian	dish soap	229.84
	powder laundry det	204.73
	soap	189.73
	Dalian total:	624.30
Others	dish soap	1073.22
	powder laundry de	2612.24
	soap	2190.67
	Others total:	5876.13

7.5 习题



习题3、使用data7.1.xlsx制作按产品分组统计的报表，要求按洗涤灵、洗衣粉、肥皂的顺序显示，如下图

Product	Amount
powder laundry detergent	3182.06
soap	2871.68
dish soap	1818.36

习题4、使用data7.4.xlsx制作右图所示的按段分组报表，按地区、订单金额大小分组，统计各组订单数。

金额60以下是微单，60-150是小单，150-300是中单，300以上是大单

Region	Amount type	Orders count
North China	Tiny	47
	Small	405
	Middle	679
	Large	216
Northeast China	Tiny	60
	Small	386
	Middle	626
	Large	218
Northwest China	Tiny	43
	Small	372
	Middle	650
	Large	219
South China	Tiny	48
	Small	404
	Middle	627
	Large	221
Southwest China	Tiny	36
	Small	416
	Middle	654
	Large	213

第8章
参数与脚本



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8.1 参数的概念及使用

8.2 动态参数

8.3 动态列报表

8.4 变量和脚本

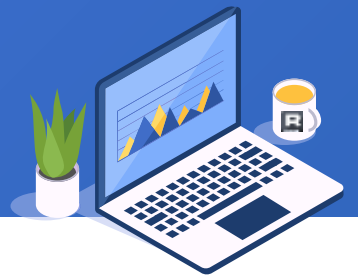
8.5 习 题

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3.1 参数的概念及使用

8.1 参数的概念及使用



查看例子1.2.rptx：

D:\tutorial\data\1\1.2.rptx

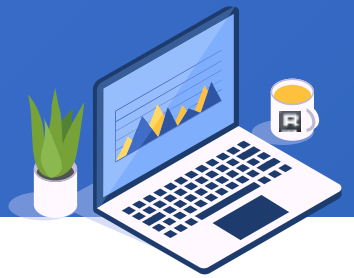
	A	B	C	D	E
1(TH)	Name	Subject	Midterm	Final	WA
2	=ds1.select(N	=ds1.Subj			
3		Total			

=ds1.select(Name,Subject=="Math")

由于表达式写死了查询条件，如果想统计物理总分，只能修改源报表的表达式，使用不方便

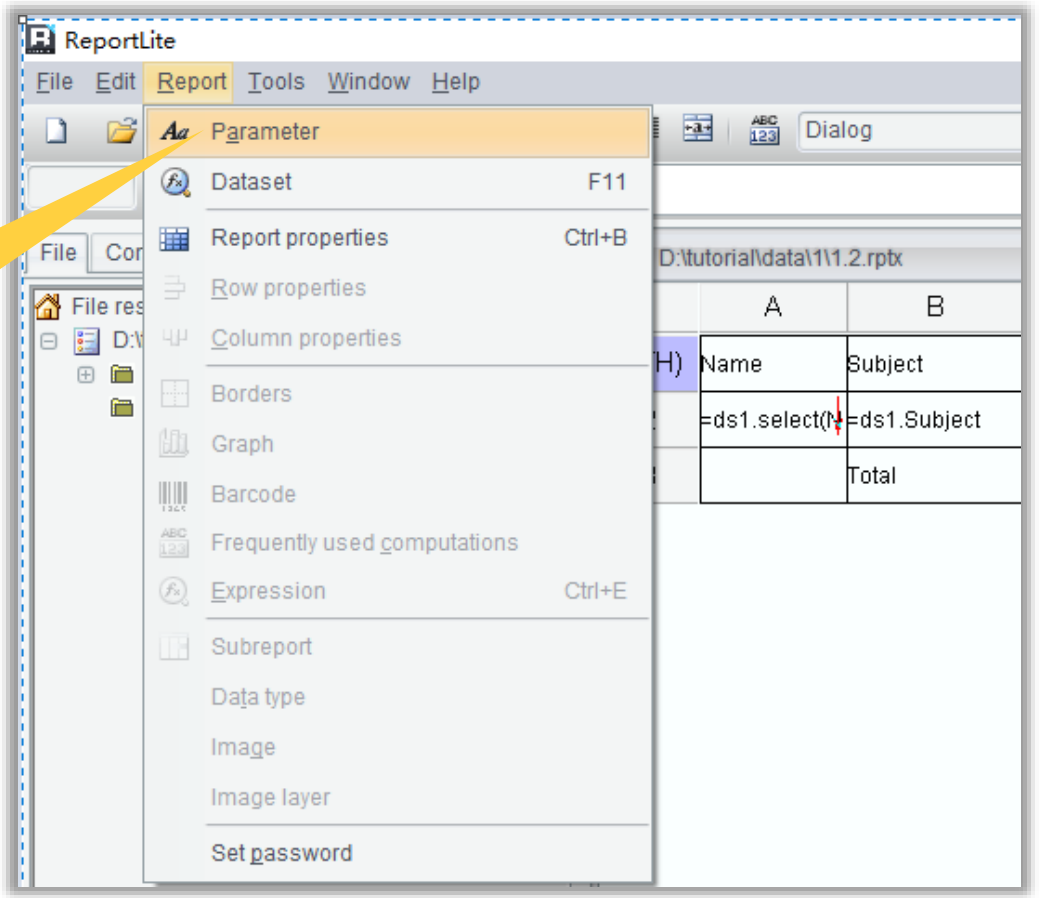
Name	Subject	Midterm	Final	WA
Rose	Math	59	80	71.6
Mike	Math	87	77	81.0
Ronald	Math	36	50	44.4
Smith	Math	25	55	43.0
Frank	Math	66	80	74.4
Aimay	Math	32	50	42.8
James	Math	37	67	55.0
Jay	Math	56	56	56.0
	Total	398	515	468.2

8.1 参数的概念及使用

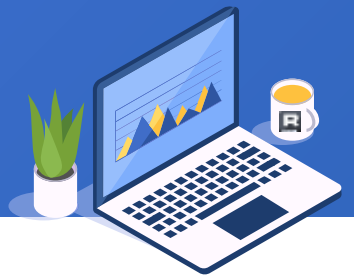


为了使报表查询条件更灵活，可以事先在报表中定义查询变量，然后在报表中引用查询变量来灵活设置报表的数据。这个查询变量便称为参数。

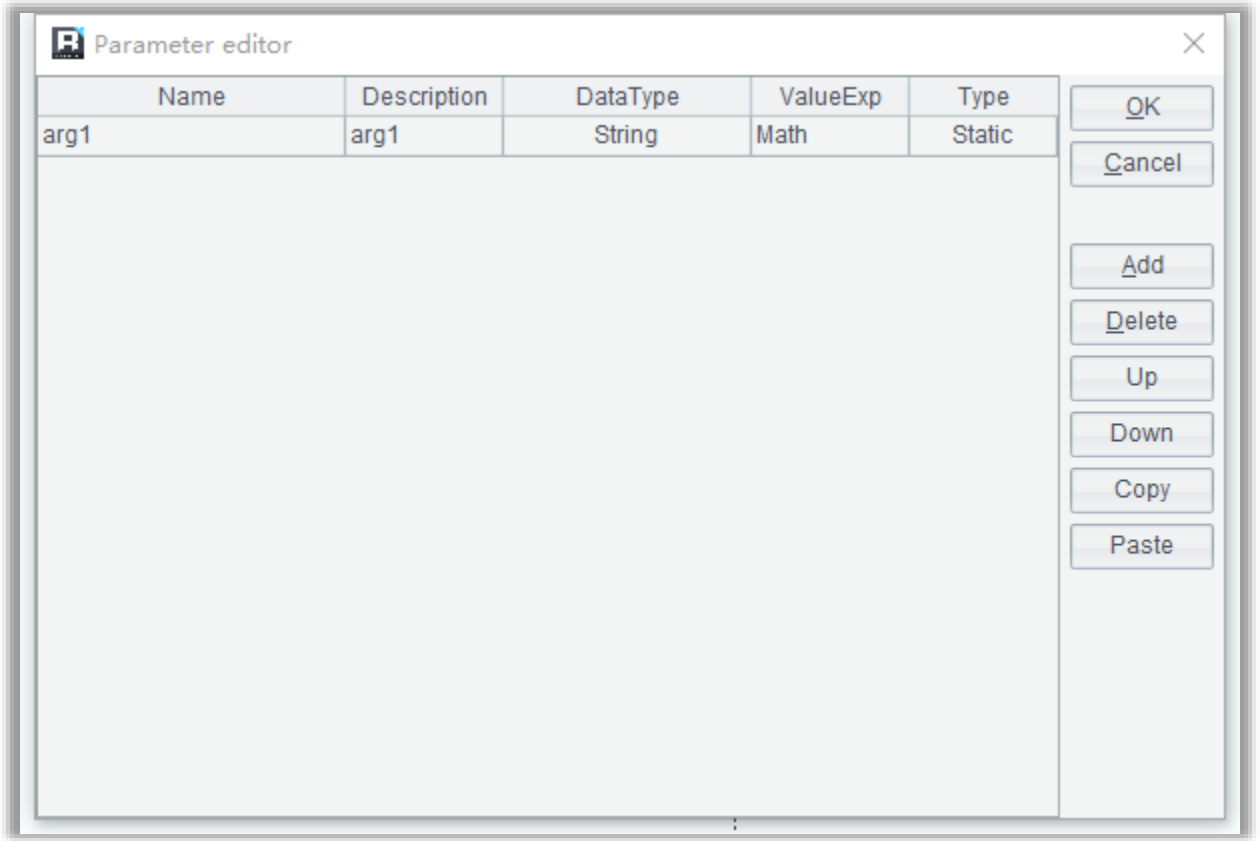
通过菜单项参数，可以为报表添加参数



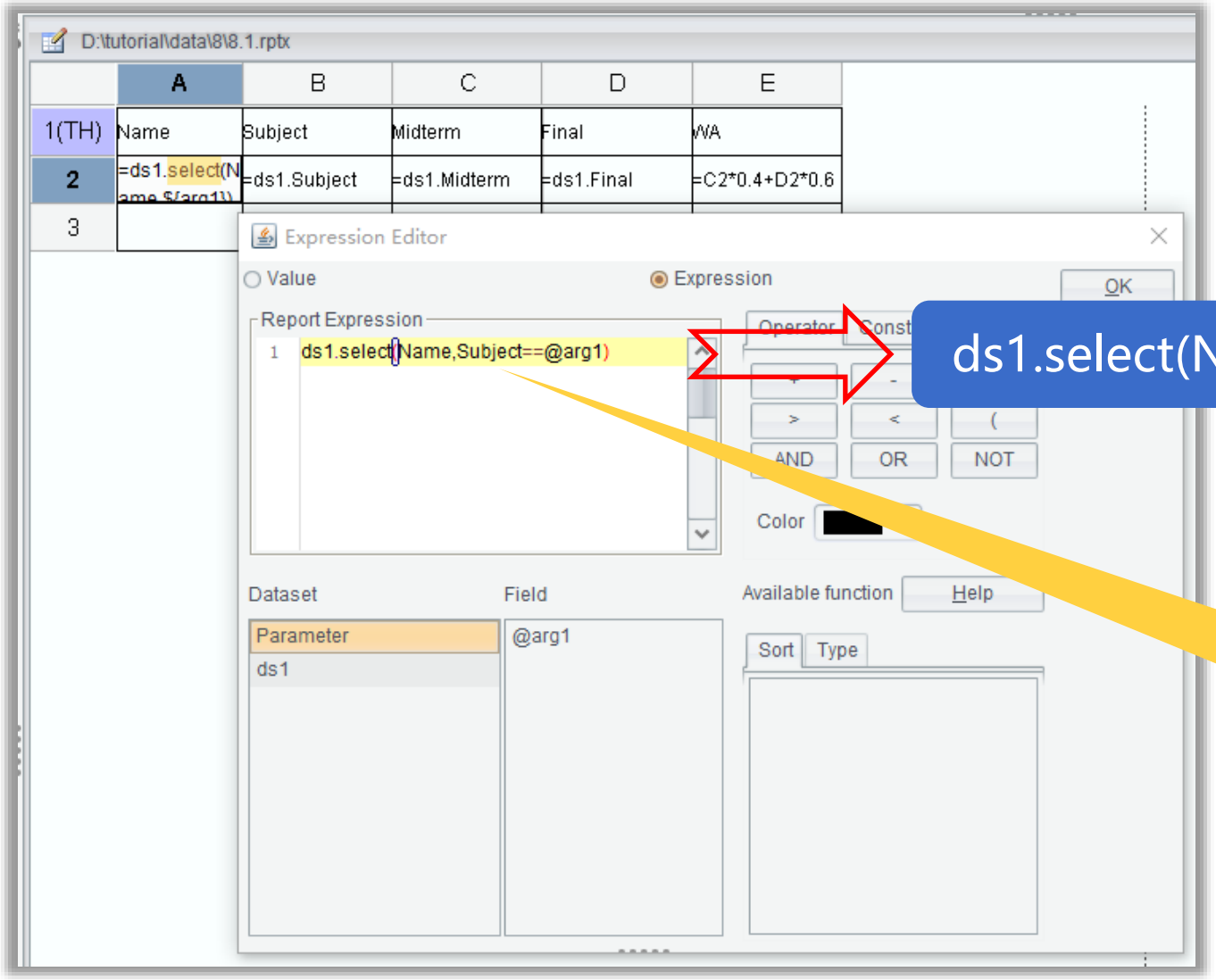
8.1 参数的概念及使用



将1.2.rptx另存为8.1.rptx,然后在参数编辑窗口中，点击增加，便可添加报表的参数，如图在值表达式中设置参数arg1的缺省查询值为Math。



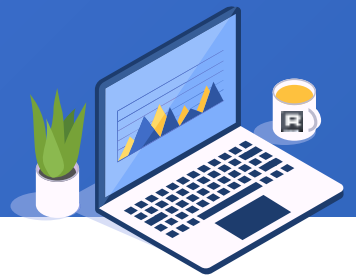
8.1 参数的概念及使用



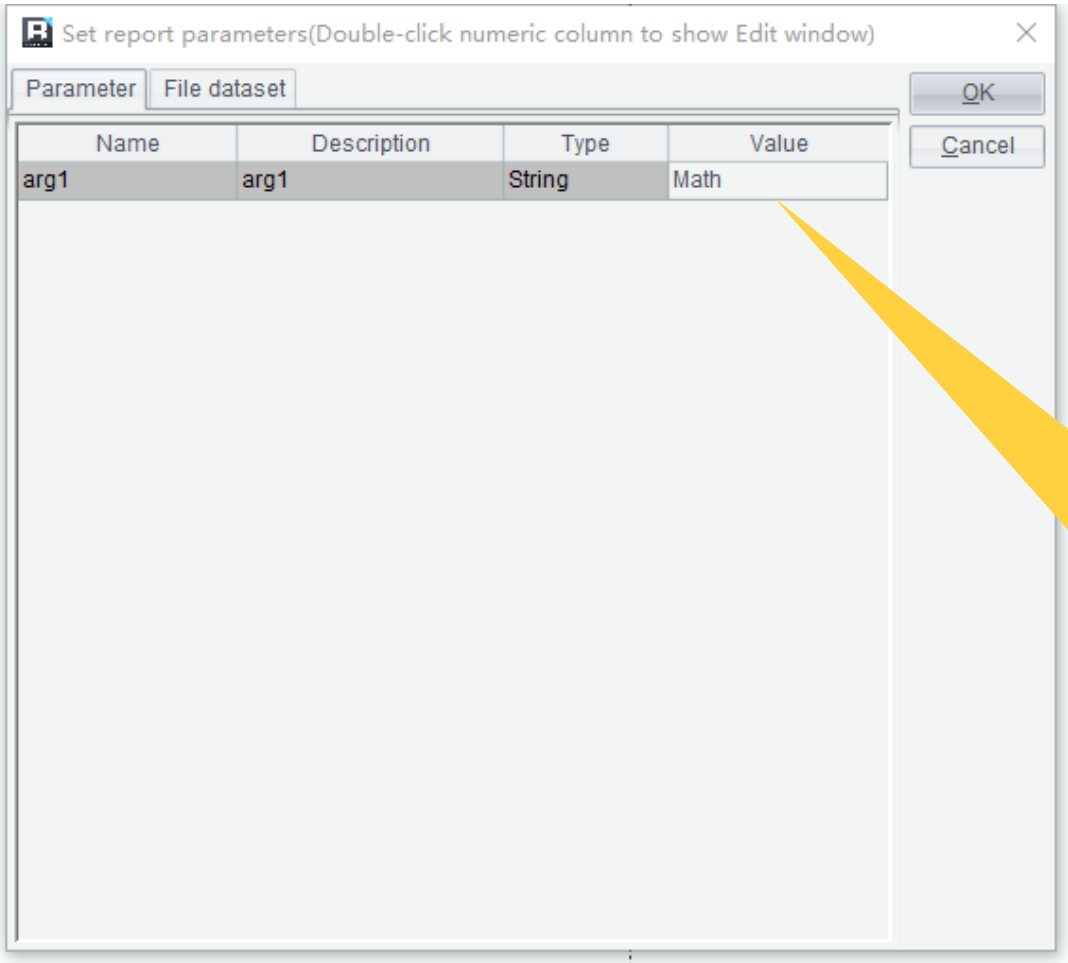
ds1.select(Name,Subject==@arg1)

如图所示修改A2的表达式，在表达式中引用参数

8.1 参数的概念及使用



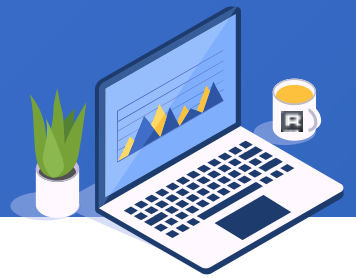
预览报表，此时会弹出设置报表参数值的窗口：



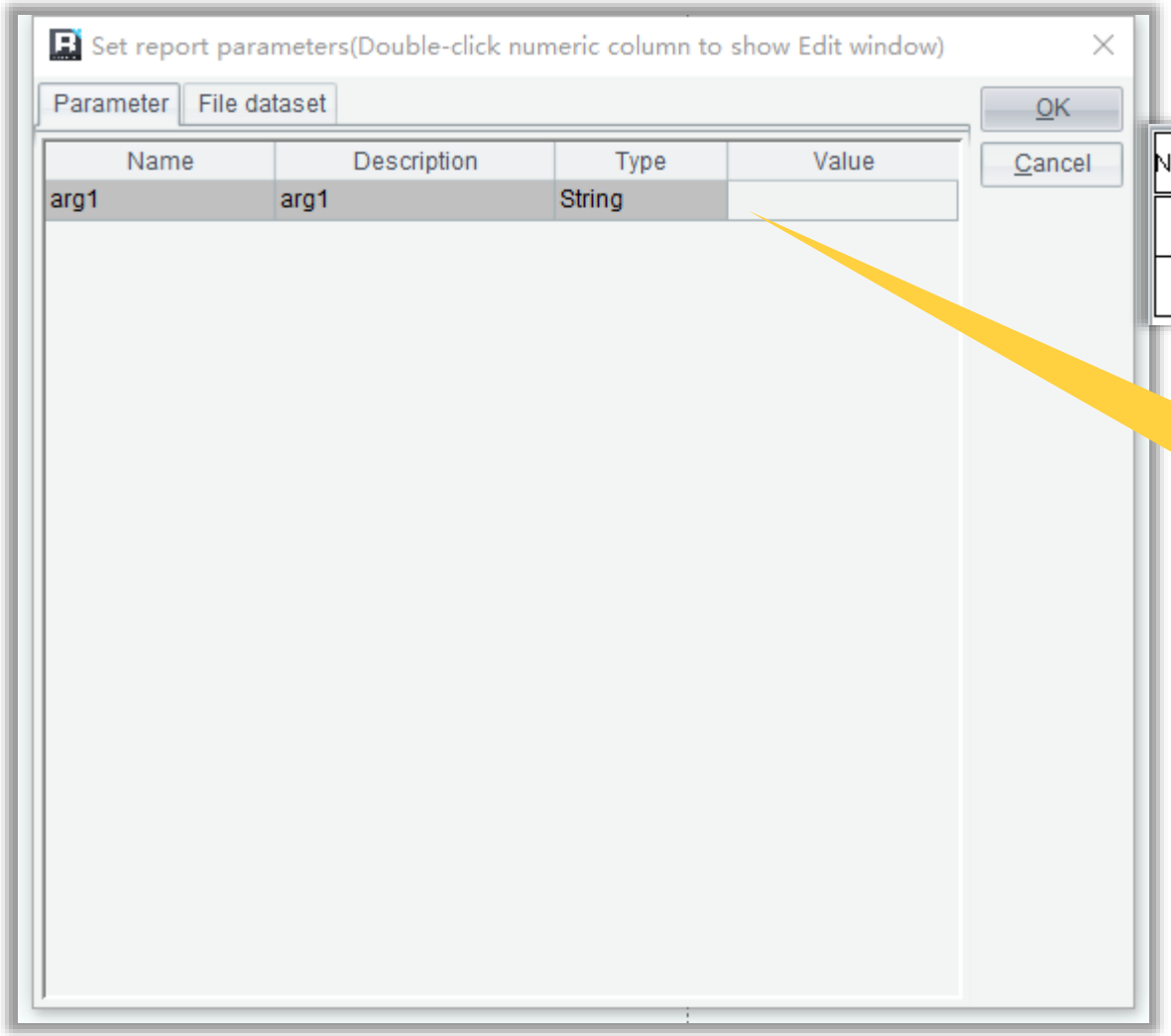
Name	Subject	Midterm	Final	WA
Rose	Math	59	80	71.6
Mike	Math	87	77	81.0
Ronald	Math	36	50	44.4
Smith	Math	25	55	43.0
Frank	Math	66	80	74.4
Aimay	Math	32	50	42.8
James	Math	7	67	55.0
Jay	Math		56	56.0
	Total		515	468.2

设置参数值为Math，按OK后便可得到Math科目统计报表

8.1 参数的概念及使用



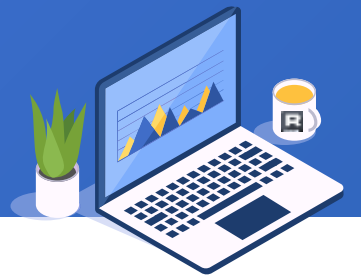
再次预览报表，清空参数值后查询：



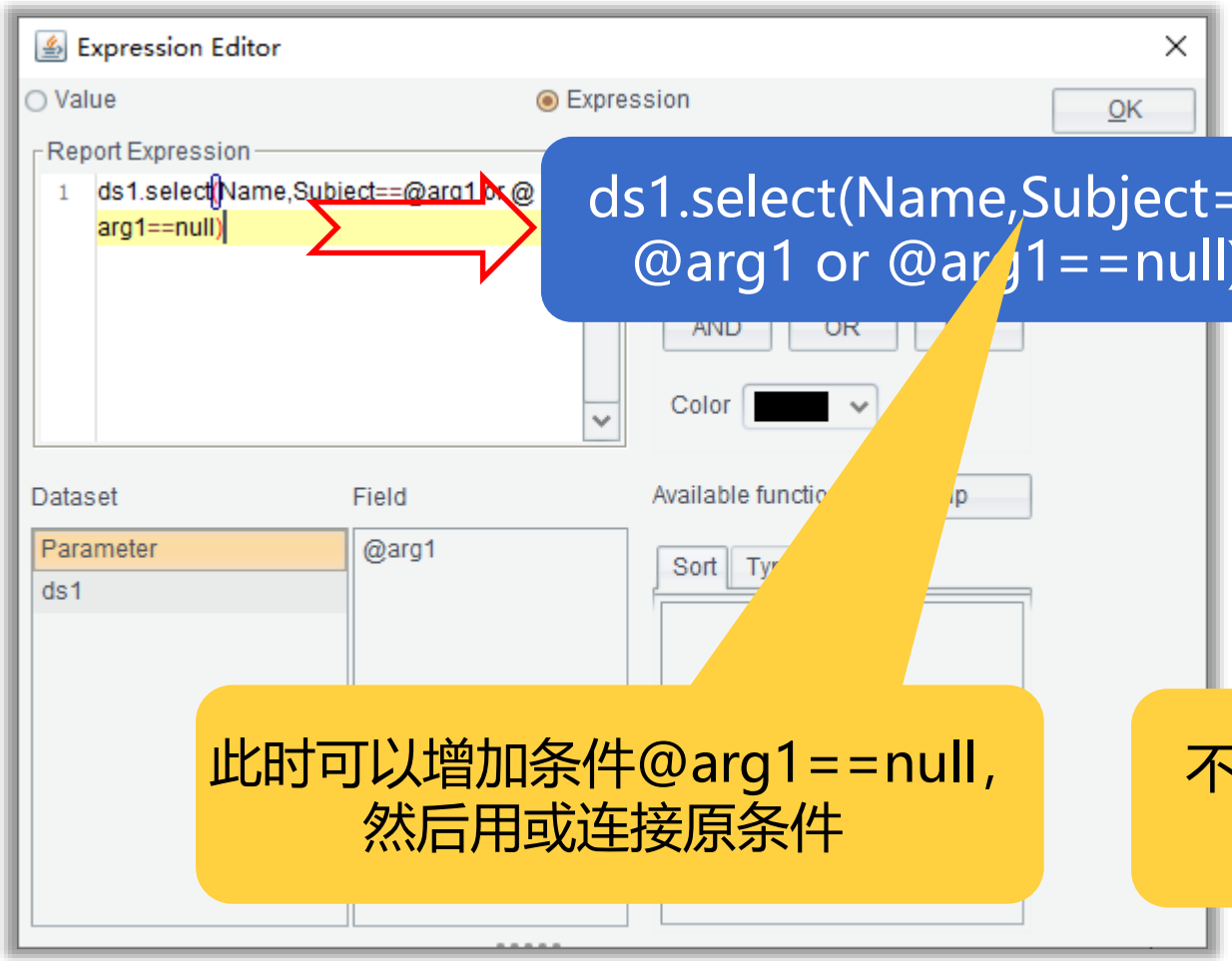
Name	Subject	Midterm	Final	WA
	Total	0.0	0.0	.00

参数值为空，查询结果也为空

8.1 参数的概念及使用



我们习惯于不输入参数值时，能查出全部记录：



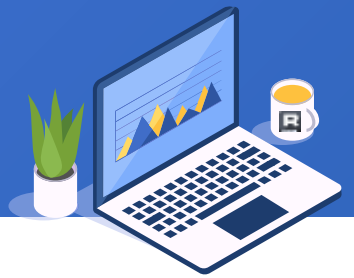
ds1.select(Name,Subject==@arg1 or @arg1==null)

此时可以增加条件@arg1==null，然后用或连接原条件

不录入参数，查询到全部记录

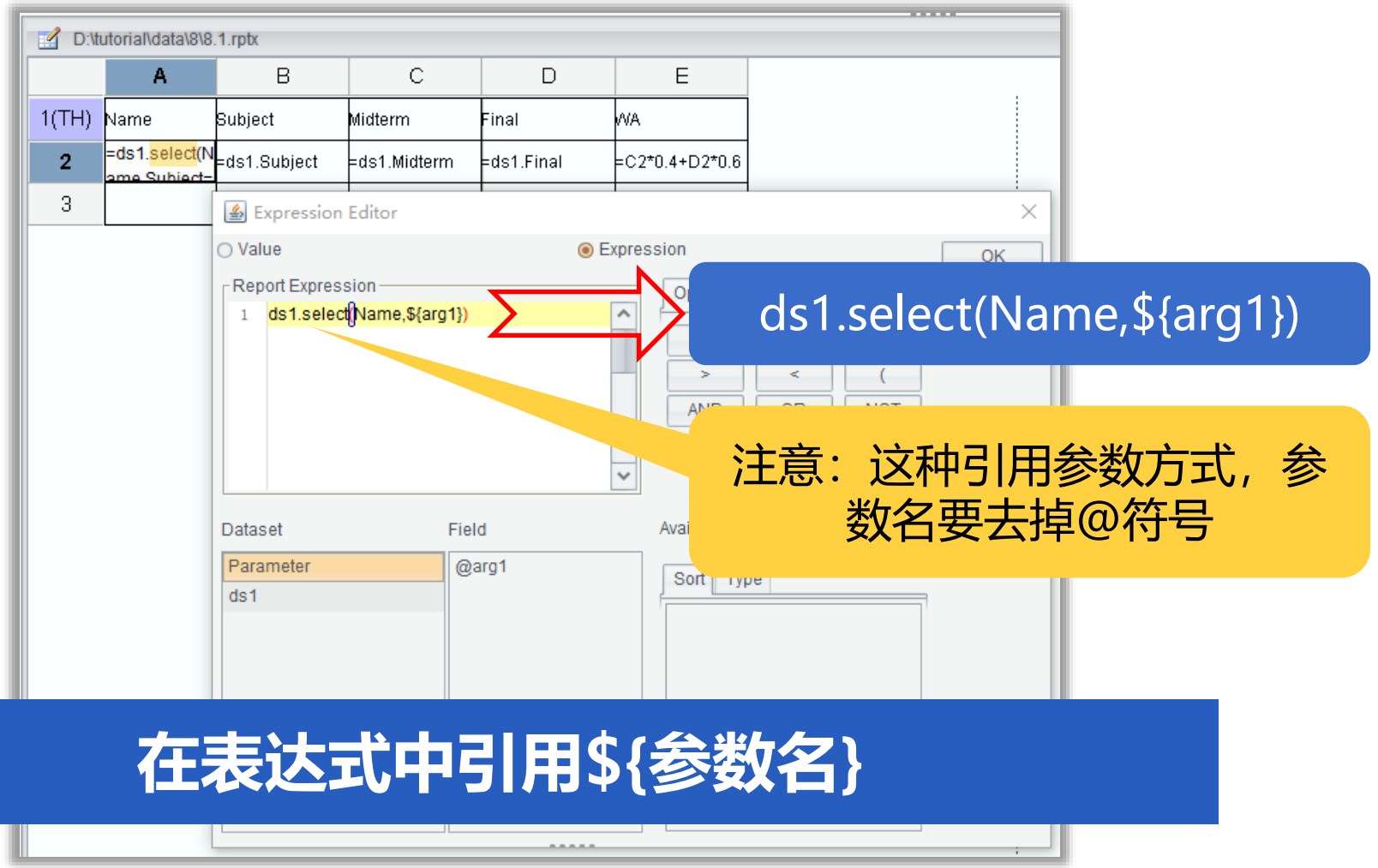
Name	Subject	Midterm	Final	WA
Rose	Math	59	80	71.6
Rose	Physics	48	65	58.2
Mike	Math	87	77	81.0
Mike	Physics	70	86	79.6
Ronald	Math	36	50	44.4
Ronald	Physics	54	75	66.6
Smith	Math	25	55	43.0
Smith	Physics	56	75	67.4
Frank	Math	66	80	74.4
Frank	P	75	65	69.0
Aimay		32	50	42.8
Aimay		42	65	55.8
		37	67	55.0
		97	88	91.6
		56	56	56.0
		88	79	82.6
	Total	928	1113	1039.00

8.1 参数的概念及使用



我们还可以用一段表达式作为参数，抄进报表中的表达式使用

写法为${参数}：

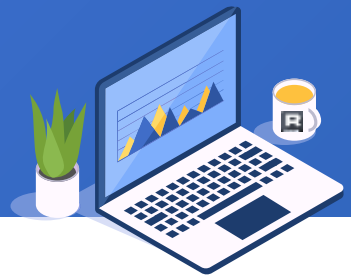


ds1.select(Name,${arg1})

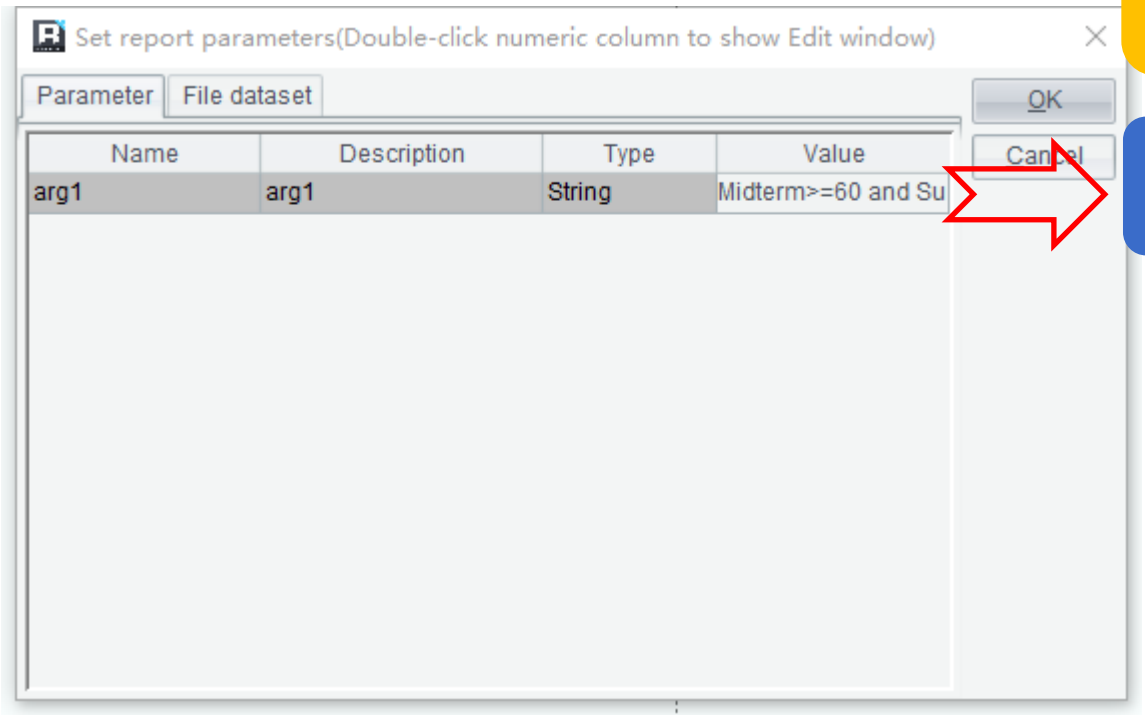
注意：这种引用参数方式，参数名要去掉@符号

在表达式中引用${参数名}

8.1 参数的概念及使用



预览报表，此时的参数值为表达式：



使用表达式参数，可以同时查询科目为数学而且期中成绩及格的同学，查询条件非常灵活

```
Midterm>=60 and Subject=="Math"
```

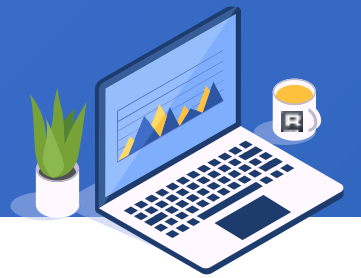
Name	Subject	Midterm	Final	WA
Mike	Math	87	77	81.0
Frank	Math	66	80	74.4
	Total	153	157	155.40

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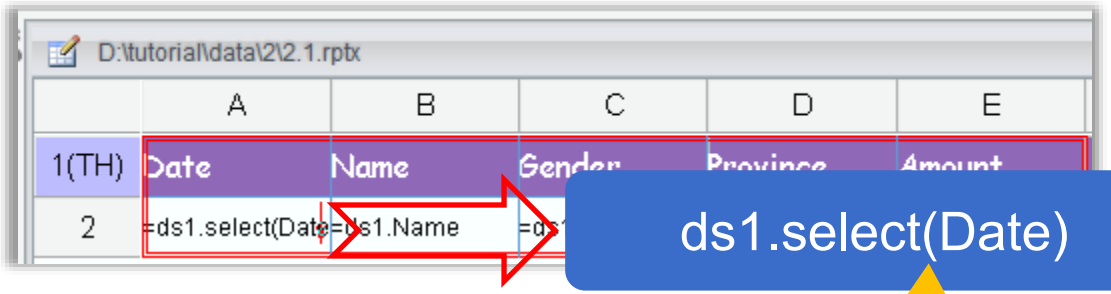


3.2 动态参数

8.2 动态参数



查看2.1.rptx：

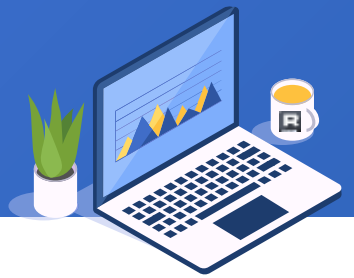


ds1.select(Date)

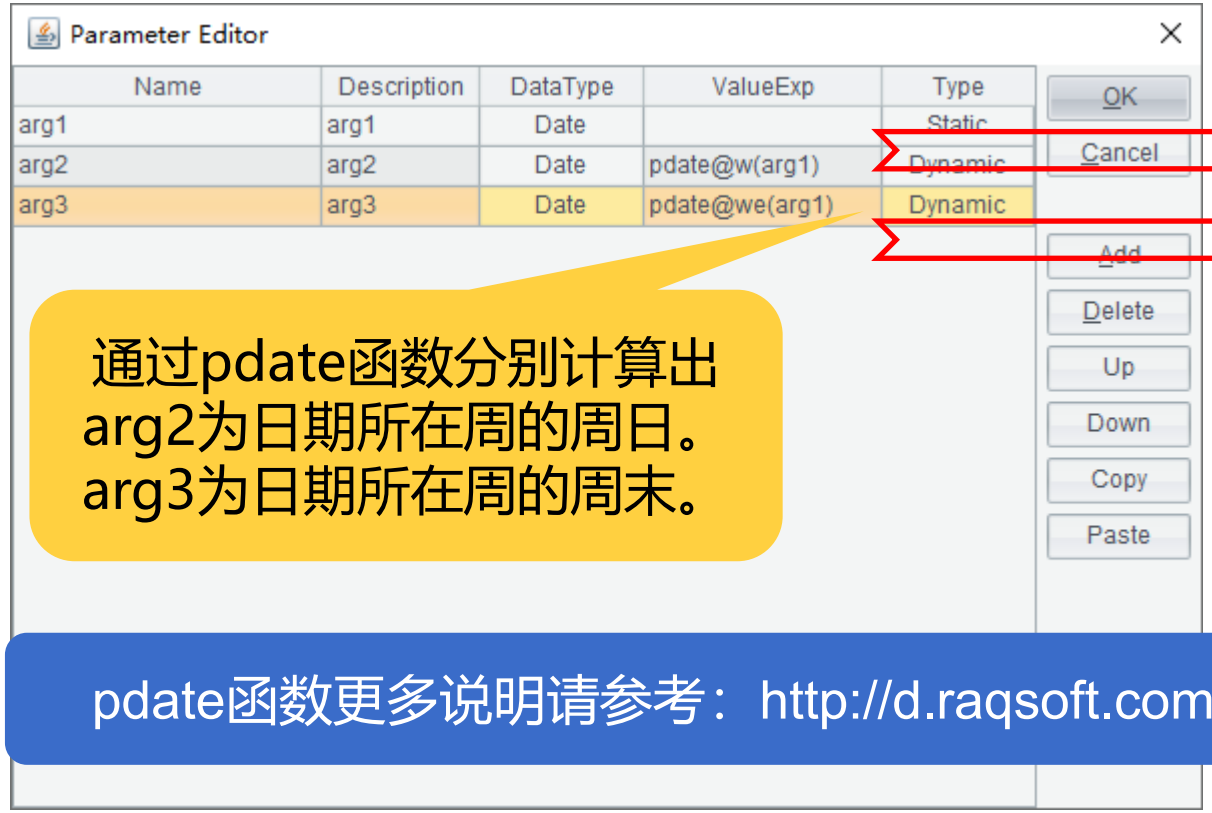
如果只查询某个日期所在周的充值记录，显然让用户记住每一周的开始日期和结束日期不可能

Date	Name	Gender	Province	Amount
Jan 12,2019	Lisa	Female	Washington	$100.00
Feb 08,2019	Ice Chan	Female	New York	$200.00
Feb 15,2019	Lisa	Female	Washington	$200.00
Apr 16,2019	Amay	Female	Los Angeles	$100.00
Apr 27,2019	Tossman	Male	Seattle	$100.00
May 09,2019	Smith	Male	Detroit	$200.00
May 11,2019	Tossman	Male	Seattle	$100.00
May 21,2019	Amay	Female	Los Angeles	$200.00
Jun 22,2019	Ice Chan	Female	New York	$100.00
Jun 23,2019	Tossman	Male	Seattle	$100.00
Jun 25,2019	Ice Chan	Female	New York	$150.00
Jun 27,2019	Smith	Male	Detroit	$100.00
Jul 01,2019	Amay	Female	Los Angeles	$200.00

8.2 动态参数



将2.1.rptx另存为8.2.rptx，再如图定义参数，arg2和arg3均为动态参数，其值可以通过arg1用表达式计算出来。



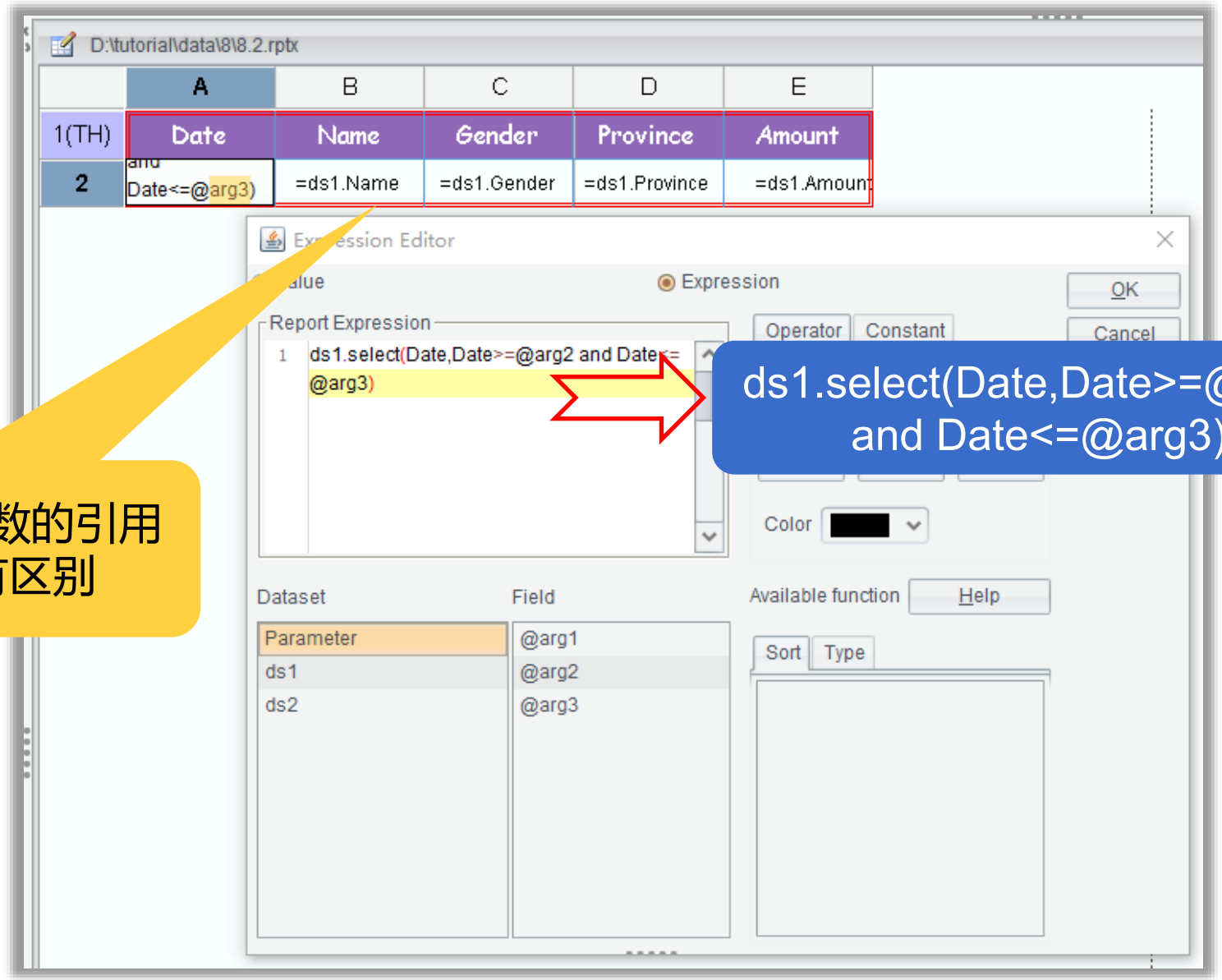
pdate@w(arg1)

pdate@we(arg1)

pdate函数更多说明请参考：http://d.raqsoft.com.cn:6999/report/preference/pdatede.html

8.2 动态参数

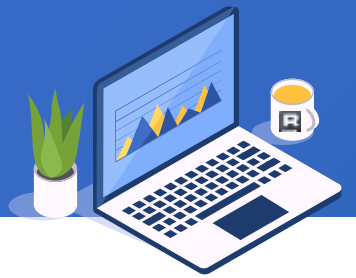
修改A2的查询表达式：



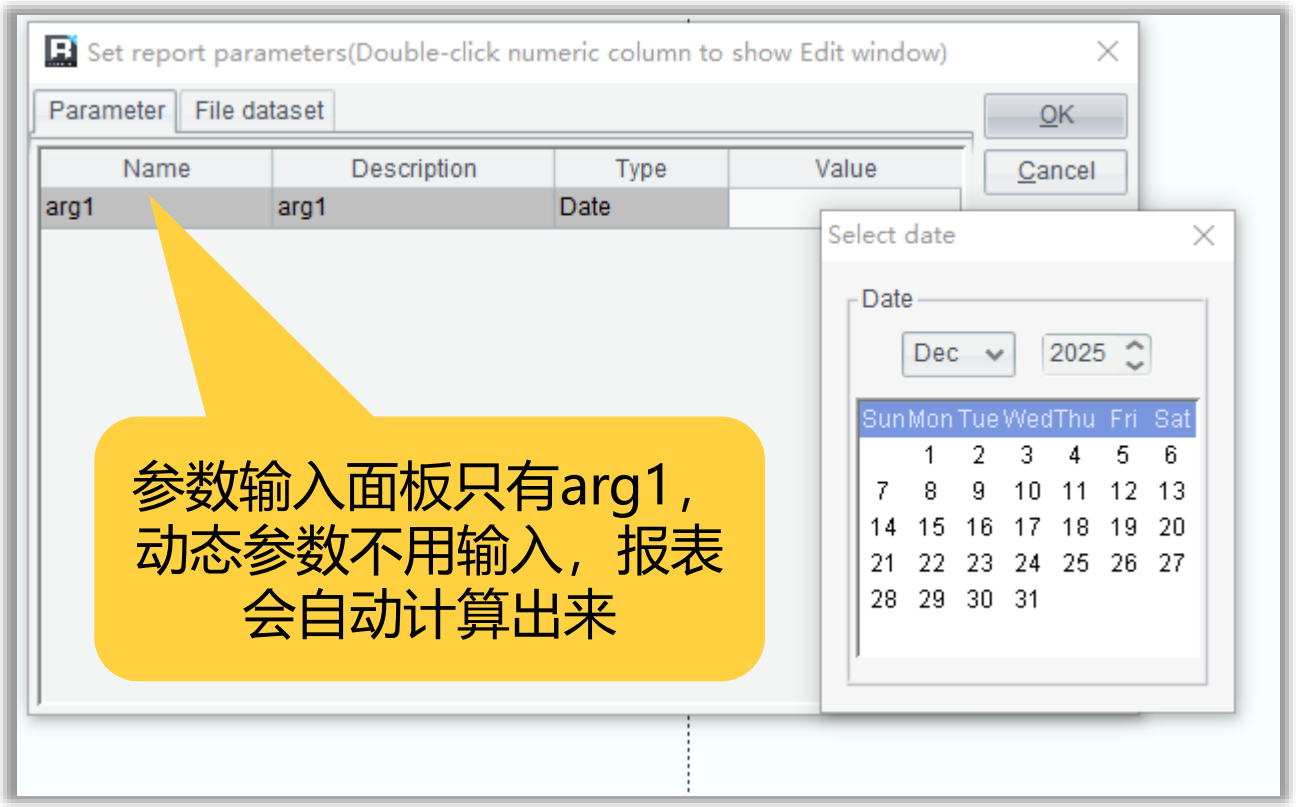
表达式中对动态参数的引用跟普通参数没有区别

ds1.select(Date,Date>=@arg2 and Date<=@arg3)

8.2 动态参数



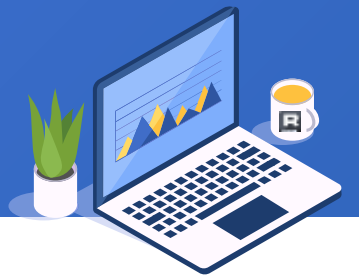
预览报表：



Date	Name	Gender	Province	Amount
Jun 23,2019	Tossman	Male	Seattle	$100.000
Jun 24,2019	Lisa	Female	Washington	$50.000
Jun 25,2019	Ice Chan	Female	New York	$150.000
Jun 26,2019	Tossman	Male	Seattle	$50.000
Jun 27,2019	Smith	Male	Detroit	$100.000
Jun 28,2019	[illegible]Chan	Female	New York	$50.000
Jun 29,2019	Tos[illegible]	Male	Seattle	$50.000

输入参数2019年6月25后，查询出本周的所有记录

8.2 动态参数



不同参数类型的应用特点：

1. 普通参数应用最广，参数值只能为常量。
2. 普通参数的${参数名}写法时，参数值只能是字符串，并且是个表达式（或部分）。表达式写法虽然稍微难用，但是非常灵活。
3. 动态参数用于普通参数不方便或者没法直接录入时，采用表达式，将普通参数计算后，再用于报表中查询。动态参数是一个能立即计算的表达式。

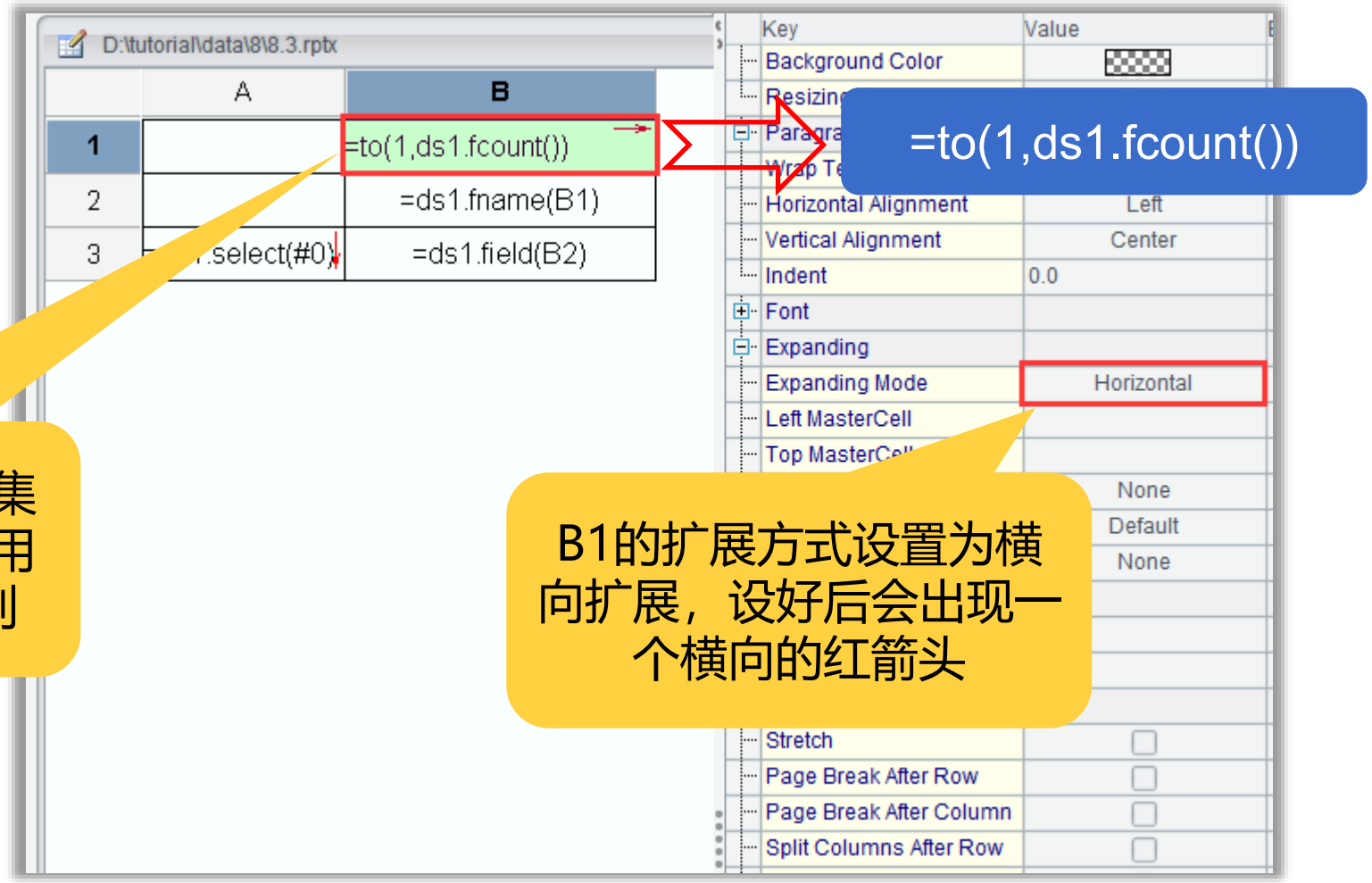
CONTENTS



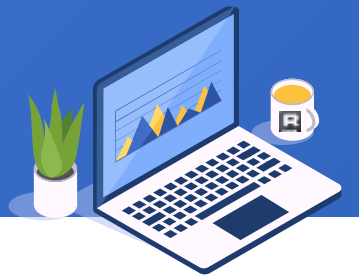
3.3 动态列报表

8.3 动态列报表

新建空白报表，添加文件数据集，数据采用data2.1.xlsx，并保存为文件名8.3.rptx 。



8.3 动态列报表



用fname函数设置B2的动态字段名：

D:\tutorial\data\8\8.3.rptx

	A	B
1		=to(1,ds1.fcount())
2		=ds1.fname(B1)
3		

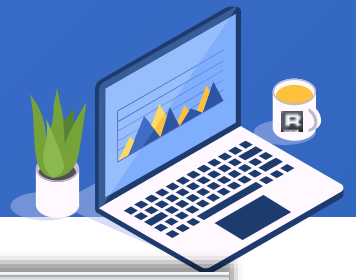
fname()函数返回给定序号的字段名

=ds1.fname(B1)

动态扩展出字段名的预览效果

	1	2	3	4	5
	Date	Name	Gender	Province	Amount

8.3 动态列报表



用field函数获取动态列名的字段值：

D:\tutorial\data\8\8.3.rptx

	A	B
1		=to(1,ds1.fcount
2		=ds1.fname(B1)
3	=ds1.select(	=ds1.field(B2)

隐藏第一行跟A列后预览报表

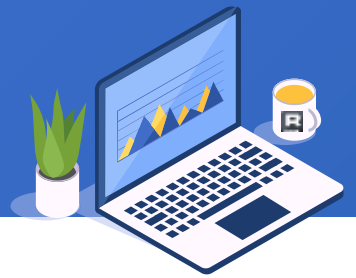
Preview report:D:\tutorial\data\8\8.3.rptx

Date	Name	Gender	Province	Amount
2019-01-12	Lisa	1	1	100
2019-02-08	Ice Chan	1	2	200
2019-02-14	Tossman	0	3	50
2019-02-15	Lisa	1	1	200
2019-04-16	Amay	1	4	100
2019-04-27	Tossman	0	3	100
2019-04-28	Lisa	1	1	50
2019-05-09	Smith	0	5	200
2019-05-11	Tossman	0	3	100
2019-05-21	Amay	1	4	200
2019-06-22	Ice Chan	1	2	100
2019-06-23	Tossman	0	3	100
2019-06-24	Lisa	1	1	50
2019-06-25	Ice Chan	1	2	150
2019-06-26	Tossman	0	3	50
2019-06-27	Smith	0	5	100
2019-06-28	Ice Chan	1	2	50
2019-06-29	Tossman	0	3	50
2019-06-30	Lisa	1	1	50
2019-07-01	Amay	1	4	200
2019-07-02	Tossman	0	3	50

“#+数字”表示按序号访问列，第0列为系统默认的序号列

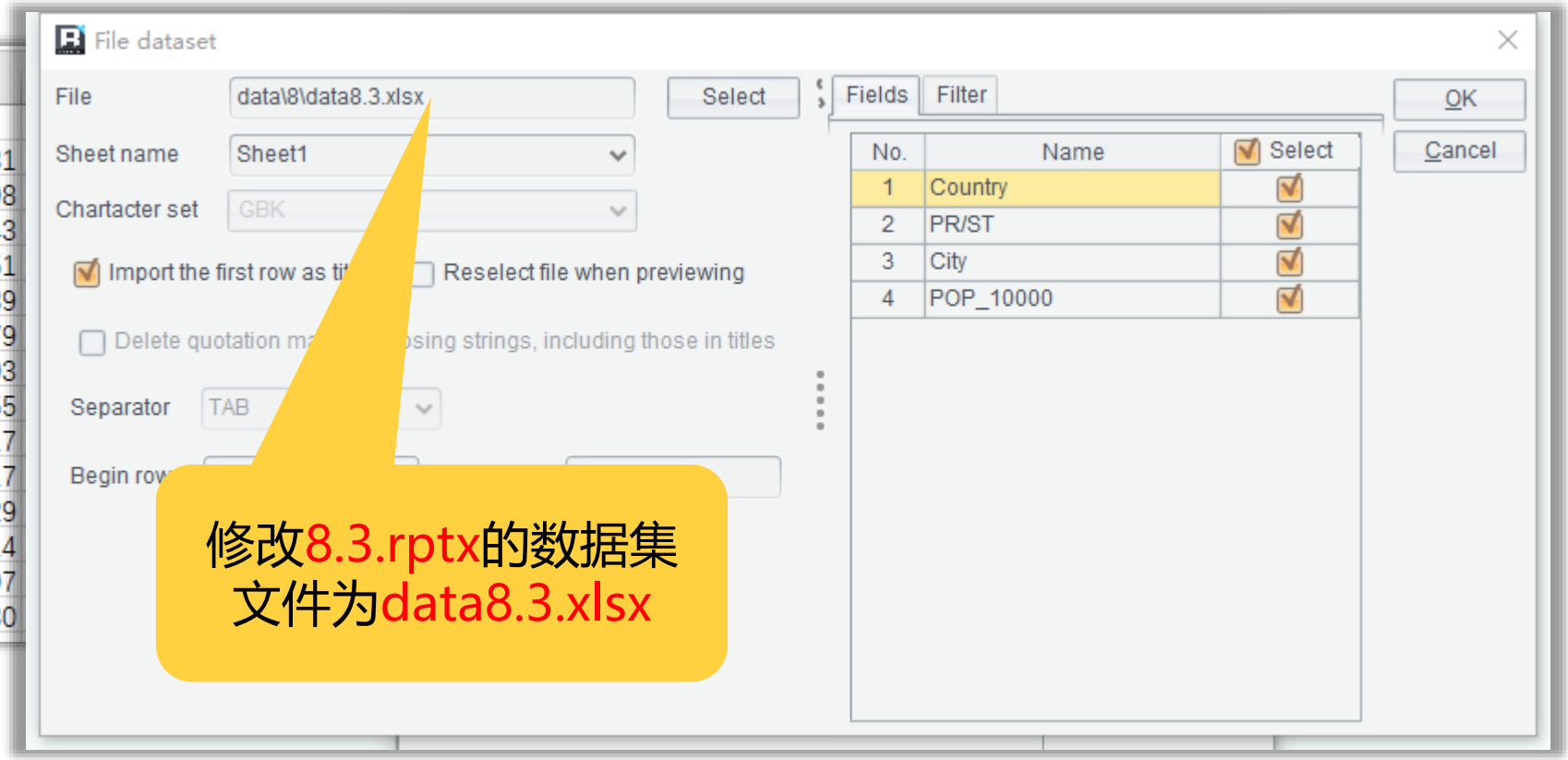
B3会自动跟随B1横向扩展出字段，同时跟随A3纵向扩展出记录行

8.3 动态列报表



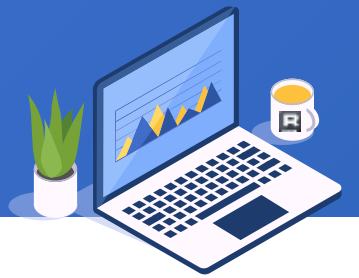
查看例子数据data8.3.xlsx：

	A	B	C	D
1	Country	PR/ST	City	POP_10000
2	China	Liaoning	Shenyang	831
3	China	Liaoning	Dalian	598
4	China	Liaoning	Yingkou	243
5	China	Liaoning	Benxi	151
6	China	Hebei	Shijiazhuang	1039
7	China	Hebei	Langfang	479
8	China	Hebei	Tangshan	793
9	China	Shandong	Jinan	655
10	China	Shandong	Qingdao	817
11	China	Shandong	Dongying	217
12	US	Alaska	Anchorage	29
13	US	Alaska	Fairbanks	5.4
14	US	California	Los Angeles	397
15	US	California	Hollywood	30



动态列报表可以随意更改数据集

8.3 动态列报表



预览报表：

Preview report:D:\tutorial\data\8\8.3.rptx

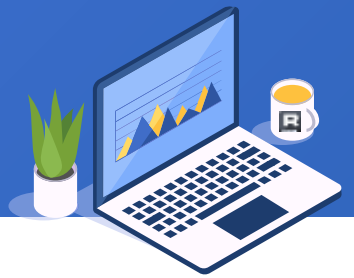
Country	PR/ST	City	POP_10000
China	Liaoning	Shenyang	831
China	Liaoning	Dalian	598
China	Liaoning	Yingkou	243
China	Liaoning	Benxi	151
China	Hebei	Shijiazhuang	1039
China	Hebei	Langfang	479
China	Hebei	Tangshan	793
China	Shandong	Jinan	655
China	Shandong	Qingdao	817
China	Shandong	Dongying	217
US	Alaska	Anchorage	29
US	Alaska	Fairbanks	5.4
US	California	Los Angeles	397
US	California	Hollywood	30

CONTENTS



3.4 变量和脚本

8.4 变量和脚本



在报表中使用表达式 “=变量名=值”定义临时变量。

D:\tutorial\data\8\8.4.1.rptx

	A	B	C	D	E	F
1(TH)	=a=0					
2(TH)	Date	Name	Gender	Province	Amount	CumAmount
3	ect(Date	=ds1.Name	=ds1.Gender	=ds1.Province	=ds1.Amount	=a=a+E3

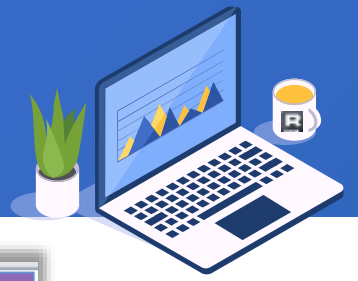
=a=a+E3

在A1定义临时变量a，并赋初值为0

在扩展格F3里，使用临时变量a对金额进行累计

使用临时变量给2.1.rptx增加累计金额,存为8.4.1.rptx

8.4 变量和脚本

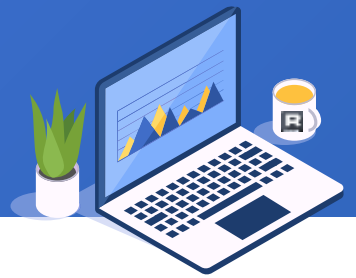


隐藏第一行后，预览报表：

Date	Name	Gender	Province	Amount	CumAmount
Jan 12,2019	Lisa	Female	Washington	$100.000	100
Feb 08,2019	Ice Chan	Female	New York	$200.000	300
Feb 14,2019	Tossman	Male	Seattle	$50.000	350
Feb 15,2019	Lisa	Female	Washington	$200.000	550
Apr 16,2019	Amay	Female	Los Angeles	$100.000	650
Apr 27,2019	Tossman	Male	Seattle	$100.000	750
Apr 28,2019	Lisa	Female	Washington	$50.000	800
May 09,2019	Smith	Male	Detroit	$200.000	1000
May 11,2019	Tossman	Male	Seattle	$100.000	1100
May 21,2019	Amay	[illegible]	Los Angeles	$200.000	1300
[illegible]	Ice Chan	Female	New York	$100.000	1400
Jun 23,2019	Tossman	Male	Seattle	$100.000	1500
Jun 24,2019	Lisa	Female	Washington	$50.000	1550
Jun 25,2019	Ice Chan	Female	New York	$150.000	1700
Jun 26,2019	Tossman	Male	Seattle	$50.000	1750
Jun 27,2019	Smith	Male	Detroit	$100.000	1850
Jun 28,2019	Ice Chan	Female	New York	$50.000	1900
Jun 29,2019	Tossman	Male	Seattle	$50.000	1950
Jun 30,2019	Lisa	Female	Washington	$50.000	2000
Jul 01,2019	Amay	Female	Los Angeles	$200.000	2200
Jul 02,2019	Tossman	Male	Seattle	$50.000	2250

通过临时变量，在扩展行时，可以自动计算出每一行的累计值

8.4 变量和脚本



例子数据data8.4.xlsx中按页面记录了两个季度的充值记录：

	A	B	C
1	Date	Name	Credit
2	2019/1/12	Luna	100
3	2019/1/18	Jackie	200
4	2019/1/20	Ethan	50
5	2019/1/25	Luna	200
6	2019/2/2	Bella	100
7	2019/2/7	Ethan	100
8	2019/2/18	Luna	50
9	2019/2/19	Finley	200
10	2019/2/21	Ethan	100
11	2019/2/21	Bella	200
12	2019/2/22	Jackie	100
13	2019/2/23	Ethan	100
14	2019/2/24	Luna	50
15	2019/2/25	Jackie	150
16	2019/3/2	Ethan	50
17	2019/3/7	Finley	100
18	2019/3/8	Jackie	50
19	2019/3/12	Ethan	50
20	2019/3/13	Luna	50
21	2019/3/21	Bella	200
22	2019/3/22	Ethan	50
23			

Q1 Q2



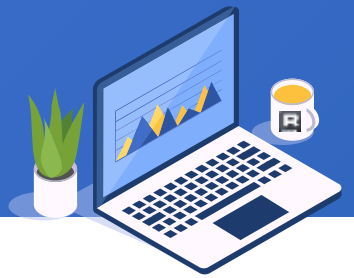
	A	B	C
1	Date	Name	Credit
2	2019/4/12	Finley	100
3	2019/4/18	Ethan	200
4	2019/5/1	Bella	350
5	2019/5/2	Jackie	200
6	2019/5/4	Ethan	100
7	2019/5/4	Luna	200
8	2019/5/8	Luna	150
9	2019/5/9	Jackie	
10	2019/5/21	Ethan	100
11	2019/6/11	Finley	300
12	2019/6/22	Jackie	100
13	2019/6/23	Ethan	100
14	2019/6/30	Luna	50
15			
16			
17			
18			
19			
20			
21			
22			
23			

Q1 Q2

现在想要统计两个季度内，每人的充值总额

在报表中计算时，一个数据集只能对应一个季度的数据，两个独立的数据集没法合并

8.4 变量和脚本

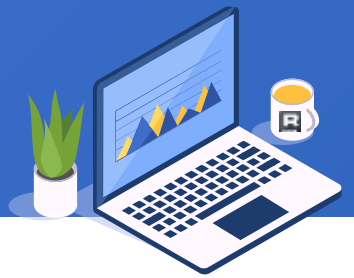


新建空白报表，添加脚本数据集：



使用脚本数据集预处理数据

8.4 变量和脚本



编辑示例脚本如下：

Script dataset

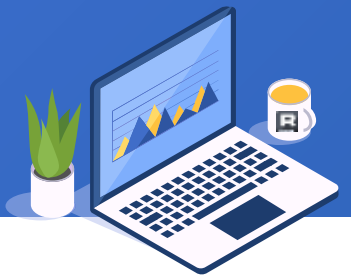
Cache file varname

	A	B	C
1	=file("data\\8\\data8.4.xlsx")		
2	=A1.xlsimport@t(;"Q1",:)		
3	=A1.xlsimport@t(;"Q2",:)		
4	=A2|A3		
5			
6			
7			
8			

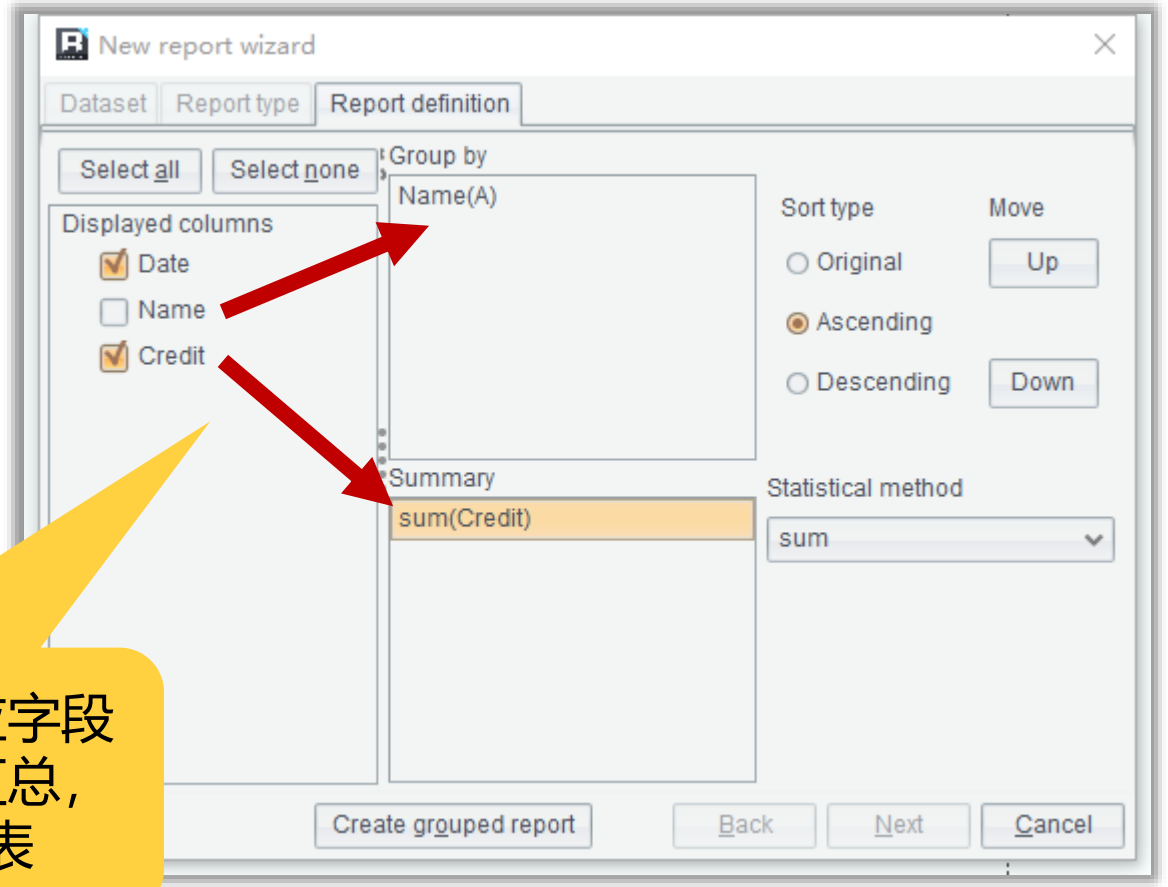
该脚本将例子数据data8.4.xlsx中的两个季度数据，合并为一个大序表

润乾报表使用SPL作为脚本，学习材料在这里：
http://www.raqsoft.com.cn/wx/SPL-programming.html

8.4 变量和脚本

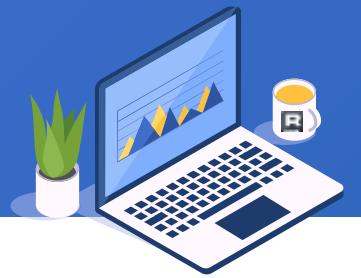


从菜单项，选择生成分组报表：



如图从显示列拖拽相应字段到分组定义以及分组汇总，再单击生成分组报表

8.4 变量和脚本



生成如下分组报表：

report_0

	A	B	C
1(TH)	Name	Date	Credit
2	=ds1.group(Nam	=ds1.select(Date	=ds1.Credit
3	sum(Name)		=ds1.sum(Credit
4	sum		=ds1.sum(Credit

保存报表后，再预览，如图为部分结果。

Preview report:D:\tutorial\data\8\8.4.2.rptx

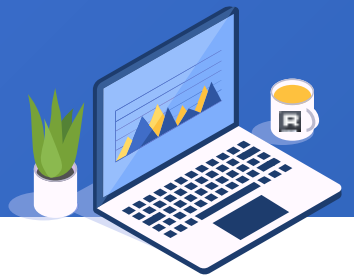
Name	Date	Credit
Bella	2019-02-02	100
	2019-02-21	200
	2019-03-21	200
	2019-05-01	350
sum(Name)		850
Ethan	2019-01-20	50
	2019-02-07	100
	2019-02-21	100
	2019-02-23	100
	2019-03-02	50
	2019-03-12	50
	2019-03-22	50
	2019-04-18	200
	2019-05-04	100
	2019-05-21	100
	2019-06-23	100
sum(Name)		1000

CONTENTS



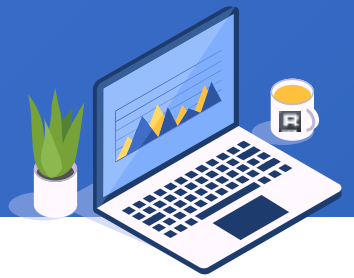
3.5 习题

8.5 习题1



请将2.4.rptx增加编号以及姓名查询的参数，方便直接查询出具体的同学卡片，（要求根据姓名查询时，可以只输入部分名字）。

8.5 习题2



请复制8.4.2.rptx中的脚本数据集代码到集算器开发环境中，运行并单步查看每一步执行完后的格子内容。并查阅相关函数文档，了解例子中用到的函数功能和作用。



可以在页面的搜索框直接输入函数的名称查找帮助页面

第9章
报表组



CONTENTS



9.1 新建一个报表组

9.2 报表组的参数

9.3 公用数据集和参数

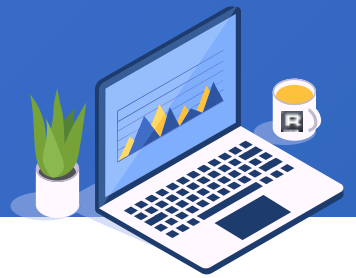
9.4 实现连续的分页码

CONTENTS



9.1 新建一个报表组

9.1 新建一个报表组

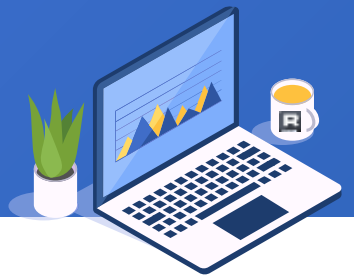


报表 2.4.rptx为卡片式，但有时列表式的浏览起来更方便，现新增列表式报表 9.1.rptx：

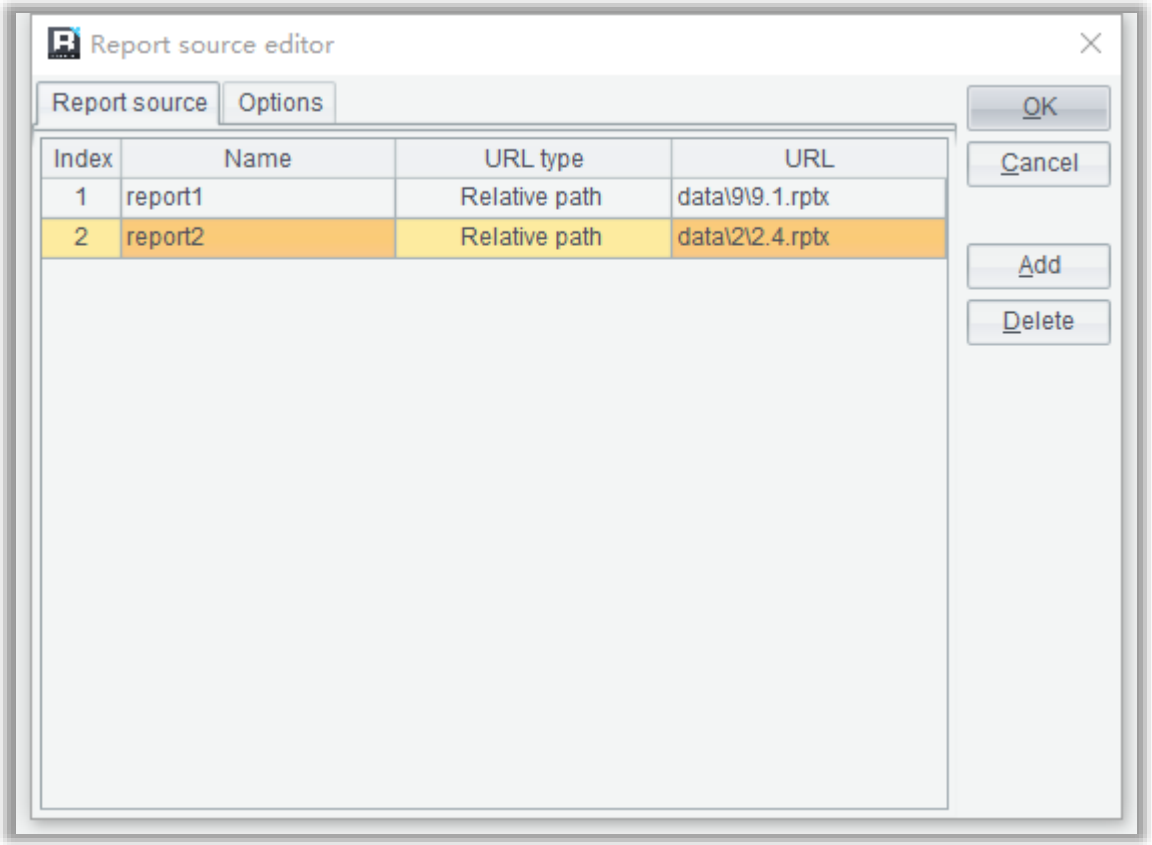
Preview report:D:/tutorial/data/9/9.1.rptx

ID	Name	Gender	Age	Address
1	Roddy	M	17	30830 Orchard Lake Road
2	Lisa	F	16	120 Wall St. 22nd Floor New York, NY
3	Shaw	M	15	4114 Sepulveda Blvd Culver City，CA
4	Dreamy	F	14	3330 S Figueroa St. Los Angeles CA
5	Alice	M	13	20 Main St.East Hampton NY

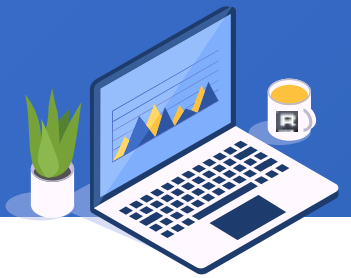
9.1 新建一个报表组



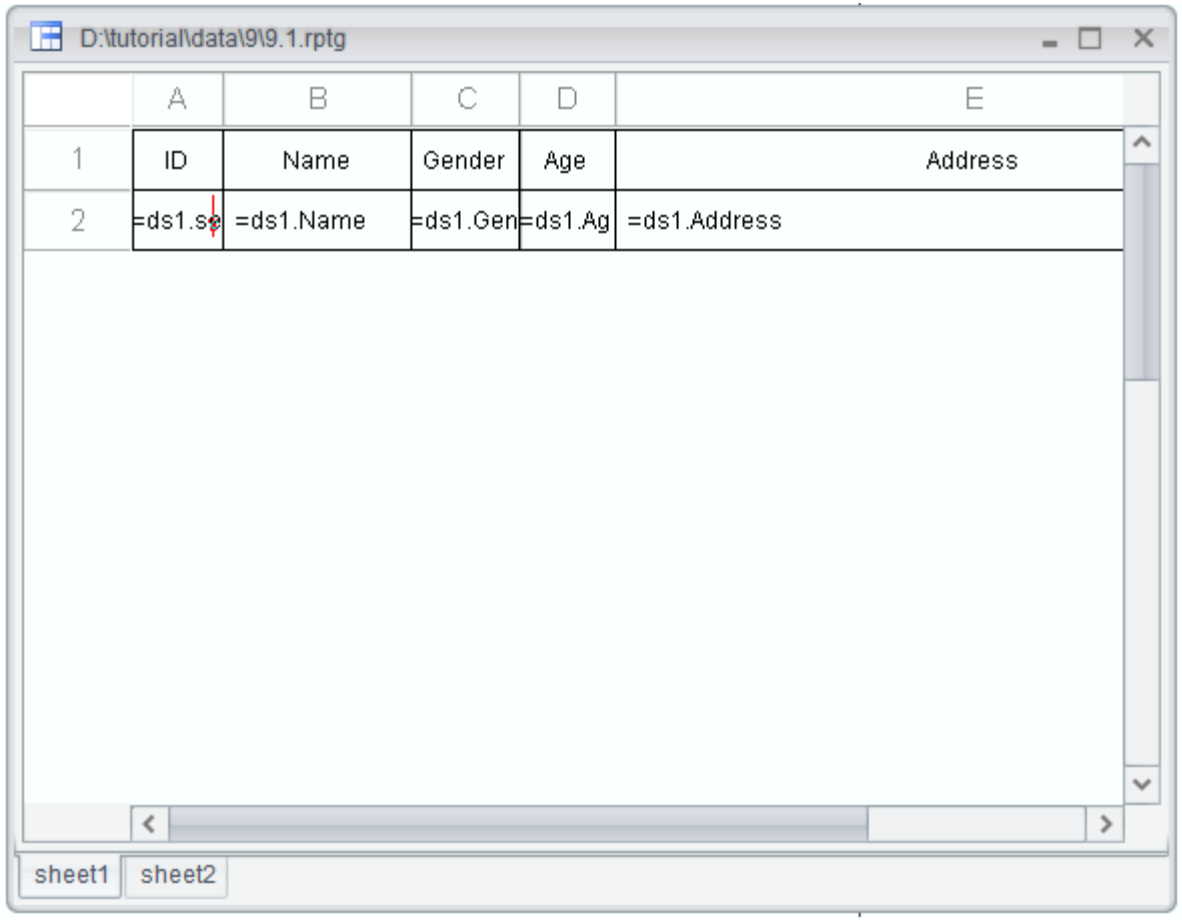
单击菜单“File”、“New Report Group”，或者直接按热键Ctrl+G，弹出Report source editor的界面，如下增加两个报表源：



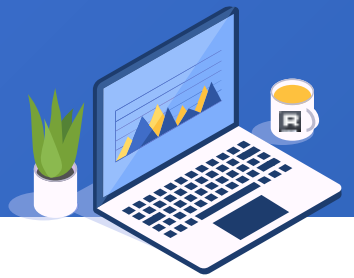
9.1 新建一个报表组



点击按钮“OK”后的界面，得到一个如下类似Excel页面的报表组：



9.1 新建一个报表组



预览该报表组，可以看到位于两个页面的报表：

Preview report group:D:\tutorial\data\9\9.1.rptg

ID	Name	Gender	Age	Address
1	Roddy	M	17	30830 Orchard Lake Road
2	Lisa	F	16	120 Wall St. 22nd Floor New York, NY
3	Shaw	M	15	4114 Sepulveda Blvd Culver City，CA
4	Dreamy	F	14	3330 S Figueroa St. Los Angeles CA
5	Alice	M	13	20 Main St.East Hampton NY

sheet1 sheet2

Preview report group:D:\tutorial\data\9\9.1.rptg

Student Information

ID	1
Name	Roddy
Gender	0
Age	17
Address	30830 Orchard Lake Road

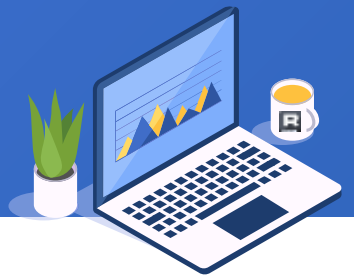


ID	2
Name	Lisa
Gender	1



sheet1 sheet2

9.1 新建一个报表组



将报表组导出为分页的PDF，可以看到是将两张报表一起输出到PDF文件：

ID	Name	Gender	Age	Address
1	Roddy	M	17	30830 Orchard Lake Road
2	Lisa	F	16	120 Wall St. 22nd Floor New York, NY
3	Shaw	M	15	4114 Sepulveda Blvd Culver City , CA
4	Dreamy	F	14	3330 S Figueroa St. Los Angeles CA
5	Alice	M	13	20 Main St.East Hampton NY

Student Information

ID	1
Name	Roddy
Gender	0
Age	17
Address	30830 Orchard Lake Road

ID	2
Name	Lisa
Gender	1
Age	16
Address	120 Wall St. 22nd Floor New York, NY

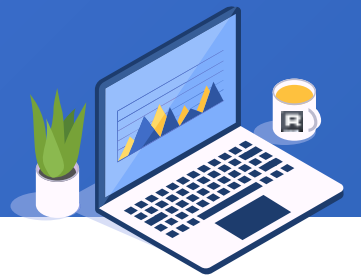
ID	3
Name	Shaw
Gender	0
Age	15
Address	4114 Sepulveda Blvd Culver City , CA

CONTENTS

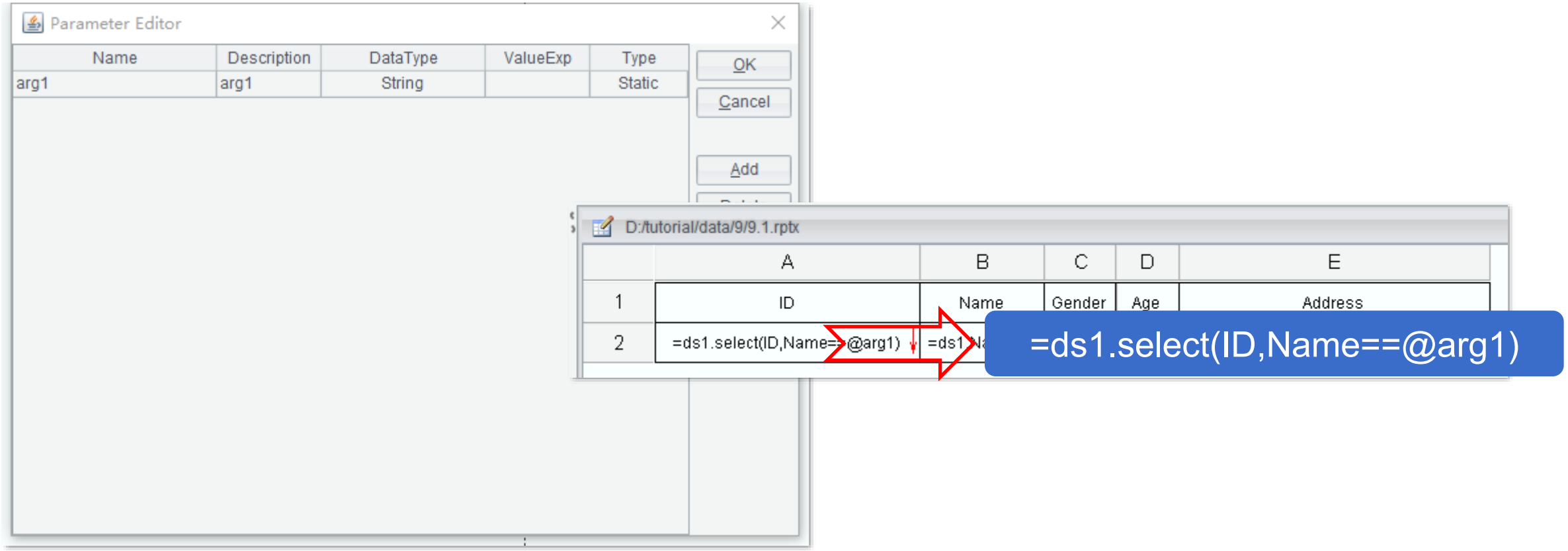


9.2 报表组的参数

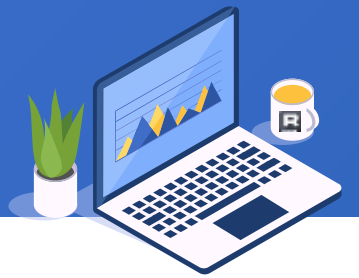
9.2 报表组的参数



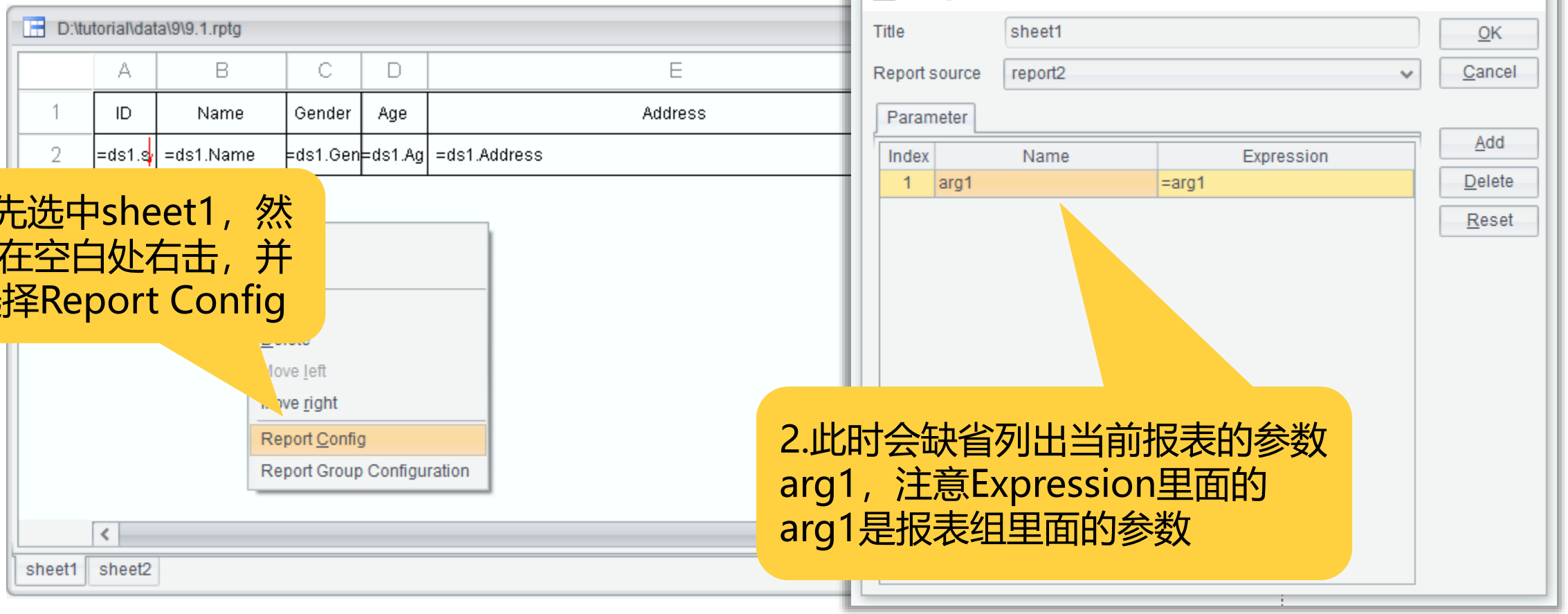
报表组也同样可以设置参数，然后应用参数同时查询多个报表项。但此时需要先将报表2.4.rptx和9.1.rptx设置好参数：



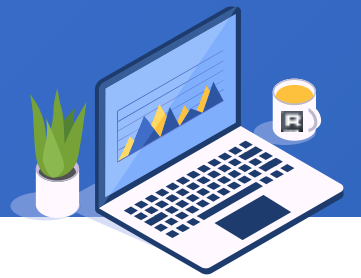
9.2 报表组的参数



报表组也同样增加一个参数arg1。注意此时每个sheet都要设置报表的参数值表达式。



9.2 报表组的参数



预览报表组，输入参数值Lisa，得到只有Lisa的报表：

Preview report group:D:\tutorial\data\9\9.1.rptg

ID	Name	Gender	Age	Address
2	Lisa	F	16	120 Wall St. 22nd Floor New York, NY

sheet1 sheet2

Preview report group:D:\tutorial\data\9\9.1.rptg

Student Information

ID	2
Name	Lisa
Gender	1
Age	16
Address	120 Wall St. 22nd Floor New York, NY



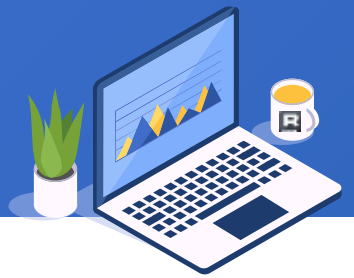
sheet1 sheet2

CONTENTS

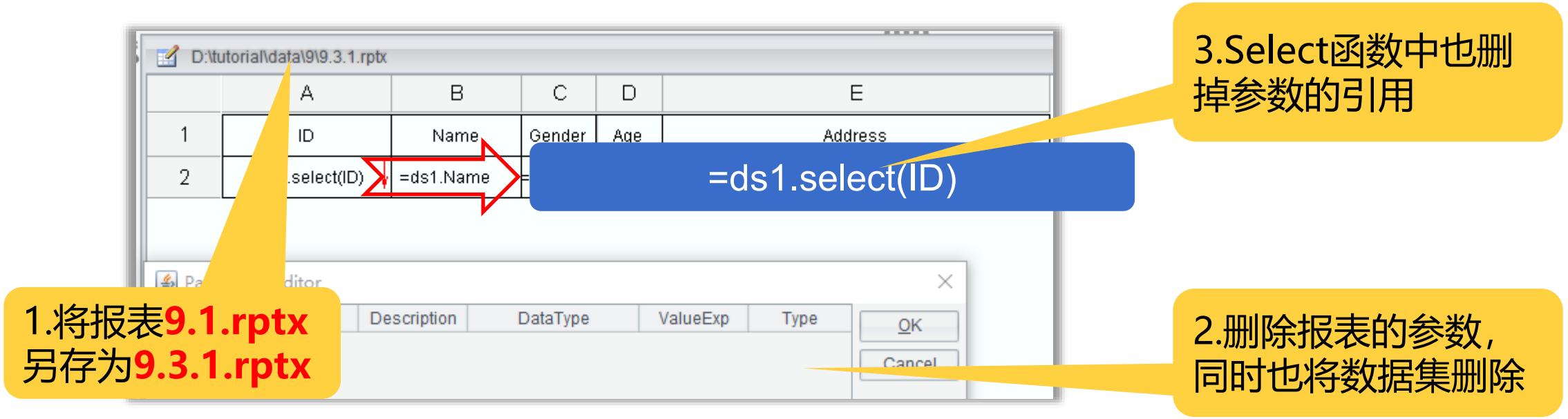


9.3 公用数据集和参数

9.3 公用数据集和参数



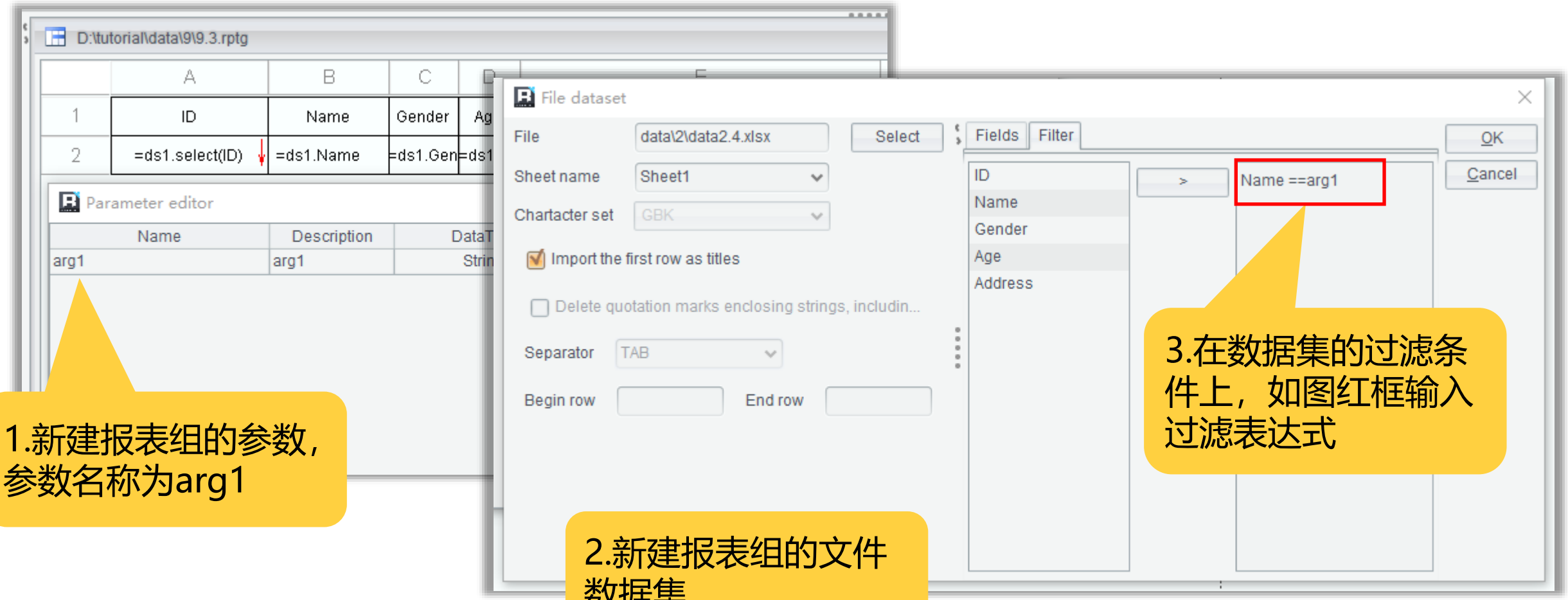
9.2节中，每个报表都定义了数据集，而这两个数据集其实是完全一样的，这会造成计算报表组时，重复计算数据集，影响性能。



另一张报表存为9.3.2.rptx，也同样修改一遍

9.3 公用数据集和参数

使用报表文件9.3.1.rptx和9.3.2.rptx，重新创建报表组，并保存为9.3.rptg

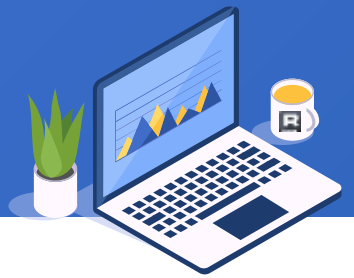


1.新建报表组的参数，参数名称为arg1

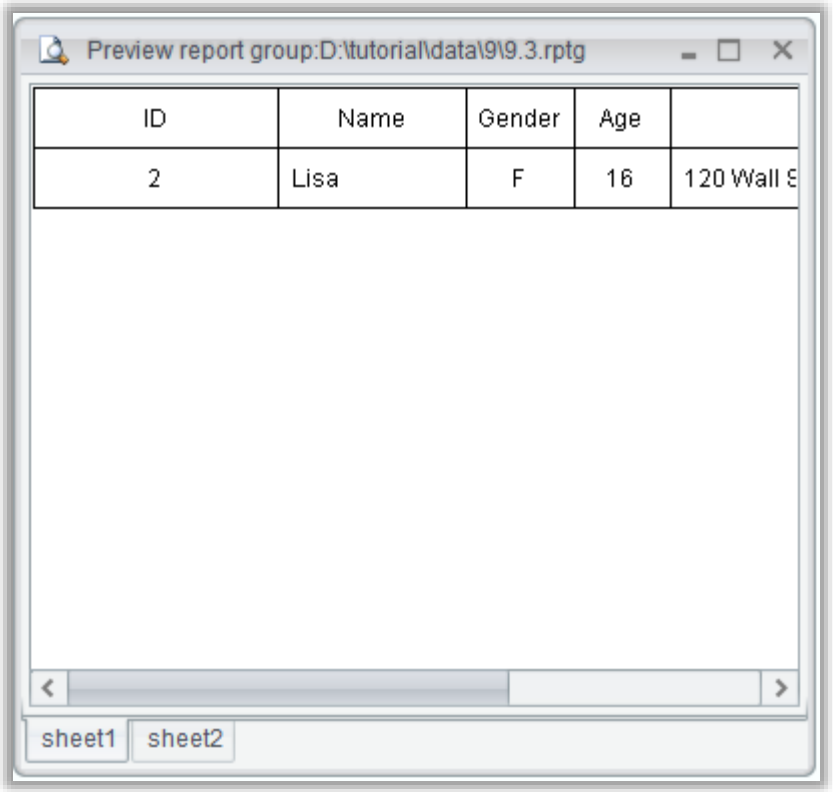
2.新建报表组的文件数据集

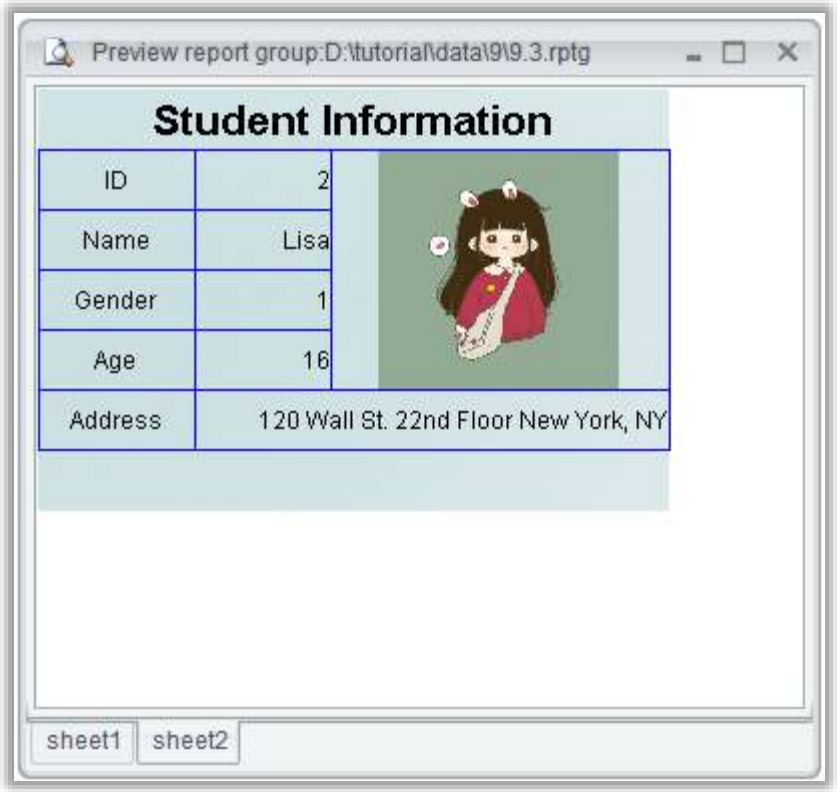
3.在数据集的过滤条件上，如图红框输入过滤表达式

9.3 公用数据集和参数



预览报表组，输入参数值Lisa，得到跟9.2节相同的报表：



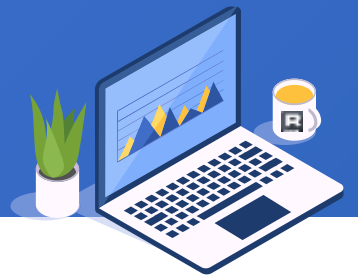


CONTENTS



9.4 实现连续的分页码

9.4 实现连续的分页码



仍然将报表文件9.3.1.rptx和9.3.2.rptx 各复制一份，另存为9.4.1.rptx和9.4.2.rptx

D:/tutorial/data/9/9.4.1.rptx

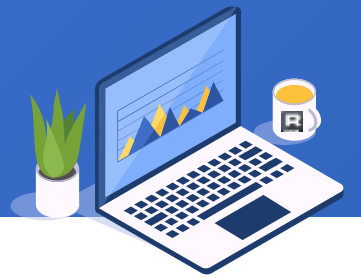
	A	B	C	D	E
1(PH)	ID	Name	Gender	Age	Address
2	=ds1.select(ID)	=ds1.Name	=ds1.Gen	=ds1.Ag	=ds1.Address
3					
4(PF			="Page "+pno()		

1.修改部分报表属性，比如页头，页脚等

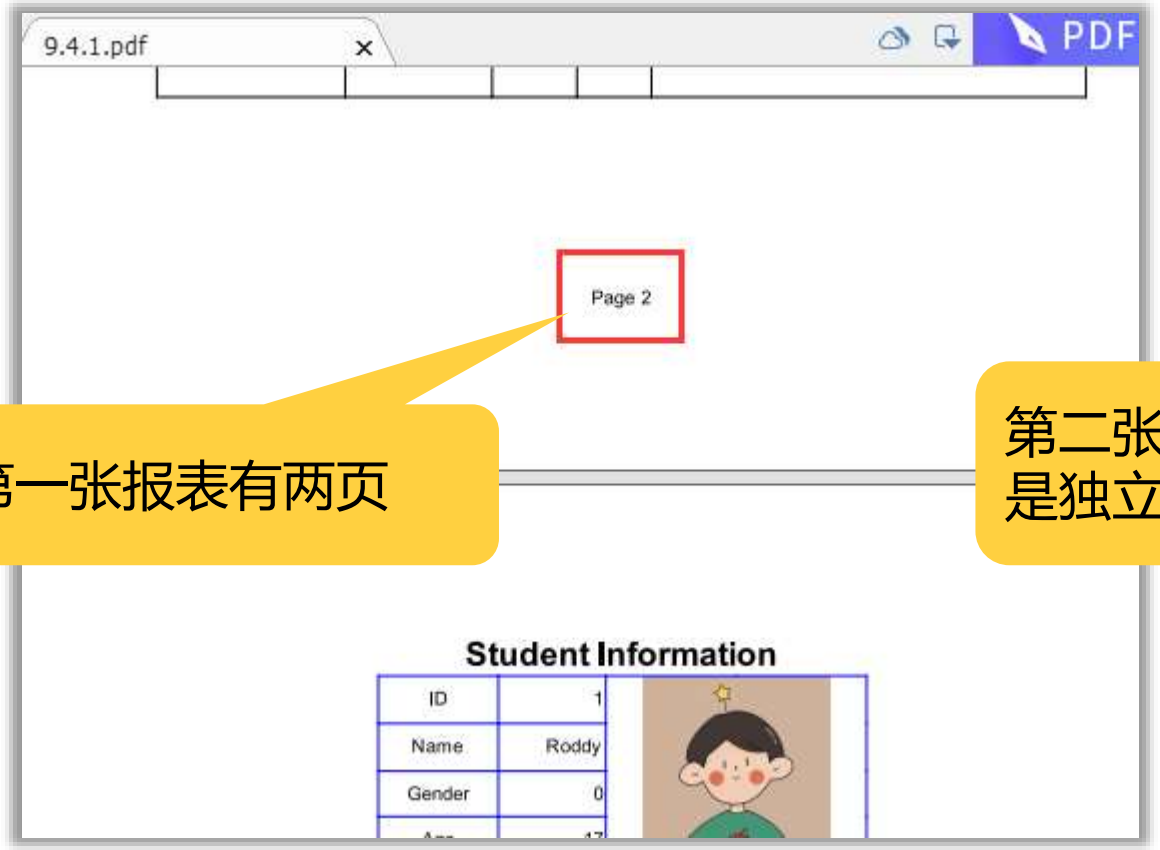
2.在页脚使用页码函数pno()

注意：
分页函数，以及页眉页脚的展现效果，只能在导出分页后的文件中生效。

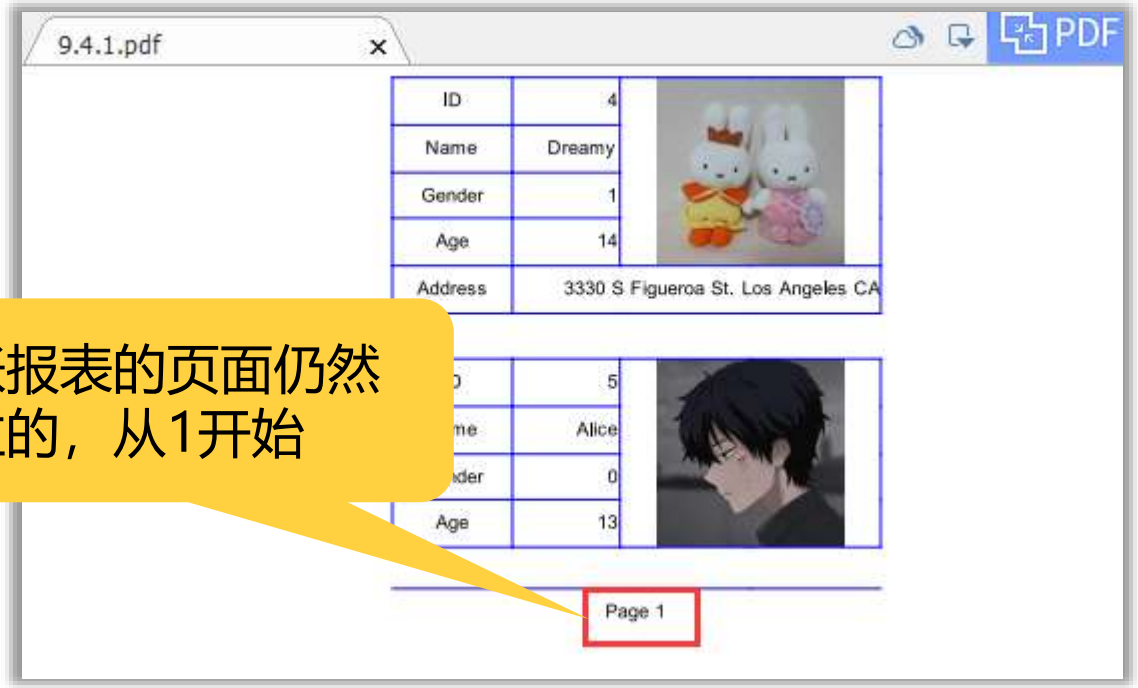
9.4 实现连续的分页码



报表文件9.4.2.rptx 同样增加页脚，使用上述报表新建报表组9.4.rptg，预览后导出分页的PDF文件到9.4.1.pdf

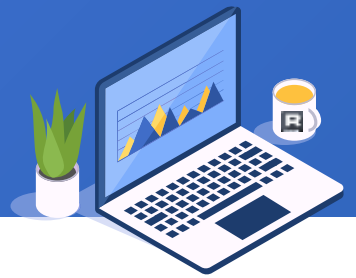


第一张报表有两页



第二张报表的页面仍然是独立的，从1开始

9.4 实现连续的分页码



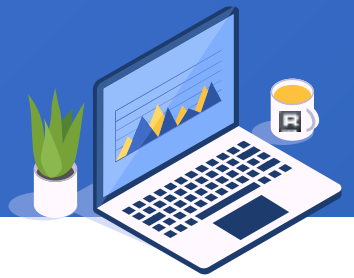
仅使用pno()函数没法得到连续的页码。需要在报表9.4.1rptx 中将该报表的总页数用pcount（）获取到，并赋值给变量arg1

D:/tutorial/data/9/9.4.1.rptx

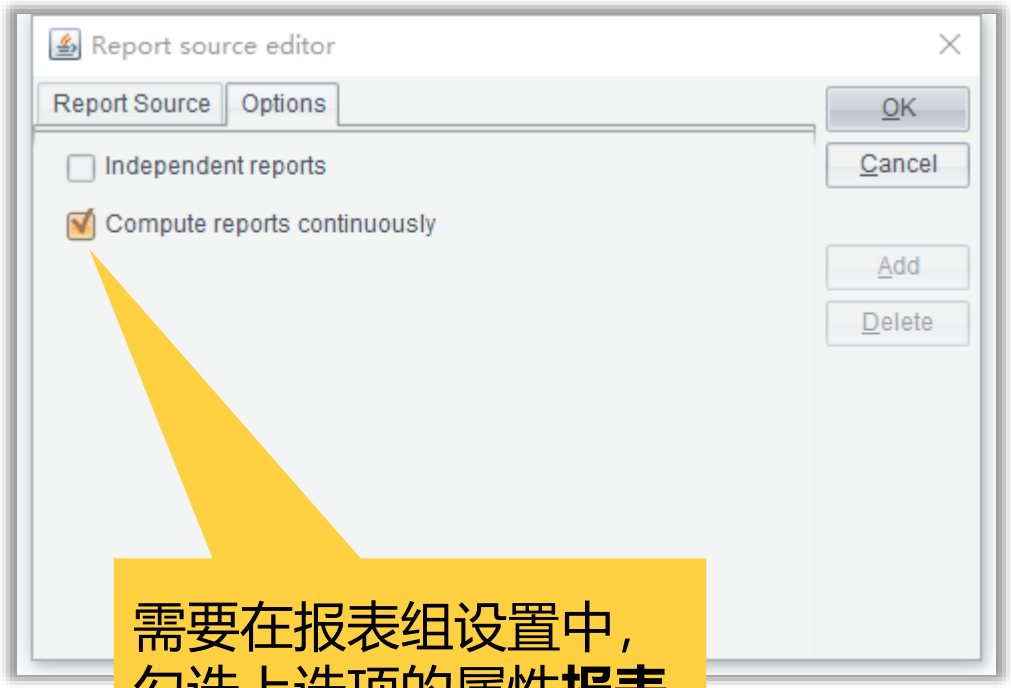
	A	B	C	D	E
1(PH)	ID	Name	Gender	Age	Address
2	=ds1.select(ID)	=ds1.Name	=ds1.Gen	=ds1.Ag	=ds1.Address
3	=arg1=pcount()				
4(PF)	="Page "+pno()				

使用表达式=arg1=pcount()将sheet1报表的总页数赋值给变量arg1

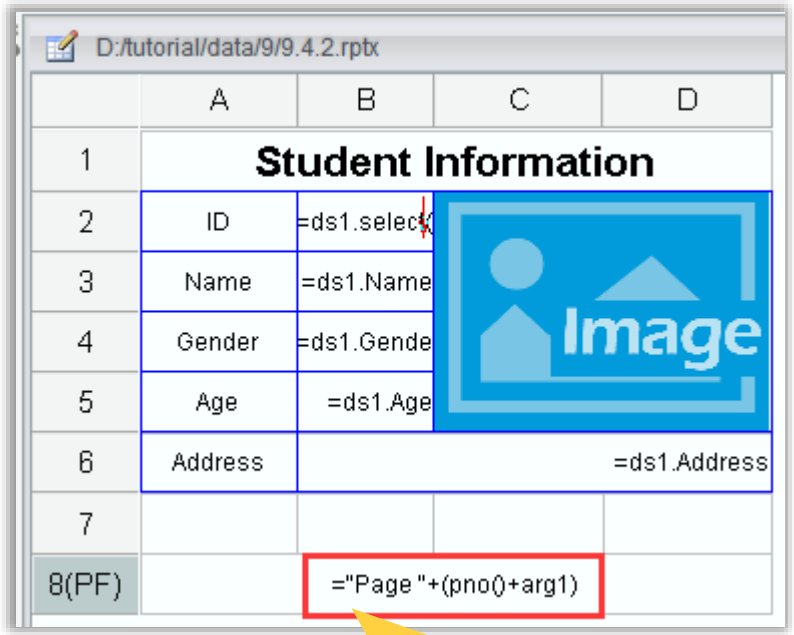
9.4 实现连续的分页码



为了能在报表9.4.2.rptx 中引用前一个报表定义的变量arg1



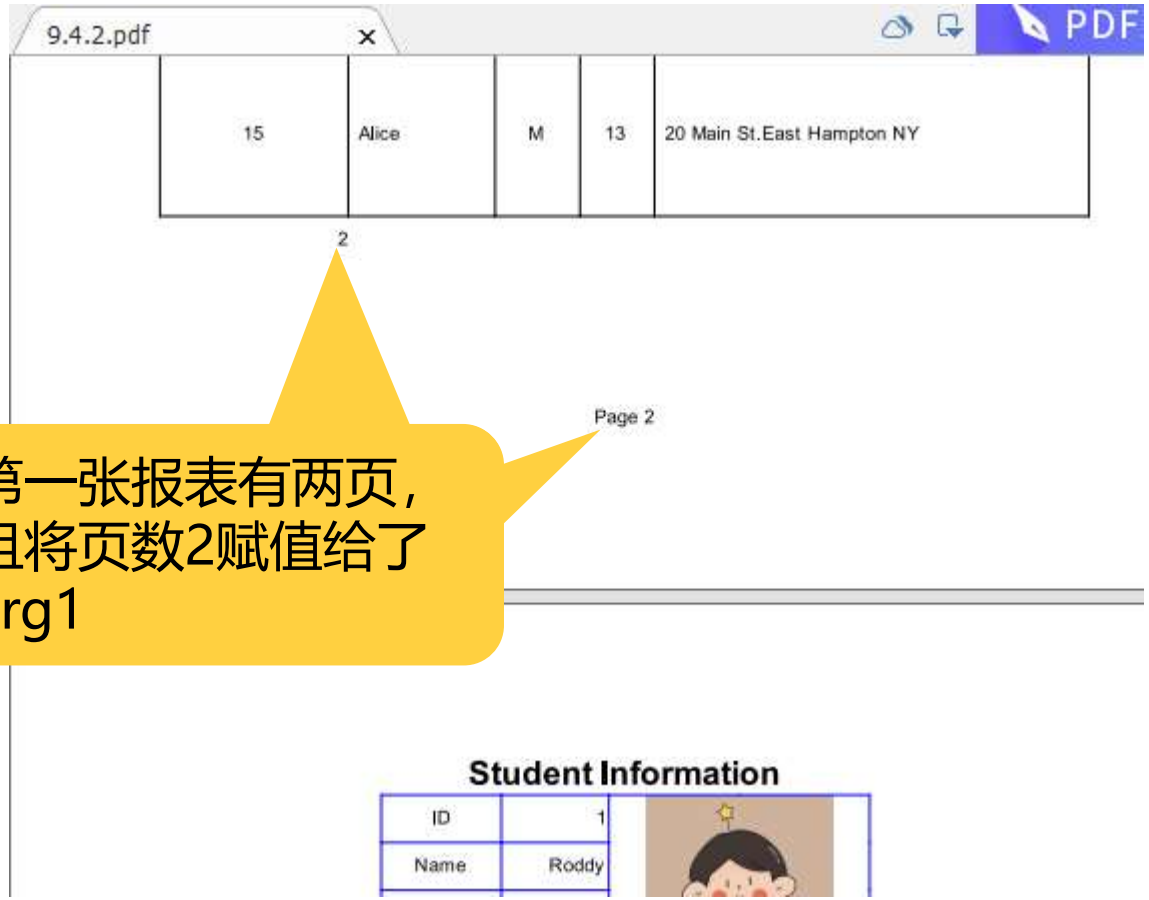
需要在报表组设置中，勾选上选项的属性**报表全部连续计算**



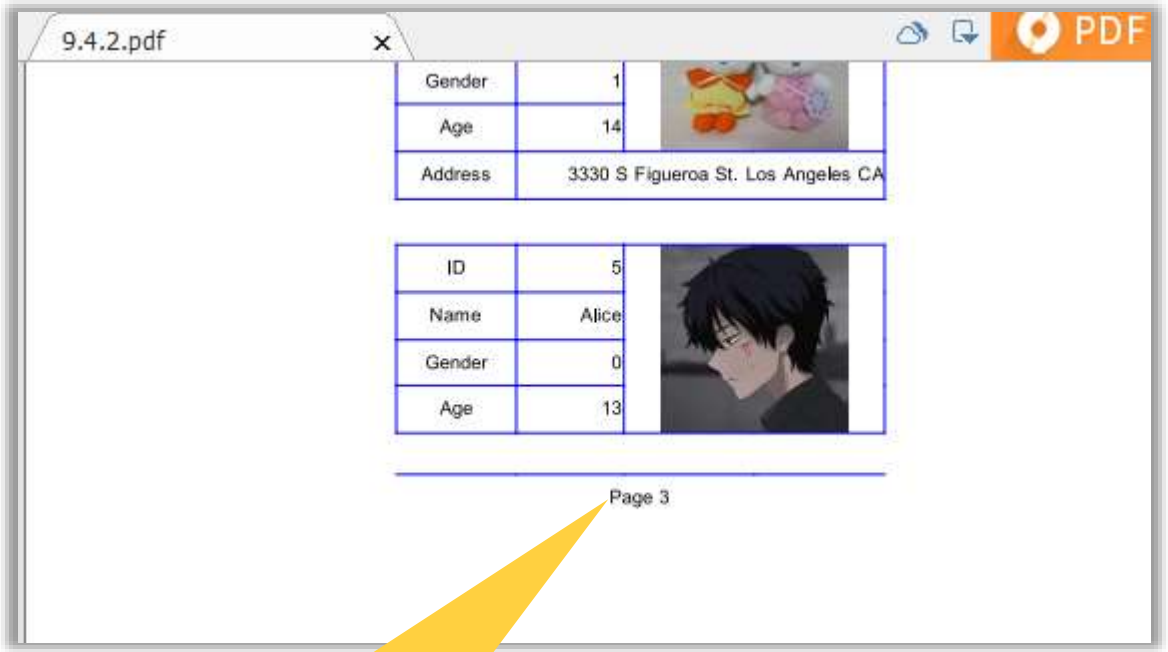
然后可以在sheet2报表的页码表达式中引用变量arg1

9.4 实现连续的分页码

预览报表组9.4.rptg， 并导出分页的PDF文件到9.4.2.pdf



第一张报表有两页，且将页数2赋值给了arg1



第二张报表的页号已经从3开始

提示
连续计算报表组时只能是后面的报表引用前面报表中产生的变量。

THANKS

感谢观看

